vSAN

Use **vSAN** as a VI administrator or storage administrator to enhance storage performance by aggregating local storage into a single shared pool across all hosts in a vSAN cluster.

In This Chapter

- Designing vSAN Network
- Planning and Configuring vSAN
- Administering VMware vSAN
- Monitoring and Troubleshooting vSAN

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Designing vSAN Network

The *Designing vSAN Network* guide describes network requirements, network design, and configuration practices for deploying a highly available and scalable VMware® vSAN™ cluster.

vSAN is a distributed storage solution. As with any distributed solution, the network is an important component of the design. For best results, you must adhere to the guidance provided in this document as improper networking hardware and designs can lead to unfavorable results.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

This guide is intended for anyone who is designing, deploying, and managing a vSAN cluster. The information in this guide is written for experienced network administrators who are familiar with network design and configuration, virtual machine management, and virtual data center operations. This guide also assumes familiarity with VMware vSphere, including VMware ESX, vCenter, and the vSphere Client.

- For more information about creating vSAN clusters, see the Planning and Configuring vSAN guide.
- For more information about vSAN features and how to configure a vSAN cluster, see the Administering VMware vSAN guide.
- For more information about monitoring a vSAN cluster and fixing problems, see the Monitoring and Troubleshooting vSAN guide.

What is vSAN Network Design

You can use vSAN to provision shared storage within vSphere. vSAN aggregates local or direct-attached storage devices of a host cluster and creates a single storage pool shared across all hosts in the vSAN cluster.

vSAN is a distributed and shared storage solution that depends on a highly available, properly configured network for vSAN storage traffic. A fast and resilient network is crucial to a successful vSAN deployment. This guide provides recommendations on how to design and configure a vSAN network.

vSAN has a distributed architecture that relies on a fast, scalable, and resilient network. All host nodes within a vSAN cluster communicate over the IP network. All hosts in the cluster must maintain IP unicast connectivity, so they can

communicate over a Layer 2 or Layer 3 network. For more information on the unicast communication, see Using Unicast in vSAN Network.

vSAN Networking Terms and Definitions

vSAN introduces specific terms and definitions that are important to understand. Before you get start designing your vSAN network, review the key vSAN networking terms and definitions.

Terms	Definitions
CMMDS	The Cluster Monitoring, Membership, and Directory Service (CMMDS) is responsible for the recovery and maintenance of a cluster of networked node members. It manages the inventory of items such as host nodes, devices, and networks. It also stores metadata information, such as policies and RAID configuration for vSAN objects.
DOM	The Distributed Object Manager (DOM) is responsible for creating the components and distributing them across the cluster. After a DOM object is created, one of the nodes (host) is nominated as the DOM owner for that object. This host handles all IOPS to that DOM object by locating the respective child components across the cluster and redirecting the I/O to respective components over the vSAN network. DOM objects include vdisk, snapshot, vmnamespace, vmswap, vmem, and so on.
NIC Teaming	Network Interface Card (NIC) teaming can be defined as two or more network adapters (NICs) that are set up as a "team" for high availability and load balancing.
NIOC	Network I/O Control (NIOC) determines the bandwidth that different network traffic types are given on a vSphere distributed switch. The bandwidth distribution is a user configurable parameter. When NIOC is enabled, distributed switch traffic is divided into predefined network resource pools: Fault Tolerance traffic, iSCSI traffic, vMotion traffic, management traffic, vSphere Replication traffic, NFS traffic, and virtual machine traffic.
Objects and Components	Each object is composed of a set of components, determined by capabilities that are in use in the VM Storage Policy. A vSAN datastore contains several object types: • **VM Home Namespace** - The VM Home Namespace is a virtual machine home directory where all virtual machine configuration files are stored. This includes files such as .vmx, log files, vmdks, and snapshot delta description files. • **VMDK** - VMDK is a virtual machine disk or .vmdk file that stores the contents of the virtual machine's hard disk drive. • **VM Swap Object** - VM Swap Objects are created when a virtual machine is powered on. • **Snapshot Delta VMDKs** - Snapshot Delta VMDKs are created when virtual machine snapshots are taken. This is applicable only for vSAN OSA cluster. • **Memory Object** - Memory Objects are created when the snapshot memory option is selected when creating or suspending a virtual machine.
RDT	The Reliable Data Transport (RDT) protocol is used for communication between hosts over the vSAN VMkernel ports. It uses TCP at the transport layer and is responsible to create and destroy TCP connections (sockets) on demand. It is optimized to send large files.

Terms	Definitions
SPBM	Storage Policy-Based Management (SPBM) provides a storage policy framework that serves as a single unified control panel across a broad range of data services and storage solutions. This framework helps you to align storage with application demands of your VMs.
VLAN	A VLAN enables a single physical LAN segment to be further segmented so that groups of ports are isolated from one another as if they were on physically different segments.
Witness Component	A witness is a component that contains only metadata and does not contain any actual application data. It serves as a tiebreaker when a decision must be made regarding the availability of the surviving datastore components, after a potential failure.

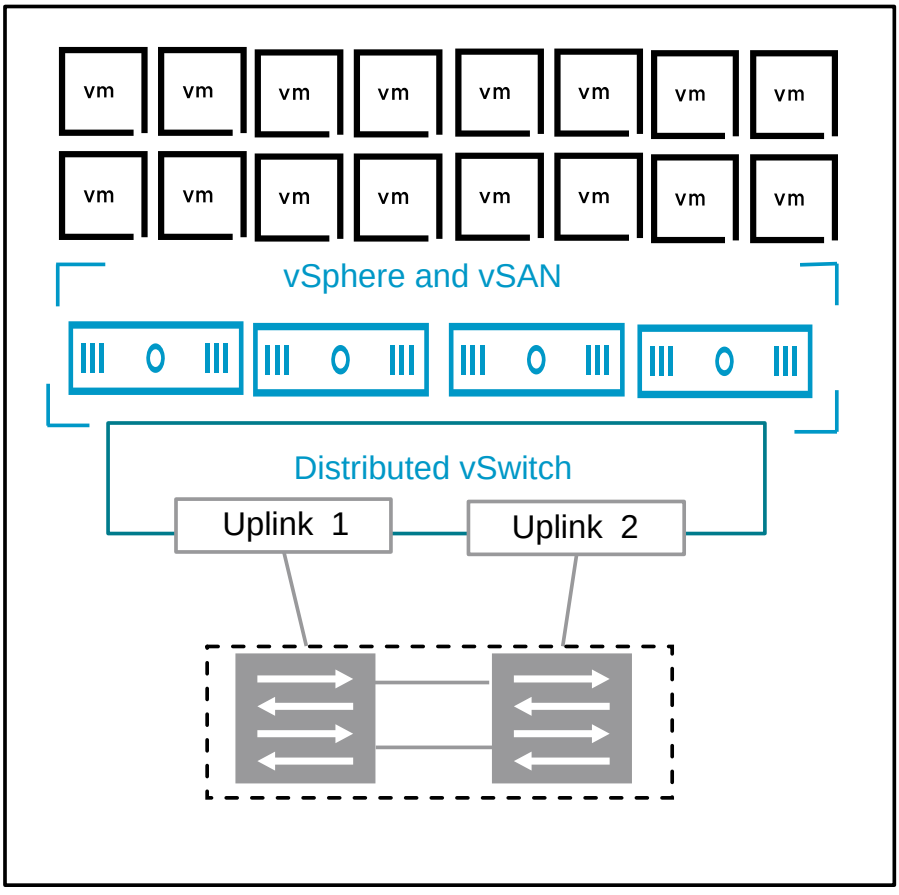
Understanding vSAN Networking

A vSAN network facilitates the communication between cluster hosts, and must be fast, resilient, and scalable.

vSAN uses the network to communicate between the ESX hosts and for virtual machine disk I/O.

Virtual machines (VMs) on vSAN datastores are made up of a set of objects, and each object can be made up of one or more components. These components are distributed across multiple hosts for resilience to drive and host failures. vSAN maintains and updates these components using the vSAN network.

The following diagram provides a high-level overview of the vSAN network:



vSAN Network Characteristics

vSAN is network-dependent. Understanding and configuring the right vSAN network settings is critical to avoiding performance and stability issues.

A reliable and robust vSAN network has the following characteristics:

vSAN Network Traffic

vSAN network traffic is primarily made up of cluster, metadata, and storage replication traffic. A VM can access its storage resources from any host within a vSAN cluster. It is not necessary for that data to be locally available, thus allowing a VM to access its data on another node.

Unicast

vSAN uses unicast for communication. Unicast traffic is a one-to-one transmission of IP packets from one point in the network to another point. Unicast transmits the heartbeat sent from the primary host to all other hosts each second. This ensures that the hosts are active and indicates the participation of hosts in the vSAN cluster. You can design a simple unicast network for vSAN. For more information on the unicast communication, see Using Unicast in vSAN Network.

> **Note:**
>
> If possible, always use the latest version of vSAN.

Layer 2 and Layer 3 Network

All hosts in the vSAN cluster must be connected through a Layer 2 (L2) or Layer 3 (L3) network.

VMkernel Network

Each ESX host in a vSAN cluster must have a network adapter for vSAN communication. All the intra cluster communication happens through the vSAN VMkernel port.

VMkernel Interface

Each ESX host in a vSAN cluster must have a VMkernel interface used for vSAN traffic to handle data replication, I/O traffic, and cluster health monitoring.

Virtual Switch

vSAN supports the following types of virtual switches:

- The Standard Virtual Switch provides connectivity from VMs and VMkernel ports to external networks. This switch is local to each ESX host.
- A vSphere Distributed Switch provides central control of the virtual switch administration across multiple ESX hosts. A distributed switch also provides networking features such as Network I/O Control (NIOC) that can help you set Quality of Service (QoS) levels on vSphere or virtual network. vSAN includes vSphere Distributed Switch irrespective of the vCenter version.

Network Interfaces

vSAN can have dedicated physical network interfaces to secure vSAN storage traffic and to provide bandwidth and resources. vSAN traffic can share physical network adapters with other system traffic types, such as vSphere vMotion traffic, vSphere HA traffic, and virtual machine traffic. To guarantee the amount of bandwidth required for vSAN, configure Network I/O control on the distributed switch.

Bandwidth

In vSAN environments that share physical resources with other traffic, use Network I/O Control to allocate bandwidth among traffic types. Assigning shares to vSAN traffic ensures that it receives sufficient bandwidth during network contention. This prioritizes vSAN traffic and prevents other traffic types from affecting its performance when the physical NIC is saturated.

Avoid using reservations or limits in vSAN environments. Reservations can leave unused bandwidth inaccessible to VMs, while limits restrict traffic even when additional bandwidth is available.

vSAN recommends the usage of shares to allocate bandwidth for vSAN traffic. This offers the most balanced and efficient use of shared physical NICs.

For information about using Network I/O Control to configure bandwidth allocation for vSAN traffic, see the vSphere Networking guide.

Using Unicast in vSAN Network

Unicast traffic refers to a one-to-one transmission from one point in the network to another. vSAN uses unicast to simplify network design and deployment.

All ESX hosts use the unicast traffic, and the vCenter becomes the source for the cluster membership. The vSAN nodes are automatically updated with the latest host membership list that vCenter provides. vSAN communicates using unicast for CMMDS updates. For more information on CMMDS updates, see Broadcom knowledge base article 385769.

DHCP Support on Unicast Network

vCenter deployed on a vSAN cluster can use IP addresses from Dynamic Host Configuration Protocol (DHCP) without reservations.

You can use DHCP without reservations as the assigned IP addresses are tied to the MAC addresses of VMkernel ports.

IPv6 Support on Unicast Network

vSAN supports IPv6 with unicast communications.

With IPv6, the link-local address is automatically configured on any interface using the link-local prefix. By default, vSAN does not add the link local address of a node to other neighboring cluster nodes. As a result, vSAN does not support IPv6 link local addresses for unicast communications.

Query Unicast with ESXCLI

You can run ESXCLI commands to determine the unicast configuration.

View the Communication Modes

Using `esxcli vsan cluster get` command, you can view the CMMDS mode of the vSAN cluster node.

Run the `esxcli vsan cluster get` command.

```
Cluster Information
   Enabled: true
   Current Local Time: 2025-05-08T09:08:05Z
   Local Node UUID: 6813878b-163d-3a33-cc63-0050569c4bac
   Local Node Type: NORMAL
   Local Node State: MASTER
   Local Node Health State: HEALTHY
   Sub-Cluster Master UUID: 6813878b-163d-3a33-cc63-0050569c4bac
   Sub-Cluster Backup UUID: 6813878b-f828-8805-6e39-0050569c8100
   Sub-Cluster UUID: 52fafd2f-885a-912b-123d-29320735129c
   Sub-Cluster Membership Entry Revision: 17
   Sub-Cluster Member Count: 16
   Sub-Cluster Member UUIDs: 6813878b-163d-3a33-cc63-0050569c4bac, 68138792-d6cb-49f1-24d4-0050569c8d7d,
 6813878c-1ae0-1798-21ba-0050569c8c1a, 68138791-7d9b-c043-b0e7-0050569c74f8, 6813878b-5ec9-
b6d7-70ce-0050569c37d1, 6813878b-d309-023b-2c66-0050569ccf58, 6813878c-9879-9875-2cf0-0050569c5f2f,
 6813878b-d7b2-c9f1-ce14-0050569ce307, 6813878b-f828-8805-6e39-0050569c8100, 6813878c-
aab3-3810-65dc-0050569c55f1, 68138797-2924-1613-be3b-0050569cfe64, 68138793-f8bc-f72f-1e0d-0050569ce100,
 68138791-58f8-8831-8403-0050569c8cfb, 6813878b-ae9c-6346-d735-0050569c73af, 68138782-
b02a-785a-4b5e-0050569ce4b4, 681512d1-f5d1-c10e-8b0c-0050569cf5e5
   Sub-Cluster Member HostNames: sfo01-m01-r01-esx01.sfo.rainpole.io, sfo01-m01-r01-esx02.sfo.rainpole.io,
 sfo01-m01-r01-esx03.sfo.rainpole.io, sfo01-m01-r01-esx04.sfo.rainpole.io, sfo01-m01-r01-
esx05.sfo.rainpole.io, sfo01-m01-r01-esx06.sfo.rainpole.io, sfo01-m01-r01-esx07.sfo.rainpole.io, sfo01-m01-
```

```
r01-esx08.sfo.rainpole.io, sfo02-m01-r01-esx08.sfo.rainpole.io, sfo02-m01-r01-esx07.sfo.rainpole.io, sfo02-
m01-r01-esx04.sfo.rainpole.io, sfo02-m01-r01-esx06.sfo.rainpole.io, sfo02-m01-r01-esx03.sfo.rainpole.io,
 sfo02-m01-r01-esx02.sfo.rainpole.io, sfo02-m01-r01-esx05.sfo.rainpole.io, sfo-m01-cl01-vsw01.sfo.rainpole.io
   Sub-Cluster Membership UUID: f89b1468-4c6d-7ebf-2a19-0050569c929a
   Unicast Mode Enabled: true
   Maintenance Mode State: OFF
   Config Generation: a90bad14-6d09-4a42-b001-5137386ff625 9 2025-05-08T03:05:15.343
   Mode: REGULAR
   vSAN ESA Enabled: true
   vSAN Max Client Network Enabled: false
```

Verify the vSAN Cluster Hosts

Use the esxcli vSAN cluster unicastagent list command to list the unicast communications details

Run the `esxcli vsan cluster unicastagent list` command.

```
NodeUuid                              IsWitness  Supports Unicast  IP Address     Port  Iface Name  Cert
 Thumbprint                                                                                    SubClusterUuid
                     Traffic Type
------------------------------------  ---------  ----------------  -------------  -----  ----------
  ------------------------------------------------------------------------------------------------------
 ------------------------------------  ------------
6813878b-f828-8805-6e39-0050569c8100          0              true  10.12.13.108  12321
  D8:14:DA:80:2F:0C:0E:0C:BD:40:89:24:70:3D:33:EE:62:0B:6E:7C:4F:8D:D3:06:88:46:39:DD:E7:15:9D:0F
 52fafd2f-885a-912b-123d-29320735129c  vsan
681512d1-f5d1-c10e-8b0c-0050569cf5e5          1              true  10.21.10.218  12321
  A7:02:80:3B:64:8F:9A:AB:29:6B:54:3E:C1:76:01:3E:00:A2:DE:80:F4:78:7B:6C:E8:B9:3A:4A:77:4B:0F:38
 52fafd2f-885a-912b-123d-29320735129c
6813878c-aab3-3810-65dc-0050569c55f1          0              true  10.12.13.107  12321
  45:2E:E7:40:9E:0B:5D:87:18:2E:8E:C6:86:86:97:B1:B5:6E:79:57:C2:9E:21:3C:5F:4B:76:D3:3B:72:64:1B
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138797-2924-1613-be3b-0050569cfe64          0              true  10.12.13.104  12321
  ED:97:72:FA:A0:52:6E:17:61:97:8C:3D:6F:CC:CB:49:98:56:97:AA:8B:9B:BE:47:D0:4A:35:A2:0E:95:FA:6C
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138793-f8bc-f72f-1e0d-0050569ce100          0              true  10.12.13.106  12321
  86:99:32:7A:20:4A:D2:B3:B3:A9:6D:BE:60:A3:26:5B:83:EB:49:34:76:FB:7A:5B:EE:0E:54:FB:DC:AC:BE:74
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138791-58f8-8831-8403-0050569c8cfb          0              true  10.12.13.103  12321
  B2:4F:11:0E:98:07:80:45:F2:F2:06:C9:D9:89:57:FA:8D:3C:2B:E3:A5:25:5C:2B:A5:D8:5A:17:6F:87:0D:7C
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878b-d309-023b-2c66-0050569ccf58          0              true  10.11.13.106  12321
  E2:D6:36:BE:C8:D5:A2:F0:42:41:EF:9F:4C:D2:AD:67:6C:54:4E:3A:50:B3:B3:7C:35:54:A2:B7:3F:C9:5B:11
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138791-7d9b-c043-b0e7-0050569c74f8          0              true  10.11.13.104  12321
  21:FC:6F:56:C9:26:33:6E:8E:1E:58:7D:A2:3F:3E:38:09:96:99:61:F2:7F:AB:F1:25:C9:6C:CC:0D:6D:BB:06
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878b-ae9c-6346-d735-0050569c73af          0              true  10.12.13.102  12321
  A5:47:6C:FE:2A:9A:A9:8C:08:03:34:AD:D1:E0:A9:FA:9E:17:4F:62:1F:15:D1:ED:8B:01:D8:CE:D4:D7:01:D8
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138782-b02a-785a-4b5e-0050569ce4b4          0              true  10.12.13.105  12321
  3C:92:51:62:F5:59:A0:73:39:E5:DE:B1:A9:1F:72:1E:F7:5D:4C:48:12:76:A5:BE:9B:16:FE:05:94:2A:EA:37
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878c-9879-9875-2cf0-0050569c5f2f          0              true  10.11.13.107  12321
  77:8E:F9:D1:21:6B:CD:3C:66:A2:4A:55:BE:78:30:00:E9:BD:96:6B:F8:6F:FB:1C:1F:5E:33:6C:DE:D3:4D:7E
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878c-1ae0-1798-21ba-0050569c8c1a          0              true  10.11.13.103  12321
  77:24:E0:DF:64:FD:D2:B1:FD:FA:D7:5B:1B:2C:FF:12:15:99:11:CE:FB:41:84:8D:3F:57:C0:67:A1:6F:12:B3
 52fafd2f-885a-912b-123d-29320735129c  vsan
```

```
6813878c-8bcb-20c9-0429-0050569c2f06          0              true  10.12.13.101  12321
  43:E4:88:EB:55:04:FF:BA:8F:BE:12:C5:52:AB:21:05:9B:2C:04:11:68:34:4B:E7:AC:BE:F7:07:A9:B3:B1:C9
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878b-5ec9-b6d7-70ce-0050569c37d1          0              true  10.11.13.105  12321
  A3:FC:61:0B:C3:37:43:47:54:EA:B2:A1:9F:B4:B2:D6:A0:C0:92:F9:79:6F:FF:B8:CD:3F:48:B0:28:FC:CA:AD
 52fafd2f-885a-912b-123d-29320735129c  vsan
68138792-d6cb-49f1-24d4-0050569c8d7d          0              true  10.11.13.102  12321
  C1:3C:C4:B1:C1:94:A6:5B:C1:17:F5:15:39:BB:C2:2D:41:2A:1E:71:B8:0F:8D:FB:DD:96:12:70:F6:82:D4:DA
 52fafd2f-885a-912b-123d-29320735129c  vsan
6813878b-d7b2-c9f1-ce14-0050569ce307          0              true  10.11.13.108  12321
  20:92:8B:1B:CE:D6:7B:5D:98:89:67:CD:D8:4B:E5:43:A5:56:F3:1A:D1:51:16:02:90:E4:50:76:02:A5:29:93
 52fafd2f-885a-912b-123d-29320735129c  vsan
```

The output includes the vSAN node UUID, whether the node is a data host (0) or a witness host (1), IPv4 address/IPv6 address, UDP port, certificate thumbprint, and vSAN subcluster ID and traffic type. If troubleshooting, you can use this output to identify the vSAN cluster nodes match to ensure what vCenter maintains.

View the vSAN Network Information

Use the `esxcli vsan network list` command to view the vSAN network information such as the VMkernel interface that vSAN uses for communication, the unicast port (12321), and the traffic type (vSAN or witness) associated with the vSAN interface.

Run the `esxcli vsan network list` command.

```
Interface
   VmkNic Name: vmk1
   IP Protocol: IP
   Interface UUID: e290be58-15fe-61e5-1043-246e962c24d0
   Agent Group Multicast Address: 224.2.3.4
   Agent Group IPv6 Multicast Address: ff19::2:3:4
   Agent Group Multicast Port: 23451
   Master Group Multicast Address: 224.1.2.3
   Master Group IPv6 Multicast Address: ff19::1:2:3
   Master Group Multicast Port: 12345
   Host Unicast Channel Bound Port: 12321
   Multicast TTL: 5
   Traffic Type: vsan
```

This output also displays the vmkernal interface used for vSAN, and vSAN traffic type. While the output displays muticast, it is no longer used for vSAN communication and can be ignored.

Intra-Cluster Traffic

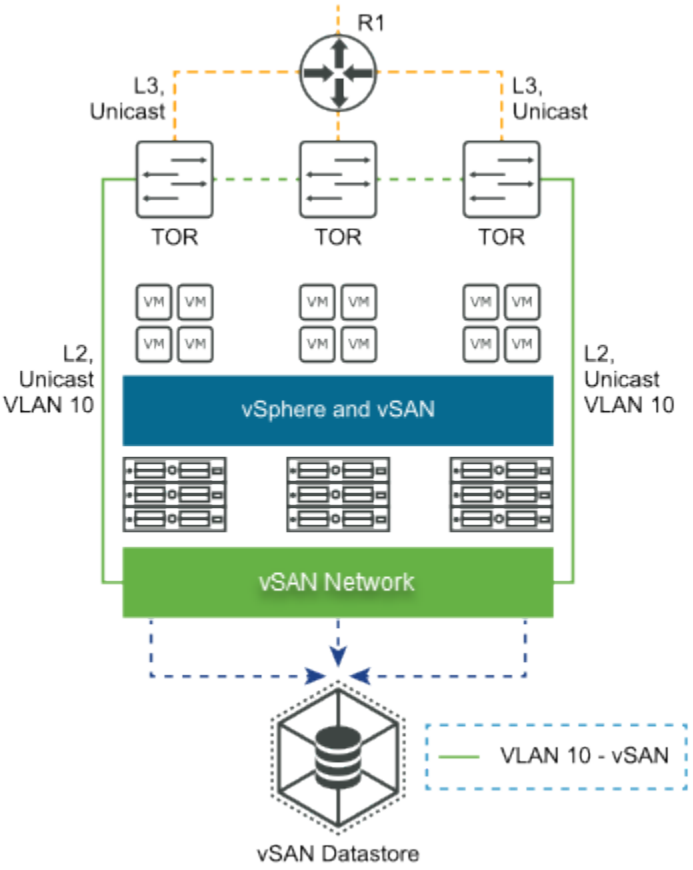
In unicast mode, the primary node addresses all the cluster nodes as it sends the same message to all the vSAN nodes in a cluster.

For example, if N is the number of vSAN nodes, then the primary node sends the messages N number of times. This results in a slight increase of vSAN CMMDS traffic. You might not notice this slight increase of traffic during normal, steady-state operations.

Intra-Cluster Traffic in a Single Rack

If all the nodes in a vSAN cluster are connected to the same top of the rack (TOR) switch, then the total increase in traffic is only between the primary node and the switch.

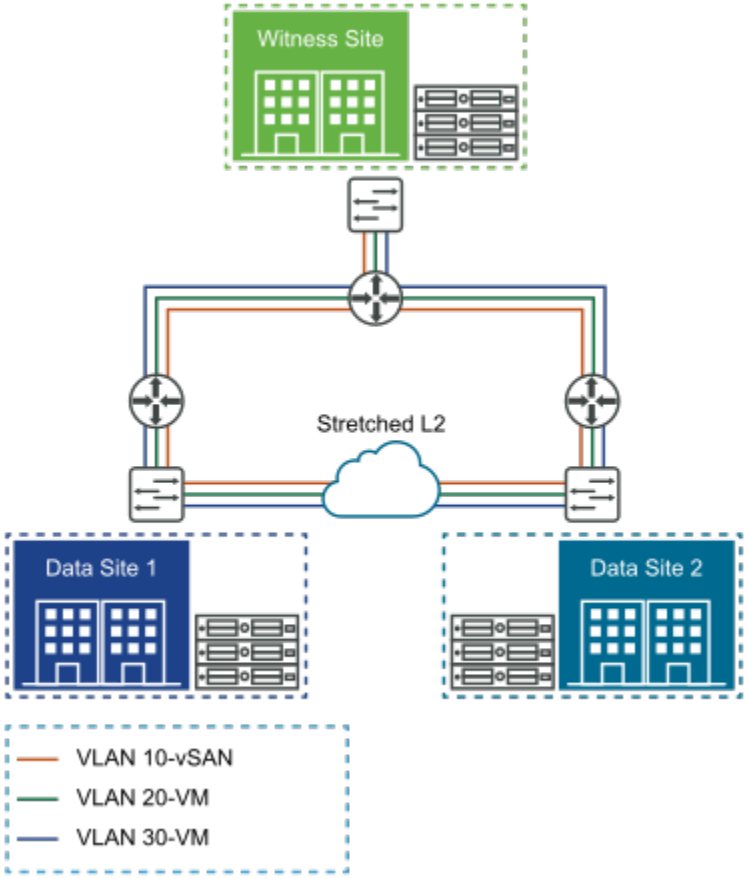
If a vSAN cluster spans more than one TOR switch, you need to monitor the network traffic bandwidth and network latency to ensure adequate resources are available to satisfy vSAN requirements. If a cluster spans many racks, multiple TORs may be involved. A single L2 can span multiple racks or L3 can be used across multiple TORs. vSAN Fault Domains could be considered with L3 to have a logical and physical fault domain at a rack boundary.



Intra-Cluster Traffic in a vSAN Stretched Cluster

In a vSAN stretched cluster, the primary node is located at the preferred site.

In a vSAN stretched cluster configuration, vSAN data is synchronously replicated bidirectionally between the preferred site and the secondary site. vSAN read I/O is local to each site. As a result write I/O bandwidth between the sites depend on many factors such as number of vSAN nodes on each site, the aggregate physical NIC bandwidth of the combined hosts between each site, and the number of vSAN objects that are using site to site replication. To calculate bandwidth requirements for vSAN OSA and vSAN ESA, see VMware vSAN Design and Sizing Guide.



With the unicast traffic, there is no change in the witness site traffic requirements.

ESX Traffic Types

ESX hosts use different network traffic types to support vSAN.

Following are the different traffic types that you need to set up for vSAN.

> **Note:**
>
> These network traffic types are not applicable to vSAN storage clusters if the VMs are not deployed on the cluster.

Table 991: Network Traffic Types

Traffic Types	Description
Management network	The management network is the primary network interface that uses a VMkernel TCP/IP stack to facilitate the host connectivity and management. It can also handle the system traffic such as vMotion, iSCSI, Network File System (NFS), Fiber Channel over Ethernet (FCoE), and fault tolerance.
Virtual Machine network	With virtual networking, you can network VMs and build complex networks within a single ESX host or across multiple ESX hosts.
vMotion network	Traffic type that facilitates migration of VM from one host to another. Migration with vMotion requires correctly configured network interfaces on source and target hosts. Ensure that the vMotion network is distinct from the vSAN network.
vSAN network	A vSAN cluster requires the VMkernel network for the exchange of data. Each ESX host in the vSAN cluster must have a VMkernel network adapter for the vSAN traffic. For more information, see Manually Enabling vSAN.

Network Requirements for vSAN

vSAN is a distributed storage solution that depends on the network for communication between hosts. Before deployment, ensure that your vSAN environment has all the networking requirements.

Physical NIC Requirements

Network Interface Cards (NICs) used in vSAN hosts must meet certain requirements. vSAN works on 10 GbE, 25 GbE, 40 GbE, 50 GbE, and 100 GbE networks.

Ensure your hosts meet the minimum NIC requirements:

Table 992: Minimum NIC Requirements and Recommendations

Deployment Type	Support for 1 GbE NIC	Support for NICs Greater than 10 GbE	Inter-Node Latency	Inter-Site Link Bandwidth or Latency	Latency Between Nodes and vSAN Witness Hosts	Bandwidth Between Nodes and vSAN Witness Hosts
vSAN HCI Cluster (Hybrid)	Yes	Yes	Less than 1 ms RTT.	NA	NA	NA
vSAN HCI Cluster (All Flash)	No	Yes	Less than 1 ms RTT.	NA	NA	NA
vSAN Stretched Cluster	No	Yes	Less than 1 ms RTT within each site.	Recommended is 25 GbE or higher (workload dependent) and 5 ms RTT.	Less than 200 ms RTT. Up to 10 hosts per site. Less than 100 ms RTT. 11–15 hosts per site.	2 Mbps per 1000 components (Maximum of 100 Mbps with 45 k components).
Two-Node vSAN Cluster	No	Yes	Less than 1 ms RTT within the same site.	Recommended is 25 GbE or higher and 5 ms RTT or less.	Less than 500 ms RTT.	2 Mbps per 1000 components (Maximum of 1.5 Mbps).
vSAN Stretched Compute Cluster	No	Yes	Less than 1 ms RTT within each site.	Minimum 10 GbE and 5 ms RTT.	NA	NA
vSAN storage cluster	No	Yes	Less than 1 ms RTT	Minimum 10 GbE and 1 ms RTT	Less than 1 ms RTT	2 Mbps per 1000 components
vSAN Stretched storage cluster	No	Yes	Less than 1 ms RTT within each site.	Recommended is 25 GbE or higher (workload dependent) and 5 ms RTT.	Less than 200 ms RTT. Up to 10 hosts per site. Less than 100 ms RTT. 11–15 hosts per site.	2 Mbps per 1000 components (Maximum of 100 Mbps with 45 k components).

Note:

vSAN recommends the use of 25GbE or higher NICs in the vSAN hosts. NIC requirements assume that the packet loss is not more than 0.0001% in vSAN. There can be a drastic impact on the vSAN performance, if any of these requirements are exceeded.

For more information about the vSAN stretched cluster NIC requirements, see vSAN Stretched Cluster Guide.

Bandwidth and Latency Requirements

To ensure high performance and availability, vSAN clusters must meet certain bandwidth and network latency requirements.

The bandwidth requirements between the primary and secondary sites of a vSAN stretched cluster depend on the vSAN workload, amount of data, and the way you want to handle failures. For more information, see vSAN Stretched Cluster Bandwidth Sizing.

Table 993: Bandwidth and Latency Requirements

Site Communication	Bandwidth	Latency
Single Site		1ms latency RTT.
Site to Site	vSAN OSA: minimum of 10 GbE vSAN ESA: minimum of 10 GbE **Note:** The bandwidth requirement is based on the number of VMs being replicated between sites.	5 ms or less latency RTT.
Site to Witness	2 Mbps per 1000 vSAN components	• Less than 500 ms latency RTT for 1 host per site. • Less than 200 ms latency RTT for up to 10 hosts per site. • Less than 100 ms latency RTT for 11-15 hosts per site.
vSAN Compute-Only Cluster to a vSAN Storage Cluster		5 ms or less latency RTT.

Layer 2 and Layer 3 Support

VMware recommends Layer 2 connectivity for vSAN deployed on a single site. For vSAN deployed across multiple racks, you can use Layer 2 or Layer 3. VMware recommends Layer 3 for data site to data site communication

vSAN also supports deployments using routed Layer 3 connectivity between vSAN hosts. You must consider the number of hops and additional latency incurred while the traffic gets routed.

Table 994: Layer 2 and Layer 3 Support

Cluster Type	L2 Supported	L3 Supported	Considerations
Hybrid Cluster	Yes	Yes	L2 for single site and single rack is recommended. L2 or L3 for single site, if vSAN cluster is deployed across multiple rack and/or using vSAN fault domains.
All-Flash Cluster	Yes	Yes	L2 is recommended and L3 is supported.
vSAN Stretched Cluster Data	Yes	Yes	Both L2 and L3 between data sites are supported. Layer 3 is recommended to isolate faults per site. Layer 3 networking is preferred for vSAN stretched clusters as it helps avoid issues with Spanning Tree Protocol (STP) redirecting the traffic across less desirable links.
vSAN Stretched Cluster Witness	No	Yes	L3 is supported. L2 between data and witness sites is not supported.

Cluster Type	L2 Supported	L3 Supported	Considerations
Two-Node vSAN Cluster	Yes	Yes	Both L2 and L3 between data sites are supported.
vSAN Stretched Compute Cluster	Yes	Yes	Both L2 and L3 between data sites are supported.
vSAN Compute Client Traffic	Yes	Yes	Both L2 and L3 between data sites are supported.
vSAN Storage Cluster	Yes	Yes	L2 is recommended and L3 is supported.
vSAN Stretched Storage Cluster	Yes	Yes	Both L2 and L3 between data sites are supported. Layer 3 is recommended to isolate faults per site. Layer 3 networking is preferred for vSAN stretched storage clusters as it helps avoid issues with Spanning Tree Protocol (STP) redirecting the traffic across less desirable links.

Routing and Switching Requirements

All three sites in a vSAN stretched cluster communicate across the management network and across the vSAN network. The VMs in all data sites communicate across a common virtual machine network.

Following are the vSAN stretched cluster routing requirements:

Table 995: ESX Host Routing Requirements

Site Communication	Deployment Model	Layer	Routing
Site to Site	Default	Layer 2	Not required
Site to Site	Default	Layer 3	Use static routes or gateway override. Recommended is gateway override. See Override the Default Gateway of a VMkernel Adapter.
Site to Witness	Default	Layer 3	Use static routes or gateway override.
Site to Witness	Witness Traffic Separation	Layer 3	Use static routes or gateway override when using an interface other than the Management (vmk0) interface.
Site to Witness	Witness Traffic Separation	Layer 2 for two-host cluster	Static routes are not required.

Virtual Switch Requirements

You can create a vSAN network with either vSphere Standard Switch or vSphere Distributed Switch. Use a distributed switch to prioritize bandwidth for vSAN traffic. vSAN uses a distributed switch with all the vCenter versions.

The following table compares the advantages and benefits of a distributed switch over a standard switch:

Table 996: Virtual Switch Types

Design Requirement	Option 1 - vSphere Distributed Switch	Option 2 - vSphere Standard Switch	Description
Availability	No impact	No impact	You can use either of the options
Manageability	Positive impact	Negative impact	The distributed switch is centrally managed across all hosts, unlike the standard switch which is managed on each host individually.
Performance	Positive impact	Negative impact	The distributed switch has added controls, such as Network I/O Control, which you can use to guarantee performance for vSAN traffic.
Recoverability	Positive impact	Negative impact	The distributed switch configuration can be backed up and restored, the standard switch does not have this functionality.
Security	Positive impact	Negative impact	The distributed switch has added built-in security controls to help protect traffic.

vSAN Network Port Requirements

vSAN deployments require specific network ports and settings to provide access and services.

vSAN sends messages on certain ports on each host in the cluster. Verify that the host firewalls allow traffic on these ports. For the list of all supported vSAN ports and protocols, see the *Broadcom Ports and Protocols Portal* at https://ports.broadcom.com/.

Network Firewall Requirements

When you configure the network firewall, consider which version of vSAN you are deploying.

When you enable vSAN on a cluster, all required ports are added to ESXfirewall rules and configured automatically. You do not need to open any firewall ports or enable any firewall services manually. You can view open ports for incoming and outgoing connections in the ESX host security profile (**Configure > Security Profile**).

vsanEncryption Firewall Rule

If your cluster uses vSAN encryption, consider the communication between hosts and the KMS server.

vSAN encryption requires a Key Management Server (KMS). vSAN can use a vCenter Native Key Provider or an external KMS. vCenter obtains the key IDs from the KMS, and distributes them to the ESX hosts. KMS servers and ESX hosts communicate directly with each other. KMS servers might use different port numbers, so the vsanEncryption firewall rule enables you to simplify communication between each vSAN host and the KMS server. This allows a vSAN host to communicate directly to any port on a KMS server (TCP port 0 through 65535).

When a host establishes communication to a KMS server, the following operations occur.

- The KMS server IP is added to the vsanEncryption rule and the firewall rule is enabled.
- Communication between vSAN node and KMS server is established during the exchange.
- After communication between the vSAN node and the KMS server ends, the IP address is removed from vsanEncryption rule, and the firewall rule is deactivatedagain.

vSAN hosts can communicate with multiple KMS hosts using the same rule.

Configuring IP Network Transport

Transport protocols provide communication services across the network. These services include the TCP/IP stack and flow control.

vSAN does not support vSphere TCP/IP stacks. You cannot isolate vSAN traffic on dedicated stack with its own routing configuration.

vSphere RDMA

vSAN supports Remote Direct Memory Access (RDMA) communication.

RDMA allows direct memory access from the memory of one computer to the memory of another computer without involving the operating system or CPU. The transfer of memory is offloaded to the RDMA-capable Host Channel Adapters (HCA).

vSAN supports the RoCE v2 protocol. RoCE v2 requires a network configured for lossless operation that is free of congestion. If your network has congestion, certain large I/O workloads might experience lower performance than TCP.

Each vSAN host must have a vSAN certified RDMA-capable NIC, as listed in the vSAN section of the *Broadcom Compatibility Guide*. Use only the same model network adapters from the same vendor on each end of the connection.

> **Note:**
>
> vSphere RDMA is not supported on vSAN stretched clusters, two-node vSAN clusters, vSAN storage clusters, or datastore sharing (HCI Mesh).

All hosts in the cluster must support RDMA. If any host loses RDMA support, the entire vSAN cluster switches to TCP.

vSAN with RDMA supports NIC failover, but does not support LACP or IP-hash-based NIC teaming.

IPv6 Support

vSAN supports IPv6.

vSAN supports the following IP versions.

- IPv4
- IPv6

For more information about using IPv6, consult with your network vendor.

Override the Default Gateway

You can override the default gateway for the vSAN VMkernel adapter on each host, and configure a gateway address for the vSAN network. You can override the default gateway when the vSAN network uses a different subnet than the management network. In vSAN stretched clusters and two-node vSAN clusters, data sites and the witness hosts can reside on separate networks. A dedicated vSAN gateway allows traffic to reach the witness host without static routes, simplifying network configuration. For more information on overriding default gateway, see Override the Default Gateway of VMKernel Adapter.

Static Routes

You can use static routes to allow vSAN network interfaces from hosts on one subnet to reach the hosts on another network.

Most organizations separate the vSAN network from the management network, so the vSAN network does not have a default gateway. In an L3 deployment, hosts that are on different subnets or different L2 segments cannot reach each other over the default gateway, which is typically associated with the management network.

Use *static routes* to allow the vSAN network interfaces from hosts on one subnet to reach the vSAN networks on hosts on the other network. Static routes instruct a host how to reach a particular network over an interface, rather than using the default gateway.

The following example shows how to add an IPv4 static route to an ESX host. Specify the gateway (-g) and the network (-n) you want to reach through that gateway:

```
esxcli network ip route ipv4 add –g 172.16.10.253 -n 192.168.10.0/24
```

When the static routes have been added, vSAN traffic connectivity is available across all networks, assuming the physical infrastructure allows it. Run the `vmkping` command to test and confirm communication between the different networks by pinging the IP address or the default gateway of the remote network. You also can check different size packets (-s) and prevent fragmentation (-d) of the packet.

```
vmkping –I vmk3 192.168.10.253
```

Jumbo Frames

vSAN fully supports jumbo frames on the vSAN network.

Jumbo frames are Ethernet frames with more than 1500 bytes of payload. Jumbo frames typically carry up to 9000 bytes of payload, but variations exist.

Using jumbo frames can reduce CPU utilization and improve throughput.

> **Note:**
>
> Enable jumbo frames support for vSAN storage cluster deployments to improve performance.

You must decide whether these gains outweigh the overhead of implementing jumbo frames throughout the network. In data centers where jumbo frames are already enabled in the network infrastructure, you can use them for vSAN. The operational cost of configuring jumbo frames throughout the network might outweigh the limited CPU and performance benefits.

Using VMware NSX with vSAN

vSAN and VMware NSX can be deployed and coexist in the same vSphere infrastructure. A distributed vSwitch can support NSX and vSphere or vSAN traffic

NSX does not support the configuration of the vSAN data network over an NSX-managed VXLAN or Geneve overlay.

vSAN and NSX are compatible. vSAN and NSX are not dependent on each other to deliver their functionalities, resources, and services.

Using Congestion Control and Flow Control

Flow control is used to manage the rate of data transfer between senders and receivers on the network. Congestion control handles congestion in the network. Flow control is enabled by default on the ESX hosts.

Flow Control

You can use flow control to manage the rate of data transfer between two devices.

Flow control is configured when two physically connected devices perform auto-negotiation. An overwhelmed network node might send a pause frame to halt the transmission of the sender for a specified period.

One reason to use pause frames is to support network interface controllers (NICs) that do not have enough buffering to handle full-speed reception. This problem is uncommon with advances in bus speeds and memory sizes.

Congestion Control

Congestion control helps you control the traffic on the network.

Congestion control applies mainly to packet switching networks. Network congestion within a switch might be caused by overloaded inter-switch links. If inter-switch links overload the capability on the physical layer, the switch introduces pause frames to protect itself.

Priority Flow Control

Priority-based flow control (PFC) helps you eliminate frame loss due to congestion.

Priority-based flow control (IEEE 802.1Qbb) is achieved by a mechanism similar to pause frames, but operates on individual priorities. PFC is also called Class-Based Flow Control (CBFC) or Per Priority Pause (PPP).

Flow Control Design Considerations

By default, flow control is enabled on all network interfaces in ESX hosts.

Flow control configuration on a NIC is done by the driver. When a NIC is overwhelmed by network traffic, the NIC sends pause frames.

Flow control mechanisms such as pause frames can trigger overall latency in the VM guest I/O due to increased latency at the vSAN network layer. Some network drivers provide module options that configure flow control functionality within the driver. For information about configuring flow control on ESX hosts, see Broadcom knowledge base article 1013413.

In deployments with 1 GbE, leave flow control enabled on ESX network interfaces (default). If pause frames are a problem, carefully plan disabling flow control in conjunction with Hardware Vendor Support or Broadcom Global Support Services.

To learn how you can recognize the presence of pause frames being sent from a receiver to an ESX host, see Troubleshooting the vSAN Network. A large number of pause frames in an environment usually indicates an underlying network or transport issue to investigate.

Basic NIC Teaming, Failover, and Load Balancing

For IP storage based solutions such as vSAN, it is best practice to implement network redundancy.

You can use NIC teaming to achieve network redundancy. You can configure two or more network adapters (NICs) as a team for high availability and load balancing. Basic NIC teaming is available with vSphere networking, and these techniques can affect vSAN design and architecture.

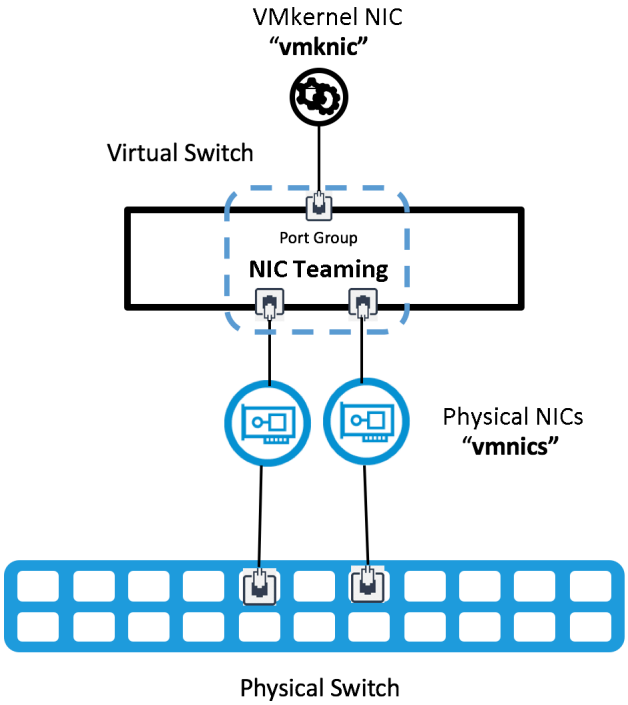
Several NIC teaming options are available. Avoid NIC teaming policies that require physical switch configurations, or that require an understanding of networking concepts such as Link Aggregation. Best results are achieved with a basic, simple, and reliable setup.

If you are not sure about NIC teaming options, use an Active/Standby configuration with explicit failover.

Basic NIC Teaming

Basic NIC teaming uses multiple physical uplinks, one vmknic, and a single switch.

vSphere NIC teaming uses multiple uplink adapters, called vmnics, which are associated with a single virtual switch to form a team. This is the most basic option, and you can configure it using a standard vSphere standard switch or a vSphere distributed switch.



Failover and Redundancy

vSAN can use the basic NIC teaming and failover policy provided by vSphere.

NIC teaming on a vSwitch can have multiple active uplinks, or an Active/Standby uplink configuration. Basic NIC teaming does not require any special configuration at the physical switch layer.

> **Note:**
> vSAN does not use NIC teaming for load balancing.

A typical NIC teaming configuration has the following settings. When working on distributed switches, edit the settings of the distributed port group used for vSAN traffic.

- Load balancing: Route based on originating virtual port
- Network failure detection: Link status only
- Notify switches: Yes
- Failback: Yes

Configure Load Balancing for NIC Teams

Several load-balancing techniques are available for NIC teaming, and each technique has its pros and cons.

Route Based on Originating Virtual Port

In Active/Active or Active/Passive configurations, use **Route based on originating virtual port** for basic NIC teaming. When this policy is in effect, only one physical NIC is used per VMkernel port.

Pros

- This is the simplest NIC teaming method that requires minimal physical switch configuration.
- This method requires only a single port for vSAN traffic, which simplifies troubleshooting.

Cons

- A single VMkernel interface is limited to a single physical NIC's bandwidth. As typical vSAN environments use one VMkernel adapter, only one physical NIC in the team is used.

Route Based on Physical NIC Load

Route Based on Physical NIC Load is based on **Route Based on Originating Virtual Port**, where the virtual switch monitors the actual load of the uplinks and takes steps to reduce load on overloaded uplinks. This load-balancing method is available only with a vSphere Distributed Switch, not on vSphere Standard Switches.

The distributed switch calculates uplinks for each VMkernel port by using the port ID and the number of uplinks in the NIC team. The distributed switch checks the uplinks every 30 seconds, and if the load exceeds 75 percent, the port ID of the VMkernel port with the highest I/O is moved to a different uplink.

Pros

- No physical switch configuration is required.
- Although vSAN has one VMkernel port, the same uplinks can be shared by other VMkernel ports or network services. vSAN can benefit by using different uplinks from other contending services, such as vMotion or management.

Cons

- As vSAN typically only has one VMkernel port configured, its effectiveness is limited.
- The ESX VMkernel reevaluates the traffic load after each time interval, which can result in processing overhead.

Settings: Network Failure Detection

Use the default setting: **Link status only**. Do not use Beacon probing for link failure detection. Beacon probing requires at least three physical NICs to avoid split-brain scenarios. There can be a network traffic failover if the ESX host detects a link down event on an ESX host network interface. For more details, see Broadcom kbnowledge ase article 1005577.

Settings: Notify Switches

Use the default setting: **Yes**. Physical switches have MAC address forwarding tables to associate each MAC address with a physical switch port. When a frame comes in, the switch determines the destination MAC address in the table and decides the correct physical port.

If a NIC failover occurs, the ESX host must notify the network switches that something has changed, or the physical switch might continue to use the old information and send the frames to the wrong port.

When you set Notify Switches to **Yes**, if one physical NIC fails and traffic is rerouted to a different physical NIC in the team, the virtual switch sends notifications over the network to update the lookup tables on physical switches.

This setting does not catch VLAN misconfigurations, or uplink losses that occur further upstream in the network. The vSAN network partitions health check can detect these issues.

Settings: Failback

This option determines how a physical adapter is returned to active duty after recovering from a failure. A failover event triggers the network traffic to move from one NIC to another. When a **link up** state is detected on the originating NIC, traffic automatically reverts to the original network adapter when Failback is set to **Yes**. When Failback is set to **No**, a manual failback is required.

Setting Failback to **No** can be useful in some situations. For example, after a physical switch port recovers from a failure, the port might be active but can take several seconds to begin forwarding traffic. Automatic Failback has been known to cause problems in certain environments that use the Spanning Tree Protocol. For more information about Spanning Tree Protocol (STP), see Broadcom knowledge base article 1003804.

Setting Failover Order

Failover order determines which links are active during normal operations, and which links are active in the event of a failover. Different supported configurations are possible for the vSAN network.

Active/Standby uplinks: If a failure occurs on an Active/Standby setup, the NIC driver notifies vSphere of a link down event on Uplink 1. The standby Uplink 2 becomes active, and traffic resumes on Uplink 2.

Active/Active uplinks: If you set the failover order to Active/Active, the virtual port used by vSAN traffic cannot use both physical ports at the same time.

If your NIC teaming configuration for both Uplink 1 and Uplink 2 is active, there is no need for the standby uplink to become active.

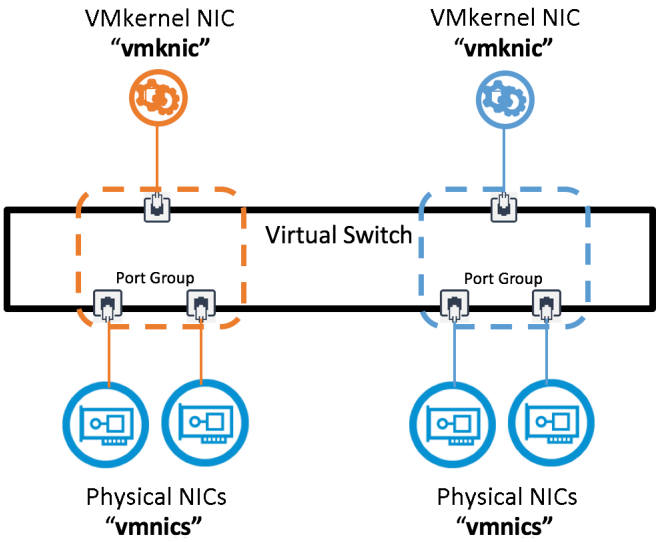
Note:
When using an Active/Active configuration, ensure that Failback is set to **No**. For more information, see Broadcom knowledge base article 2072928.

Advanced NIC Teaming

You can use advanced NIC teaming methods with multiple VMkernel adapters having two dedicated subnets to configure the vSAN network. If you use Link Aggregation Protocol (LAG/LACP), the vSAN network can be configured with a single VMkernel adapter.

You can use advanced NIC teaming to implement an air gap, so a failure that occurs on one network path does not impact the other network path. If any part of one network path fails, the other network path can carry the traffic. Configure multiple VMkernel NICs for vSAN on different subnets, such as another VLAN or separate physical network fabric.

vSphere and vSAN do not support multiple VMkernel adapters (vmknics) on the same subnet. For more details, see Broadcom knowledge base article 2010877.



Link Aggregation Group Overview

By using the LACP protocol, a network device can negotiate an automatic bundling of links by sending LACP packets to a peer.

A Link Aggregation Group (LAG) states that Link Aggregation allows one or more links to be aggregated together to form a Link Aggregation Group.

LAG can be configured as either static (manual) or dynamic by using LACP to negotiate the LAG formation. LACP can be configured as follows:

Active	Devices immediately send LACP messages when the port comes up. End devices with LACP enabled (for example, ESX hosts and physical switches) send and receive frames called LACP messages to each other to negotiate the creation of a LAG.
Passive	Devices place a port into a passive negotiating state, in which the port only responds to received LACP messages, but do not initiate negotiation.

Note:
If the host and switch are both in passive mode, the LAG does not initialize, because an active part is required to trigger the linking. At least one must be Active.

For more information about LACP support on a vSphere Distributed Switch, see the vSphere Networking guide.

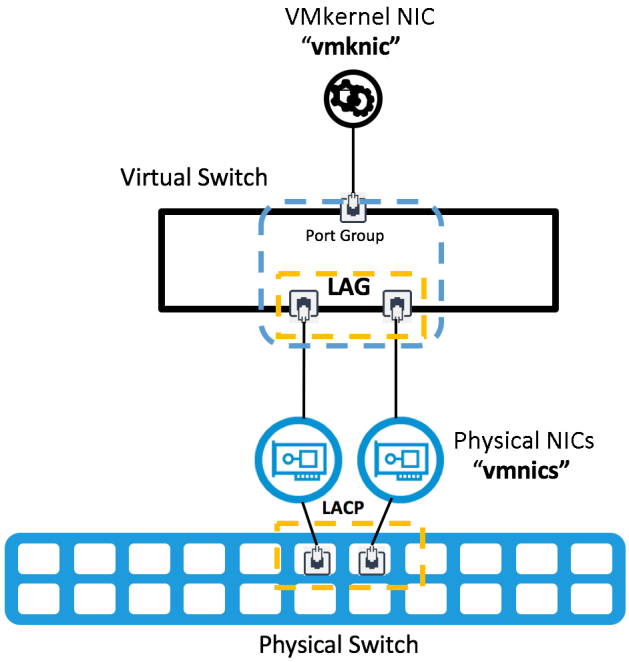
Note:
The number of LAGs you can use depends on the capabilities of the underlying physical environment and the topology of the virtual network.

Static and Dynamic Link Aggregation

You can use LACP to combine and aggregate multiple network connections.

When LACP is in **active** or **dynamic** mode, a physical switch sends LACP messages to network devices, such as ESX hosts, to negotiate the creation of a Link Aggregation Group (LAG).

To configure Link Aggregation on hosts using vSphere Standard Switches (and pre-5.5 vSphere Distributed Switches), configure a static channel-group on the physical switch. See your vendor documentation for more details.



Pros and Cons of Dynamic Link Aggregation

Consider the tradeoffs to using Dynamic Link Aggregation.

Pros

Improves performance and bandwidth. One vSAN host or VMkernel port can communicate with many other vSAN hosts using many different load-balancing options.

Provides network adapter redundancy. If a NIC fails and the link-state fails, the remaining NICs in the team continue to pass traffic.

Improves traffic balancing. Balancing of traffic after failures is automatic and fast.

Cons

Less flexible. Physical switch configuration requires that physical switch ports be configured in a port-channel configuration.

More complex. Use of multiple switches to produce full physical redundancy configuration is complex. Vendor-specific implementations add to the complexity.

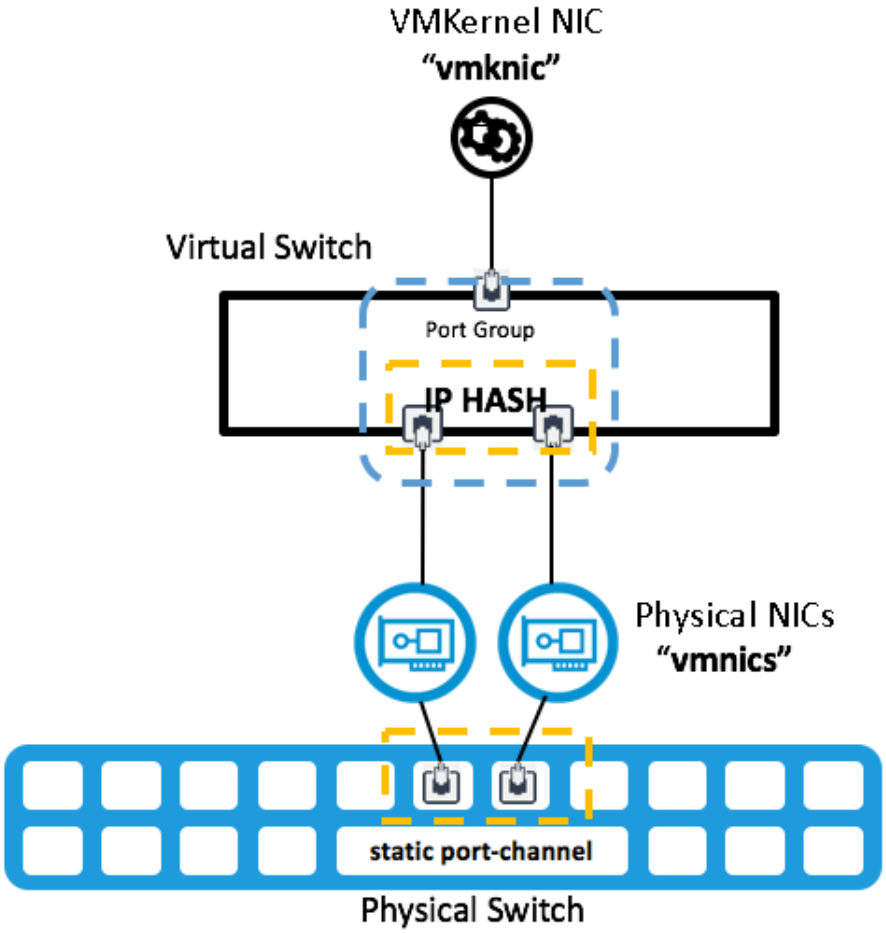
Static LACP with Route Based on IP Hash

You can create a vSAN cluster using static LACP with an IP-hash policy. This section focuses on vSphere Standard Switches, but you also can use vSphere Distributed Switches.

You can use the Route based on IP Hash load balancing policy.

Select **Route based on IP Hash** load balancing policy at a vSwitch or port-group level. Set all uplinks assigned to static channel group to the Active Uplink position on the Teaming and Failover Policies at the virtual switch or port-group level.

When IP Hash is configured on a vSphere port group, the port group uses the **Route based on IP Hash** policy. The number of ports in the port-channel must be same as the number of uplinks in the team.



Pros and Cons of Static LACP with IP Hash

Consider the tradeoffs to using Static LACP with IP Hash.

Pros

- **Improves performance and bandwidth**. One vSAN host or VMkernel port can communicate with many other vSAN hosts using the IP Hash algorithm.
- **Provides network adapter redundancy**. If a NIC fails and the link-state fails, the remaining NICs in the team continue to pass traffic.
- **Adds flexibility**. You can use IP Hash with both vSphere Standard Switches and vSphere Distributed Switches.

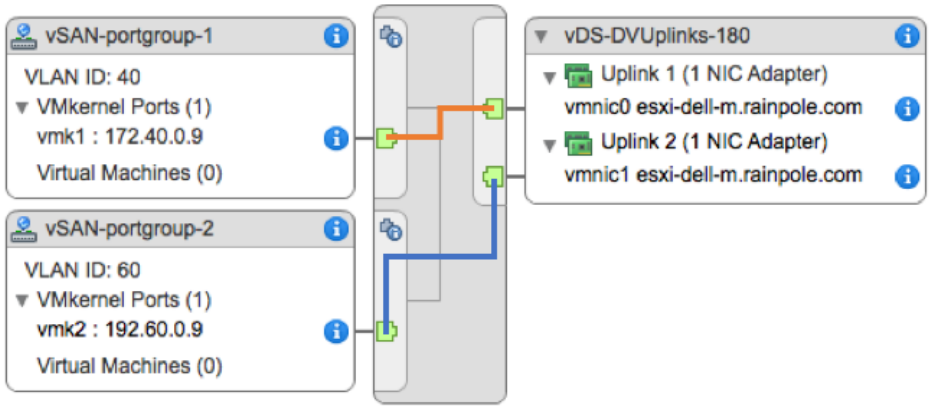
Cons

- **Physical switch configuration is less flexible**. Physical switch ports must be configured in a static port-channel configuration.
- **Increased chance of misconfiguration**. Static port-channels form without any verification on either end (unlike LACP dynamic port-channel).
- **More complex**. Introducing full physical redundancy configuration increases complexity when multiple switches are used. Implementations can become quite vendor specific.
- **Limited load balancing**. If your environment has only a few IP addresses, the virtual switch might consistently pass the traffic through one uplink in the team. This can be especially true for small vSAN clusters.

Understanding Network Air Gaps

You can use advanced NIC teaming methods to create an air-gap storage fabric. Two storage networks are used to create a redundant storage network topology, with each storage network physically and logically isolated from the other by an air gap.

You can configure a network air gap for vSAN in a vSphere environment. Configure multiple VMkernel ports per vSAN host. Associate each VMkernel port to dedicated physical uplinks, using either a single vSwitch or multiple virtual switches, such as vSphere Standard Switch or vSphere Distributed Switch.



Typically, each uplink must be connected to fully redundant physical infrastructure.

This topology is not ideal. The failure of components such as NICs on different hosts that reside on the same network can lead to interruption of storage I/O. To avoid this problem, implement physical NIC redundancy on all hosts and all network segments. Configuration example 2 addresses this topology in detail.

These configurations are applicable to both L2 and L3 topologies, with unicast configuration.

Pros and Cons of Air Gap Network Configurations with vSAN

Network air gaps can be useful to separate and isolate vSAN traffic. Use caution when configuring this topology.

Pros

- Physical and logical separation of vSAN traffic.

Cons

- vSAN does not support multiple VMkernel adapters (vmknics) on the same subnet. For more information, see Broadcom knowledge base article 2010877.
- Setup is complex and error prone, so troubleshooting is more complex.
- Network availability is not guaranteed with multiple vmknics in some asymmetric failures, such as one NIC failure on one host and another NIC failure on another host.
- Load-balanced vSAN traffic across physical NICs is not guaranteed.
- Costs increase for vSAN hosts, as you might need multiple VMkernel adapters (vmknics) to protect multiple physical NICs (vmnics). For example, 2 x 2 vmnics might be required to provide redundancy for two vSAN vmknics.
- Required logical resources are doubled, such as VMkernel ports, IP addresses, and VLANs.

- vSAN does not implement port binding. This means that techniques such as multi-pathing are not available.
- Layer 3 topologies are not suitable for vSAN traffic with multiple vmknics. These topologies might not function as expected.

Dynamic LACP combines, or aggregates, multiple network connections in parallel to increase throughput and provide redundancy. When NIC teaming is configured with LACP, load balancing of the vSAN network across multiple uplinks occurs. This load balancing happens at the network layer, and is not done through vSAN.

> **Note:**
> Other terms sometimes used to describe link aggregation include port trunking, link bundling, Ethernet/network/NIC bonding, EtherChannel.

This section focuses on Link Aggregation Control Protocol (LACP). The IEEE standard is 802.3ad, but some vendors have proprietary LACP features, such as PAgP (Port Aggregation Protocol). Follow the best practices recommended by your vendor.

> **Note:**
> The LACP support introduced in vSphere Distributed Switch 5.1 only supports IP-hash load balancing. vSphere Distributed Switch 5.5 and later fully support LACP.

LACP is an industry standard that uses port-channels. Many hashing algorithms are available. The vSwitch port-group policy and the port-channel configuration must agree and match.

NIC Teaming Configuration Examples

The following NIC teaming configurations illustrate typical vSAN networking scenarios.

Configuration 1: Single vmknic, Route Based on Physical NIC Load

You can configure basic Active/Active NIC Teaming with the **Route based on Physical NIC Load** policy for vSAN hosts. Use a vSphere Distributed Switch (vDS).

For this example, the vDS must have two uplinks configured for each host. A distributed port group is designated for vSAN traffic and isolated to a specific VLAN. Jumbo frames are already enabled on the vDS with an MTU value of 9000.

Configure teaming and failover for the distributed port group for vSAN traffic as follows:

- Load balancing policy set to **Route Based on Physical Nic Load**.
- Network failure detection set to **Link status only**.
- Notify Switches set to **Yes**.
- Failback set to **No**. You can set Failback to **yes**, but not for this example.
- Ensure both uplinks are in the **Active uplinks** position.

Network Uplink Redundancy Lost

When the link down state is detected, the workload switches from one uplink to another. There is no noticeable impact to the vSAN cluster and VM workload.

Recovery and Failback

When you set **Failback** to **No**, traffic is not promoted back to the original vmnic. If **Failback** is set to **Yes**, traffic is promoted back to the original vmnic on recovery.

Load Balancing

Since this is a single VMkernel NIC, there is no performance benefit to using **Route based on physical load**.

Only one physical NIC is in use at any time. The other physical NIC is idle.

Configuration 2: Multiple vmknics, Route Based on Originating Port ID

You can use two non-routable VLANs that are logically and physically separated, to produce an air-gap topology.

This example provides configuration steps for a vSphere distributed switch, but you also can use vSphere standard switches. It uses two 10 GbE physical NICs and logically separates them on the vSphere networking layer.

Create two distributed port groups for each vSAN VMkernel vmknic. Each port group has a separate VLAN tag. For vSAN VMkernel configuration, two IP addresses on both VLANs are required for vSAN traffic.

> **Note:**
>
> Practical implementations typically use four physical uplinks for full redundancy.

For each port group, the teaming and failover policy use the default settings.

- Load balancing set to **Route based on originating port ID**
- Network failure detection set to **Link Status Only**
- Notify Switches set to the default value of **Yes**
- Failback set to the default value of **Yes**
- The uplink configuration has one uplink in the **Active** position and one uplink in the **Unused** position.

One network is completely isolated from the other network.

vSAN Port Group 1

This example uses a distributed port group called **vSAN-DPortGroup-1**. **VLAN 3266** is tagged for this port group with the following Teaming and Failover policy:

- Traffic on the port group tagged with VLAN 3266
- Load balancing set to **Route based on originating port ID**
- Network failure detection set to **Link Status Only**
- Notify Switches set to default value of **Yes**
- Failback set to default value of **Yes**
- The uplink configuration has **Uplink 1** in the **Active** position and **Uplink 2** in the **Unused** position.

vSAN Port Group 2

To complement vSAN port group 1, configure a second distributed port group called **vSAN-portgroup-2**, with the following differences:

- Traffic on the port group tagged with VLAN 3265
- The uplink configuration has **Uplink 2** in the **Active** position and **Uplink 1** in the **Unused**position.

vSAN VMkernel Port Configuration

Create two vSAN VMkernel interfaces and on both port groups. In this example, the port groups are named **vmk1** and **vmk2**.

- **vmk1** is associated with VLAN 3266 (172.40.0.xx), and as a result port group **vSAN-DPortGroup-1**.
- **vmk2** is associated with VLAN 3265 (192.60.0.xx), and as a result port group **vSAN-DPortGroup-2**.

Load Balancing

vSAN has no load balancing mechanism to differentiate between multiple vmknics, so the vSAN I/O path chosen is not deterministic across physical NICs. The vSphere performance charts show that one physical NIC is often more utilized than the other. A simple I/O test performed in our labs, using 120 VMs with a 70:30 read/write ratio with a 64K block size on a four-host all flash vSAN cluster, revealed an unbalanced load across NICs.

vSphere performance graphs show an unbalanced load across NICs.

Network Uplink Redundancy Lost

Consider a network failure introduced in this configuration. vmnic1 is not enabled on a given vSAN host. As a result, port **vmk2** is impacted. A failing NIC triggers both network connectivity alarms and redundancy alarms.

For vSAN, this failover process triggers approximately **10 seconds** after CMMDS (Cluster Monitoring, Membership, and Directory Services) detects a failure. During failover and recovery, vSAN stops any active connections on the failed network, and attempts to re-establish connections on the remaining functional network.

Since two separate vSAN VMkernel ports communicate on isolated VLANs, vSAN health check failures might be triggered. This is expected as **vmk2** can no longer communicate to its peers on VLAN 3265.

The performance charts show that the affected workload has restarted on vmnic0, since vmnic1 has a failure. This test illustrates an important distinction between vSphere NIC teaming and this topology. vSAN attempts to re-establish or restart connections on the remaining network.

However, in some failure scenarios, recovering the impacted connections might require up to **90 seconds** to complete, due to ESX TCP connection timeout. Subsequent connection attempts might fail, but connection attempts time out at 5 seconds, and the attempts rotate through all possible IP addresses. This behavior might affect virtual machine guest I/O. As a result, application and virtual machine I/O might have to be retried.

For example, on Windows Server 2012 VMs, Event IDs 153 (device reset) and 129 (retry events) might be logged during the failover and recovery process. In the example, event ID 129 was logging for approximately 90 seconds until the I/O was recovered.

You might have to modify disk timeout settings of some guest OSes to ensure that they are not severely impacted. Disk timeout values might vary, depending on the presence of VMware Tools, and the specific guest OS type and version.

Recovery and Failback

When the network is repaired, workloads are not automatically rebalanced unless another failure to force workload occurs, due to another failure. As soon as the impacted network is recovered, it becomes available for new TCP connections.

Configuration 3: Dynamic LACP

You can configure a two-port LACP port channel on a switch and a two-uplink Link Aggregation Group on a vSphere distributed switch.

In this example, use 10 GbE networking with two physical uplinks per server.

> **Note:**
>
> vSAN over RDMA does not support this configuration.

Configure the Network Switch

Configure the vSphere distributed switch with the following settings.

- Identify the ports in question where the vSAN host will connect.
- Create a port channel.
- If using VLANs, then trunk the correct VLAN to the port channel.
- Configure the desired distribution or load-balancing options (hash).
- Setting LACP mode to active/dynamic.
- Verify MTU configuration.

Configure vSphere

Configure the vSphere network with the following settings.

- Configure vDS with the correct MTU.
- Add hosts to vDS.
- Create a LAG with the correct number of uplinks and matching attributes to port channel.

- Assign physical uplinks to the LAG.
- Create a distributed port group for vSAN traffic and assign correct VLAN.
- Configure VMkernel ports for vSAN with correct MTU.

Set Up the Physical Switch

Configure the physical switch with the following settings. For guidance about how to set up this configuration on Dell servers based on Dell Networking PowerConnect Switch, refer to: http://www.dell.com/Support/Article/us/en/19/HOW10364.

Configure a two uplink LAG:

- Use switch ports 36 and 18.
- This configuration uses VLAN trunking, so port channel is in VLAN trunk mode, with the appropriate VLANs trunked.
- Use the following method for load-balancing or load distribution: **Source and destination IP addresses, TCP/UDP port and VLAN**
- Verify that the LACP mode is **Active** (Dynamic).

Use the following commands to configure an individual port channel on a Dell switch:

- Create a port-channel.

```
# interface port-channel 1
```

- Set port-channel to VLAN trunk mode.

```
# switchport mode trunk
```

- Allow VLAN access.

```
# switchport trunk allowed vlan 3262
```

- Configure the load balancing option.

```
# hashing-mode 6
```

- Assign the correct ports to the port-channel and set the mode to Active.
- Verify that the port channel is configured correctly.

```
# show interfaces port-channel 1

Channel Ports Ch-Type Hash Type Min-links Local Prf
------- ----------------------------- -------- --------- --------- ---------
Po1 Active:  Te1/0/36, Te1/0/18  Dynamic   6 1 Disabled
Hash Algorithm Type
1 - Source MAC, VLAN, EtherType, source module and port Id
2 - Destination MAC, VLAN, EtherType, source module and port Id
3 - Source IP and source TCP/UDP port
4 - Destination IP and destination TCP/UDP port
5 - Source/Destination MAC, VLAN, EtherType, source MODID/port
6 - Source/Destination IP and source/destination TCP/UDP port
7 - Enhanced hashing mode
```

```
# interface range Te1/0/36, Te1/0/18
```

```
# channel-group 1 mode active
```

Full configuration:

```
# interface port-channel 1
```

```
# switchport mode trunk
```

```
# switchport trunk allowed vlan 3262
```

```
# hashing-mode 6
```

```
# exit
# interface range Te1/0/36,Te1/018
# channel-group 1 mode active
# show interfaces port-channel 1
```

Note:
Repeat this procedure on all participating switch ports that are connected to vSAN hosts.

Set Up vSphere Distributed Switch

Before you begin, make sure that the vDS is upgraded to a version that supports LACP. To verify, right click the vDS, and check if the Upgrade option is available. You might have to upgrade the vDS to a version that supports LACP.

Create LAG on vDS

To create a LAG on a distributed switch, select the vDS, click the **Configure** tab, and select **LACP**. Add a new LAG.

Configure the LAG with the following properties:

- LAG name: **lag1**
- Number of ports: **2** (to match port channel on switch)
- Mode: **Active**, to match the physical switch.
- Load balancing mode: **Source and destination IP addresses, TCP/UDP port and VLAN**

Add Physical Uplinks to LAG

vSAN hosts have been added to the vDS. Assign the individual vmnics to the appropriate LAG ports.

- Right click the vDS, and select **Add and Manage Hosts…**
- Select **Manage Host Networking**, and add your attached hosts.
- On **Manage Physical Adapters**, select the appropriate adapters and assign them to the LAG port.
- Migrate vmnic0 from Uplink 1 position to port 0 on LAG1.

Repeat the procedure for vmnic1 to the second LAG port position, lag1-1.

Configure Distributed Port Group Teaming and Failover Policy

Assign the LAG group as an **Active uplink** on distributed port group teaming and failover policy. Select or create the designated distributed port group for vSAN traffic. This configuration uses a vSAN port group called **vSAN** with VLAN ID 3262 tagged. Edit the port group, and configure Teaming and Failover Policy to reflect the new LAG configuration.

Ensure the LAG group **lag1** is in the active uplinks position, and ensure the remaining uplinks are in the **Unused** position.

Note:
When a link aggregation group (LAG) is selected as the only active uplink, the load-balancing mode of the LAG overrides the load-balancing mode of the port group. Therefore, the following policy plays no role: **Route based on originating virtual port**.

Create the VMkernel Interfaces

The final step is to create the VMkernel interfaces to use the new distributed port group, ensuring that they are tagged for vSAN traffic. Observe that each vSAN vmknic can communicate over vmnic0 and vmnic1 on a LAG group to provide load balancing and failover.

Configure Load Balancing

From a load balancing perspective, there is not a consistent balance of traffic across all hosts on all vmnics in this LAG setup, but there is more consistency compared to **Route based on physical NIC load** used in Configuration 1 and the air-gapped/multiple vmknics method used in Configuration 2.

The individual hosts' vSphere performance graph shows improved load balancing.

Network Uplink Redundancy Lost

When vmnic1 is not enabled on a given vSAN host, a Network Redundancy alarm is triggered.

No vSAN health alarms are triggered, and the impact to Guest I/O is minimal compared to the air-gapped, multi-vmknics configuration. This configuration does not have to stop any TCP sessions with LACP configured.

Recovery and Failback

In a failback scenario, the behavior differs between Load Based Teaming, multiple vmknics, and LACP in a vSAN environment. After vmnic1 recovers, traffic is automatically balanced across both active uplinks. This behavior can be advantageous for vSAN traffic.

Failback Set to Yes or No?

A LAG load-balancing policy overrides the Teaming and Failover policy for vSphere distributed port groups. Also consider the guidance on Failback value. Lab tests show no discernable behavior differences between Failback set to **yes** or **no** with LACP. LAG settings takes priority over the port-group settings.

> **Note:**
> Network failure detection values remain as **link status only**, since beacon probing is not supported with LACP. See Broadcom knowledge base article Understanding IP Hash load balancing (2006129).

Configuration 4: Static LACP – Route Based on IP Hash

You can use a two-port LACP static port-channel on a switch, and two active uplinks on a vSphere Standard Switch.

In this configuration, use 10 GbE networking with two physical uplinks per server. A single VMkernel interface (vmknic) for vSAN exists on each host.

For more information about host requirements and configuration examples, see the following Broadcom knowledge base articles:

- Host requirements for link aggregation for ESX (1001938)
- Sample configuration of EtherChannel / Link Aggregation Control Protocol (LACP) with ESX and Cisco/HP switches (KB 1004048)

> **Note:**
>
> vSAN over RDMA does not support this configuration.

Configure the Physical Switch

Configure a two-uplink static port-channel as follows:

- Switch ports 43 and 44
- VLAN trunking, so port-channel is in VLAN trunk mode, with the appropriate VLANs trunked.
- Do not specify the load-balancing policy on the port-channel group.

These steps can be used to configure an individual port-channel on the switch:

Step 1: Create a port-channel.

```
#interface port-channel 13
```

Step 2: Set port-channel to VLAN trunk mode.

```
#switchport mode trunk
```

Step 3: Allow appropriate VLANs.

```
#switchport trunk allowed vlan 3266
```

Step 4: Assign the correct ports to the port-channel and set mode to active.

```
#interface range Te1/0/43, Te1/0/44
#channel-group 1 mode on
```

Step 5: Verify that the port-channel is configured as a static port-channel.

```
#show interfaces port-channel 13


Channel Ports Ch-Type Hash Type Min-links Local Prf
------- --------------------------- -------- --------- --------- --
Po13 Active: Te1/0/43, Te1/0/44  Static   7 1 Disabled
Hash Algorithm Type
1 - Source MAC, VLAN, EtherType, source module and port Id
2 - Destination MAC, VLAN, EtherType, source module and port Id
3 - Source IP and source TCP/UDP port
4 - Destination IP and destination TCP/UDP port
5 - Source/Destination MAC, VLAN, EtherType, source MODID/port
6 - Source/Destination IP and source/destination TCP/UDP port
7 - Enhanced hashing mode
```

Configure vSphere Standard Switch

This example assumes you understand the configuration and creation of vSphere Standard Switches.

This example uses the following configuration:

- Uplinks named vmnic0 and vmnic1
- VLAN 3266 trunked to the switch ports and port-channel
- Jumbo frames

On each host, create a **vSwitch1** with MTU set to 9000, and vmnic0 and vmnic1 added to the vSwitch. On the Teaming and Failover Policy, set both adapters to the **Active** position. Set the Load Balancing Policy to **Route Based on IP Hash**.

Configure teaming and failover for the distributed port group for vSAN traffic as follows:

- Load balancing policy set to **Route Based on IP hash**.
- Network failure detection set to **Link status only**.
- Notify Switches set to **Yes**.
- Failback set to **Yes**.
- Ensure both uplinks are in the **Active uplinks** position.

Use defaults for network detection, Notify Switches and Failback. All port groups inherit the Teaming and Failover Policy that was set at the vSwitch level. You can override individual port group teaming and failover polices to differ from the parent vSwitch, but make sure you use the same set of uplinks for IP hash load balancing for all port groups.

Configure Load Balancing

Although both physical uplinks are utilized, there is not a consistent balance of traffic across all physical vmnics. The figure shows that only active traffic is vSAN traffic, which was essentially four vmknics or IP addresses. The behavior might be caused by the low number of IP addresses and possible hashes. However, in some situations, the virtual switch might consistently pass the traffic through one uplink in the team. For further details on the IP Hash algorithm, see the vSphere Networking guide.

Network Redundancy

In this example, vmnic1 is connected to a port that has been disabled from the switch, to focus on failure and redundancy behavior. Note that a network uplink redundancy alarm has triggered.

No vSAN health alarms were triggered. Cluster and VM components are not affected and Guest Storage I/O is not interrupted by this failure.

Recovery and Failback

Once vmnic1 recovers, traffic is automatically balanced across both active uplinks.

Network I/O Control

Use vSphere Network I/O Control to set Quality of Service (QoS) levels on network traffic.

vSphere Network I/O Control is a feature available with vSphere Distributed Switches. Use it to implement Quality of Service (QoS) on network traffic. This can be useful for vSAN when vSAN traffic must share the physical NIC with other traffic types, such as vMotion, management, and VMs.

Reservations, Shares, and Limits

You can set a **reservation** so that Network I/O Control guarantees minimum bandwidth is available on the physical adapter for vSAN.

Reservations can be useful when *bursty* traffic, such as vMotion or full host evacuation, might impact vSAN traffic. Reservations are only invoked if there is contention for network bandwidth. One disadvantage with reservations in Network I/O Control is that unused reservation bandwidth cannot be allocated to virtual machine traffic. The total bandwidth reserved among all system traffic types cannot exceed 75 percent of the bandwidth provided by the physical network adapter with the lowest capacity.

vSAN **best practices for reservations**. Traffic reserved for vSAN cannot be allocated to virtual machine traffic, so avoid using NIOC reservations in vSAN environments.

Setting **shares** makes a certain bandwidth available to vSAN when the physical adapter assigned for vSAN becomes saturated. This prevents vSAN from consuming the entire capacity of the physical adapter during rebuild and synchronization operations. For example, the physical adapter might become saturated when another physical adapter in the team fails and all traffic in the port group is transferred to the remaining adapters in the team. The **shares** option ensures that no other traffic impacts the vSAN network.

vSAN **recommendation on shares**. This is the fairest bandwidth allocation technique in NIOC, and is preferred for use in vSAN environments.

Setting **limits** defines the maximum bandwidth that a certain traffic type can consume on an adapter. If no one else is using the additional bandwidth, the traffic type with the limit also cannot consume it.

vSAN **recommendation on limits**. As traffic types with limits cannot consume additional bandwidth, avoid using NIOC limits in vSAN environments.

Network Resource Pools

You can view all system traffic types that can be controlled with Network I/O Control. If you have multiple virtual machine networks, you can assign certain bandwidth to virtual machine traffic. Use network resource pools to consume parts of that bandwidth based on the virtual machine port group.

Enabling Network I/O Control

You can enable Network I/O Control in the configuration properties of the vDS. Right-click the vDS in the vSphere Client, and choose menu **Settings > Edit Settings**.

> **Note:**
> Network I/O Control is only available on vSphere distributed switches, not on standard vSwitches.

You can use Network I/O Control to reserve bandwidth for network traffic based on the capacity of the physical adapters on a host. For example, if vSAN traffic uses 10 GbE physical network adapters, and those adapters are shared with other system traffic types, you can use vSphere Network I/O Control to guarantee a certain amount of bandwidth for vSAN. This can be useful when traffic such as vSphere vMotion, vSphere HA, and virtual machine traffic share the same physical NIC as the vSAN network.

Network I/O Control Configuration Example

You can configure Network I/O Control for a vSAN cluster.

Consider a vSAN cluster with a single 10 GbE physical adapter. This NIC handles traffic for vSAN, vSphere vMotion, and VMs. To change the shares value for a traffic type, select that traffic type from the System Traffic view (**VDS > Configure > Resource Allocation > System Traffic**), and click **Edit**. The shares value for vSAN traffic has been changed from the default of Normal/50 to High/100.

Edit the other traffic types to match the share values shown in the table.

Table 997: Sample NIOC Settings

Traffic Type	Shares	Value
vSAN	High	100
vSphere vMotion	Low	25
Virtual machine	Normal	50
iSCSI/NFS	Low	25

For example. if a 10GbE adapter is saturated, Network I/O Control allocates 5 GbE to vSAN on the physical adapter, 3.5 GbE to virtual machine traffic, and 1.5 GbE to vMotion. Use these values as a starting point to configure NIOC configuration on your vSAN network. Ensure that vSAN has the highest priority of any protocol.

For more details about the various parameters for bandwidth allocation, see the vSphere Networking guide.

With each of the vSphere editions for vSAN, VMware provides a vSphere Distributed Switch as part of the edition. Network I/O Control can be configured with any vSAN edition.

Understanding vSAN Network Topologies

vSAN architecture supports different network topologies. These topologies impact on the overall deployment and management of vSAN.

The introduction of unicast support in vSAN simplifies the network design.

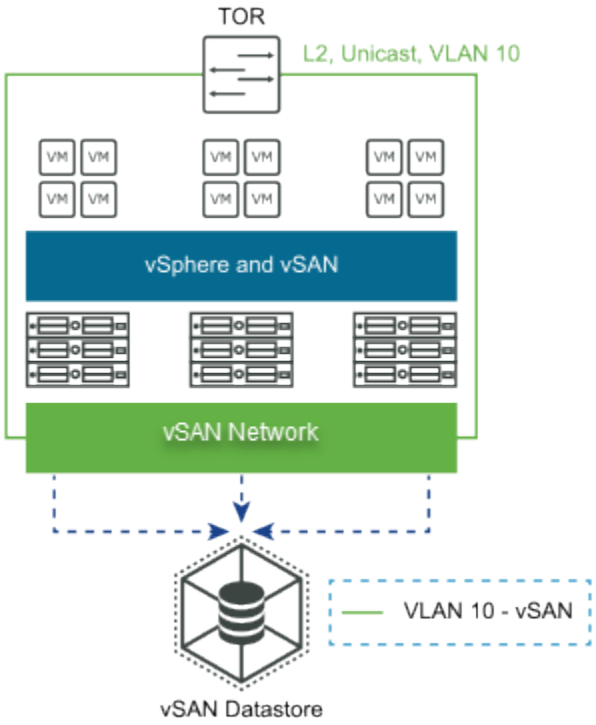
Standard Deployments

vSAN supports several single-site deployment types.

Layer-2, Single Site, Single Rack

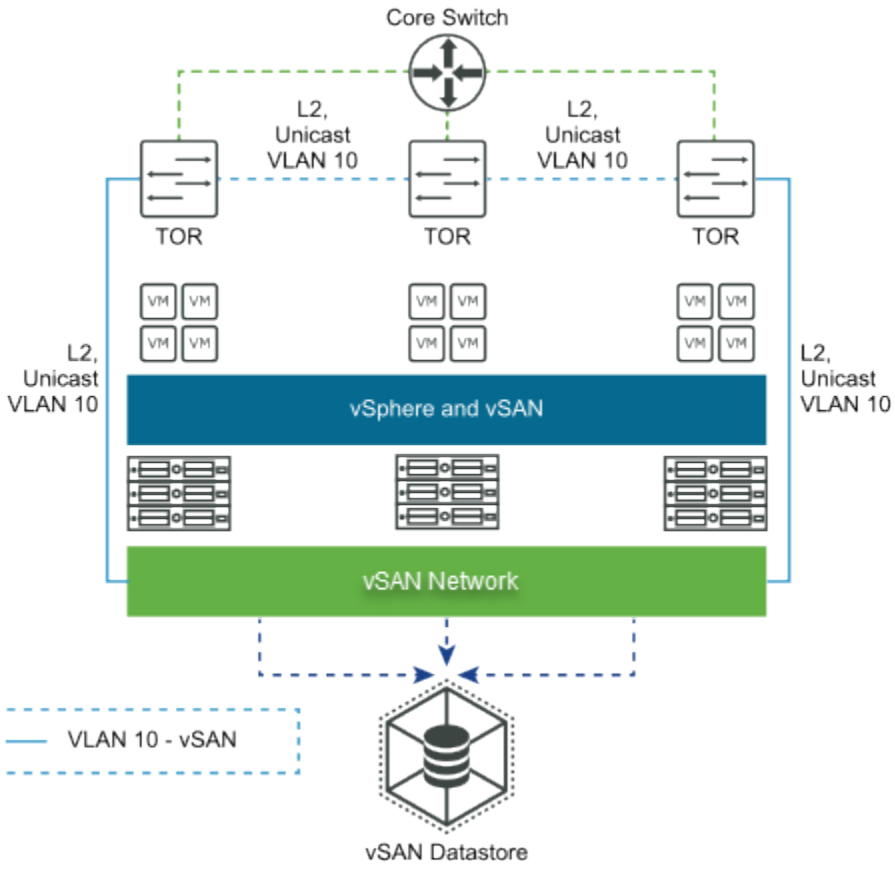
This network topology is responsible for forwarding packets through intermediate Layer 2 devices such as hosts, bridges, or switches.

Layer 2 implementations are simplified even further and introduces unicast support. IGMP Snooping is not required.



Layer 2, Single Site, Multiple Racks

This network topology works with the Layer 2 implementation where there are multiple racks, and multiple top-of-rack switches, or TORs, connected to a core switch.

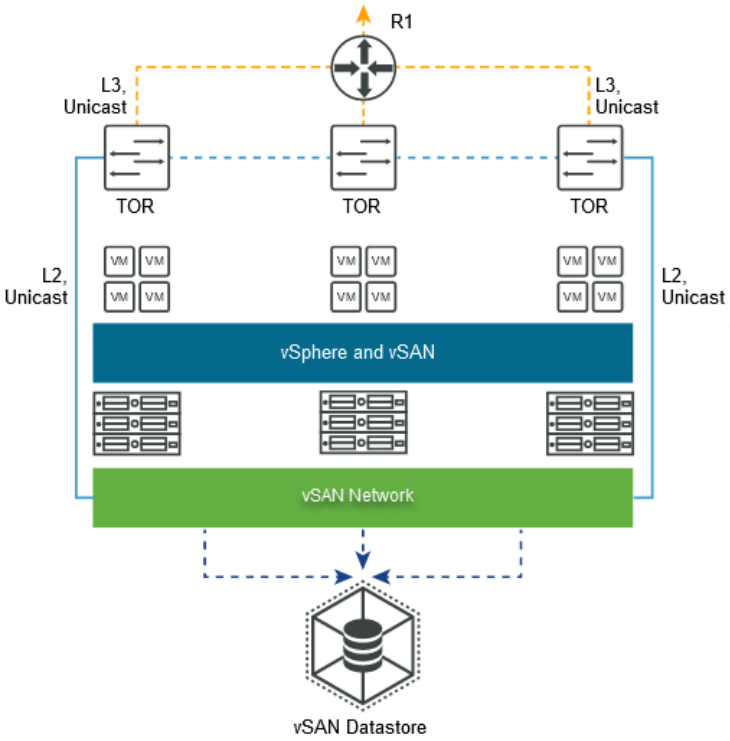


Layer 3, Single Site, Multiple Racks

This network topology works for vSAN deployments where Layer 3 is used to route vSAN traffic.

This simple Layer 3 network topology uses multiple racks in the same data center, each with its own TOR switch. Route the vSAN network between the different racks over L3, to allow all the hosts in the vSAN cluster to communicate. Place the vSAN VMkernel ports on different subnets or VLANs, and use a separate subnet or VLAN for each rack.

This network topology routes packets through intermediate Layer 3 capable devices, such as routers and Layer 3 capable switches. Whenever hosts are deployed across different Layer 3 network segments, the result is a routed network topology.



vSAN Stretched Cluster Deployments

vSAN supports stretched cluster deployments that span two locations.

Supported vSAN Stretched Cluster Configurations

vSAN supports stretched cluster configurations.

The following configuration prevent traffic from Site 1 being routed to Site 2 through the witness host, in the event of a failure on either of the data sites' network. This configuration avoids performance degradation. To ensure that data traffic is not switched through the witness host, use the following network topology.

Between Site 1 and Site 2, implement a stretched Layer 2 switched configuration or a Layer 3 routed configuration. Both configurations are supported. VMware recommends Layer 3 for fault isolation and easier troubleshooting

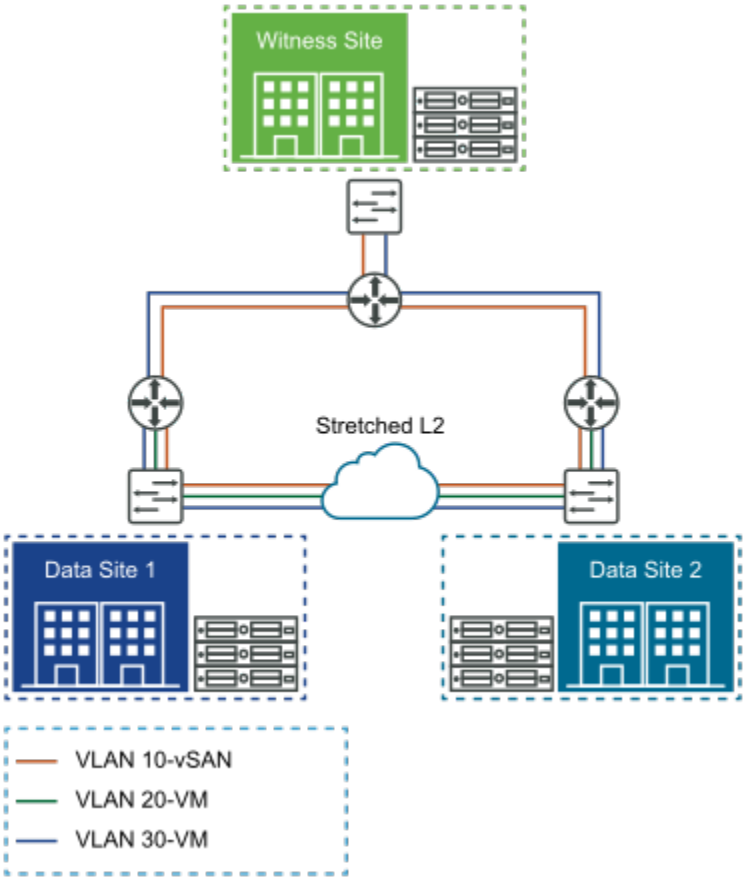
Between Site 1 and the witness host, implement a Layer 3 routed configuration.

Between Site 2 and the witness host, implement a Layer 3 routed configuration.

We shall examine a stretched Layer 2 network between the data sites and a Layer 3 routed network to the witness site. To demonstrate a combination of Layer 2 and Layer 3 as simply as possible, use a combination of switches and routers in the topologies.

Stretched Layer 2 Between Data Sites, Layer 3 to Witness Host

vSAN supports stretched Layer 2 configurations between data sites.

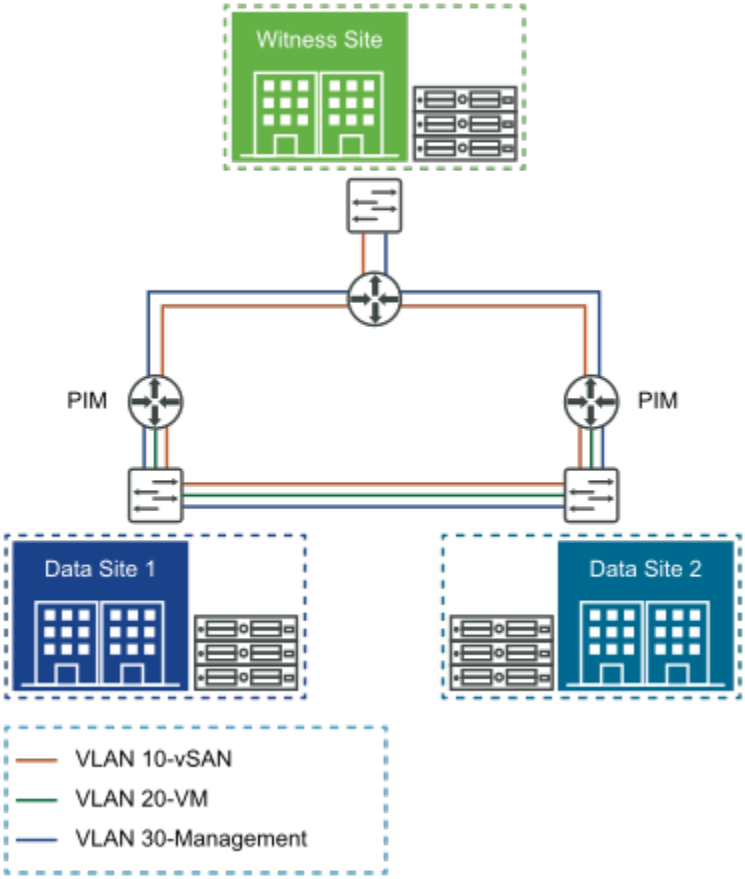


Layer 3 Everywhere

In this vSAN stretched cluster configuration, the data traffic is routed between the data sites and the witness host.

To implement Layer 3 everywhere as simply as possible, use routers or routing switches in the topologies.

vSAN does not require IGMP snooping or PIM because all the routed traffic is unicast.



Note:

You can configure a vSAN stretched cluster in a Layer 2 network, but this configuration is not recommended.

Separating Witness Traffic on vSAN Stretched Clusters

vSAN supports separating witness traffic on stretched clusters.

You can separate witness traffic from vSAN traffic in two-node configurations. This means that the two vSAN hosts can be directly connected without a 10 GbE switch.

This witness traffic separation is only supported on two-node deployments in vSAN. Separating the witness traffic on vSAN stretched clusters is supported in vSAN.

Two Node vSAN Deployments

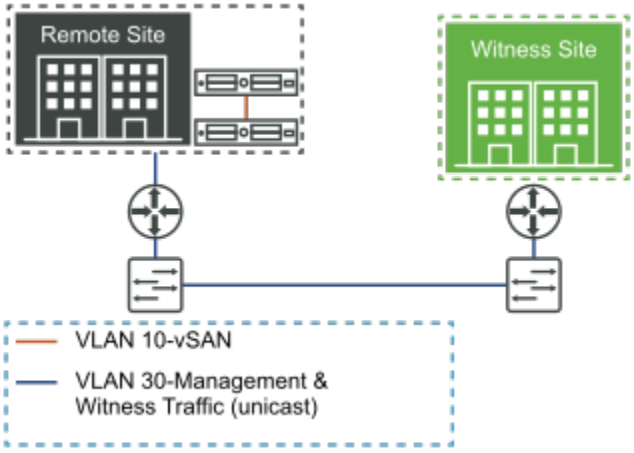
vSAN supports two-node deployments. Two-node vSAN deployments are typically used for remote offices/branch offices (ROBO) that have a small number of workloads, but require high availability.

vSAN two-node deployments use a third witness host, which can be located remotely from the branch office. Often the witness is maintained in the branch office, along with the management components, such as the vCenter.

Two Node Deployments

vSAN supports two-node deployments.

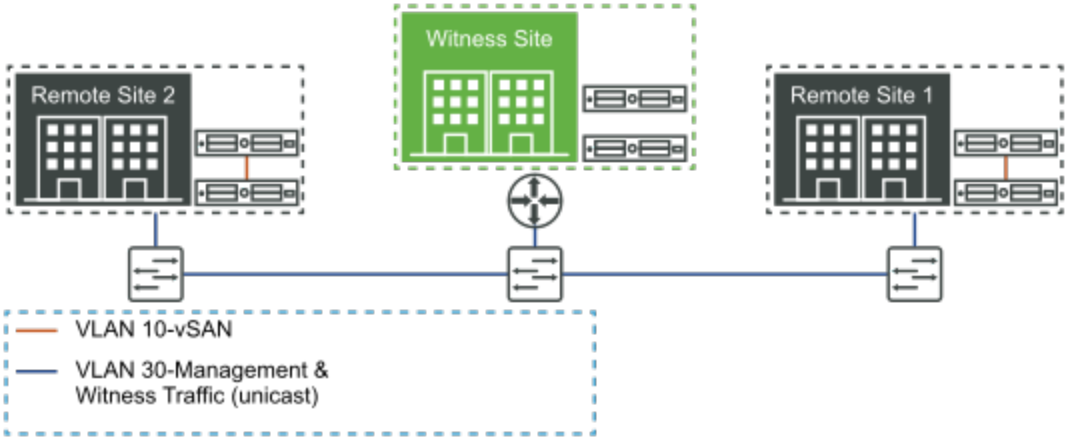
With vSAN, the two-node vSAN implementation is much simpler. vSAN allows the two hosts at the data site to be directly connected.



To enable this functionality, the witness traffic is separated completely from the vSAN data traffic. The vSAN data traffic can flow between the two nodes on the direct connect, while the witness traffic can be routed to the witness site over the management network.

The witness appliance can be located remotely from the branch office. For example, the witness might be running back in the main data center, alongside the management infrastructure (vCenter, vROps, Log Insight, and so on). Another supported place where the witness can reside remotely from the branch office is in vCloud Air.

Multiple remote office/branch office two-node deployments are supported on shared witness.



Common Considerations for Two Node vSAN Deployments

Two-node vSAN deployments provide support to other topologies. This section describes common configurations.

> **Note:**
>
> vSAN recommends a minimum of 1500 MTU between the witness host and data hosts and the MTUs on the witness and data hosts must match.

For more information about two-node configurations and detailed deployment considerations outside of network, see Creating a vSAN Stretched Cluster or Two-Node vSAN Cluster.

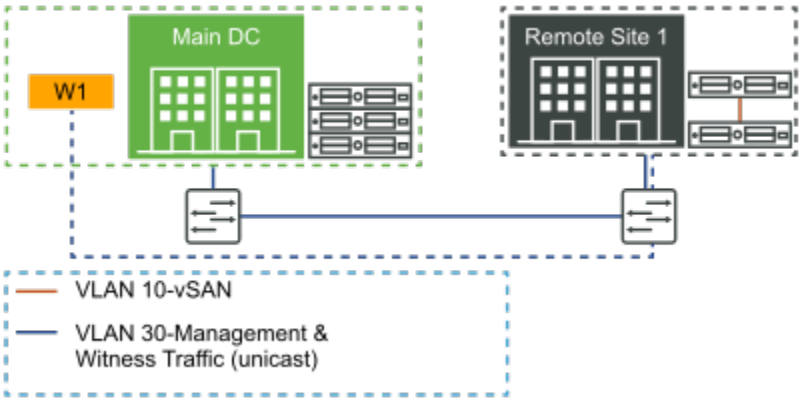
Running the Witness on Another Two Node vSAN Cluster

vSAN does not support cross hosting the witness on another two-node cluster.

Witness Running on Another Standard vSAN Deployment

vSAN supports witness running on another standard vSAN deployment.

This configuration is supported. Any failure on the two-node vSAN at the remote site does not impact the availability of the standard vSAN environment at the main data center.



Configuration of Network from Data Sites to Witness Host

The host interfaces in the data sites communicate to the witness host over the vSAN network. There are different configuration options available.

This topic discusses how to implement these configurations. It addresses how the interfaces on the hosts in the data sites, which communicate to each other over the vSAN network, communicate with the witness host.

Option 1: Physical ESX Witness Connected over L3 with Static Routes

The data sites can be connected over a stretched L2 network. Use this also for the data sites' management network, vSAN network, vMotion network, and virtual machine network.

The physical network router in this network infrastructure does not automatically transfer traffic from the hosts in the data sites (site 1 and site 2) to the host in the witness site (site 3). To configure the vSAN stretched cluster successfully, all hosts in the cluster must communicate. It is possible to deploy a vSAN stretched cluster in this environment.

The solution is to use *static routes* configured on the ESX hosts, so that the vSAN traffic from site1 and site 2 can reach the witness host in site 3. In the case of the ESX hosts on the data sites, add a static route to the vSAN interface, which redirects traffic to the witness host on site 3 over a specified gateway for that network. In the case of the witness host, the vSAN interface must have a static route added, which redirects vSAN traffic destined for the hosts in the data sites. Use the following command to add a static route on each ESX host in the vSAN stretched cluster: `esxcli network ip route ipv4 add -g<gateway>-n <network>`

> **Note:**
>
> The vCenter must be able to manage the ESX hosts at both the data sites and the witness site. As long as there is direct connectivity from the witness host to vCenter, there are no additional concerns regarding the management network.

There is no need to configure a vMotion network or a VM network, or add any static routes for these networks in the context of a vSAN stretched cluster. VMs are never migrated or deployed to the vSAN witness host. Its purpose is to maintain witness objects only, and does not require either of these networks for this task.

Option 2: Virtual ESX Witness Appliance Connected over L3 with Static Routes

Since the witness host is a virtual machine that gets deployed on a physical ESX host, which is not part of the vSAN cluster, that physical ESX host must have a minimum of one VM network pre-configured. This VM network must reach both the management network and the vSAN network shared by the ESX hosts on the data sites.

> **Note:**
> The witness host does not need to be a dedicated host. It can be used for many other VM workloads, while simultaneously hosting the witness.

An alternative option is to have two preconfigured VM networks on the underlying physical ESX host, one for the management network and one for the vSAN network. When the virtual ESX witness is deployed on this physical ESX host, the network needs to be attached and configured accordingly.

Once you have deployed the virtual ESX witness host, configure the static route. Assume that the data sites are connected over a stretched L2 network. Use this also for the data sites' management network, vSAN network, vMotion network, and virtual machine network. vSAN traffic is not routed from the hosts in the data sites (site 1 and site 2) to the host in the witness site (site 3) over the default gateway. To configure the vSAN stretched cluster successfully, all hosts in the cluster require static routes, so that the vSAN traffic from site 1 and site 2 can reach the witness host in site 3. Use the `esxcli network ip route` command to add a static route on each ESX host.

Corner Case Deployments

It is possible to deploy vSAN in unusual, or corner-case configurations.

These unusual topologies require special considerations.

Three Locations and No vSAN Stretched Cluster

You can deploy vSAN across multiple rooms, buildings or sites, rather than deploy a stretched cluster configuration.

This configuration is supported. The one requirement is that the latency between the sites must be at the same level as the latency expected for a normal vSAN deployment in the same data center. The latency must be **<1ms** between all hosts. If latency is greater than this value, consider a vSAN stretched cluster which tolerates latency of 5ms RTT.

For best results, maintain a uniform configuration across all sites in such a topology. To maintain availability of VMs, configure fault domains, where the hosts in each room, building, or site are placed in the same fault domain. Avoid asymmetric partitioning of the cluster, where host A cannot communicate to host B, but host B can communicate to host A.

Two-Node Deployed as 1+1+Witness Stretched Cluster

You can deploy a two-node configuration as a vSAN stretched cluster configuration, placing each host in different rooms, buildings, or sites.

Attempt to increase the number of hosts at each site fail with an error related to licensing. For any cluster that is larger than two hosts and that uses the dedicated witness appliance/host feature (N+N+Witness, where N > 1), the configuration is considered a vSAN stretched cluster.

Troubleshooting the vSAN Network

vSAN allows you to examine and troubleshoot the different types of issues that arise from a misconfigured vSAN network.

vSAN operations depend on the network configuration, reliability, and performance. Many support requests stem from an incorrect network configuration, or the network not performing as expected.

Use the vSAN health service to resolve network issues. Network health checks can direct you to an appropriate knowledge base article, depending on the results of the health check. The knowledge base article provides instructions to solve the network problem.

Network Health Checks

The health service includes a category for networking health checks.

Each health check has an **Ask Broadcom** link. If a health check fails, click **Ask Broadcom** and read the associated Broadcom knowledge base article for further details, and guidance on how to address the issue at hand.

The following networking health checks provide useful information about your vSAN environment.

- vSAN**: Basic (unicast) connectivity check**. This check verifies that IP connectivity exists among all ESX hosts in the vSAN cluster, by pinging each ESX host on the vSAN network from each other ESX host.

- **vMotion: Basic (unicast) connectivity check**. This check verifies that IP connectivity exists among all ESX hosts in the vSAN cluster that have vMotion configured. Each ESX host on the vMotion network pings all other ESX hosts.
- **All hosts have a vSAN vmknic configured**. This check ensures each ESX host in the vSAN cluster has a VMkernel NIC configured for vSAN traffic.
- **All hosts have matching subnets**. This check tests that all ESX hosts in a vSAN cluster have been configured so that all vSAN VMkernel NICs are on the same IP subnet.
- **Hosts disconnected from VC**. This check verifies that the vCenter has an active connection to all ESX hosts in the vSAN cluster.
- **Hosts with connectivity issues**. This check refers to situations where vCenter lists the host as connected, but API calls from vCenter to the host are failing. It can highlight connectivity issues between a host and the vCenter.
- **Network latency**. This check performs a network latency check of vSAN hosts. If the threshold exceeds 5 ms, a warning is displayed.
- **vMotion: MTU checks (ping with large packet size)**. This check complements the basic vMotion ping connectivity check. Maximum Transmission Unit size is increased to improve network performance. Incorrectly configured MTUs might not appear as a network configuration issue, but can cause performance issues.
- **vSAN cluster partition**. This health check examines the cluster to see how many partitions exist. It displays an error if there is more than a single partition in the vSAN cluster.

Commands to Check the Network

When the vSAN network has been configured, use these commands to check its state. You can check which VMkernel Adapter (vmknic) is used for vSAN, and what attributes it contains.

Use ESXCLI to verify that the network is fully functional, and to troubleshoot any network issues with vSAN.

You can verify that the vmknic used for the vSAN network is uniformly configured correctly across all hosts, and verify that hosts participating in the vSAN cluster can successfully communicate with one another.

esxcli vsan network list

This command enables you to identify the VMkernel interface used by the vSAN network.

The output below shows that the vSAN network is using vmk1. This command continues to work even if vSAN has been turned off and the hosts no longer participate in vSAN.

```
[root@esxi-dell-m:~] esxcli vsan network list
Interface
   VmkNic Name: vmk1
   IP Protocol: IP
   Interface UUID: 32efc758-9ca0-57b9-c7e3-246e962c24d0
   Agent Group Multicast Address: 224.2.3.4
   Agent Group IPv6 Multicast Address: ff19::2:3:4
   Agent Group Multicast Port: 23451
   Master Group Multicast Address: 224.1.2.3
   Master Group IPv6 Multicast Address: ff19::1:2:3
   Master Group Multicast Port: 12345
   Host Unicast Channel Bound Port: 12321
   Multicast TTL: 5
   Traffic Type: vsan
```

This provides useful information, such as which VMkernel interface is being used for vSAN traffic. In this case, it is **vmk1**. Port 23451 is used for the heartbeat, sent every second by the primary, and is visible on every other host in the cluster. Port 12345 is used for the CMMDS updates between the primary and backup. vSAN no longer supports multicast for any network communication.

esxcli network ip interface list

This command enables you to verify items such as vSwitch or distributed switch.

Use this command to check which vSwitch or distributed switch that it is attached to, and the MTU size, which can be useful if jumbo frames have been configured in the environment. In this case, MTU is at the default of 1500.

```
[root@esxi-dell-m:~] esxcli network ip interface list
vmk0
   Name: vmk0
   <<truncated>>
vmk1
   Name: vmk1
   MAC Address: 00:50:56:69:96:f0
   Enabled: true
   Portset: DvsPortset-0
   Portgroup: N/A
   Netstack Instance: defaultTcpipStack
   VDS Name: vDS
   VDS UUID: 50 1e 5b ad e3 b4 af 25-18 f3 1c 4c fa 98 3d bb
   VDS Port: 16
   VDS Connection: 1123658315
   Opaque Network ID: N/A
   Opaque Network Type: N/A
   External ID: N/A
   MTU: 9000
   TSO MSS: 65535
   Port ID: 50331814
```

The Maximum Transmission Unit size is shown as 9000, so this VMkernel port is configured for jumbo frames, which require an MTU of about 9,000. VMware does not make any recommendation around the use of jumbo frames. However, jumbo frames are supported for use with vSAN.

esxcli network ip interface ipv4 get –i vmk1

This command displays information such as IP address and netmask of the vSAN VMkernel interface.

With this information, an administrator can now begin to use other commands available at the command line to check that the vSAN network is working correctly.

```
[root@esxi-dell-m:~] esxcli network ip interface ipv4 get -i vmk1
Name  IPv4 Address  IPv4 Netmask   IPv4 Broadcast  Address Type  Gateway  DHCP DNS
----  ------------  -------------  --------------  ------------  -------  --------
vmk1  172.40.0.9   255.255.255.0   172.40.0.255   STATIC         0.0.0.0 false
```

vmkping

The `vmkping` command verifies whether all the other ESX hosts on the network are responding to your ping requests.

```
~ # vmkping -I vmk2 172.32.0.3 -s 1472 -d
 PING 172.32.0.3 (172.32.0.3): 56 data bytes
 64 bytes from 172.32.0.3: icmp_seq=0 ttl=64 time=0.186 ms
 64 bytes from 172.32.0.3: icmp_seq=1 ttl=64 time=2.690 ms
 64 bytes from 172.32.0.3: icmp_seq=2 ttl=64 time=0.139 ms

 --- 172.32.0.3 ping statistics ---
 3 packets transmitted, 3 packets received, 0% packet loss
 round-trip min/avg/max = 0.139/1.005/2.690 ms
```

While it does not verify multicast functionality, it can help identify a rogue ESX host that has network issues. You can also examine the response times to see if there is any abnormal latency on the vSAN network.

If jumbo frames are configured, this command does not report any issues if the jumbo frame MTU size is incorrect. By default, this command uses an MTU size of 1500. If there is a need to verify if jumbo frames are successfully working end-to-end, use vmkping with a larger packet size (-s) option as follows:

```
~ # vmkping -I vmk2 172.32.0.3 -s 8972 -d
PING 172.32.0.3 (172.32.0.3): 8972 data bytes
9008 bytes from 172.32.0.3: icmp_seq=0 ttl=64 time=0.554 ms
9008 bytes from 172.32.0.3: icmp_seq=1 ttl=64 time=0.638 ms
9008 bytes from 172.32.0.3: icmp_seq=2 ttl=64 time=0.533 ms

--- 172.32.0.3 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.533/0.575/0.638 ms
~ #
```

Consider adding -d to the vmkping command to test if packets can be sent without fragmentation.

esxcli network ip neighbor list

This command helps to verify if all vSAN hosts are on the same network segment.

In this configuration, we have a four-host cluster, and this command returns the ARP (Address Resolution Protocol) entries of the other three hosts, including their IP addresses and their vmknic (vSAN is configured to use vmk1 on all hosts in this cluster).

```
[root@esxi-dell-m:~] esxcli network ip neighbor list -i vmk1
Neighbor     Mac Address        Vmknic   Expiry  State  Type
-----------  -----------------  ------  -------  -----  -------
172.40.0.12  00:50:56:61:ce:22  vmk1    164 sec         Unknown
172.40.0.10  00:50:56:67:1d:b2  vmk1    338 sec         Unknown
172.40.0.11  00:50:56:6c:fe:c5  vmk1    162 sec         Unknown
[root@esxi-dell-m:~]
```

esxcli network diag ping

This command checks for duplicates on the network, and round-trip times.

To get even more detail regarding the vSAN network connectivity between the various hosts, ESXCLI provides a powerful network diagnostic command. Here is an example of one such output, where the VMkernel interface is on vmk1 and the remote vSAN network IP of another host on the network is 172.40.0.10

```
[root@esxi-dell-m:~] esxcli network diag ping -I vmk1 -H 172.40.0.10
   Trace:
         Received Bytes: 64
         Host: 172.40.0.10
         ICMP Seq: 0
         TTL: 64
         Round-trip Time: 1864 us
         Dup: false
         Detail:

         Received Bytes: 64
         Host: 172.40.0.10
         ICMP Seq: 1
         TTL: 64
         Round-trip Time: 1834 us
         Dup: false
         Detail:

         Received Bytes: 64
         Host: 172.40.0.10
         ICMP Seq: 2
         TTL: 64
         Round-trip Time: 1824 us
```

```
        Dup: false
        Detail:
    Summary:
        Host Addr: 172.40.0.10
        Transmitted: 3
        Recieved: 3
        Duplicated: 0
        Packet Lost: 0
        Round-trip Min: 1824 us
        Round-trip Avg: 1840 us
        Round-trip Max: 1864 us
[root@esxi-dell-m:~]
```

Checking vSAN Network Performance

Make that there is sufficient bandwidth between your ESX hosts. This tool can assist you in testing whether your vSAN network is performing optimally.

To check the performance of the vSAN network, you can use `iperf` tool to measure maximum TCP bandwidth and latency. It is located in `/usr/lib/vmware/vsan/bin/iperf.copy`. Run it with `--help` to see the various options. Use this tool to check network bandwidth and latency between ESX hosts participating in a vSAN cluster.

Broadcom knowledge base article 2001003 can assist with setup and testing.

This is most useful when a vSAN cluster is being commissioned. Running **iperf** tests on the vSAN network when the cluster is already in production can impact the performance of the VMs running on the cluster.

Checking vSAN Network Limits

The `vsan.check.limits` command verifies that none of the vSAN thresholds are being breached.

```
> ls
0 /
1 vcsa-04.rainpole.com/
> cd 1
/vcsa-04.rainpole.com> ls
0 Datacenter (datacenter)
/vcsa-04.rainpole.com> cd 0
/vcsa-04.rainpole.com/Datacenter> ls
0 storage/
1 computers [host]/
2 networks [network]/
3 datastores [datastore]/
4 vms [vm]/
/vcsa-04.rainpole.com/Datacenter> cd 1
/vcsa-04.rainpole.com/Datacenter/computers> ls
0 Cluster (cluster): cpu 155 GHz, memory 400 GbE
1 esxi-dell-e.rainpole.com (standalone): cpu 38 GHz, memory 123 GbE
2 esxi-dell-f.rainpole.com (standalone): cpu 38 GHz, memory 123 GbE
3 esxi-dell-g.rainpole.com (standalone): cpu 38 GHz, memory 123 GbE
4 esxi-dell-h.rainpole.com (standalone): cpu 38 GHz, memory 123 GbE
/vcsa-04.rainpole.com/Datacenter/computers> vsan.check_limits 0
2017-03-14 16:09:32 +0000: Querying limit stats from all hosts ...
2017-03-14 16:09:34 +0000: Fetching vSAN disk info from esxi-dell-m.rainpole.com (may take a moment) ...
2017-03-14 16:09:34 +0000: Fetching vSAN disk info from esxi-dell-n.rainpole.com (may take a moment) ...
2017-03-14 16:09:34 +0000: Fetching vSAN disk info from esxi-dell-o.rainpole.com (may take a moment) ...
2017-03-14 16:09:34 +0000: Fetching vSAN disk info from esxi-dell-p.rainpole.com (may take a moment) ...
2017-03-14 16:09:39 +0000: Done fetching vSAN disk infos
+-------------------------+--------------------
+-----------------------------------------------------------------+
```

```
| Host                     | RDT               | Disks
     |
+--------------------------+-------------------
+-------------------------------------------------------------------+
| esxi-dell-m.rainpole.com | Assocs: 1309/45000 | Components: 485/9000
     |
|                          | Sockets: 89/10000  | naa.500a075113019b33: 0% Components: 0/0
     |
|                          | Clients: 136       | naa.500a075113019b37: 40% Components: 81/47661
     |
|                          | Owners: 138        | t10.ATA_____Micron_P420m2DMTFDGAR1T4MAX_____ 0% Components:
 0/0 |
|                          |                    | naa.500a075113019b41: 37% Components: 80/47661
     |
|                          |                    | naa.500a07511301a1eb: 38% Components: 81/47661
     |
|                          |                    | naa.500a075113019b39: 39% Components: 79/47661
     |
|                          |                    | naa.500a07511301a1ec: 41% Components: 79/47661
     |
<<truncated>>
```

From a network perspective, it is the RDT associations (Assocs) and sockets count that are important. There are 45,000 associations per host in vSAN 6.0 and later. An RDT association is used to track peer-to-peer network state within vSAN. vSAN is sized so that it never runs out of RDT associations. vSAN also limits how many TCP sockets it is allowed to use, and vSAN is sized so that it never runs out of its allocation of TCP sockets. There is a limit of 10,000 sockets per host.

A **vSAN client** represents object's access in the vSAN cluster. The client typically represents a virtual machine running on a host. The client and the object might not be on the same host. There is no hard defined limit, but this metric is shown to help understand how clients balance across hosts.

There is only one **vSAN owner** for a given vSAN object, typically co-located with the vSAN client accessing this object. vSAN owners coordinate all access to the vSAN object and implement functionality, such as mirroring and striping. There is no hard defined limit, but this metric is once again shown to help understand how owners balance across hosts.

Networking Considerations for vSAN File Service

vSAN File Service is a layer that sits on top of vSAN to provide file shares. It currently supports SMB, NFSv3, and NFSv4.1 file shares.

Following are the network considerations for vSAN File Service:

- You must allocate static IP addresses as file server IPs from vSAN File Service network, each IP is the access point to vSAN file shares.
  - For best performance, the number of IP addresses must be equal to the number of hosts in the vSAN cluster.
  - All the static IP addresses should be from the same subnet.
  - Every static IP address has a corresponding FQDN, which should be part of the Forward lookup and Reverse lookup zones in the DNS server.
- You must ensure to prepare the network as vSAN File Service network:
  - If using standard switch based network, the Promiscuous Mode and Forged Transmits are enabled as part of the vSAN File Services enablement process.
  - If using DVS based network, vSAN File Services are supported on DVS. Create a dedicated port group for vSAN File Services in the DVS. MacLearning and Forged Transmits are enabled as part of the vSAN File Services enablement process for a provided DVS port group.

Note:
If using NSX-based network, ensure that MacLearning is enabled for the provided network entity from the NSX admin console, and all the hosts and File Services nodes are connected to the desired NSX-T network.

- For SMB share and NFS share with Kerberos security, you must provide information about your AD domain and organizational unit (optional). In addition, a user account with sufficient privileges to create and delete objects is required.
- Ensure that the file server can access AD server and DNS server. The file server must be able to access all the ports required by AD service.
  Following are the ports that vSAN File Service uses for network connectivity. Ensure that these ports are not blocked by the firewall.
  For more information on ports and protocols for vSAN File Service, see ports.broadcom.com.

Networking Considerations for iSCSI on vSAN

vSAN iSCSI target service allows hosts and physical workloads that reside outside the vSAN cluster to access the vSAN datastore. This feature enables an iSCSI initiator on a remote host to transport block-level data to an iSCSI target on a storage device within the vSAN cluster.

The iSCSI targets on vSAN are managed using Storage Policy Based Management (SPBM) similar to other vSAN objects. For the iSCSI LUNs, this space savings the space through deduplication and compression, and provides security through encryption. For enhanced security, vSAN iSCSI target service uses Challenge Handshake Authentication Protocol (CHAP) and Mutual CHAP authentication.

vSAN identifies each iSCSI target by a unique iSCSI qualified Name (IQN). The iSCSI target is presented to a remote iSCSI initiator using the IQN, so that the initiator can access the LUN of the target. vSAN iSCSI target service allows creating iSCSI initiator groups. The iSCSI initiator group restricts access to only those initiators that are members of the group.

Characteristics of vSAN iSCSI Network

Following are the characteristics of a vSAN iSCSI network:

- iSCSI Routing - iSCSI initiators can make routed connections to vSAN iSCSI targets over an L3 network.
- IPv4 and IPv6 - vSAN iSCSI network supports both IPv4 and IPv6.
- IP Security - IPSec on the vSAN iSCSI network provides increased security.

  Note:

  ESX hosts support IPsec using IPv6 only.
- Jumbo Frames - Jumbo Frames are supported on the vSAN iSCSI network.
- NIC Teaming - All NIC teaming configurations are supported on the vSAN iSCSI network.
- Multiple Connections per Session (MCS) - vSAN iSCSI implementation does not support MCS.

Migrating from Standard to Distributed vSwitch

You can migrate from a vSphere Standard Switch to a vSphere Distributed Switch, and use Network I/O Control. This enables you to prioritize the QoS (Quality of Service) on vSAN traffic.

Warning:

It is best to have access to the ESX hosts, although you might not need it. If something goes wrong, you can access the console of the ESX hosts.

Make a note of the original vSwitch setup. In particular, note the load-balancing and NIC teaming settings on the source. Make sure the destination configuration matches the source.

Create a Distributed Switch

Create the distributed vSwitch and give it a name.

1. In the vSphere Client Host and Clusters view, right-click a data center and select menu **New Distributed Switch**.
2. Enter a name.
3. Select the version of the vSphere Distributed Switch.
4. Add the settings. Determine how many uplinks you are currently using for networking. This example has six: management, vMotion, VMs, and three for vSAN (a LAG configuration). Enter 6 for the number of uplinks. Your environment might be different, but you can edit it later.
   You can create a default port group at this point, but additional port groups are needed.
5. Finish the configuration of the distributed vSwitch.

The next step is to configure and create the additional port groups.

Create Port Groups

A single default port group was created for the management network. Edit this port group to make sure it has all the characteristics of the management port group on the standard vSwitch, such as VLAN and NIC teaming, and failover settings.

Configure the management port group.

1. In the vSphere Client Networking view, select the distributed port group, and click **Edit**.
2. For some port groups, you must change the VLAN. Since VLAN 51 is the management VLAN, tag the distributed port group accordingly.
3. Click **OK**.

Create distributed port groups for vMotion, virtual machine networking, and vSAN networking.

1. Right-click the vSphere Distributed Switch and select menu **Distributed Port Group > New Distributed Port Group**.
2. For this example, create a port group for the vMotion network.

Create all the distributed port groups on the distributed vSwitch. Then migrate the uplinks, VMkernel networking, and virtual machine networking to the distributed vSwitch and associated distributed port groups.

> **Warning:**
> Migrate the uplinks and networks in step-by-step fashion to proceed smoothly and with caution.

Migrate Management Network

Migrate the management network (vmk0) and its associated uplink (vmnic0) from the standard vSwitch to the distributed vSwitch (vDS).

1. Add hosts to the vDS.
   a. Right-click the vDS and select menu **Add and Manage Hosts**.
   b. Add hosts to the vDS. Click the green Add icon (+), and add all hosts from the cluster.
2. Configure the physical adapters and VMkernel adapters.
   a. Click **Manage physical adapters** to migrate the physical adapters and VMkernel adapters, vmnic0 and vmk0 to the vDS.
   b. Select an appropriate uplink on the vDS for physical adapter vmnic0. For this example, use Uplink1. The physical adapter is selected and an uplink is chosen.
3. Migrate the management network on vmk0 from the standard vSwitch to the distributed vSwitch. Perform these steps on each host.
   a. Select vmk0, and click **Assign port group**.
   b. Assign the distributed port group created for the management network earlier.
4. Finish the configuration.
   a. Review the changes to ensure that you are adding four hosts, four uplinks (vmnic0 from each host), and four VMkernel adapters (vmk0 from each host).
   b. Click **Finish**.

When you examine the networking configuration of each host, review the switch settings, with one uplink (vmnic0) and the vmk0 management port on each host.

Repeat this process for the other networks.

Migrate vMotion

To migrate the vMotion network, use the same steps used for the management network.

Before you begin, ensure that the distributed port group for the vMotion network has the same attributes as the port group on the standard vSwitch. Then migrate the uplink used for vMotion (vmnic1), with the VMkernel adapter (vmk1).

Migrate vSAN Network

If you have a single uplink for the vSAN network, then use the same process as before. However, if you are using more than one uplink, there are additional steps.

If the vSAN network is using Link Aggregation (LACP), or it is on a different VLAN to the other VMkernel networks, place some of the uplinks into an unused state for certain VMkernel adapters.

For example, VMkernel adapter vmk2 is used for vSAN. However, uplinks vmnic3, 4 and 5 are used for vSAN and they are in a LACP configuration. Therefore, for vmk2, all other vmnics (0, 1 and 2) must be placed in an unused state. Similarly, for the management adapter (vmk0) and vMotion adapter (vmk0), place the vSAN uplinks/vmnics in an unused state.

Modify the settings of the distributed port group and change the path policy and failover settings. On the **Manage physical network adapter** page, perform the steps for multiple adapters.

Assign the vSAN VMkernel adapter (vmk2) to the distributed port group for vSAN.

> **Note:**
> If you are only now migrating the uplinks for the vSAN network, you might not be able to change the distributed port group settings until after the migration. During this time, vSAN might have communication issues. After the migration, move to the distributed port group settings and make any policy changes and mark any uplinks to be unused. vSAN networking then returns to normal when this task is finished. Use the vSAN health service to verify that everything is functional.

Migrate VM Network

The final task needed to migrate the network from a standard vSwitch to a distributed vSwitch is to migrate the VM network.

Manage host networking.

1. Right-click the vDS and choose menu **Add and Manage Hosts**.
2. Select all the hosts in the cluster, to migrate virtual machine networking for all hosts to the distributed vSwitch.
   Do not move any uplinks. However, if the VM networking on your hosts used a different uplink, then migrate the uplink from the standard vSwitch.
3. Select the VMs to migrate from a virtual machine network on the standard vSwitch to the virtual machine distributed port group on the distributed vSwitch. Click **Assign port group**, and select the distributed port group.
4. Review the changes and click **Finish**. In this example, you are moving to VMs. Any templates using the original standard vSwitch virtual machine network must be converted to VMs, and edited. The new distributed port group for VMs must be selected as the network. This step cannot be achieved through the migration wizard.

Since the standard vSwitch no longer has any uplinks or port groups, it can be safely removed.

This completes the migration from a vSphere Standard Switch to a vSphere Distributed Switch.

Checklist Summary for vSAN Network

Use the checklist summary to verify your vSAN network requirements.

- Check if you use shared 10 GbE NIC or dedicated 1GbE NIC. All-flash clusters require 10 GbE NICs.
- Verify that redundant NIC teaming connections are configured.
- Verify that flow control is enabled on the ESX host NICs.

- Verify that VMkernel port for vSAN network traffic is configured on each host.
- Verify that you have identical VLAN, MTU and subnet across all interfaces.
- Verify that you can run `vmkping` successfully between all hosts. Use the health service to verify.
- If you use jumbo frames, verify that you can run `vmkping` successfully with 9000 packet size between all hosts. Use the health service to verify.
- Ensure that the physical switch can meet vSAN requirements (flow control and feature interoperability).
- Verify that the network does not have performance issues, such as excessive dropped packets or pause frames.
- Verify that network limits are within acceptable margins.
- Test vSAN network performance with `iperf`, and verify that it meets expectations.

Planning and Configuring vSAN

Planning and Configuring vSAN describes how to design and configure a VMware® vSAN™ cluster in a vSphere environment, including system requirements, sizing guidelines, and suggested best practices.

Intended Audience

This planning and configuring guide is intended for anyone who wants to deploy a vSAN cluster in a VMware vSphere environment. This information is written for experienced system administrators who are familiar with virtual machine (VM) technology and virtual datacenter operations. This manual assumes familiarity with VMware vSphere, including VMware ESX, vCenter, and the vSphere Client.

- For more information about network requirements and network design, see the Designing vSAN Network guide.
- For more information about vSAN features and how to configure a vSAN cluster, see the Administering VMware vSAN guide.
- For more information about monitoring a vSAN cluster and fixing problems, see the Monitoring and Troubleshooting vSAN guide.

What is vSAN

VMware vSAN is a software-defined, enterprise storage solution that supports hyper-converged infrastructure (HCI) systems.

vSAN aggregates local or direct-attached capacity devices of all ESX hosts in a cluster and creates a single storage pool shared across all ESX hosts in the vSAN cluster.

While supporting vSphere features that require shared storage, such as High Availability (HA), vMotion, and Distributed Resource Scheduler (DRS), vSAN eliminates the need for external shared storage and simplifies storage configuration and VM provisioning activities.

vSAN Concepts

VMware vSAN is an easy to manage, object-based enterprise software-distributed storage (SDS) system that leverages the capabilities of modern hardware.

vSAN pools disk space from multiple ESX hosts in a cluster to create a single shared datastore. You can configure vSAN to work as either a hybrid or an all-flash cluster. In hybrid clusters, flash devices are used for the cache layer and magnetic disks are used for the storage capacity layer. In all-flash clusters, flash devices are used for both cache and capacity.

You can enable vSAN on existing vSphere clusters, or when you create a new cluster. vSAN aggregates the storage capacity of eligible devices from multiple ESX hosts in a cluster into a single, shared datastore.

You can expand capacity in vSAN by doing the following:

- **Expand storage capacity only**: Add storage devices to existing ESX hosts in a vSAN cluster.
- **Expand compute capacity only**: Add new ESX hosts without local storage devices to an existing vSAN cluster.
- **Expand compute and storage capacity**: Add new ESX hosts with local storage to an existing vSAN cluster.

vSAN works best when all ESX hosts in the vSphere cluster share similar or identical configurations across all cluster members, including similar or identical storage configurations. ESX hosts without any local devices also can participate and run their VMs on the vSAN datastore.

With the availability of vSAN Express Storage Architecture (ESA), all storage devices claimed by vSAN contribute to capacity and performance. Each ESX host's storage devices claimed by vSAN form a storage pool. The storage pool represents the amount of caching and capacity provided by the ESX host to the vSAN datastore.

For vSAN Original Storage Architecture (OSA), each ESX host that contributes storage devices to the vSAN datastore must provide at least one device for flash cache and at least one device for capacity. The devices on the contributing ESX host form one or more disk groups. Each disk group contains one flash cache device, and one or more capacity devices for persistent storage. Each ESX host can be configured to use multiple disk groups.

vSAN ESA also provides performance, efficiency, and scalability improvements. The hardware requirements for vSAN ESA are different than vSAN OSA. For more information on vSAN ESA and vSAN OSA, see Comparing vSAN OSA to vSAN ESA.

vSAN storage cluster is a distributed storage system that provides vSAN ESA capabilities while functioning as a storage-only cluster. It pools storage devices across dedicated ESX hosts, separating storage resources from compute hosts. For more information, see Introducing vSAN Storage Cluster.

For best practices, capacity considerations, and general recommendations about designing and sizing a vSAN cluster, see the VMware vSAN Design Guide.

Characteristics of vSAN

The following characteristics apply to vSAN, its clusters, and datastores.

vSAN includes features to add resiliency and efficiency to your data computing and storage environment.

Table 998: vSAN Features

Supported Features	Description
Shared storage support	vSAN supports vSphere features that require shared storage, such as HA, vMotion, and DRS. For example, if a ESX host becomes overloaded, DRS can migrate VMs to other ESX hosts in the cluster.
On-disk format	vSAN on-disk virtual file format provides highly scalable snapshot and clone management support per vSAN ESA cluster. For information about the number of VM snapshots and clones supported per vSAN cluster, see the vSphere Configuration Maximums guide.
All-flash configurations	vSAN can be configured for all-flash clusters.
Fault domains	vSAN supports configuring fault domains or rack awareness to protect ESX hosts from rack or chassis failures when the vSAN cluster spans across multiple racks or blade server chassis in a data center.
File service	The vSAN file service enables you to create file shares in the vSAN datastore that client workstations or VMs can access.
iSCSI target service	The vSAN iSCSI target service enables ESX hosts and physical workloads that reside outside the vSAN cluster to access the vSAN datastore.
vSAN Stretched clusters	vSAN supports stretched clusters that span across two availability zones.
Remote Branch Office (ROBO) or Two node vSAN clusters	vSAN supports two node clusters that consists of two ESX hosts and a witness host. This configuration is ideal for environments with a limited number of workloads, such as remote or branch offices.

Supported Features	Description
Support for Windows Server Failover Clusters (WSFC)	vSAN supports SCSI-3 Persistent Reservations (SCSI-3 PR) on a virtual disk level required by WSFC to arbitrate access to a shared disk between nodes. Support of SCSI-3 PRs enables configuration of WSFC with a disk resource shared between VMs natively on vSAN datastores. Currently the following configurations are supported: • Up to 6 application nodes per cluster. • Up to 64 shared virtual disks per node. For more information on the supported versions, see Windows Server Release Information.
vSAN health service	The vSAN health service includes preconfigured health check tests to monitor, troubleshoot, and diagnose the cause of cluster component problems, and identify any potential risk.
vSAN performance service	The vSAN performance service includes statistical charts used to monitor IOPS, throughput, latency, and congestion. You can monitor the performance of a vSAN cluster, ESX host, disk group, disk, and VMs.
Integration with vSphere storage features	vSAN integrates with vSphere data management features traditionally used with VMFS and NFS storage. These features include snapshots, linked clones, and vSphere Replication.
Storage Policy Based Management for Virtual Machine Storage	vSAN works with VM storage policies that control the type of storage provided for the VM, the placement of the VM within storage, and the data services the VM uses. If you do not assign a storage policy to the VM during deployment, the vSAN datastore default storage policy or vSAN Default Storage policy is automatically assigned to the VM.
Rapid provisioning	vSAN rapid provisioning simplifies storage management by enabling rapid storage creation and deployment within vCenter, as part of VM creation and deployment. vSAN through its distributed shared storage, streamlines storage allocation for VMs, allowing storage provisioning without the need for manual allocation or pre-provisioning of storage. For more information, see vSAN Space Efficiency Technologies.
Space Efficency: Compression	vSAN ESA enables compression by default. The data compression occurs at the top of the vSAN stack. By handling compression at this level, vSAN ESA eliminates the need to compress the data on other ESX hosts with object copies. This reduces the amount of data transmitted over the network. You can use the Compression-only setting for vSAN OSA clusters with demanding workloads that cannot take advantage of deduplication techniques.
Space Efficiency: Deduplication	When using vSAN OSA, you can enable deduplication and compression. Deduplication is enabled at the cluster level and the data is destaged to the capacity tier. The data in the disk group initially goes through deduplication followed by compression. In vSAN ESA, you can enable deduplication so that the data is deduplicated across the entire cluster.
Space Efficently: Trim and Unmap	vSAN has full awareness of TRIM/UNMAP commands sent from the guest operating system and can reclaim the previously allocated storage as free space. You can enable Trim and Unmap on vSAN OSA. Trim and Unmap is enabled by default on vSAN ESA.

Supported Features	Description
Data-at-rest encryption	vSAN provides data-at-rest encryption. In vSAN ESA, the data is encrypted in the upper layers of vSAN and it receives incoming writes. The data goes through compression before it is encrypted resulting in lower overhead or latency. In vSAN OSA, the data is encrypted after all other processing, such as deduplication, is performed. Data-at-rest encryption protects data on storage devices, in case a device is removed from the cluster. **Note:** Ensure that you have a key management server to perform data-at-rest encryption.
Data-in-transit encryption	vSAN provides data-in-transit encryption across ESX hosts in the cluster. When you enable data-in-transit encryption, vSAN encrypts all data and metadata traffic between ESX hosts. vSAN OSA and vSAN ESA support multiple key rotation options, and uses AES-256 bit encryption for data-in-transit. **Note:** You do not need a key management server to perform data-in-transit encryption.
Cloud native storage	Cloud native storage integrates storage directly into the VMware vSphere platform enabling VMs and containers to use shared storage.
Container storage interface	The Container storage interface (CSI) is an open standard that enables container orchestration platforms like Kubernetes. It manages storage resources through a consistent plugin-based architecture.
Data persistence platform	The vSAN Data Persistence platform provides a framework for software technology partners to integrate with VMware infrastructure. Each partner must develop their own plug-in for VMware customers to receive the benefits of the vSAN Data Persistence platform.
Disaggregated HCI	Disaggregated HCI in vSAN decouples compute and storage resources, allowing compute and storage resources to scale independently within a VMware environment. It enables the use of dedicated storage-only clusters.
vSAN storage clusters	vSAN storage clusters are groups of ESX hosts that pool their local storage devices to create a shared, distributed datastore managed by VMware vSAN.
RAID5/6	RAID 5/6 in vSAN OSA provides data protection through erasure coding. This allows effecient storage utilization while maintaining fault tolerance. • RAID 5: Protects against one failure. The data is split into three data block and one parity block. It requires a minimum of four hosts. • RAID 6: Protects against two simultaneous failures. The data is split into four data blocks and two parity blocks. It requires a minimum of six hosts. For vSAN ESA, the host requirement for RAID 5 or 6 depends on the cluster size.
SDK support	The VMware vSAN SDK is an extension of the VMware vSphere Management SDK. It includes documentation, libraries and code examples that help developers automate installation, configuration, monitoring, and troubleshooting of vSAN.

For more information, see vSAN Feature Matrix.

vSAN Terms and Definitions

vSAN introduces specific terms and definitions that are important to understand.

Before you get started with vSAN, review these key vSAN terms and definitions.

Storage Pool (vSAN Express Storage Architecture)

vSAN ESA replaces disk groups with a new logical construct called a storage pool. It organizes eligible NVMe devices into a single storage pool that handles both caching and capacity, eliminating the need for separate cache and capacity tiers. This single-tier design improves performance and simplifies management.

Disk Group (vSAN Original Storage Architecture)

A disk group is a logical construct used to manage the relationship between capacity devices and their cache tier.

- Each ESX host that contributes storage in a vSAN cluster must have at least one disk group.
- A disk group contains one cache device and between one to seven capacity devices.
- An ESX host can have up to five disk groups.
- Each disk group can include up to seven capacity devices, allowing a maximum of 35 capacity devices per host.
- The cache device must be a flash device in both hybrid and all-flash configurations.

For information about creating and managing disk groups, see Create a Disk Group or Storage Pool in vSAN Cluster.

Consumed Capacity

Consumed capacity is the amount of physical capacity consumed by one or more VMs at any point in time. Many factors determine consumed capacity, including the consumed size of your `.vmdk` files, protection replicas, and more. When calculating for cache sizing, do **not** consider the capacity used for protection replicas.

Object-Based Storage

vSAN stores and manages data in the form of flexible data containers called objects. An object is a logical volume that has its data and metadata distributed across the cluster. VMDKs, VM home namespace, VM swap areas, snapshot delta disks, and snapshot memory maps are all examples of storage objects in vSAN. When you provision a VM on a vSAN datastore, vSAN creates a set of objects comprised of multiple components for each virtual disk or any other data types being stored. It also creates the VM home namespace, which is a container object that stores all metadata files of your VM. Based on the assigned VM storage policy, vSAN provisions and manages each object individually, which might also involve creating a RAID configuration for every object.

> **Note:**
>
> If vSAN ESA is enabled, all native snapshots of a virtual disk are contained in the same vSAN object for the virtual disk.

When vSAN creates an object for a virtual disk and determines how to distribute the object in the cluster, it considers the following factors:

- vSAN verifies that the virtual disk requirements are applied according to the specified VM storage policy settings.
- vSAN verifies that the correct cluster resources are used at the time of provisioning. For example, based on the protection policy, vSAN determines how many replicas to create. The performance policy determines the amount of flash read cache allocated for each replica and how many stripes to create for each replica and where to place them in the cluster.
- vSAN continually monitors and reports the policy compliance status of the virtual disk. If you find any noncompliant policy status, you must troubleshoot and resolve the underlying problem.

  > **Note:**
  >
  > When required, you can edit VM storage policy settings. Changing the storage policy settings does not affect VM access. vSAN actively throttles the storage and network resources used for reconfiguration to minimize the impact of object reconfiguration to normal workloads. When you change VM storage policy settings, vSAN might initiate an object recreation process and subsequent resynchronization. For more information, see About vSAN Cluster Resynchronization.
- vSAN verifies that the required protection components, such as mirrors and witnesses, are placed on separate ESX hosts or fault domains. For example, to rebuild components during a failure, vSAN looks for ESX hosts that satisfy the

placement rules where protection components of VM objects must be placed on two different ESX hosts, or across fault domains.

vSAN Datastore

After you enable vSAN on a cluster, a single vSAN datastore is created. It appears as another type of datastore in the list of datastores that might be available, including VMFS and NFS. A single vSAN datastore can provide different service levels for each VM or each virtual disk. In vCenter, storage characteristics of the vSAN datastore appear as a set of capabilities. You can reference these capabilities when defining a storage policy for VMs. When you later deploy VMs, vSAN uses this policy to place VMs storage objects in the optimal manner based on the requirements of each VM. For general information about using storage policies, see the vSphere Storage guide.

A vSAN datastore has specific characteristics to consider.

- vSAN provides a single vSAN datastore accessible to all ESX hosts in the cluster, whether or not they contribute storage to the cluster. Each ESX host can also mount any other datastores, including VMFS or NFS.
- You can use Storage vMotion to move VMs between vSAN datastores, NFS datastores, and VMFS datastores.
- Only magnetic disks and flash devices used for capacity can contribute to the datastore capacity. The devices used for flash cache are not counted as part of the datastore.

vSAN Objects and Components

Each vSAN object is composed of a set of components, determined by capabilities that are in use in the VM storage policy. For example, with **Failures to tolerate** set to 1, vSAN ensures that the protection components, such as replicas and witnesses, are placed on separate ESX hosts in the vSAN cluster, where each replica is an object component. In addition, in the same policy, if the **Number of disk stripes per object** configured to two or more, vSAN also stripes the object across multiple capacity devices and each stripe is considered a component of the specified object. When needed, vSAN might also break large objects into multiple components.

A vSAN datastore contains the following object types:

VM home namespace	The VM home directory where all VM configuration files are stored, such as `.vmx`, log files, `.vmdk` files, and snapshot delta description files.
VMDK	A VM disk or `.vmdk` file that stores the contents of the VM's hard disk drive.
VM swap object	Created when a VM is powered on.
Snapshot delta VMDKs	Created when VM snapshots are taken. Such delta disks are not created for vSAN ESA.
Memory object	Created when the snapshot memory option is selected when creating or suspending a VM.
vSAN durability component	Provides a mechanism to maintain the required availability for VMs while there is an ESX host failure or maintenance. This ensures that vSAN still maintains the availability level specified within the storage policy.

For more information on vSAN objects and components, see vSAN Objects and Components Revisited.

Virtual Machine Compliance Status: Compliant and Noncompliant

A virtual machine is considered noncompliant when one or more of its objects fail to meet the requirements of its assigned storage policy. For example, the status might become noncompliant when one of the mirror copies is inaccessible. If your virtual machines are in compliance with the requirements defined in the storage policy, the status of your virtual machines is compliant.

From the **Physical Disk Placement** tab on the **Virtual Disks** page, you can verify the virtual machine object compliance status. For information about troubleshooting a vSAN cluster, see Monitor Datastore Sharing with vSphere Client.

Component State: Degraded, Absent, and Out of date States

vSAN acknowledges the following failure states for components:

- **Degraded**. A component is *Degraded* when vSAN detects a permanent component failure and determines that the failed component cannot recover to its original working state. As a result, vSAN starts to rebuild the degraded components immediately. This state might occur when a component is on a failed device.

- **Absent**. A component is *Absent* when vSAN detects a temporary component failure where components, including all its data, might recover and return vSAN to its original state. This state might occur when you are restarting ESX hosts or if you unplug a device from a vSAN host. vSAN starts to rebuild the components in absent status after waiting for 60 minutes.
- **Out of date**. A component is *Out of date* when vSAN detects that the assigned storage policy is outdated. When you click **Reapply VM Storage Policy**, all the objects of the VM are back to the compliant state.

Object State: Healthy and Unhealthy

An unhealthy object is also called as inaccessible object as these objects are virtual. vSAN cannot access these objects as its components are no longer available. Depending on the type and number of failures in the cluster, an object might be in one of the following states:

- Healthy. When at least one full RAID 1 mirror is available, or the minimum required number of data segments are available, the object is considered healthy.
- Unhealthy. An object is considered unhealthy when no full mirror is available or the minimum required number of data segments are unavailable for RAID 5 or RAID 6 objects. If fewer than 50 percent of an object's votes are available, the object is unhealthy. Multiple failures in the cluster can cause objects to become unhealthy. When the operational status of an object is considered unhealthy, it impacts the availability of the associated virtual machine.

Witness

A witness is a component that contains only metadata and does not contain any actual application data. It serves as a tiebreaker when a decision must be made regarding the availability of the surviving datastore components, after a potential failure. A witness consumes approximately 4 MB of space for metadata on the vSAN datastore when using on-disk format version 2.0 and later.

> **Note:**
>
> In a vSAN stretched cluster or Two node vSAN cluster configuration, a witness component gets created on the dedicated witness host.

vSAN maintains a quorum by using an asymmetrical voting system where each component might have more than one vote to decide the availability of objects. Greater than 50 percent of the votes that make up a virtual machines storage object must be accessible at all times for the object to be considered available. When 50 percent or fewer votes are accessible to all hosts, the object is no longer accessible to the vSAN datastore. Inaccessible objects can impact the availability of the associated virtual machine.

Storage Policy-Based Management (SPBM)

When you use vSAN, you can define virtual machine storage requirements, such as performance and availability, in the form of a policy. vSAN ensures that virtual machines deployed to vSAN datastores are assigned at least one virtual machine storage policy. When you know the storage requirements of your virtual machines, you can define storage policies and assign the policies to your virtual machines.

If you do not apply a storage policy when deploying virtual machines, vSAN automatically assigns a default vSAN storage policy with **Failures to tolerate** set to 1, a single disk stripe for each object, and thin provisioned virtual disk. When you enable vSAN ESA Auto-Policy Management, vSAN automatically creates an optimized, cluster-specific default storage policy that helps you to run workloads on an ESA cluster using the optimal level of resilience and efficiency. For information about working with vSAN storage policies, see Using vSAN Policies.

vSphere PowerCLI

VMware vSphere PowerCLI adds command-line scripting support for vSAN, to help you automate configuration and management tasks. vSphere PowerCLI provides a Windows PowerShell interface to the vSphere API. PowerCLI includes cmdlets for administering vSAN components. For information about using vSphere PowerCLI, see the vSphere PowerCLI guide.

How vSAN Differs from Traditional Storage

Although vSAN shares many characteristics with traditional storage arrays, the overall behavior and function of vSAN is different.

For example, vSAN can manage and work only with ESX hosts, and a single vSAN instance provides a single datastore for the cluster.

vSAN and traditional storage also differ in the following key ways:

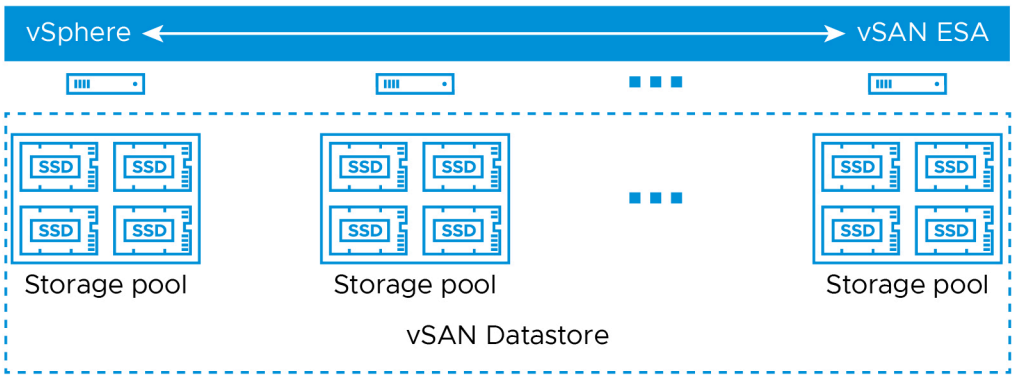
- vSAN does not require external networked storage for storing virtual machine files remotely, such as on a Fibre Channel (FC) or Storage Area Network (SAN).
- Using traditional storage, the storage administrator preallocates storage space on different storage systems. vSAN automatically turns the local physical storage resources of the ESX hosts into a single pool of storage. These pools can be divided and assigned to virtual machines and applications according to their quality-of-service requirements.
- vSAN does not behave like traditional storage volumes based on LUNs or NFS shares. The iSCSI target service uses LUNs to enable an initiator on a remote ESX host to transport block-level data to a storage device in the vSAN cluster.
- Some standard storage protocols, such as Fiber Channel Protocol (FCP), do not apply to vSAN.
- vSAN is highly integrated with vSphere. You do not need dedicated plug-ins or a storage console for vSAN, compared to traditional storage. You can deploy, manage, and monitor vSAN by using the vSphere Client.
- A dedicated storage administrator does not need to manage vSAN. Instead a vSphere administrator can manage a vSAN environment.
- With vSAN, virtual machine storage policies are automatically assigned when you deploy new virtual machines. The storage policies can be changed dynamically as needed.

Building a vSAN Cluster

You can choose the storage architecture and deployment option when creating a vSAN cluster.

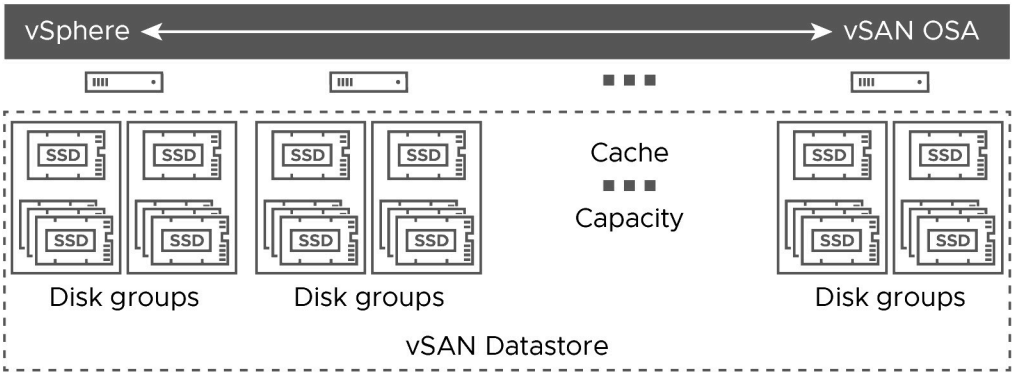
vSAN Express Storage Architecture

vSAN ESA is designed for high-performance NVMe based TLC flash devices and high performance networks. Each ESX host that contributes storage contains a single storage pool of one or more flash devices. Each flash device provides caching and capacity to the cluster.



vSAN Original Storage Architecture

vSAN OSA is designed for a wide range of storage devices, including flash solid state drives (SSD) and magnetic disk drives (HDD). Each ESX host that contributes storage contains one or more disk groups. Each disk group contains one flash cache device and one or more capacity devices.



vSAN ReadyNode

The vSAN ReadyNode is a preconfigured solution of the vSAN software provided by Broadcom partners, such as Cisco, Dell, HPE, Fujitsu, IBM, and Supermicro. This solution includes validated server configuration in a tested, certified hardware form factor for vSAN deployment that is recommended by the server OEM and VMware. For information about the vSAN ReadyNode solution for a specific partner, visit the *Broadcom Partner Portal* at https://partnerportal.broadcom.com/web/partner-portal.

For more information on vSAN ReadyNode, see the Broadcom knowledge base article 326476.

User-Defined vSAN Cluster

You can build a vSAN cluster by selecting individual software and hardware components, such as drivers, firmware, and storage I/O controllers that are listed in the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/. You can choose any servers, storage I/O controllers, capacity and flash cache devices, memory, any number of cores you must have per CPU, that are certified and listed on the *Broadcom Compatibility Guide*. Review the compatibility information on the *Broadcom Compatibility Guide* before choosing software and hardware components, drivers, firmware, and storage I/O controllers that vSAN supports.

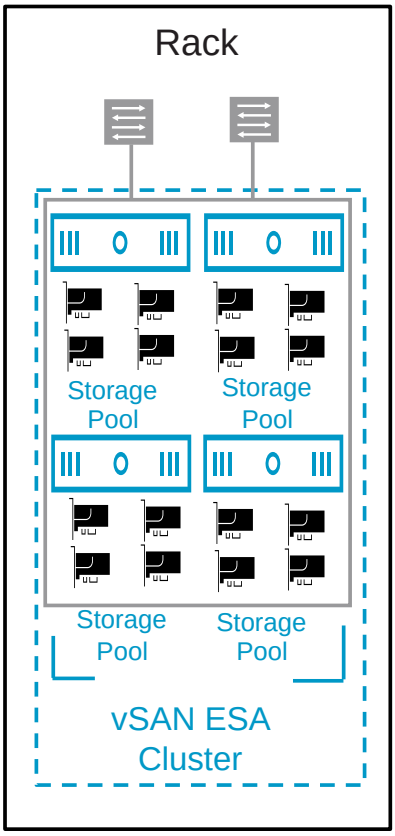
When designing a vSAN cluster, use only devices, firmware, and drivers that are listed on the *Broadcom Compatibility Guide*. Using software and hardware versions that are not listed in the *Broadcom Compatibility Guide* might cause cluster failure or unexpected data loss. For information about designing a vSAN cluster, see Designing and Sizing a vSAN Cluster. For vSAN ESA requirements, see vSAN ESA ReadyNode Hardware Guidance.

vSAN Deployment Options

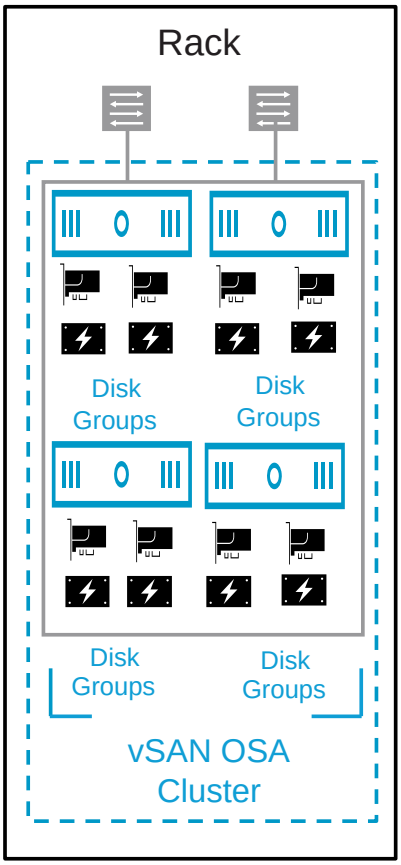
This section covers the supported deployment options for vSAN clusters.

vSAN HCI Cluster

vSAN HCI cluster or a single site vSAN cluster consists of a minimum of three ESX hosts. Typically, all ESX hosts in a single site vSAN cluster reside at a single availability zone, and may be connected on the same Layer 2 network. All-flash configurations including vSAN ESA require network connection of minimum 10 GbE and a network latency of minimum 1 millisecond.



For more information, see Creating a Single Site vSAN Cluster.

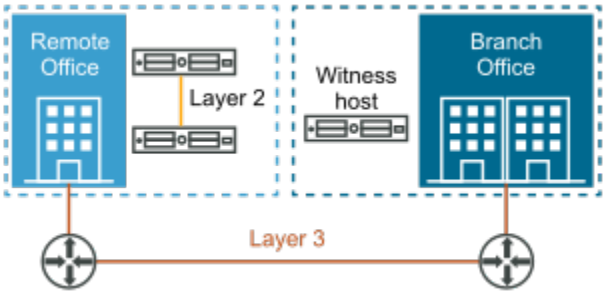


Two-Node vSAN Cluster

Two-node vSAN clusters are often used for remote office/branch office environments, typically running a small number of workloads that require high availability. A two-node vSAN cluster consists of two ESX hosts at the same location, connected to the same network switch or directly connected.

A two-node vSAN cluster requires a third ESX host to function as a witness. This witness host can be located remotely or locally to the two data hosts. Usually the witness resides at the main site, along with the vCenter. vSAN witness hosts can be shared by multiple vSAN two-node clusters.

For more information, see Creating a vSAN Stretched Cluster or Two-Node vSAN Cluster and vSAN 2-node Cluster Guide.

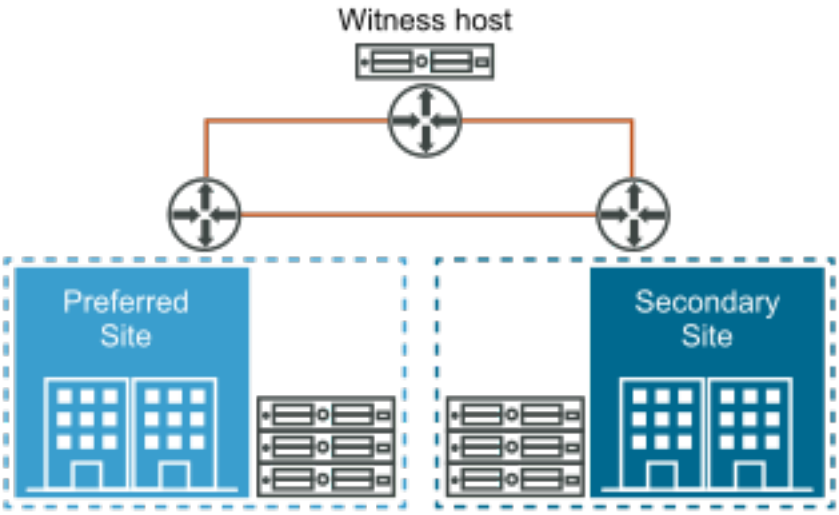


vSAN Stretched Cluster

A vSAN stretched cluster provides resiliency against the loss of an availability zone. The ESX hosts in a vSAN stretched cluster can be distributed evenly across two sites. The two sites must have a network latency of no more than five milliseconds (5 ms) round trip (RTT).

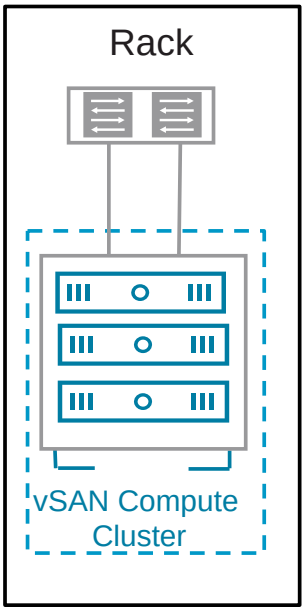
A vSAN witness ESX host resides at a third site to provide the witness function. The witness acts as tie-breaker in scenarios where a network partition occurs between the two data sites. Only metadata such as witness components is stored on the witness. The witness and the data sites must have a witness latency of less than 200 ms RTT. Broadcom recommends a witness latency of 100 ms one way.

For more information, see Creating a vSAN Stretched Cluster or Two-Node vSAN Cluster.



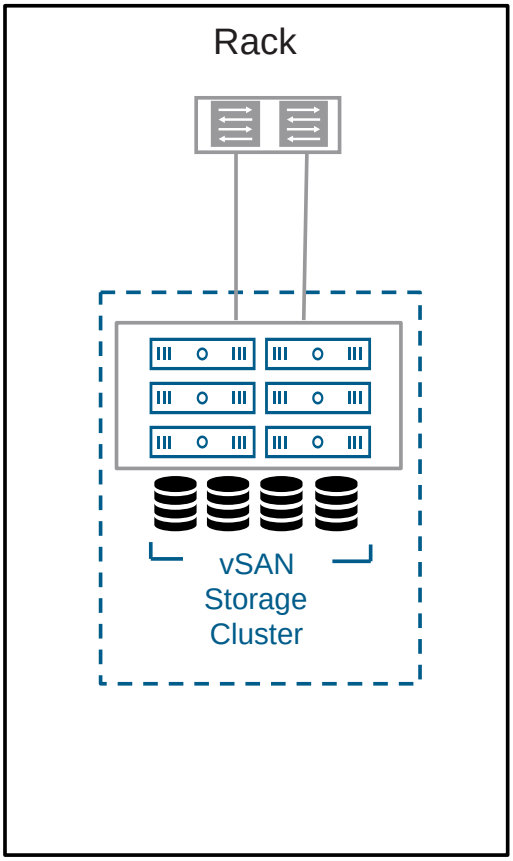
Compute Cluster

In a compute cluster, ESX hosts provide only compute resources and access shared storage from a vSAN storage cluster. While the compute cluster does not need local storage for vSAN, it must be connected to the vSAN storage cluster through a 10 GbE or higher network to ensure low latency access to the storage. The compute cluster can contain two or three hosts based on the workload needs and vCenter requirements.



vSAN Storage Cluster

A vSAN storage cluster consists of ESX hosts that contribute only storage resources and do not run virtual machine workloads. With vSAN ESA architecture, it serves as a storage pool for one or more compute clusters. It requires a minimum of four hosts to maintain storage availability and fault tolerance. A vSAN storage cluster requires a minimum of 10 GbE network. Broadcom recommends a dedicated 25 GbE or higher network for the vSAN traffic to ensure consistent performance between ESX hosts. For more information, see vSAN ESA ReadyNode Hardware Guidance.



Integrate vSAN with Other VMware Software

After you have vSAN up and running, it is integrated with the rest of the VMware software stack.

You can do most of what you can do with traditional storage by using vSphere components and features including vSphere vMotion, snapshots, clones, vSphere DRS, vSphere HA, Protection and Recovery, and more.

vSphere HA

You can enable vSphere HA and vSAN on the same cluster. As with traditional datastores, vSphere HA provides the same level of protection for virtual machines on vSAN datastores. This level of protection imposes specific restrictions when vSphere HA and vSAN interact. For specific considerations about integrating vSphere HA and vSAN, see Using vSAN and vSphere HA.

Limitations of vSAN

It is important to know the limitations of vSAN before you begin deploying the software.

When working with vSAN, consider the following limitations:

- vSAN does not support ESX hosts participating in multiple vSAN clusters. However, a vSAN host can access other external storage resources that are shared across clusters.
- vSAN does not support vSphere Distributed Power Management (DPM) and Storage I/O Control.
- vSAN does not support SE Sparse disks.
- vSAN does not support Raw Device Mappings (RDM), VMFS, diagnostic partition, and other device access features.

Requirements for Enabling vSAN

Before you deploy a vSAN cluster, verify that your environment meets the requirements for running vSAN.

Hardware Requirements for vSAN

Verify that your ESX hosts and storage devices meet the vSAN hardware requirements.

Storage Device Requirements

All capacity devices, drivers, and firmware versions in your vSAN configuration must be certified and listed in the vSAN section of the *Broadcom Compatibility Guide* available at: https://compatibilityguide.broadcom.com/.

Table 999: vSAN OSA storage device requirements

Storage Component	Requirements
Cache	• One SAS or SATA solid-state disk (SSD) or PCIe flash device. • Before calculating the **Failures to tolerate**, check the size of the flash caching device in each disk group. For hybrid clusters, it must provide at least 10 percent of the anticipated storage consumed on the capacity devices, not including replicas such as mirrors. • Sphere Flash Read Cache must not use any of the flash devices reserved for vSAN cache. • The cache flash devices must not be formatted with VMFS or another file system.
Capacity	• Hybrid group configuration must have at least one SAS or NL-SAS magnetic disk. • All-flash disk group configuration must have at least one SAS, or SATA solid-state disk (SSD), or PCIe flash device.

Storage Component	Requirements
Storage controllers	One SAS or SATA host bus adapter (HBA), or a RAID controller that is in passthrough mode or RAID 0 mode. To avoid issues, consider these points when the same storage controller is backing both vSAN and non-vSAN disks: • Do not mix the controller mode for vSAN and non-vSAN disks to avoid handling the disks inconsistently, which can negatively impact vSAN operations. If the vSAN disks are in RAID mode, the non-vSAN disks must also be in RAID mode. • When you use non-vSAN disks for VMFS, use the VMFS datastore only for scratch, logging, and core dumps. • Do not run virtual machines from a disk or RAID group that shares its controller with vSAN disks or RAID groups. • Do not pass through non-vSAN disks to virtual machine guests as Raw Device Mappings (RDMs). To learn about controller supported features, such as passthrough and RAID, refer to the *Broadcom Compatibility Guide*.

Table 1000: vSAN ESA storage device requirements

Storage Component	Requirements
Cache and capacity	Each storage pool must have at least one NVMe TLC devices.

Host Memory

The memory requirements for vSAN OSA depend on the number of disk groups and devices that the ESX hypervisor must manage. For more information, see vSAN Sizer tool.

vSAN ESA requires at least 128 GB host memory. The memory needed for your environment depends on the number of devices in the host's storage pool. For more information on guidelines to use with vSAN ESA, see vSAN ESA ReadyNode Hardware Guidance.

Flash Boot Devices

During installation, the ESX installer creates a coredump partition on the boot device. The default size of the coredump partition satisfies most installation requirements.

During installation, the ESX installer creates an ESX-OSData partition on the boot device to store infrequently written persistent data, including core dumps. The partition size is automatically determined by the capacity of the boot media to satisfy installation requirements.

- If the memory of the ESX host has 512 GB of memory or less, you can boot the host from a USB, SD, or SATADOM device. When you boot a vSAN host from a USB device or SD card, the size of the boot device must be at least 32 GB.
- If the memory of the ESX host has more than 512 GB, consider the following guidelines.
  - You can boot the host from a SATADOM or disk device with a size of at least 16 GB. When you use a SATADOM device, use a single-level cell (SLC) device.
  - If you are using vSAN, you must resize the coredump partition on ESX hosts to boot from USB/SD devices.

When you boot an ESX host from USB device or from SD card, vSAN trace logs are written to RAMDisk. These logs are automatically offloaded to persistent media during shutdown or system crash (panic). This is the only support method for handling vSAN traces when booting an ESX from a USB stick or SD card. If a power failure occurs, vSAN trace logs are not preserved.

When you boot an ESX host from a SATADOM device, vSAN trace logs are written directly to the SATADOM device. It is important that the SATADOM device meets the specifications outlined in this guide. For more information, see ESX Hardware Requirements.

Cluster Requirements for vSAN

Verify that a host cluster meets the requirements for enabling vSAN.

- All capacity devices, drivers, and firmware versions in your vSAN configuration must be certified and listed in the vSAN section of the *Broadcom Compatibility Guide* at: https://compatibilityguide.broadcom.com/.
- A standard vSAN cluster must contain a minimum of three ESX hosts that contribute capacity to the cluster. A two host vSAN cluster consists of two data hosts and an external witness host. For information about the considerations for a three-host cluster, see Design Considerations for a vSAN Cluster.
- An ESX host that resides in a vSAN cluster must not participate in other clusters.

Software Requirements for vSAN

Verify that the vSphere components in your environment meet the software version requirements for using vSAN.

To use the full set of vSAN capabilities, the ESX hosts that participate in vSAN clusters must be version 9.1 or later. During the vSAN upgrade from previous versions, you can keep the current on-disk format version, but you cannot use many of the new features. vSAN 9.1 and later software supports all on-disk formats.

Networking Requirements for vSAN

Verify that the network infrastructure and the networking configuration on the ESX hosts meet the minimum networking requirements for vSAN.

Table 1001: Networking Requirements for vSAN

Networking Component	Requirement
Host Bandwidth	Each host must have minimum bandwidth dedicated to vSAN. • vSAN OSA: Dedicated 1 GbE for hybrid configurations, dedicated or shared 10 GbEs for all-flash configurations • vSAN ESA: Dedicated or shared 10 GbEs minimum, recommended 25GbEs or higher. See vSAN Compatibility Guide. For information about networking considerations in vSAN, see Designing the vSAN Network.
Connection between hosts	Each host in the vSAN cluster, regardless of whether it contributes capacity, must have a VMkernel network adapter for vSAN traffic. See Set Up a VMkernel Network for vSAN.
Host network	All hosts in your vSAN cluster must be connected to a vSAN Layer 2 or Layer 3 network.
IPv4 and IPv6 support	The vSAN network supports both IPv4 and IPv6.
Network latency	• Maximum of 1 ms RTT for single site (non-stretched) vSAN clusters between all hosts in the cluster • Maximum of 5 ms RTT between the two main sites for vSAN stretched clusters • Maximum of 200 ms RTT from a main site to the vSAN witness host • Maximum of 5ms RTT between the vSAN client cluster and vSAN storage cluster

License Requirements

Starting with version 9.0, product components are licensed automatically after you license the vCenter instance connected to the component. You no longer license vSAN manually. For more information, see the VCF Licensing guide.

Designing and Sizing a vSAN Cluster

For best performance and use, plan the capabilities and configuration of your hosts and their storage devices before you deploy vSAN in a vSphere environment. Carefully consider certain host and networking configurations within the vSAN cluster.

The Administering VMware vSAN documentation examines the key points about designing and sizing a vSAN cluster. For detailed instructions about designing and sizing a vSAN cluster, see VMware vSAN Design Guide.

Designing and Sizing vSAN Storage

Plan capacity and cache based on the expected data storage consumption. Consider your requirements for availability and endurance.

Planning Capacity in vSAN

You can calculate the capacity of a vSAN datastore to accommodate the virtual machines (VMs) files in the cluster, and to handle failures and maintenance operations.

Raw Capacity

Use this formula to determine the raw capacity of a vSAN datastore. Multiply the total number of disk groups in the cluster by the size of the capacity devices in those disk groups. Subtract the overhead required by the vSAN on-disk format.

Failures to Tolerate

When you plan the capacity of the vSAN datastore, not including the number of virtual machines and the size of their VMDK files, you must consider the **Failures to tolerate** of the virtual machine storage policies for the cluster.

FTT (Failures To Tolerate) is a vSAN storage policy parameter that defines the number of host, disk, or network failures a virtual machine's storage object can withstand without data loss. Higher FTT values provide greater redundancy but consume more storage capacity.

The **Failures to tolerate** has an important role when you plan and size storage capacity for vSAN. Based on the availability requirements of a virtual machine, the setting might result in doubled consumption or more, compared with the consumption of a virtual machine and its individual devices.

For example, if the **Failures to tolerate** is set to **1 failure - RAID-1 (Mirroring)**, virtual machines can use about 50 percent of the raw capacity. If the FTT is set to 2, the usable capacity is about 33 percent. If the FTT is set to 3, the usable capacity is about 25 percent.

But if the **Failures to tolerate** is set to **1 failure - RAID-5 (Erasure Coding)**, virtual machines can use about 75 percent of the raw capacity. If the FTT is set to **2 failures - RAID-6 (Erasure Coding)**, the usable capacity is about 67 percent. For more information about RAID 5/6, see Selecting the Best RAID Configuration for a vSAN Storage Cluster.

For information about the attributes in a vSAN storage policy, see What are vSAN Policies?

Capacity Sizing Guidelines

- Keep some unused space to prevent vSAN from rebalancing the storage load. vSAN rebalances the components across the cluster whenever the consumption on a single capacity device reaches 80 percent or more. The rebalance operation might impact the performance of applications. To avoid these issues, keep storage consumption to less than 80 percent. vSAN enables you to manage unused capacity using operations reserve and host rebuild reserve.
- Plan extra capacity to handle any potential failure or replacement of capacity devices, disk groups, and hosts. When a capacity device is not reachable, vSAN recovers the components from another device in the cluster. When a flash cache device fails or is removed, vSAN recovers the components from the entire disk group.
- Reserve extra capacity to make sure that vSAN recovers components after a host failure or when a host enters maintenance mode. For example, provision hosts with enough capacity so that you have sufficient free capacity left for components to rebuild after a host failure or during maintenance. This extra space is important when you have more than three hosts, so you have sufficient free capacity to rebuild the failed components. If a host fails, the rebuilding

takes place on the storage available on another host, so that another failure can be tolerated. However, in a three-host cluster, vSAN does not perform the rebuild operation if the **Failures to tolerate** is set to 1 because when one host fails, only two hosts remain in the cluster. To tolerate a rebuild after a failure, you must have at least three surviving hosts.

- Provide enough temporary storage space for changes in the vSAN VM storage policy. When you dynamically change a VM storage policy, vSAN might create a new RAID tree layout of the object. When vSAN instantiates and synchronizes a new layout, the object may consume extra space temporarily. Keep some temporary storage space in the cluster to handle such changes.
- If you plan to use advanced features, such as software checksum or deduplication and compression, reserve extra capacity to handle the operational overhead.
- You can use vSAN Sizer tool to assist with capacity requirements, operations reserve, and host rebuild reserve, and to determine how vSAN can meet your performance requirements.

Considerations for Virtual Machine Objects

When you plan the storage capacity in the vSAN datastore, consider the space required in the datastore for the VM home namespace objects, snapshots, and swap files.

- VM Home Namespace. You can assign a storage policy specifically to the home namespace object for a virtual machine. To prevent unnecessary allocation of capacity and cache storage, vSAN applies only the **Failures to tolerate** and the **Force provisioning** settings from the policy on the VM home namespace. Plan storage space to meet the requirements for a storage policy assigned to a VM Home Namespace whose **Failures to tolerate** is greater than 0.
- Snapshots. Delta devices inherit the policy of the base VMDK file. Plan extra space according to the expected size and number of snapshots, and to the settings in the vSAN storage policies. If vSAN ESA is enabled, every snapshot is not a new object. A base VMDK and its snapshots are contained in one vSAN object.

  The space that is required might be different. Its size depends on how often the virtual machine changes data and how long a snapshot is attached to the virtual machine.
- Swap files. Virtual machine swap files inherit the storage policy of the VM Namespace.

Design Considerations for Flash Caching Devices in vSAN

Plan the configuration of flash devices for vSAN cache and all-flash capacity to provide high performance and required storage space, and to accommodate future growth.

Flash Devices as vSAN Cache

Design the configuration of flash cache for vSAN for write endurance, performance, and potential growth based on these considerations.

Table 1002: Sizing vSAN Cache

Storage Configuration	Considerations
All-flash and hybrid configurations	• A higher cache-to-capacity ratio eases future capacity growth. Oversizing cache enables you to add more capacity to an existing disk group without the need to increase the size of the cache. • Flash caching devices must have high write endurance. • Replacing a flash caching device is more complicated than replacing a capacity device because such an operation impacts the entire disk group. • If you add more flash devices to increase the size of the cache, you must create more disk groups. The ratio between flash cache devices and disk groups is always 1:1. A configuration of multiple disk groups provides the following advantages: – Reduced risk of failure. If a single caching device fails, fewer capacity devices are affected. – Improved performance if you deploy multiple disk groups that contain smaller flash caching devices. However, when you configure multiple disk groups, the memory consumption of the hosts increases.
All-flash configurations	In all-flash configurations, vSAN uses the cache layer for write caching only. The write cache must be able to handle high write activities. This approach extends the life of capacity flash that might be less expensive and might have lower write endurance.
Hybrid configurations	The flash caching device must provide at least 10 percent of the anticipated storage that virtual machines are expected to consume, not including replicas such as mirrors. The **Primary level of failures to tolerate** attribute from the VM storage policy does not impact the size of the cache. If the read cache reservation is configured in the active VM storage policy, the hosts in the vSAN cluster must have sufficient cache to satisfy the reservation during a post-failure rebuild or maintenance operation. If the available read cache is not sufficient to satisfy the reservation, the rebuild or maintenance operation fails. Use read cache reservation only if you must meet a specific, known performance requirement for a particular workload. The use of snapshots consumes cache resources. If you plan to use several snapshots, consider dedicating more cache than the conventional 10 percent cache-to-consumed-capacity ratio.

Design Considerations for Flash Capacity Devices in vSAN

Plan the configuration of flash capacity devices for vSAN all-flash configurations to provide high performance and required storage space, and to accommodate future growth.

Flash Devices as vSAN Capacity

In all-flash configurations, vSAN does not use cache for read operations and does not apply the read-cache reservation setting from the VM storage policy. For cache, you can use a small amount of more expensive flash that has high write endurance. For capacity, you can use flash that is less expensive and has lower write endurance.

Plan a configuration of flash capacity devices by following these guidelines:

- For better performance of vSAN, use more disk groups of smaller flash capacity devices.
- For balanced performance and predictable behavior, use the same type and model of flash capacity devices.

Design Considerations for Magnetic Disks in vSAN

Plan the size and number of magnetic disks for capacity in hybrid configurations by following the requirements for storage space and performance.

SAS and NL-SAS Magnetic Devices

Use SAS or NL-SAS magnetic devices by following the requirements for performance, capacity, and cost of the vSAN storage.

- Compatibility. The model of the magnetic disk must be certified and listed in the vSAN section of the *Broadcom Compatibility Guide* available at: https://compatibilityguide.broadcom.com/.
- Performance. SAS and NL-SAS devices have faster performance.
- Capacity. The capacity of SAS or NL-SAS magnetic disks for vSAN is available in the vSAN section of the *Broadcom Compatibility Guide*. Consider using a larger number of smaller devices instead of a smaller number of larger devices.
- Cost. SAS and NL-SAS devices can be expensive.

Magnetic Disks as vSAN Capacity

Plan a magnetic disk configuration by following these guidelines:

- For better performance of vSAN, use many magnetic disks that have smaller capacity.
  You must have enough magnetic disks that provide adequate aggregated performance for transferring data between cache and capacity. Using more small devices provides better performance than using fewer large devices. Using multiple magnetic disk spindles can speed up the destaging process.
  In environments that contain many virtual machines, the number of magnetic disks is also important for read operations when data is not available in the read cache and vSAN reads it from the magnetic disk. In environments that contain a small number of virtual machines, the disk number impacts read operations if the **Number of disk stripes per object** in the active VM storage policy is greater than one.
- For balanced performance and predictable behavior, use the same type and model of magnetic disks in a vSAN datastore.
- Dedicate a high enough number of magnetic disks to satisfy the value of the **Failures to tolerate** and the **Number of disk stripes per object** attributes in the defined storage policies. For information about the VM storage policies for vSAN, see What are vSAN Policies?

Design Considerations for Storage Controllers in vSAN

Use storage controllers on the hosts of a vSAN cluster that best satisfy your requirements for performance and availability.

- Use storage controller models, and driver and firmware versions that are listed in the *Broadcom Compatibility Guide*. Search for vSAN in the *Broadcom Compatibility Guide*.
- Use multiple storage controllers, if possible, to improve performance and to isolate a potential controller failure to only a subset of disk groups.
- Use storage controllers that have the highest queue depths in the *Broadcom Compatibility Guide*. Using controllers with high queue depth improves performance. For example, when vSAN is rebuilding components after a failure or when a host enters maintenance mode.
- Use storage controllers in passthrough mode for best performance of vSAN. Storage controllers in RAID 0 mode require higher configuration and maintenance efforts compared to storage controllers in passthrough mode.
- Deactivate caching on the controller, or set caching to 100 percent Read.

Designing and Sizing vSAN Hosts

Plan the configuration of the hosts in your vSAN cluster for best performance and availability.

Memory and CPU

Calculate the memory and the CPU requirements of the hosts in the vSAN cluster based on the following considerations.

Table 1003: Sizing Memory and CPU of vSAN Hosts

Compute Resource	Considerations
Memory	• Memory per virtual machine • Memory per host, based on the expected number of virtual machines • vSAN OSA must have at least 32 GB memory to support 5 disk groups per host and 7 capacity devices per disk group. • vSAN ESA requires at least 128 GB memory. Hosts that have 512 GB memory or less can boot from a USB, SD, or SATADOM device. If the memory of the host is greater than 512 GB, boot the host from a SATADOM or disk device. For more information, see the Broadcom knowledge base article 2113954.
CPU	• Sockets per host • Cores per socket • Number of vCPUs based on the expected number of virtual machines • vCPU-to-core ratio **Note:** vSAN ESA requires at least 16 CPU cores per host.

Host Networking

Provide more bandwidth for vSAN traffic to improve performance.

- vSAN OSA
  - If you plan to use hosts that have 1 GbE adapters, dedicate adapters for vSAN hybrid. For all-flash configurations, plan hosts that have dedicated or shared 10 GbE adapters. Use dedicated or shared 25 GbE physical adapters or higher is recommended.
  - If you plan to use 10 GbE adapters, they can be shared with other traffic types for both hybrid and all-flash configurations.
- vSAN ESA
  - Support the use of dedicated or shared 10 GbE physical adapters. Use dedicated or shared 25 GbE physical adapters or higher is recommended.
  - Network adapters can be shared with other traffic types.
- If a network adapter is shared with other traffic types, use a vSphere Distributed Switch to isolate vSAN traffic by using Network I/O Control to manage and prioritize traffic and dedicated VLAN for vSAN traffic
- Create a team of physical adapters to provide redundancy for vSAN traffic.

Disk Groups vs. Storage Pools

vSAN OSA uses disk groups to balance performance and reliability. If a flash cache or storage controller stops responding and a disk group fails, vSAN rebuilds all components from another location in the cluster.

Using multiple disk groups, with each disk group providing a portion of datastore capacity, provides advantages but also has disadvantages.

- Advantages of multiple disk groups
  - Performance is improved because the datastore has more aggregated cache, and I/O operations are faster.
  - Risk of failure is spread among multiple disk groups.
  - If a disk group fails, vSAN rebuilds fewer components, so performance is improved.
- Disadvantages of multiple disk groups
  - Costs are increased because two or more caching devices are required.

- More memory is required to handle more disk groups.
- Multiple storage controllers are required to reduce the risk of a single point of failure.

vSAN ESA uses storage pools, where each device provides both performance and capacity. The number of storage tier devices has an impact on the performance of vSAN ESA. Broadcom recommends configuring more devices for better performance. Any single device can fail without impacting the availability of data on any of the other devices in the storage pool. This design reduces the size of a failure domain.

Drive Bays

For easy maintenance, consider hosts whose drive bays and PCIe slots are at the front of the server body.

Hot Plug and Swap of Devices

Consider the storage controller passthrough mode support for easy hot plugging or replacement of magnetic disks and flash capacity devices on a host. If a controller works in RAID 0 mode, you must perform additional steps before the host can discover the new drive. vSAN ESA requires all NVMe drives and supports hot plugging of these drives.

Design Considerations for a vSAN Cluster

Design the configuration of hosts and management nodes for best availability and tolerance to consumption growth.

Sizing the vSAN Cluster for Failures to Tolerate

You configure the **Failures to tolerate** (FTT) attribute in the VM storage policies to handle host failures. The number of hosts required for the cluster is calculated as follows: `2 * FTT + 1`. The more failures the cluster is configured to tolerate, the more capacity hosts are required.

If the cluster hosts are connected across multiple racks, you can organize the hosts into fault domains to improve resilience against issues such as top-of-rack switch failures and loss of server rack power. See Designing and Sizing vSAN Fault Domains .

Limitations of a Two-Host or Three-Host Cluster Configuration

In a three-host configuration, you can tolerate only one host failure by setting the number of failures to tolerate to 1. vSAN saves each of the two required replicas of virtual machine data on separate hosts. The witness object is on a third host. Because of the small number of hosts in the cluster, the following limitations exist:

- When a host fails, vSAN cannot rebuild data on another host to protect against another failure.
- If a host must enter maintenance mode, vSAN cannot evacuate data from the host to maintain policy compliance. While the host is in maintenance mode, data is exposed to a potential failure or inaccessibility if an additional failure occurs.

  You can use only the **Ensure data accessibility** data evacuation option. **Ensure data accessibility** guarantees that the object remains available during data migration, although it might be at risk if another failure occurs. vSAN objects on two-host or three-host clusters are not policy compliant. When the host exits maintenance mode, objects are rebuilt to ensure policy compliance.

  In any situation where two-host or three-host cluster has an inaccessible host or disk group, vSAN objects are at risk of becoming inaccessible should another failure occur.

Nested Fault Domains for Two-Node Clusters

In a two-node configuration, the nested fault domains for two-node clusters allows you to tolerate multiple drive failures within a host. Each two-node cluster must have at least three disk groups and use a vSAN storage policy that leverages the nested fault domains capability. The nested fault domains provide protection against disk group failures within a host by mirroring data within each host in addition to the mirroring between the two hosts.

Balanced and Unbalanced Cluster Configuration

vSAN works best on hosts with uniform configurations, including storage configurations.

Using hosts with different configurations has the following disadvantages in a vSAN cluster:

- Reduced predictability of storage performance because vSAN does not store the same number of components on each host.
- Different maintenance procedures.
- Reduced performance on hosts in the cluster that have smaller or different types of cache devices.

Deploying vCenter on vSAN

If the vCenter becomes unavailable, vSAN continues to operate normally and virtual machines continue to run.

If vCenter is deployed on the vSAN datastore, and a problem occurs in the vSAN cluster, you can use a Web browser to access each ESX host and monitor vSAN through the vSphere Host Client. vSAN health information is visible in the Host Client, and also through esxcli commands.

Designing the vSAN Network

Consider networking features that can provide availability, security, and bandwidth guarantee in a vSAN cluster.

For details about the vSAN network configuration, see Understanding vSAN Networking.

Networking Failover and Load Balancing

vSAN uses the teaming and failover policy that is configured on the backing virtual switch for network redundancy only. vSAN does not use NIC teaming for load balancing.

If you plan to configure a NIC team for availability, consider these failover configurations.

Teaming Algorithm	Failover Configuration of the Adapters in the Team
Route based on originating virtual port	Active/Passive
Route based on IP hash	Active/Active with static EtherChannel for the standard switch and LACP port channel for the distributed switch
Route based on physical network adapter load	Active/Active

vSAN supports IP-hash load balancing, but cannot guarantee improvement in performance for all configurations. You can benefit from IP hash when vSAN is among its many consumers. In this case, IP hash performs load balancing. If vSAN is the only consumer, you might observe no improvement. This behavior specifically applies to 1 GbE environments. For example, if you use four 1 GbE physical adapters with IP hash for vSAN, you might not be able to use more than 1 GbEs. This behavior also applies to all NIC teaming policies that VMware supports.

vSAN does not support multiple VMkernel adapters on the same subnet. You can use different VMkernel adapters on different subnets, such as another VLAN or separate physical fabric. Providing availability by using several VMkernel adapters has configuration costs that involve vSphere and the network infrastructure. You can increase network availability by teaming physical network adapters.

Using Unicast in vSAN Network

You can design a simple unicast network for vSAN.

> **Note:**
>
> You can use DHCP with reservations, because the assigned IP addresses are bound to the MAC addresses of VMkernel ports.

Using RDMA

vSAN can use Remote Direct Memory Access (RDMA). RDMA typically has lower CPU utilization and less I/O latency. If your hosts support the RoCE v2 protocol, you can enable RDMA through the vSAN network service in vSphere Client.

Consider the following guidelines when designing vSAN over RDMA:

- Each vSAN host must have a vSAN certified RDMA-capable NIC, as listed in the vSAN section of the *Broadcom Compatibility Guide* available at: https://compatibilityguide.broadcom.com/. Use only the same model network adapters from the same vendor on each end of the connection. Configure the DCBx mode to IEEE.
- All hosts must support RDMA. If any host loses RDMA support, the entire vSAN cluster switches to TCP.
- The network must be lossless. Configure network switches to use Data Center Bridging with Priority Flow Control. Configure a lossless traffic class for vSAN traffic marked at priority level 3.
- vSAN with RDMA does not support LACP, IP-hash-based NIC teaming, or route based on physical network adapter load. vSAN with RDMA does support NIC failover.
- All hosts must be on the same subnet. vSAN with RDMA supports up to 32 hosts.
- vSAN with RDMA does not support vSAN stretched clusters and vSAN storage clusters.

Allocating Bandwidth for vSAN by Using Network I/O Control

vSAN traffic can share physical network adapters with other system traffic types, such as vSphere vMotion traffic, vSphere HA traffic, and virtual machine traffic. To guarantee the amount of bandwidth required for vSAN, use vSphere Network I/O Control in the vSphere Distributed Switch.

In vSphere Network I/O Control, you can configure reservation and shares for the vSAN outgoing traffic.

- Set a reservation so that Network I/O Control guarantees that minimum bandwidth is available on the physical adapter for vSAN.
- Set shares so that when the physical adapter assigned for vSAN becomes saturated, certain bandwidth is available to vSAN and to prevent vSAN from consuming the entire capacity of the physical adapter during rebuild and synchronization operations. For example, the physical adapter might become saturated when another physical adapter in the team fails and all traffic in the port group is transferred to the other adapters in the team.

For example, on a 10 GbE physical adapter that handles traffic for vSAN, vSphere vMotion, and virtual machines, you can configure certain bandwidth and shares.

Table 1004: Example Network I/O Control Configuration for a Physical Adapter That Handles vSAN

Traffic Type	Shares
vSAN	100
vSphere vMotion	70
Virtual machine	30

For information about using vSphere Network I/O Control to configure bandwidth allocation for vSAN traffic, see the vSphere Networking guide.

Marking vSAN Traffic

Priority tagging is a mechanism to indicate to the connected network devices that vSAN traffic has high Quality of Service (QoS) demands. You can assign vSAN traffic to a certain class and mark the traffic accordingly with a Class of Service (CoS) value from 0 (low priority) to 7 (high priority). Use the traffic filtering and marking policy of vSphere Distributed Switch to configure priority levels.

For more information, see What is Traffic Filtering and Marking Policy in the *vSphere Networking* guide.

Segmenting vSAN Traffic in a VLAN

Consider isolating vSAN traffic in a VLAN for enhanced security and performance, especially if you share the capacity of the backing physical adapter among several traffic types.

Jumbo Frames

If you plan to use jumbo frames with vSAN to improve CPU performance, verify that jumbo frames are enabled on all network devices and hosts in the cluster.

By default, the TCP segmentation offload (TSO) and large receive offload (LRO) features are enabled on ESX. Consider whether using jumbo frames improves the performance enough to justify the cost of enabling them on all nodes on the network.

Creating Static Routes for vSAN Networking

In traditional configurations, where vSphere uses a single default gateway, all routed traffic attempts to reach its destination through this gateway.

In such cases, you might need to create static routes in your vSAN environment.

vSAN enables you to override the default gateway for the vSAN VMkernel adapter on each host, and configure a gateway address for the vSAN network. This offers greater flexibility than static routes and simplifies management.

However, certain vSAN deployments might require static routing. For example, deployments where the witness is on a different network, or the vSAN stretched cluster deployment, where both the data sites and the witness host are on different networks.

To configure static routing on your ESX hosts, use the esxcli command:

```
esxcli network ip route ipv4 add -g gateway-to-use -n remote-network
```

remote-network is the remote network that your host must access, and *gateway-to-use* is the interface to use when traffic is sent to the remote network.

For information about network design for vSAN stretched clusters, see vSAN Stretched Clusters Network Design.

Best Practices for vSAN Networking

Consider networking best practices for vSAN to improve performance and throughput.

- vSAN OSA: For hybrid configurations, dedicate at least 1 GbE physical network adapter. Place vSAN traffic on a dedicated or shared 10 GbE physical adapter for best networking performance. For all-flash configurations, use a dedicated or shared 10 GbE physical network adapter.
- vSAN ESA: Use a dedicated or shared 25 GbE physical network adapter or higher
- Provision one additional physical NIC as a failover NIC.
- If you use a shared network adapter, place the vSAN traffic on a distributed switch and configure Network I/O Control to guarantee bandwidth to vSAN.

Designing and Sizing vSAN Fault Domains

vSAN fault domains can spread redundancy components across the servers in separate computing racks. In this way, you can protect the environment from a rack-level failure such as loss of power or connectivity.

Fault Domain Constructs

vSAN requires at least three fault domains to support **Failures to tolerate** (FTT) of 1. Each fault domain consists of one or more hosts. Fault domain definitions must acknowledge physical hardware constructs that might represent a potential zone of failure, for example, an individual computing rack enclosure.

If possible, use at least four fault domains. Three fault domains do not support certain data evacuation modes, and vSAN is unable to reprotect data after a failure. In this case, you need an additional fault domain with capacity for rebuilding, which you cannot provide with only three fault domains.

If fault domains are enabled, vSAN applies the active virtual machine storage policy to the fault domains instead of the individual hosts.

Calculate the number of fault domains in a cluster based on the **FTT** attribute from the storage policies that you plan to assign to virtual machines.

```
number of fault domains = 2 * FTT + 1
```

If a host is not a member of a fault domain, vSAN interprets it as a stand-alone fault domain.

Using Fault Domains Against Failures of Several Hosts

Consider a cluster that contains four server racks, each with two hosts. If the **Failures to tolerate** is set to one and fault domains are not enabled, vSAN might store both replicas of an object with hosts in the same rack enclosure. In this way, applications might be exposed to a potential data loss on a rack-level failure. When you configure hosts that could potentially fail together into separate fault domains, vSAN ensures that each protection component (replicas and witnesses) is placed in a separate fault domain.

If you add hosts and capacity, you can use the existing fault domain configuration or you can define fault domains.

For balanced storage load and fault tolerance when using fault domains, consider the following guidelines:

- Provide enough fault domains to satisfy the **Failures to tolerate** that are configured in the storage policies. Define at least three fault domains. Define a minimum of four domains for best protection.
- Assign the same number of hosts to each fault domain.
- Use hosts that have uniform configurations.
- Dedicate one fault domain of free capacity for rebuilding data after a failure, if possible.

Using Boot Devices and vSAN

Starting an ESX installation that is a part of a vSAN cluster from a flash device imposes certain restrictions.

When you boot a vSAN host from a USB/SD device, you must use a high-quality USB or SD flash drive of 4 GbE or larger.

When you boot a vSAN host from a SATADOM device, you must use single-level cell (SLC) device. The size of the boot device must be at least 16 GB.

During installation, the ESX installer creates a coredump partition on the boot device. The default size of the coredump partition satisfies most installation requirements.

- If the memory of the ESX host has 512 GB of memory or less, you can boot the host from a USB, SD, or SATADOM device.
- If the memory of the ESX host has more than 512 GB, consider the following guidelines.
  - You can boot the host from a SATADOM or disk device with a size of at least 16 GbE. When you use a SATADOM device, use a single-level cell (SLC) device.
  - If you are using vSAN, you must resize the coredump partition on ESX hosts to boot from USB/SD devices.

Hosts that boot from a disk have a local VMFS. If you have a disk with VMFS that runs virtual machines, you must separate the disk for an ESX boot that is not for vSAN. In this case you need separate controllers.

Log Information and Boot Devices in vSAN

When you boot ESX from a USB or SD device, log information and stack traces are lost on host reboot. They are lost because the scratch partition is on a RAM drive. Use persistent storage for logs, stack traces, and memory dumps.

Do not store log information on the vSAN datastore. This configuration is not supported because a failure in the vSAN cluster could impact the accessibility of log information.

Consider the following options for persistent log storage:

- Use a storage device that is not used for vSAN and is formatted with VMFS or NFS.
- Configure the ESX Dump Collector and vSphere Syslog Collector on the host to send memory dumps and system logs to vCenter.

Persistent Logging in a vSAN Cluster

Provide storage for persistence of the logs from the hosts in the vSAN cluster.

If you install ESX on a USB or SD device and you allocate local storage to vSAN, you might not have enough local storage or datastore space left for persistent logging.

To avoid potential loss of log information, configure the ESX Dump Collector and vSphere Syslog Collector to redirect ESX memory dumps and system logs to a network server.

Preparing a New or Existing Cluster for vSAN

Before you deploy a vSAN cluster and start using it as virtual machine storage, you must provide the infrastructure that is required for correct operation of vSAN.

Preparing Storage

Provide enough disk space for vSAN and for the virtualized workloads that use the vSAN datastore.

Verify the Compatibility of Storage Devices

Consult the *Broadcom Compatability Guide* to verify that your storage devices, drivers, and firmware are compatible with vSAN.

By using the vSAN section of the *Broadcom Compatibility Guide* available at Broadcom Comapatibility Guide, you can select a magnetic disk, flash device, and storage controllers, or check the compatibility of your storage devices for vSAN.

> **Note:**
>
> vSAN 9.1 does not support ARM servers as data nodes, witness nodes, or ARM compute-only clients from remotely mounting X86 vSAN Max storage.

You can choose from several options for vSAN compatibility.

- Use a vSAN ReadyNode server, a physical server that OEM vendors and VMware validate for vSAN compatibility.
- Assemble a node by selecting individual components from validated device models.

***Broadcom Compatibility Guide* Section**	**Component Type for Verification**
Systems	Physical server that runs ESX.
vSAN	• Magnetic disk SAS model for hybrid configurations. • Flash device model that is listed in the *Broadcom Compatibility Guide*. Certain models of PCIe flash devices can also work with vSAN. Consider also write endurance and performance class. • Storage controller model that supports passthrough. vSAN can work with storage controllers that are configured for RAID 0 mode if each storage device is represented as an individual RAID 0 group. • For vSAN ESA, NVMe TLC drives on Broadcom HCL with capacity 1.6 TB or higher, performance class F higher, and an endurance class of 1 Drive Writes Per Day (DWPD). For more information see vSAN ESA ReadyNode Hardware Guidance.

Preparing Storage Devices

Use flash devices and magnetic disks based on the requirements for vSAN.

Verify that the cluster has the capacity to accommodate anticipated virtual machine consumption and the **Failures to tolerate** in the storage policy for the virtual machines.

The storage devices must meet the following requirements so that vSAN can claim them:

- The storage devices are local to the ESX hosts. vSAN cannot claim remote devices.
- The storage devices do not have any existing partition information.
- On the same host, you cannot have both all-flash and hybrid disk groups.
- For vSAN ESA , NVME drives are only supported.

Prepare Devices for Storage Pool

vSAN ESA ready nodes requires a minimum one NVMe drive of 1.6 TB or higher for a storage pool. vSAN storage cluster ready nodes requires a minimum of two NVMe drives.

Prepare Devices for Disk Groups

Each disk group provides one flash caching device and at least one magnetic disk or one flash capacity device. For hybrid clusters, the capacity of the flash caching device must be at least 10 percent of the anticipated consumed storage on the capacity device, without the protection copies.

vSAN requires at least one disk group on a host that contributes storage to a cluster that consists of at least three hosts. Use hosts that have uniform configuration for best performance of vSAN.

Raw and Usable Capacity

Provide raw storage capacity that is greater than the capacity for virtual machines to handle certain cases.

- Do not include the size of the flash caching devices as capacity. These devices do not contribute storage and are used as cache unless you have added flash devices for storage.
- Provide enough space to handle the **Failures to tolerate** (FTT) value in a virtual machine storage policy. A FTT that is greater than 0 extends the device footprint. If the FTT is set to 1, the footprint is double. If the FTT is set to 2, the footprint is triple, and so on.
- Verify whether the vSAN datastore has enough space for an operation by examining the space on the individual hosts rather than on the consolidated vSAN datastore object. For example, when you evacuate a host, all free space in the datastore might be on the host that you are evacuating. The cluster is not able to accommodate the evacuation to another host.
- Provide enough space to prevent the datastore from running out of capacity, if workloads that have thinly provisioned storage start consuming a large amount of storage.
- Verify that the physical storage can accommodate the reprotection and maintenance mode of the hosts in the vSAN cluster.
- Consider the vSAN overhead to the usable storage space.
  - On-disk format version 3.0 and later adds an extra overhead, typically no more than 1-2 percent capacity per device. Deduplication and compression with software checksum enabled require extra overhead of approximately 6.2 percent capacity per device.

For more information about planning the capacity of vSAN datastores, see vSAN Sizer tool.

vSAN Policy Impact on Capacity

The vSAN storage policy for virtual machines affects the capacity devices in several ways.

Table 1005: vSAN VM Policy and Raw Capacity

Aspects of Policy Influence	Description
Policy changes	• The **Failures to tolerate** (FTT) influences the physical storage space that you must supply for virtual machines. The greater the FTT is for higher availability, the more space you must provide. When FTT is set to 1, it imposes two replicas of the VMDK file of a virtual machine. With FTT set to 1, a VMDK file that is 50 GbE requires 100 GbE space on different hosts. If the FTT is changed to 2, you must have enough space to support three replicas of the VMDK across the hosts in the cluster, or 150 GbE. • Some policy changes, such as a new number of disk stripes per object, require temporary resources. vSAN recreates the objects affected by the change. For a certain time, the physical storage must accommodate the old and new objects.
Available space for reprotecting or maintenance mode	When you place a host in maintenance mode or you clone a virtual machine, the datastore might not be able to evacuate the virtual machine objects, although the vSAN datastore indicates that enough space is available. This lack of space can occur if the free space is on the host that is being placed in maintenance mode.

Preparing Storage Controllers

Configure the storage controller on each host according to the requirements of vSAN.

Verify that the storage controllers on the vSAN hosts satisfy certain requirements for mode, driver, and firmware version, queue depth, caching, and advanced features.

Table 1006: Examining Storage Controller Configuration for vSAN OSA

Storage Controller Feature	Storage Controller Requirement
Required mode	• Review the vSAN requirements in the *Broadcom Compatibility Guide* for the required mode, passthrough or RAID 0, of the controller. • If both passthrough and RAID 0 modes are supported, configure passthrough mode instead of RAID0. RAID 0 introduces complexity for disk replacement.
RAID mode	• In the case of RAID 0, create one RAID volume per physical disk device. • Do not enable a RAID mode other than the mode listed in the *Broadcom Compatibility Guide*. • Do not enable controller spanning.
Driver and firmware version	• Use the latest driver and firmware version for the controller according to *Broadcom Compatibility Guide*. • If you use the in-box controller driver, verify that the driver is certified for vSAN. OEM ESX releases might contain drivers that are not certified and listed in the *Broadcom Compatibility Guide*.
Queue depth	Verify that the queue depth of the controller is 256 or higher. Higher queue depth provides improved performance.
Cache	Deactivate the storage controller cache, or set it to 100 percent read if disabling cache is not possible.
Advanced features	Deactivate advanced features, for example, HP SSD Smart Path.

Mark Flash Devices as Capacity Using ESXCLI

You can manually mark the flash devices on each host as capacity devices using esxcli.

Verify that you are using vSAN 9.1 or later.

1. To learn the name of the flash device that you want to mark as capacity, run the following command on each host.
   a) In the ESX Shell, run the `esxcli storage core device list` command.
   b) Locate the device name at the top of the command output and write the name down.

   The command takes the following options:

Table 1007: Command Options

Options	Description
`-d|--disk=str`	The name of the device that you want to tag as a capacity device. For example, `mpx.vmhba1:C0:T4:L0`
`-t|--tag=str`	Specify the tag that you want to add or remove. For example, the `capacityFlash` tag is used for marking a flash device for capacity.

   The command lists all device information identified by ESX.
2. In the output, verify that the `Is SSD` attribute for the device is `true`.
3. To tag a flash device as capacity, run the `esxcli vsan storage tag add -d <device name> -t capacityFlash` command.

   For example, the `esxcli vsan storage tag add -t capacityFlash -d mpx.vmhba1:C0:T4:L0` command, where `mpx.vmhba1:C0:T4:L0` is the device name.
4. Verify whether the flash device is marked as capacity.
   a) In the output, identify whether the `IsCapacityFlash` attribute for the device is set to `1`.

   Command Output

   You can run the `vdq -q -d <device name>` command to verify the `IsCapacityFlash` attribute. For example, running the `vdq -q -d mpx.vmhba1:C0:T4:L0` command, returns the following output.

```
\{
"Name"     : "mpx.vmhba1:C0:T4:L0",
"VSANUUID" : "",
"State"    : "Eligible for use by VSAN",
"ChecksumSupport": "0",
"Reason"   : "None",
"IsSSD"    : "1",
"IsCapacityFlash": "1",
"IsPDL"    : "0",
    \},
```

Untag Flash Devices Used as Capacity Using ESXCLI

You can untag flash devices that are used as capacity devices, so that they are available for caching.

1. To untag a flash device marked as capacity, run the `esxcli vsan storage tag remove -d <device name> -t capacityFlash` command. For example, the `esxcli vsan storage tag remove -t capacityFlash -d mpx.vmhba1:C0:T4:L0` command, where `mpx.vmhba1:C0:T4:L0` is the device name.
2. Verify whether the flash device is untagged.
   a) In the output, identify whether the `IsCapacityFlash` attribute for the device is set to `0`.

   Command Output

   You can run the `vdq -q -d <device name>` command to verify the `IsCapacityFlash` attribute. For example, running the `vdq -q -d mpx.vmhba1:C0:T4:L0` command, returns the following output.

```
[
    \{
"Name"     : "mpx.vmhba1:C0:T4:L0",
"VSANUUID" : "",
```

```
"State"    : "Eligible for use by VSAN",
"ChecksumSupport": "0",
"Reason"   : "None",
"IsSSD"    : "1",
"IsCapacityFlash": "0",
"IsPDL"    : "0",
    \},
```

Providing Memory for vSAN

Provision hosts with memory to support to the maximum number of devices and disks that you intend to use for vSAN.

For vSAN OSA to satisfy the combination of devices and disk groups, you must provision hosts with 32 GB of memory for system operations. For information about the maximum device configuration, refer to the vSphere Configuration Maximums guide and see the Broadcom knowledge base article 2113954. vSAN ESA requires a minimum of 128 GB of memory. To calculate vSAN memory overhead, see vSAN Sizer tool.

Preparing Your Hosts for vSAN

As a part of the preparation for enabling vSAN, review the requirements and recommendations about the configuration of hosts for the cluster.

- Verify that the storage devices on the hosts, and the driver and firmware versions for them, are listed in the vSAN section of the *Broadcom Compatibility Guide* available at: https://compatibilityguide.broadcom.com/.
- Make sure that a minimum of three hosts contribute storage to the vSAN datastore.
- For maintenance and remediation operations on failure, add at least four hosts to the cluster.
- Designate hosts that have uniform configuration for best storage balance in the cluster.
- Do not add hosts that have only compute resources to the cluster to avoid unbalanced distribution of storage components on the hosts that contribute storage. Virtual machines that require much storage space and run on compute-only hosts might store a great number of components on individual capacity hosts. As a result, the storage performance in the cluster might be lower.
- Do not configure aggressive CPU power management policies on the hosts for saving power. Certain applications that are sensitive to CPU speed latency might have low performance.
- If your cluster contains blade servers, consider extending the capacity of the datastore with an external storage enclosure that is connected to the blade servers. Make sure the storage enclosure is listed in the vSAN section of the *Broadcom Compatibility Guide*.
- Consider the configuration of the workloads that you place on vSAN:
  - For high levels of predictable performance, use vSAN ESA based clusters.
  - For balance between performance and older hardware, consider using vSAN OSA

vSAN and vCenter Compatibility

Synchronize the versions of vCenter and ESX to avoid potential faults caused by mismatched software.

For best integration between vSAN components on vCenter and ESX, deploy the latest version of the two vSphere components. See the following:

- vCenter Deployment and Setup guide
- VMware ESX Upgrade guide
- vCenter Upgrade guide

Configuring the vSAN Network

Before you enable vSAN on a cluster of ESX hosts, you must provide the necessary network infrastructure to carry vSAN communication.

vSAN provides a distributed storage solution, which implies exchanging data across the ESX hosts that participate in the cluster. Preparing the network for installing vSAN includes certain configuration aspects.

For information about network design guidelines, see Designing the vSAN Network.

Placing Hosts in the Same Subnet

Hosts must be connected in the same subnet for best networking performance. vSAN can also connect hosts in the same Layer 3 network if necessary.

Dedicating Network Bandwidth on a Physical Adapter

Allocate at least 1 GbE bandwidth for vSAN. You might use one of the following configuration options:

- vSAN OSA: Dedicate 1 GbE physical adapters for a hybrid host configuration, or use dedicated or shared 10 GbE physical adapters if possible. Use dedicated or shared 10 GbE physical adapters for all-flash configurations.
- vSAN ESA: Support the use of dedicated or shared 10 GbE physical adapters. Use dedicated or shared 25 GbE physical adapters or higher is recommended.
- Direct vSAN traffic on a physical adapter that handles other system traffic and use vSphere Network I/O Control on a distributed switch to configure shares for vSAN.

Configuring a Port Group on a Virtual Switch

Configure a port group on a virtual switch for vSAN.

- Assign the physical adapter for vSAN to the port group as an active uplink.
  When you need a NIC team for network availability, select a teaming algorithm based on the connection of the physical adapters to the switch.
- If designed, assign vSAN traffic to a VLAN by enabling tagging in the virtual switch.

Examining the Firewall on a Host for vSAN

vSAN sends messages on certain ports on each host in the cluster. Verify that the host firewalls allow traffic on these ports.

When you enable vSAN on a cluster, all required ports are added to ESX firewall rules and configured automatically. There is no need for an administrator to open any firewall ports or enable any firewall services manually.

You can view open ports for incoming and outgoing connections. Select the ESX host, and click **Configure** > **Security Profile**.

Creating a Single Site vSAN Cluster

You can enable vSAN when you create a vSphere cluster, or you can or enable vSAN on an existing clusters.

Characteristics of a vSAN Cluster

Before working on a vSAN environment, be aware of the characteristics of a vSAN cluster.

A vSAN cluster includes the following characteristics:

- You can have multiple vSAN clusters for each vCenter instance. You can use a single vCenter to manage more than one vSAN cluster.
- vSAN consumes all devices, including flash cache and capacity devices, and does not share devices with other features.
- vSAN clusters can include ESX hosts with or without capacity devices. The minimum requirement is three ESX hosts with capacity devices. For best results, create a vSAN cluster with uniformly configured ESX hosts.
- If an ESX host contributes capacity, it must have at least one flash cache device and one capacity device. vSAN ESA requires a minimum of one capacity device.

- In hybrid clusters, the magnetic disks are used for capacity and flash devices for read and write cache. vSAN allocates 70 percent of all available cache for read cache and 30 percent of available cache for the write buffer. In a hybrid configuration, the flash devices serve as a read cache and a write buffer.
- In all-flash clusters, one designated flash device is used as a write cache, additional flash devices are used for capacity. In all-flash clusters, all read requests come directly from the flash pool capacity.
- For vSAN ESA clusters, capacity drives contribute to cache and capacity.
- Only local or direct-attached capacity devices can participate in a vSAN cluster. vSAN cannot consume other external storage, such as SAN or NAS, attached to cluster.

To learn about the characteristics of a vSAN cluster configured through Quickstart, see Using Quickstart to Configure and Expand a vSAN Cluster .

For best practices about designing and sizing a vSAN cluster, see Designing and Sizing a vSAN Cluster.

Before Creating a vSAN Cluster

This topic provides a checklist of software and hardware requirements for creating a vSAN cluster. You can also use the checklist to verify that the cluster meets the guidelines and basic requirements.

Requirements for vSAN Cluster

Before you get started, verify specific models of hardware devices, and specific versions of drivers and firmware in the *Broadcom Compatibility Guide* available at https://compatibilityguide.broadcom.com/. The following table lists the key software and hardware requirements supported by vSAN.

> **CAUTION:**
> Using uncertified software and hardware components, drivers, controllers, and firmware might cause unexpected data loss and performance issues.

Table 1008: vSAN Cluster Requirements

Requirements	Description
ESX hosts	• Verify that you are using the latest version of ESX on your hosts. • Verify that there are at least three ESX hosts with supported storage configurations available to be assigned to the vSAN cluster. For best results, configure the vSAN cluster with four or more ESX hosts.
Memory	• Verify that each ESX host has a minimum of 32 GB of memory. • Verify that vSAN ESA has a minimum of 128 GB of memory. • For larger configurations, better performance, and to calculate required memory, see vSAN Sizer tool.
Storage I/O controllers, drivers, firmware	• Verify that the storage I/O controllers, drivers, and firmware versions are certified and listed in the *Broadcom Compatibility Guide* available at https://compatibilityguide.broadcom.com/. • Verify that the controller is configured for passthrough or RAID 0 mode. • Verify that the controller cache and advanced features are deactivated. If you cannot deactivate the cache, you must set the read cache to 100 percent. • Verify that you are using controllers with higher queue depths. Using controllers with queue depths less than 256 can significantly impact the performance of your virtual machines during maintenance and failure. **Note:** vSAN ESA supports NVMe drives and does not support storage controllers.

Requirements	Description
Cache and capacity	• For vSAN OSA, verify that vSAN hosts contributing storage to the cluster have at least one cache and one capacity device. vSAN requires exclusive access to the local cache and capacity devices of the ESX hosts in the vSAN cluster. They cannot share these devices with other uses, such as Virtual Flash File System (VFFS), VMFS partitions, or an ESX boot partition. • For vSAN ESA, verify that ESX hosts contributing storage have compatible flash storage devices. • For best results, create a vSAN cluster with uniformly configured ESX hosts.
Network connectivity	• Verify that each ESX host is configured with at least one network adapter. • For hybrid configurations, verify that vSAN hosts have a minimum dedicated bandwidth of 1 GbE. • For all-flash configurations, verify that vSAN hosts have a minimum bandwidth of 10 GbE. For best practices and considerations about designing the vSAN network, see Designing the vSAN Network and Networking Requirements for vSAN.
vSAN and vCenter compatibility	Verify that you are using the latest version of the vCenter.

For detailed information about vSAN cluster requirements, see Requirements for Enabling vSAN.

For in-depth information about designing and sizing the vSAN cluster, see the VMware vSAN Design and Sizing Guide.

Using Quickstart to Configure and Expand a vSAN Cluster

You can use the Quickstart workflow to quickly create, configure, and expand a vSAN cluster.

Quickstart consolidates the workflow to enable you to quickly configure a new vSAN cluster that uses recommended default settings for common functions such as networking, storage, and services. Quickstart groups common tasks and uses configuration wizards that guide you through the process. Once you enter the required information on each wizard, Quickstart configures the cluster based on your input.

Quickstart uses the vSAN health service to validate the configuration and help you correct configuration issues. Each Quickstart card displays a configuration checklist. You can click a green message, yellow warning, or red failure to display details.

Hosts added to a Quickstart cluster are automatically configured to match the cluster settings. The ESX software and patch levels of new hosts must match those in the cluster. Hosts cannot have any networking or vSAN configuration when added to a cluster using the Quickstart workflow. For more information about adding hosts, see Expanding a vSAN Cluster.

> **Note:**
> If you modify any network settings outside of QuickStart, this hampers your ability to add and configure more hosts to the cluster using the QuickStart workflow.

Characteristics of a Quickstart Cluster

A vSAN cluster configured using Quickstart has the following characteristics.

- Hosts must have ESX 9.1 or later.
- Host all have similar configuration, including network settings. Quickstart modifies network settings on each host to match the cluster requirements.
- Cluster configuration is based on recommended default settings for networking and services.
- Licenses are allocated to vCenter and gets automatically assigned to the vSAN clusters.

Managing and Expanding a Quickstart Cluster

Once you complete the Quickstart workflow, you can manage the cluster through vCenter, using the vSphere Client or command-line interface.

You can use the Quickstart workflow to add hosts to the cluster and claim additional disks. But once the cluster is configured through Quickstart, you cannot use Quickstart to modify the cluster configuration.

The Quickstart workflow is available only through the HTML5-based vSphere Client.

Skipping Quickstart

You can use the **Skip Quickstart** button to exit the Quickstart workflow, and continue configuring the cluster and its hosts manually. You can add new hosts individually, and manually configure those hosts. Once skipped, you cannot restore the Quickstart workflow for the cluster.

The Quickstart workflow is designed for new clusters. When you upgrade an existing vSAN cluster, the Quickstart workflow appears. Skip the Quickstart workflow and continue to manage the cluster through vCenter.

Use Quickstart to Configure a vSAN Cluster

You can use the Quickstart workflow to quickly configure a vSAN cluster.

- Verify that hosts are running ESX 9.1 or later.
- Verify that ESX hosts in the cluster do not have any existing vSAN or networking configuration.

  Note:
  If you perform network configuration through Quickstart, then modify those parameters from outside of Quickstart, you cannot use Quickstart to add or configure additional hosts.

1. Navigate to the new cluster in the vSphere Client.
2. Click the Configure tab, and select **Configuration > Quickstart**.
3. (optional) On the Cluster basics card, click **Edit** to open the Cluster basics wizard.
   a) (Optional) Enter a cluster name.
   b) Select basic services, such as DRS, vSphere HA, and vSAN.
      - Check **Enable vSAN ESA** to use vSAN ESA. vSAN ESA is optimized for high-performance flash storage devices that provide greater performance and efficiency.
      - Check **Cyber recovery** to recover your workloads from ransomware attacks and reduce downtime.
   c) Select any option to **Choose how to set up the cluster's image**.
   d) Select **Manage configuration at a cluster level** check box to ensure that all the hosts in the cluster have the same settings.
   e) Click **Next**.
   f) Select an image from the Image Library of Lifecycle Manager.
   g) Click **Next** and review the cluster details.
   h) Click **Finish**.
4. On the Add hosts card, click **Add** to open the Add hosts wizard.
   a) On the Add hosts page, enter information for new hosts, or click Existing hosts and select from hosts listed in the inventory.
   b) On the Host summary page, verify the host settings.
   c) On the Ready to complete page, click **Finish**.

   Note:

   If you are running vCenter on a host, the host cannot be placed into maintenance mode as you add it to a cluster using the Quickstart workflow. All other virtual machines on the host must be powered off.
5. On the Cluster configuration card, click **Configure** to open the Cluster configuration wizard.
   a) (vSAN ESA clusters) On the Cluster Type page, enter the HCI cluster type:
      - **vSAN HCI** provides compute resources and storage resources. The datastore can be shared across data centers and vCenter instances.

- **vSAN storage cluster** provides storage resources, but not compute resources. The datastore can be mounted by remote vSAN clusters across data centers and vCenter instances.

b) On the Configure the distributed switches page, enter networking settings, including distributed switches, port groups, and physical adapters.

- In the **Distributed switches** section, enter the number of distributed switches to configure from the drop-down menu. Enter a name for each distributed switch. Click **Use Existing** to select an existing distributed switch.
  If the host has a standard virtual switch with the same name as the selected distributed switch, the standard switch is migrated to the corresponding distributed switch.
  Network resource control is enabled and set to version 3. Distributed switches with network resource control version 2 cannot be used.
- In the **Port Groups** section, select a distributed switch to use for vMotion and a distributed switch to use for the vSAN network.
- In the **Physical adapters** section, select a distributed switch for each physical network adapter. You must assign each distributed switch to at least one physical adapter.
  If the physical adapters chosen are attached to a standard virtual switch with the same name across hosts, the standard switch is migrated to the distributed switch. If the physical adapters chosen are unused, there is no migration from standard switch to distributed switch.
  Network resource control is enabled and set to version 3. Distributed switches with network resource control version 2 cannot be used.

c) On the vMotion traffic page, enter IP address information for vMotion traffic.

d) On the Storage traffic page, enter IP address information for storage traffic.

e) On the Advanced options page, enter information for cluster settings, including DRS, HA, vSAN, host options, and EVC. Use the toggle button to enable **Apply Auto-RAID to all objects** even if the storage policy does not have vSAN ESA Auto-RAID. This overrides the individual storage policy rules such as site disaster tolerance, failure to tolerate, and stripe width configured on the vSAN cluster without impacting other storage policy rules applied. You can disable Apply Auto-RAID to all objects if you want to use site disaster tolerance and failure to tolerate rules specified in the storage policy instead of vSAN ESA Auto-RAID. With vSAN 9.1 release, vSAN ESA Global Deduplication is a cluster setting that is available only after you configure a cluster. For vSAN ESA, compression is enabled by default.

f) On the Claim disks page, select storage devices on each host. For clusters with vSAN OSA, select one cache device and one or more capacity devices. For clusters with vSAN ESA, select flash devices for the host's storage pool.

Note:

Only the vSAN Data Persistence platform can consume vSAN Direct storage. The vSAN Data Persistence platform provides a framework for software technology partners to integrate with VMware infrastructure. Each partner must develop their own plug-in for VMware customers to receive the benefits of the vSAN Data Persistence platform. The platform is not operational until the partner solution running on top is operational. For more information, see the *vSphere Supervisor Concepts* guide.

g) (Optional) On the Create fault domains page, define fault domains for hosts that can fail together.

For more information about fault domains, see Managing Fault Domains in vSAN Clusters.

h) (Optional) On the Proxy setting page, configure the proxy server if your system uses one.

i) On the Review page, verify the cluster settings, and click **Finish**.

You can manage the cluster through your vCenter.

You can add hosts to the cluster through Quickstart. For more information, Expanding a vSAN Cluster.

Manually Enabling vSAN

To create a vSAN cluster, you create a vSphere host cluster and enable vSAN on the cluster.

A vSAN cluster can include hosts with capacity and hosts without capacity. Follow these guidelines when you create a vSAN cluster.

- A vSAN cluster must include a minimum of three ESX hosts. For a vSAN cluster to tolerate host and device failures, at least three hosts that join the vSAN cluster must contribute capacity to the cluster. For best results, consider adding four or more hosts contributing capacity to the cluster.
- Only ESX hosts can join the vSAN cluster.
- Before you move a host from a vSAN cluster to another cluster, make sure that the destination cluster is vSAN enabled.
- To be able to access the vSAN datastore, an ESX host must be a member of the vSAN cluster.

After you enable vSAN, the vSAN storage provider is automatically registered with vCenter and the vSAN datastore is created. For information about storage providers, see the vSphere Storage guide.

Set Up a VMkernel Network for vSAN

To enable the exchange of data in the vSAN cluster, you must provide a VMkernel network adapter for vSAN traffic on each ESX host.

1. Right-click the host, and select **Add Networking**.
2. On the **Select connection type** page, select **VMkernel Network Adapter** and click **Next**.
3. On the **Select target device** page, configure the target switching device.
4. On the **Port properties** page, select **vSAN** service. The configuration of vSAN or vSAN witness network interface with vSAN Storage cluster client network interface is not supported.
5. Complete the VMkernel adapter configuration.
6. On the **Ready to complete** page, verify that vSAN is Enabled in the status for the VMkernel adapter, and click **Finish**.

vSAN network is enabled for the host.

You can enable vSAN on the host cluster.

Create a vSAN Cluster

You can create a cluster, and then configure the cluster for vSAN.

1. Right-click a data center and select **New Cluster**.
2. Type a name for the cluster in the **Name** text box.
3. Turn on DRS, vSphere HA, and vSAN for the cluster.
   - Check **Enable vSAN ESA** to use vSAN ESA. vSAN ESA is optimized for high-performance flash storage devices that provide greater performance and efficiency.
   - Check **Cyber recovery** to recover your workloads from ransomware attacks and reduce downtime.
4. Select any option to **Choose how to set up the cluster's image**. For more information on setting up cluster image, see Create a Cluster That Uses an Image by Composing a New Image in the *Managing Host and Cluster Lifecycle* guide.
5. (Optional) Select **Manage configuration at a cluster level** check box to ensure that all the ESX hosts in the cluster have the same settings.
6. Click **Next**.
7. Select an image from the Image Library of Lifecycle Manager.
8. Click **Next** and review the cluster details.
9. Click **Finish**.
   The cluster appears in the inventory.
10. Add hosts to the vSAN cluster.
    vSAN clusters can include hosts with or without capacity devices. For best results, add hosts with capacity.

Configure services for the vSAN cluster. See Configure a Cluster for vSAN Using the vSphere Client.

Configure a Cluster for vSAN Using the vSphere Client

You can use the vSphere Client to configure vSAN on an existing cluster.

- Verify that your environment meets all requirements. See Requirements for Enabling vSAN.
- Create a vSphere cluster and add ESX hosts to the vSphere cluster before enabling and configuring vSAN. Configure the port properties on each ESX host to add the vSAN service.

  Note:

  You can use Quickstart to quickly create and configure a vSAN cluster. For more information, see Use Quickstart to Configure a vSAN Cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Select an HCI configuration type.
      - **vSAN HCI** provides compute resources and storage resources. The datastore can be shared across clusters in the same data center, and across clusters managed by remote vCenter instances.
      - **Compute Cluster** is a vSphere cluster that can mount a remote datastore from a vSAN storage cluster. These clusters are ESX hosts in a vSphere cluster that only complies with the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
      - **vSAN Storage Cluster** (deployment model based on vSAN ESA) provides storage resources, but not compute resources. The datastore can be mounted by client vSphere clusters and vSAN clusters in the same data center and from remote vCenter instances.
   b) Select a deployment option (**Single site vSAN cluster**, **Two node vSAN cluster**, or **vSAN stretched cluster**).
   c) Click **Configure** to open the Configure vSAN wizard.
4. Select vSAN ESA if your cluster is compatible, and click **Next**.
   - Select **vSAN managed disk claim** to claim all compatible disks on the existing host cluster. Ensure that the vSAN disks are compatible with the vSAN Hardware Compatibility List.
   - Select **Auto-Policy management** to optimize capacity utilization based on the cluster size and type.
   - Select **Apply Auto-RAID to all objects** to override the individual storage policy rules such as site disaster tolerance, failure to tolerate, and stripe width configured on the vSAN cluster without impacting other storage policy rules applied.
5. Configure the vSAN services to use, and click **Next**.
   - Configure data management features, including data-at-rest encryption and data-in-transit encryption. With vSAN 9.1 release, vSAN ESA Global Deduplication is a cluster setting that is available only after you configure a cluster.
   - Select RDMA (remote direct memory access) if your network supports it.
   - If you configure a vSAN storage cluster, you get the option to **Use Storage cluster client network**. You can select this option to separate the external VM traffic from the internal storage traffic by utilizing the dedicated VMkernel ports for different traffic types. After enabling this option, you cannot modify the network configuration.
   - Create and assign vSAN ESA Auto-RAID policy as a default datastore policy for vSAN ESA. This auto-policy management allows best capacity utilization after the cluster configuration is complete.
6. Claim disks for the vSAN cluster, and click **Next**.

   For vSAN OSA, each ESX host that contribute storage requires at least one flash device for cache, and one or more devices for capacity. For vSAN ESA, each ESX host that contributes storage requires one or more flash devices.

7. (Optional) Create fault domains to group ESX hosts that can fail together.
8. Review the configuration, and click **Finish**.

Enabling vSAN creates a vSAN datastore and registers the vSAN storage provider. vSAN storage providers are built-in software components that communicate the storage capabilities of the datastore to vCenter.

- Verify that the vSAN datastore has been created. See View vSAN Datastore.
- Verify the health of the new vSAN cluster created. See Check vSAN Health.
- Verify that the vSAN storage provider is registered. See View vSAN Storage Providers.

Edit vSAN Settings

You can edit the settings of your vSAN cluster to configure data management features and enable services provided by the cluster.

Edit the settings of an existing vSAN cluster if you want to enable deduplication and compression, compression only, or to enable encryption. If you enable deduplication and compression, or if you enable encryption, the on-disk format of the cluster is automatically upgraded to the latest version.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Click the **Edit** or **Enable** button for the service you want to configure.
   - Configure Storage. Using the vSAN **ESA Options** dialog, you can enable vSAN managed disk claim and Auto-Policy management, and Apply Auto-RAID to all objects. vSAN displays the number of failures to tolerate provided by the vSAN ESA Auto-RAID objects.
   - Configure Performance Service. For more information, see Monitoring vSAN Performance.
   - Enable file service. For more information, see vSAN File Service.
   - Configure vSAN Network options. Enable RDMA if your network supports it.
   - Configure vSAN Data Protection. Before you can use vSAN Data Protection, you must deploy the VMware Live Recovery appliance.
   - Configure iSCSI target service. For more information, see Using the vSAN iSCSI Target Service.
   - Configure Data Reduction and Encryption for vSAN ESA, including global deduplication, data-at-rest encryption, and data-in-transit encryption. With vSAN 9.1 release, vSAN ESA Global Deduplication is a cluster setting that you can enable on a vSAN ESA cluster using the toggle button. After enabling global deduplication, you can pause deduplication. The deduplicated data cannot be reverted when you pause global deduplication. For vSAN ESA, compression is enabled by default.

     Note:

     If you enable global deduplication, you cannot convert a single site vSAN cluster to a vSAN stretched cluster. If you have a vSAN stretched cluster, you cannot enable global deduplication.
   - Configure capacity reservations and alerts. For more information, see About Reserved Capacity in vSAN Cluster.
   - Configure advanced options:
     – Object repair timer. For more information, see Monitor the Resynchronization Tasks in vSAN Cluster.
     – Site read locality. In the stretched cluster environments, the reads to vSAN objects occur to the local VM object location. When you enable the site read locality, the reads occur on the local and the remote sites. You can disable the site read locality for two-node vSAN clusters.
     – Thin swap. When you enable thin swap, the VM swap objects does not reserve 100 percentage of the swap capacity.
     – Guest Trim/Unmap. This option is enabled by default for vSAN ESA cluster. For more information, see Reclaiming Storage Space in vSAN with SCSI Unmap.
     – Automatic rebalance. For more information, see Configure Automatic Rebalance in vSAN Cluster.
     .
   - Configure vSAN historical health service. For more information, see About vSAN Health.

5. Modify the settings to match your requirements.
6. Click **Apply** to confirm your selections.

Enable vSAN on an Existing Cluster

You can enable vSAN on an existing cluster, and configure features and services.

- Verify that your environment meets all requirements. See Requirements for Enabling vSAN.
- Ensure that the vSAN network on the existing vSAN cluster has a VMkernel NIC, configured with a static or DHCP IP address and the VMkernel port tagged with the vSAN traffic.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Select a configuration type (**vSAN HCI**, **Two node vSAN cluster**, **vSAN Stretched cluster, Compute Cluster, or vSAN Storage Cluster**).
   b) Select **I need local vSAN Datastore** if you plan to add disk groups or storage pools to the cluster ESX hosts.
   c) Click **Configure** to open the **Configure vSAN** wizard.

   If the vSAN HCL configuration is compatible with the vSAN ESA cluster, vSAN ESA gets auto enabled.
4. Configure the vSAN services to use, and click **Next**.
   - Configure data management features, including deduplication and compression, data-at-rest encryption, data-in-transit encryption.
     – Select Remote Direct Memory Access (RDMA) if your network supports it. For more information, see Monitor vSAN Capacity.
   - Create and assign a default datastore policy for vSAN ESA. This auto-policy management allows best capacity utilization after the cluster configuration is complete.
5. Claim disks for the vSAN cluster, and click **Next**.
   - For vSAN ESA, if the disks are compatible, the auto-claim get enabled. vSAN can claim all the eligible disks and add additional hosts.
   - For vSAN OSA, each ESX host that contribute storage requires at least one flash device for cache, and one or more devices for capacity. For vSAN ESA, each ESX host that contributes storage requires one or more flash devices.
6. Create fault domains to groupESX hosts that can fail together.
7. Review the configuration, and click **Finish**.

View vSAN Datastore

After you enable vSAN, a single datastore is created. You can review the capacity of the vSAN datastore. Configure vSAN and disk groups or storage pools.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Capacity**.

   vSAN provides the capacity overview of the vSAN datastore and the vSAN system overheads. For more information, see .

Using vSAN and vSphere HA

You can enable vSphere HA and vSAN on the same cluster. vSphere HA provides the same level of protection for virtual machines on vSAN datastores as it does on traditional datastores. This level of protection imposes specific restrictions when vSphere HA and vSAN interact.

ESX Host Requirements

You can use vSAN with a vSphere HA cluster only if the following conditions are met:

- The cluster's ESX hosts all must be version 9.1 or later.
- The cluster must have a minimum of three ESX hosts, unless it is a vSAN two-host cluster. For best results, configure the vSAN cluster with four or more hosts.

> **Note:**
> vSAN supports Proactive HA. Select the following remediation method: **Maintenance mode for all failures**. Quarantine mode is supported, but it does not protect against data loss if the host in quarantine mode fails, and there are objects with FTT=0 or objects with FTT=1 that are degraded.

Networking Differences

vSAN uses its own logical network. When vSAN and vSphere HA are enabled for the same cluster, the HA interagent traffic flows over this storage network rather than the management network. vSphere HA uses the management network only when vSAN is turned off. vCenter chooses the appropriate network when vSphere HA is configured on a host.

> **Note:**
> Make sure vSphere HA is not enabled when you enable vSAN on the cluster. Then you can re-enable vSphere HA.

When a virtual machine is only partially accessible in all network partitions, you cannot power on the virtual machine or fully access it in any partition. For example, if you partition a cluster into P1 and P2, the VM namespace object is accessible to the partition named P1 and not to P2. The VMDK is accessible to the partition named P2 and not to P1. In such cases, the virtual machine cannot be powered on and it is not fully accessible in any partition .

The following table shows the differences in vSphere HA networking whether or not vSAN is used.

Table 1009: vSphere HA Networking Differences

	vSAN On	vSAN Off
Network used by vSphere HA	vSAN storage network	Management network
Heartbeat datastores	Any datastore mounted to more than one host, but not vSAN datastores	Any datastore mounted to more than one host
Host declared isolated	Isolation addresses not pingable and vSAN storage network inaccessible	Isolation addresses not pingable and management network inaccessible

If you change the vSAN network configuration, the vSphere HA agents do not automatically acquire the new network settings. To change the vSAN network, you must re-enable host monitoring for the vSphere HA cluster:

1. Deactivate Host Monitoring for the vSphere HA cluster.
2. Make the vSAN network changes.
3. Right-click all hosts in the cluster and select **Reconfigure HA**.
4. Reactivate Host Monitoring for the vSphere HA cluster.

> **Note:**
>
> You can disable the default gateway by setting `das.usedefaultisolationaddresss` and `das.isolationaddress0` to an address on the vSAN network. For more information, see Using an Isolation address to ensure HA functionality in a vSAN environment.

Capacity Reservation Settings

When you reserve capacity for your vSphere HA cluster with an admission control policy, this setting must be coordinated with the corresponding **Failures to tolerate** policy setting in the vSAN rule set. It must not be lower than the capacity reserved by the vSphere HA admission control setting. For example, if the vSAN rule set allows for only two failures, the vSphere HA admission control policy must reserve capacity that is equivalent to only one or two host failures. If you are

using the Percentage of Cluster Resources Reserved policy for a cluster that has eight hosts, you must not reserve more than 25 percent of the cluster resources. In the same cluster, with the **Failures to tolerate** policy, the setting must not be higher than two hosts. If vSphere HA reserves less capacity, failover activity might be unpredictable. Reserving too much capacity overly constrains the powering on of virtual machines and inter-cluster vSphere vMotion migrations. For information about the Percentage of Cluster Resources Reserved policy, see the vSphere Availability guide.

vSAN and vSphere HA Behavior in a Multiple Host Failure

After a vSAN cluster fails with a loss of failover quorum for a virtual machine object, vSphere HA might not be able to restart the virtual machine even when the cluster quorum has been restored. vSphere HA guarantees the restart only when it has a cluster quorum and can access the most recent copy of the virtual machine object. The most recent copy is the last copy to be written.

Consider an example where a vSAN virtual machine is provisioned to tolerate one host failure. The virtual machine runs on a vSAN cluster that includes three hosts, H1, H2, and H3. All three hosts fail in a sequence, with H3 being the last host to fail.

After H1 and H2 recover, the cluster has a quorum (one host failure tolerated). Despite this quorum, vSphere HA is unable to restart the virtual machine because the last host that failed (H3) contains the most recent copy of the virtual machine object and is still inaccessible.

In this example, either all three hosts must recover at the same time, or the two-host quorum must include H3. If neither condition is met, HA attempts to restart the virtual machine when host H3 is online again.

Deploying vSAN with vCenter

With vSphere Foundation 9.1, you can use an installer to deploy a vCenter instance and host the vCenter on that cluster.

When you use the vCenter installer, ensure that you do not select **Install on a new vSAN cluster** option. Follow the steps in the installer wizard to complete the deployment. You can activate vSAN from the vSphere Client after deploying vCenter. For more information on vSAN bootstrap not working with the installer, see Broadcom knowledge base article 389238.

Turn Off vSAN

You can turn off vSAN for a vSphere cluster when the vCenter managing the vSphere cluster is deployed outside the vSAN cluster.

- Disable vSphere High Availability (HA), if HA is enabled.
- Shutdown all the VMs, unless vCenter is on the cluster. If vCenter is on the cluster, you must migrate vCenter from the cluster that you want to turn off.
- Verify that the ESX hosts are in maintenance mode with 'No Data Migration' selected. For more information, see Place a Member of vSAN Cluster in Maintenance Mode.

When you turn off vSAN for a cluster, all virtual machines and data services located on the vSAN datastore become inaccessible. If you have consumed storage on the vSAN cluster using vSAN Direct, then the vSAN Direct monitoring services, such as health checks, space reporting, and performance monitoring, are not available. If you intend to use virtual machines while vSAN is off, make sure you migrate the virtual machines from vSAN datastore to another datastore before turning off the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Click **Turn Off** vSAN.
5. On the Turn Off vSAN dialog, confirm your selection.

Creating a vSAN Stretched Cluster or Two-Node vSAN Cluster

You can create a vSAN stretched cluster that spans two availability zones. vSAN stretched clusters enable you to extend the vSAN datastore across two availability zones to use it as stretched storage. The vSAN stretched cluster continues to function if a failure or scheduled maintenance occurs at one zone.

What Are vSAN Stretched Clusters

vSAN stretched clusters extend the vSAN cluster from a single data site to two availability zones for a better level of availability and intersite load balancing. vSAN stretched clusters are typically deployed in environments where the distance between data centers is limited, such as metropolitan or campus environments.

You can use vSAN stretched clusters to manage planned maintenance and avoid disaster scenarios, because maintenance or loss of one site does not affect the overall operation of the cluster. In a vSAN stretched cluster configuration, both data sites are active sites. If either site fails, vSAN uses the storage on the other site. vSphere HA restarts any VM that must be restarted on the remaining active site.

You must designate one site as the preferred site. The other site becomes a secondary or nonpreferred site. If the network connection between the two active sites is lost, vSAN continues operation with the preferred site. The site designated as preferred typically is the one that remains in operation, unless it is resyncing or has another issue. The site that leads to maximum data availability is the one that remains in operation.

A vSAN stretched cluster can tolerate one intersite link failure at a time without data becoming unavailable. A link failure is a loss of network connection between the two sites or between one site and the witness host. During a site failure or loss of network connection, vSAN automatically switches to fully functional sites.

vSAN stretched clusters can tolerate a witness host failure when one site is unavailable. Configure the storage policy Site disaster tolerance rule to Site mirroring - stretched cluster. If one site is down due to maintenance or failure and the witness host fails, objects become non-compliant but remain accessible.

For more information about working with vSAN stretched clusters, see the vSAN Stretched Cluster Guide.

Witness Host

Each vSAN stretched cluster consists of two data sites and one witness host. The witness host resides at a third site and contains the witness components of virtual machine objects. The witness host does not store customer data, only metadata, such as the size and UUID of vSAN object and components.

The witness host serves as a tiebreaker when a decision must be made regarding availability of datastore components when the network connection between the two availability zones is lost. In this case, the witness host forms a vSAN cluster with the preferred site. But if the preferred site becomes isolated from the secondary site and the witness, the witness host forms a cluster using the secondary site. When the preferred site is online again, data is resynchronized to ensure that both sites have the latest copies of all data.

If the witness host fails, all corresponding objects become noncompliant but are fully accessible.

The witness host has the following characteristics:

- The witness host can use low bandwidth/high latency links.
- The witness host cannot run virtual machines.
- A single witness host can support only one vSAN stretched cluster. Two-node vSAN clusters can share a single witness host.

- The witness host must have one VMkernel adapter with vSAN traffic enabled, with connections to all hosts in the cluster. The witness host uses one VMkernel adapter for management and one VMkernel adapter for vSAN data traffic. The witness host can have only one VMkernel adapter dedicated to vSAN.
- The witness host must be a standalone host dedicated to the vSAN stretched cluster. It cannot be added to any other cluster or moved in inventory through vCenter.

The witness host can be a physical host or an ESX host running inside a VM. The VM witness host does not provide other types of functionality, such as storing or running virtual machines. Multiple witness hosts can run as virtual machines on a single physical server. For patching and basic networking and monitoring configuration, the VM witness host works in the same way as a typical ESX host. You can manage it with vCenter, patch it and update it by using `esxcli` or vSphere Lifecycle Manager, and monitor it with standard tools that interact with ESX hosts.

You can use a witness virtual appliance as the witness host in a vSAN stretched cluster. The witness virtual appliance is an ESX host in a VM, packaged as an OVF or OVA. The appliance is available in different options, based on the size of the deployment. You can use a witness virtual appliance as the witness host in a vSAN stretched cluster. The witness virtual appliance is an ESX host in a VM, packaged as an OVF or OVA. Different appliances and different options are available, based on the vSAN architecture and the size of the deployment.

vSAN Stretched Clusters and Fault Domains

vSAN stretched clusters use fault domains to provide redundancy and failure protection across sites. Each site in a vSAN stretched cluster resides in a separate fault domain.

A vSAN stretched cluster requires three fault domains: the preferred site, the secondary site, and a witness host. Each fault domain represents a separate site. When the witness host fails or enters maintenance mode, vSAN considers it a site failure.

vSAN can provide an extra level of local fault protection for virtual machine objects in vSAN stretched clusters. When you configure a vSAN stretched cluster, the following policy rules are available for objects in the cluster:

- **Site disaster tolerance**. For vSAN stretched clusters, this rule defines the failure tolerance method. Select **Site mirroring - stretched cluster**.
- **Failures to tolerate (FTT)**. For vSAN stretched clusters, **FTT** defines the number of additional host failures that a virtual machine object can tolerate.
- **None**. Configure vSAN Storage Policy for SMP-FT VMs.
  - vSAN stretched clusters support enabling Symmetric Multiprocessing Fault Tolerance (SMP-FT) VMs only when site disaster tolerance storage policy is set to None with either Preferred or Secondary site. vSAN does not support SMP-FT VMs on a stretched cluster when site disaster tolerance storage policy is set to Site mirroring - stretched cluster.
  - vSAN ROBO clusters support enabling Symmetric Multiprocessing Fault Tolerance (SMP-FT) with FTT set to 1 only when both the data nodes are in the same physical site.

In a vSAN stretched cluster with local fault protection, even when one site is unavailable, the cluster can perform repairs on missing or broken components in the available site.

vSAN continue to serve I/O if any disks or disks on one site reach 96 percent full or 5 GbE free capacity (whichever is less) while disks on the other site have free space available. Components on the affected site are marked absent, and vSAN continues to perform I/O to healthy object copies on the other site. When disks on the affected site disk reach 94 percent capacity or 10 GbE (whichever is less), the absent components become available. vSAN resyncs the available components and all objects become policy compliant.

vSAN Stretched Cluster Design Considerations

Consider these guidelines when working with a vSAN stretched cluster.

- Configure DRS settings for the vSAN stretched cluster.
  - DRS must be enabled on the cluster. If you place DRS in partially automated mode, you can control which virtual machines to migrate to each site. vSAN enables you to operate DRS in automatic mode, and recover gracefully from network partitions.

  - Create two host groups, one for the preferred site and one for the secondary site. Associate the hosts in the vSAN preferred fault domain to the preferred site host group, and associate the hosts in the vSAN secondary fault domain to the secondary site host group.
  - Create two VM groups, one to hold the virtual machines on the preferred site and one to hold the virtual machines on the secondary site.
  - Create two VM-Host affinity rules that map VMs-to-host groups, and specify which virtual machines and hosts reside in the preferred site and which virtual machines and hosts reside in the secondary site.
  - Configure VM-Host affinity rules to perform the initial placement of virtual machines in the cluster.
- Configure HA settings for the vSAN stretched cluster.
  - HA rule settings should respect VM-Host affinity rules during failover.
  - Disable HA datastore heartbeats.
  - Use HA with Host Failure Monitoring, Admission Control, Isolation Response, and set FTT to the number of hosts in each site.
- Configure a vSAN storage policy that has a stretched cluster site failure tolerance rule. With vSAN ESA and auto-policy management enabled, Broadcom recommends storage policies to optimize capacity utilization based on the cluster size and topology.
- vSAN stretched clusters support enabling Symmetric Multiprocessing Fault Tolerance (SMP-FT) virtual machines only when **Site Disaster Tolerance** is set to **None** with either Preferred or Secondary. vSAN does not support SMP-FT virtual machines on a vSAN stretched cluster with **Site Disaster Tolerance** set to 1 or more. vSAN two-host clusters support enabling SMP-FT with **FTT** set to 1 only when both data nodes are in the same site.
- When a host is disconnected or not responding, you cannot add or remove the witness host. This limitation ensures that vSAN collects enough information from all hosts before initiating reconfiguration operations.
- Using `esxcli` to add or remove hosts is not supported for vSAN stretched clusters.
- Do not create snapshots of the witness host or backup the witness host. If the witness host fails, see Replace the Witness Host.

To learn how the stretched cluster storage policy rules impact the vSAN capacity requirements, see vSAN Stretched Cluster Guide.

Best Practices for Working with vSAN Stretched Clusters

When working with vSAN stretched clusters, follow these recommendations for proper performance.

- If one of the sites (fault domains) in a vSAN stretched cluster is inaccessible, new virtual machines can still be provisioned in the subcluster containing the operational site. These new virtual machines are implicitly force provisioned if at least two or three sites are available and are non-compliant until the partitioned site rejoins the cluster. A site here refers to either a data site or the witness host. The objects have no failure tolerance and are susceptible to data loss on any additional failure. They create new virtual machines only if necessary. With the vSAN 9.1 release, the new virtual machines are created to comply with the lower failures to tolerate, specified in the policy. This improves failure tolerance as the policy specifies a lower failures to tolerate of one or more.
- If an entire site goes offline due to a power outage or loss of network connection, restart the site immediately, without much delay. Instead of restarting vSAN hosts one by one, bring all hosts online approximately at the same time, ideally within a span of 10 minutes. By following this process, you avoid resynchronizing a large amount of data across the sites.
- If a host is permanently unavailable, remove the host from the cluster before you perform any reconfiguration tasks.

vSAN Stretched Clusters Network Design

All three sites in a vSAN stretched cluster communicate across the management network and across the vSAN network. The virtual machines in both data sites communicate across a common virtual machine network.

A vSAN stretched cluster must meet certain basic networking requirements.

- Management network requires connectivity across all three sites, using a Layer 2 stretched network or a Layer 3 network.

- The vSAN network requires connectivity across all three sites. It must have independent routing and connectivity between the data sites and the witness host. vSAN supports both Layer 2 and Layer 3 between the two data sites, and Layer 3 between the data sites and the witness host.
- VM network requires connectivity between the data sites, but not the witness host. Use a Layer 2 stretched network or Layer 3 network between the data sites. In the event of a failure, the virtual machines do not require a new IP address to work on the remote site.
- vMotion network requires connectivity between the data sites, but not the witness host. Use a Layer 2 stretched or a Layer 3 network between data sites.

Note:

vSAN over RDMA is not supported on vSAN stretched clusters or two-node vSAN clusters.

Override Default Gateway for a vSAN VMKernel Adapater

If you use a single default gateway on ESX hosts, each ESX host contains a default TCP/IP stack that has a single default gateway. The default route is typically associated with the management network TCP/IP stack.

The management network and the vSAN network might be isolated from one another. For example, the management network might use vmk0 on physical NIC 0, while the vSAN network uses vmk2 on physical NIC 1 (separate network adapters for two distinct TCP/IP stacks). This configuration implies that the vSAN network has no default gateway.

You can override the default gateway for the vSAN VMkernel adapter on each host, and configure a gateway address for the vSAN network. You can override the default gateway when the vSAN network uses a different subnet than the management network. In vSAN stretched clusters, data sites and the witness hosts can reside on separate networks. A dedicated vSAN gateway allows traffic to reach the witness host without static routes, simplifying network configuration. For more information on overriding default gateway, see Override the Default Gateway of VMKernel Adapter.

You also can use static routes to communicate across networks. Consider a vSAN network that is stretched over two data sites on a Layer 2 broadcast domain (for example, 172.10.0.0) and the witness host is on another broadcast domain (for example, 172.30.0.0). If the VMkernel adapters on a data site try to connect to the vSAN network on the witness host, the connection fails because the default gateway on the ESX host is associated with the management network. There is no route from the management network to the vSAN network.

Define a new routing entry that indicates which path to follow to reach a particular network. For a vSAN network on a vSAN stretched cluster, you can add static routes to ensure proper communication across all hosts.

For example, you can add a static route to the hosts on each data site, so requests to reach the 172.30.0.0 witness network are routed through the 172.10.0.0 interface. Also add a static route to the witness host so that requests to reach the 172.10.0.0 network for the data sites are routed through the 172.30.0.0 interface.

Note:
If you use static routes, you must manually add the static routes for new ESX hosts added to either site before those hosts can communicate across the cluster. If you replace the witness host, you must update the static route configuration.

Use the `esxcli network ip route` command to add static routes.

What Are Two-Node vSAN Clusters

A two-node vSAN cluster has two hosts at the same location. The witness function is performed at a second site on a dedicated virtual appliance.

Two-node vSAN clusters are often used for remote office/branch office environments, typically running a small number of workloads that require high availability. A two-node vSAN cluster consists of two hosts at the same location, connected to the same network switch or directly connected. A third host acts as a witness host, which can be located remotely from the branch office. Usually the witness host resides at the main site, with the vCenter.

A single witness host can support up to 64 two-node vSAN clusters. The number of clusters supported by a shared witness host is based on the host memory.

When you configure a two-node vSAN cluster in Quickstart or with the Configure vSAN wizard, you can select a witness host. To assign a new witness host for your cluster, right-click the cluster in the vSphere Client and select menu **vSAN** > **Assign Shared Witness**.

Use Quickstart to Configure a vSAN Stretched Cluster or Two-Node vSAN Cluster

You can use the Quickstart workflow to quickly configure a vSAN stretched cluster or two-node vSAN cluster.

- Deploy a host outside of any cluster to use as a witness host.
- Verify that hosts are running ESX 9.1 or later. For a two-node vSAN cluster, verify that hosts are running ESX 9.1 or later.
- Verify that ESX hosts in the cluster do not have any existing vSAN or networking configuration.

When you create a cluster in the vSphere Client, the Quickstart workflow appears. You can use Quickstart to perform basic configuration tasks, such as adding hosts and claiming disks.

1. Navigate to the cluster in the vSphere Client.
2. Click the Configure tab, and select **Configuration** > **Quickstart**.
3. On the Cluster basics card, click **Edit** to open the Cluster basics wizard.
   a) Enter the cluster name.
   b) Enable the vSAN slider.
      Select **vSAN ESA** if your cluster is compatible. You also can enable other features, such as DRS or vSphere HA.
   c) Select any option to **Choose how to set up the cluster's image**.
   d) (Optional) Select Manage configuration at a cluster level check box to ensure that all the hosts in the cluster have the same settings.
   e) Click **Next**.
   f) Select an image from the Image Library of Lifecycle Manager.
   g) Click **Next** and review the cluster details.
   h) Click **Finish**.
4. On the Add hosts card, click **Add** to open the Add hosts wizard.
   a) On the Add hosts page, enter information for new hosts, or click Existing hosts and select from hosts listed in the inventory.
   b) On the Host summary page, verify the host settings.
   c) On the Ready to complete page, click **Finish**.
5. On the Cluster configuration card, click **Configure** to open the Cluster configuration wizard.
   a) (vSAN ESA clusters) On the Cluster Type page, enter the HCI cluster type:
      - **vSAN HCI** provides compute resources and storage resources. The datastore can be shared across data centers and vCenter instances.
      - **vSAN storage clusters** provides storage resources, but not compute resources. The datastore can be mounted by remote vSAN clusters across data centers and vCenter instances.
   b) On the Configure the distributed switches page, enter networking settings, including distributed switches, port groups, and physical adapters.
      - In the **Distributed switches** section, enter the number of distributed switches to configure from the drop-down menu. Enter a name for each distributed switch. Click **Use Existing** to select an existing distributed switch.
        If the physical adapters chosen are attached to a standard virtual switch with the same name across hosts, the standard switch is migrated to the distributed switch. If the physical adapters chosen are unused, the standard switch is migrated to the distributed switch.
        Network resource control is enabled and set to version 3. Distributed switches with network resource control version 2 cannot be used.

- In the **Port Groups** section, select a distributed switch to use for vMotion and a distributed switch to use for the vSAN network.
- In the **Physical adapters** section, select a distributed switch for each physical network adapter. You must assign each distributed switch to at least one physical adapter.

  This mapping of physical NICs to the distributed switches is applied to all hosts in the cluster. If you are using an existing distributed switch, the physical adapter selection can match the mapping of the distributed switch.

c) On the vMotion traffic page, enter IP address information for vMotion traffic.

d) On the Storage traffic page, enter IP address information for storage traffic.

e) On the Advanced options page, enter information for cluster settings, including DRS, HA, vSAN, host options, and EVC.

   In the **vSAN options** section, select vSAN Stretched cluster or Two node vSAN cluster as the **Deployment type**. If you select vSAN Storage, you get the option to **Use Storage cluster client network**. Once configured, you cannot change the network configuration.

f) On the Claim disks page, select storage devices to create the vSAN datastore.

   For vSAN OSA, select devices for cache and for capacity. vSAN uses those devices to create disk groups on each host.

   For vSAN ESA, select compatible flash devices or enable **I want vSAN to manage the disks**. vSAN uses those devices to create storage pools on each host.

g) (Optional) On the Proxy settings, page, configure the proxy server if your system uses one.

h) On the Configure fault domains page, define fault domains for the hosts in the Preferred site and the Secondary site.

   For more information about fault domains, see Managing Fault Domains in vSAN Clusters.

i) On the Select witness host page, select a host to use as a witness host. The witness host cannot be part of the vSAN stretched cluster, and it can have only one VMkernel adapter configured for vSAN data traffic.

   Before you configure the witness host, verify that it is empty and does not contain any components. A two-node vSAN cluster can share a witness with other two-node vSAN clusters.

j) On the Claim disks for witness host page, select disks on the witness host.

k) On the Review page, verify the cluster settings, and click **Finish**.

You can manage the cluster through vCenter.

You can add hosts to the cluster and modify the configuration through Quickstart. You also can modify the configuration manually with the vSphere Client.

Manually Configure vSAN Stretched Cluster

Configure a vSAN cluster that stretches across two geographic locations or sites.

- Verify that you have a minimum of three hosts: one for the preferred site, one for the secondary site, and one host to act as a witness.
- Verify that you have configured one host to serve as the witness host for the vSAN stretched cluster. Verify that the witness host is not part of the vSAN cluster, and that it has only one VMkernel adapter configured for vSAN data traffic.
- Verify that the witness host is empty and does not contain any components. To configure an existing vSAN host as a witness host, first remove the witness from an existing two-node cluster or vSAN stretched cluster and remove any vSAN disk groups or storage pools on the witness host.

1. Navigate to the vSAN cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click **Configure Stretched Cluster** to open the vSAN stretched cluster configuration wizard.
5. Select the hosts that you want to assign to the secondary fault domain and click **>>**.

   The hosts that are listed under the Preferred fault domain are in the preferred site.

6. Click **Next**.
7. Select a witness host that is not a member of the vSAN stretched cluster and click **Next**.
8. Claim storage devices on the witness host and click **Next**.

   For vSAN OSA, select devices for cache and for capacity.

   For vSAN ESA, select compatible flash devices or enable **I want vSAN to manage the disks**.
9. On the **Ready to complete** page, review the configuration and click **Finish**.

Change the Preferred Fault Domain

You can configure the Secondary site as the Preferred site. The current Preferred site becomes the Secondary site.

> **Note:**
>
> Objects with **Data locality=Preferred** policy setting always move to the Preferred fault domain. Objects with **Data locality=Secondary** always move to the Secondary fault domain. If you change the Preferred domain to Secondary, and the Secondary domain to Preferred, these objects move from one site to the other. This action might cause an increase in resynchronization activity. To avoid unnecessary resynchronization, you can change the Data locality setting to **None** before you swap the Preferred and Secondary domains. Once you swap the domains back, you can reset the Data locality.

1. Navigate to the vSAN cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Select the secondary fault domain and click the **Change Preferred Fault Domain** icon.
5. Click **Yes** or **Apply** to confirm.

   The selected fault domain is marked as the preferred fault domain.

Deploying a vSAN Witness Appliance

Specific vSAN configurations, such as a stretched cluster, require a witness host. Instead of using a dedicated physical ESX host as a witness host, you can deploy the vSAN witness appliance. The appliance is a preconfigured virtual machine that runs ESX and is distributed as an OVA file.

You can cross-host the witness for multiple stretched clusters only if the clusters run in four geographical locations. vSAN does not support cross-hosting with clusters available only in two locations.

Unlike a general purpose ESX host, the witness appliance does not run virtual machines. Its only purpose is to serve as a vSAN witness, and it can contain only witness components.

The workflow to deploy and configure the vSAN witness appliance includes this process.

vSAN Witness Appliance CPU and Memory Requirements

The amount of CPU and memory that is required for the vSAN witness appliance is determined by the profile size you select, from XL to Tiny.

vSAN ESA Witness Appliance Size	Memory and CPU Required
XL	8 vCPUs and 64 GB RAM
L	4 vCPUs and 32 GB RAM
M	4 vCPUs and 16 GB RAM

vSAN OSA Witness Appliance Size	Memory and CPU Required
XL	6 vCPUs and 32 GB memory
L	2 vCPUs and 32 GB memory

vSAN OSA Witness Appliance Size	Memory and CPU Required
Normal	2 vCPUs and 16 GB memory
Tiny	2 vCPUs and 8 GB memory

vSAN Witness Appliance Sizing

When you deploy the vSAN witness appliance, you must configure the size of the witness supported by the vSAN stretched cluster. Choose one of the following options:

- Tiny supports up to 750 components (10 virtual machines or fewer).

  Note:

  vSAN ESA does not support tiny witness appliance.
- Medium supports up to 21,833 components (500 virtual machines). As a shared witness, the Medium witness appliance supports up to 21,000 components and up to 21 two-node vSAN clusters.
- Large supports up to 45,000 components (more than 500 virtual machines). As a shared witness, the Large witness appliance supports up to 24,000 components and up to 24 two-node vSAN clusters.
- Extra Large supports up to 64,000 components (more than 500 virtual machines). As a shared witness, the Extra Large witness appliance supports up to 64,000 components and up to 64 two-node vSAN clusters.

  Note:

  These estimates are based on standard VM configurations. The number of components that make up a VM can vary, depending on the number of virtual disks, policy settings, snapshot requirements, and so on. For more information about witness appliance sizing for two-node vSAN clusters, see the vSAN 2 Node Guide.

You also must select a datastore for the vSAN witness appliance. You must deploy the witness appliance on a third site independent of the two sites where it is intended to be stretched. The witness appliance must not share any physical resources with either data sites.

1. Download the appliance from the Broadcom Support Portal. Ensure that you download the correct witness appliance for vSAN ESA clusters and vSAN OSA clusters.
2. Deploy the appliance to a vSAN host or cluster. For more information, see Deploy and Export OVF and OVA Templates in the *vSphere Virtual Machine Administration* guide.
3. Specify a unique VM name and select a datacenter and folder to deploy the VM and click **Next**.
4. Select a cluster or ESX host that acts as a compute resource to deploy the VM and click **Next**.
5. Review the details and click **Next**.
6. Read and accept the license agreement and click **Next**
7. Select medium, large, or extra large as the size of the vSAN witness appliance and click **Next**. When you deploy the vSAN witness appliance, you must configure the size of the witness.
8. Select the virtual disk format and the VM storage policy.
9. Select the datastore of the vSAN witness appliance and click **Next**. You can batch configure or configure per disk group, as required.
10. Browse and select the port group for the witness appliance management interface and a secondary port group for the secondary network for vSAN VMKernel port. Click **Next**. Broadcom recommends setting the management network for vSAN traffic.

    Note:

    If you select the management network for vSAN traffic, you do not require to configure the secondary network. The secondary network port group can have the same name as the management network port group. If you are using the secondary network for vSAN traffic, select an appropriate port group.
11. Perform the following to customize the vSAN witness appliance and click **Next**:
    a. Set a root password for the vSAN witness appliance.
    b. Select the adapter to tag vSAN traffic type. You can select the management network or the secondary network. Broadcom recommends setting the management network for vSAN traffic. If you select the management network for vSAN traffic, you do not require to configure the secondary network and network settings can remain blank.

c. Customize the network settings for the management network (vmk0). You can select DHCP or set a static IP address. If you select DHCP, you can leave the management network settings blank.
d. Customise the network settings for the secondary network (vmk1), if required. You can use DHCP or set a static IP address. If you select DHCP, you can leave the secondary network settings blank.

Note:

If you select the management network for vSAN traffic, you do not require to configure the secondary network and the network settings can remain unset.

12. Review the details and click **Finish**. After deployment of the vSAN witness appliance, power on the witness VM.
You can use the management IP from the vSAN witness VM and can add it as a standalone host in the vSAN stretched cluster.

Set Up the vSAN Network on the Witness Appliance

The vSAN witness appliance includes two preconfigured network adapters. You must change the configuration of the second adapter so that the appliance can connect to the vSAN network.

1. Navigate to the virtual appliance that contains the witness host.
2. Right-click the appliance and select **Edit Settings**.
3. On the **Virtual Hardware** tab, expand the second Network adapter.
4. From the drop-down menu, select the vSAN port group and click **OK**.

   You can use the first VMkernel interface for vSAN traffic when using witness traffic separation.

Configure Management Network on the Witness Appliance

Configure the witness appliance, so that it is reachable on the network.

By default, the appliance can automatically obtain networking parameters if your network includes a DHCP server. If not, you must configure appropriate settings.

1. Power on your witness appliance and open its console.
   Because your appliance is an ESX host, you see the Direct Console User Interface (DCUI).
2. Press F2 and navigate to the Configure Management Network > Network Adapters page.
3. On the Network Adapters page, verify that at least one vmnic is selected for transport.
4. Configure the IPv4 parameters for the management network.
   a) Navigate to the IPv4 Configuration section and change the default DHCP setting to static.
   b) Enter the following settings:
      - IP address
      - Subnet mask
      - Default gateway
5. Configure DNS parameters.
   - Primary DNS server
   - Alternate DNS server
   - Hostname

Configure Network Interface for Witness Traffic

You can separate data traffic from witness traffic in two-node vSAN clusters and vSAN stretched clusters.

- Verify that the data site to witness traffic connection has a minimum bandwidth of 2 Mbps for every 1,000 vSAN components.
- Verify the latency requirements:
  – Two-node vSAN clusters must have less than 500 ms RTT.

  - vSAN stretched clusters with less than 11 hosts per site must have less than 200 ms RTT.
  - vSAN stretched clusters with 11 or more hosts per site must have less than 100 ms RTT.
- Verify that the vSAN data connection meets the following requirements.
  - For hosts directly connected in a two-node vSAN cluster, use a 10 GbE direct connection between hosts. Hybrid clusters also can use a 1 GbE crossover connection between hosts.
  - For hosts connected to a switched infrastructure, use a 10 GbE shared connection (required for all-flash clusters), or a 1 GbE dedicated connection.
- Verify that data traffic and witness traffic use the same IP version.

vSAN data traffic requires a low-latency, high-bandwidth link. Witness traffic can use a high-latency, low-bandwidth and routable link. To separate data traffic from witness traffic, you can configure a dedicated VMkernel network adapter for vSAN witness traffic on each data host on both sites

You can add support for a direct network cross-connection to carry vSAN data traffic in a vSAN stretched cluster. You can configure a separate network connection for witness traffic. On each data host in the cluster, configure the management VMkernel network adapter to also carry witness traffic. Do not configure the witness traffic type on the witness host.

Note:
Network Address Translation (NAT) is not supported between vSAN data hosts and the witness host.

1. In the vSphere Client, navigate to the host in the cluster.
2. Click the **Configure** tab.
3. Under **Networking**, click **VMkernel adapters**.
4. Click
   
   and click **Edit**.
5. Select **vSAN Witness** to separate vSAN witness traffic from other traffic types such as vMotion and management traffic.

The management VMkernel network interface is not selected for vSAN traffic. Do not re-enable the interface in the vSphere Client.

Configure Network Interface for Witness Traffic Using Command Line

You can use the command line to configure a vmkernel network interface to separate vSAN data traffic from vSAN witness traffic on the hosts of two-node vSAN clusters and vSAN stretched clusters.

1. Open an SSH connection to the ESX host.
2. Use the `esxcli network ip interface list` command to determine which VMkernel network adapter is used for management traffic.

   For example:

```
esxcli network  ip interface list
vmk0
   Name: vmk0
   MAC Address: e4:11:5b:11:8c:16
   Enabled: true
   Portset: vSwitch0
   Portgroup: Management Network
   Netstack Instance: defaultTcpipStack
   VDS Name: N/A
   VDS UUID: N/A
   VDS Port: N/A
   VDS Connection: -1
   Opaque Network ID: N/A
```

```
   Opaque Network Type: N/A
   External ID: N/A
   MTU: 1500
   TSO MSS: 65535
   Port ID: 33554437

vmk1
   Name: vmk1
   MAC Address: 00:50:56:6a:3a:74
   Enabled: true
   Portset: vSwitch1
   Portgroup: vsandata
   Netstack Instance: defaultTcpipStack
  VDS Name: N/A
   VDS UUID: N/A
   VDS Port: N/A
   VDS Connection: -1
   Opaque Network ID: N/A
   Opaque Network Type: N/A
   External ID: N/A
   MTU: 9000
   TSO MSS: 65535
   Port ID: 50331660
```

3. Use the `esxcli vsan network ip add` command to configure the management VMkernel network adapter to support witness traffic.

   ```
   esxcli vsan network ip add -i vmkx -T witness
   ```

4. Use the `esxcli vsan network list` command to verify the new network configuration.

Modifying a Witness Host

If you want to modify a witness host, follow the guidelines listed here.

As a virtual appliance, you can modify generic VM hardware settings on a witness host such as the number of vCPUs, memory size, and disk sizes. These settings cannot be modified during OVA deployment.

After the OVA deployment, the witness host appears as a regular VM in the cluster inventory. If you modify its settings, ensure that the witness host VM settings are at or above the minimum requirements:

Component	OSA	ESA
vCPU number	2	4
vMemory size	8 GB	16 GB
ESX system disk	12 GB	32 GB
vSAN cache disk	10 GB	N/A
vSAN capacity disk	15 GB	350 GB

Note:

If the witness host is shared by multiple ROBO clusters (a "shared witness"), then the resources parameters must increase accordingly.

Replace the Witness Host

You can replace or change the witness host for a vSAN stretched cluster.

Verify that the witness host is not in use by another cluster, has a VMkernel configured for vSAN traffic, and has no vSAN partitions on its disks.

Change the ESX host used as a witness host for your vSAN stretched cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click the **Change** button. The Change Witness Host wizard opens.
5. Select a new host to use as a witness host, and click **Next**.
6. Claim disks on the new witness host, and click **Next**.
7. On the Ready to complete page, review the configuration, and click **Finish**.

Convert a vSAN Stretched Cluster to a Single Site vSAN Cluster

You can decommission a vSAN stretched cluster and convert it to a single site vSAN cluster.

- Back up all running virtual machines, and verify that all virtual machines are compliant with their current storage policy.
- Ensure that no health issues exist, and that all resync activities are complete.
- Change the associated storage policy to move all VM objects to one site. Use the Data locality rule to restrict virtual machine objects to the selected site.
- Replace the stretched cluster storage policies applied to the VMs with standard vSAN cluster policies or create a new vSAN storage policy that you can apply to a standard vSAN cluster.

When you decommission a vSAN stretched cluster, you must manually delete the fault domain configuration and remove the witness host. Because the witness host is not available, all witness components are missing for your virtual machines. To ensure full availability for your virtual machines, repair the cluster objects immediately.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Change the storage policy to the default vSAN storage policy.
5. Disable the vSAN stretched cluster.
   a) Click **Disable**. The Remove Witness Host dialog opens.
   b) Click **Remove** to confirm.
6. Remove the hosts available at the secondary site from the cluster.
7. Remove the fault domain configuration.
   a) Select a fault domain, and choose menu **Actions > Delete**. Click **Yes** to confirm.
   b) Select the other fault domain, and choose menu **Actions > Delete**. Click **Yes** to confirm.
8. Remove the witness host from inventory.
9. Repair the objects in the cluster.
   a) Click the **Monitor** tab.
   b) Under vSAN, click **Health** and click **vSAN object health**.
   c) Click **Repair object immediately**.

   vSAN recreates the witness components within the cluster.

Creating a Stretched Compute Cluster

You can create a stretched compute cluster where compute resources are stretched across two geographic locations (or sites).

Stretched compute clusters enable you to extend the compute resources, hosts, and workloads running on different sites to work as a single cluster. The stretched compute cluster continues to function if a failure or scheduled maintenance occurs at a site.

What is a Stretched Compute Cluster?

Stretched compute clusters are cluster with compute resources that are distributed across two different sites or data centers. These clusters provide high availability, disaster recovery, and resource management by enabling the cluster to operate in a stretched mode across the sites.

Each site has its fault domain. The fault domain in the stretched compute cluster groups host based on their location and network topology. This ensures that the cluster operates optimally across sites. When configured with two distinct fault domains, stretched compute cluster enters stretched mode allowing the cluster to span two locations and improve redundancy.

Stretched Compute Clusters and Fault Domains

In a stretched compute cluster, the fault domain organizes the hosts-based on the site attributes and the network topologies. You can group hosts based on their physical location. This ensures that the compute resources are distributed across different fault domains and helps you manage the workload during site failure.

With hosts available in two sites, you can configure a stretched compute cluster. You can add additional hosts, as required. Use **Disable stretched compute cluster** to disable stretched compute cluster that removes the sites and converts a stretched compute cluster to a compute cluster.

Stretched Compute Clusters Design Considerations

Consider the following guidelines when working on a stretched compute cluster:

- Ensure that the hosts in a compute cluster are placed in separate sites.
- Allow only the stretched compute cluster to mount the datastores from a vSAN stretched cluster.

Stretched Compute Clusters Network Design

All the sites in a stretched compute cluster communicate across the management network and across the vSAN network. The virtual machines in both the data sites communicate across a common virtual machine network.

A stretched compute cluster must meet certain basic networking requirements:

- Management network requires connectivity across all the sites, using a Layer 3 network.
- The vSAN network requires connectivity across all the sites of the server cluster.
- VM network requires connectivity between the data sites. Use a Layer 3 network between the data sites. In the event of a failure, the virtual machines do not require a new IP address to work on the remote site.
- vMotion network requires connectivity between the data sites. Use a Layer 3 network between the data sites.

Configure a Stretched Compute Cluster

You can configure a stretched compute cluster from a vSphere cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. Select **Compute Cluster** as an HCI configuration type.
5. Select **Configure a stretched compute cluster without vSAN datastore** as the deployment option.
6. Click **Configure**. The Configure Stretched Compute Cluster dialog opens.
7. Click and drag the hosts from the fault domain or use arrow keys to move hosts from one fault domain to the other. You can rename the fault domain name, if required.
8. Click **Apply**.

Convert a Compute Cluster to a Stretched Compute Cluster

If you have an existing compute cluster, you can convert the compute cluster to a stretched compute cluster. Once converted, the stretched compute cluster can mount remote datastores from a vSAN stretched cluster.

1. In the vSphere Client, navigate to the existing compute cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Fault Domains**.
4. Click **Configure** to open the Configure Stretched Compute Cluster dialog.
5. Select the hosts that you want to be part of the first and the second fault domain. You can move hosts between fault domains or rename the fault domain name.
6. In the Configure site coupling page, select the compute client cluster that you want to couple with the server cluster. If the server cluster has multiple remote datastores mounted to the compute client cluster, vSAN applies site coupling to all the datastores in the server cluster.
7. Select **Skip site coupling** to skip site coupling for server clusters and click **Next**.
8. Review the details and click **Finish**.

Rename a Fault Domain in a Stretched Compute Cluster

You can change the name of an existing fault domain in your stretched compute cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Select the fault domain and click the **Actions** icon on the right side of the fault domain.
5. Choose **Edit** and the Edit Fault Domain dialog opens.
6. Enter a new fault domain name and click **Apply**.

   The new name appears for the fault domain.

Move Host into Selected Fault Domain in a Stretched Compute Cluster

You can move a host into a selected fault domain in the stretched compute cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click the host of a fault domain that you want to move to another fault domain.
5. Click the Actions icon on the right side of the fault domain, and select **Move**.

   The selected host moves to the fault domain.

Disable a Stretched Compute Cluster

Depending on your requirement, you can disable a stretched compute cluster.

If you disable a stretched compute cluster, you convert the cluster to a compute cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Fault Domains**.
4. Click **Disable Stretched Compute Cluster** to open the Disable stretched compute cluster dialog.
5. Select the server cluster site to couple with the client cluster.
6. Select **Skip site coupling** to skip site coupling for server clusters and click **Next**.
7. Review the details and click **Finish**.

   When you disable a stretched compute cluster, the existing fault domains configuration on the cluster gets removed.

Creating a vSAN ESA Storage Cluster for Cyber Recovery

You can configure a vSAN ESA storage cluster for storing immutable snapshots that can be used for cyber recovery.

If you have already configured a vSAN ESA storage cluster and you want to use it for cyber recovery, you can enable the cyber recovery option for the cluster. Cyber recovery allows you to recover your workloads from ransomware attacks, avoid re-infection, and reduce downtime.

Deploy a vSAN ESA Storage Cluster for Cyber Recovery

Deploying a vSAN ESA storage cluster that is enabled for cyber recovery provides a storage environment to safely inspect, analyze, and recover VMs after a ransomware attack.

Enabling cyber recovery for a vSAN ESA cluster also allows the system to report Cyber Recovery Health Checks for vSAN Storage Clusters.

A cyber recovery site is the target site of a protection and recovery server pair, which you set up after you deploy the vSAN ESA storage cluster enabled for cyber recovery. When you run a ransomware recovery plan, an isolated network on the target site serves as the ransomware recovery clean room.

A vSAN ESA storage cluster that is configured for cyber recovery is recommended for performing ransomware recovery operations. You should only use the cyber recovery cluster for ransomware recovery workflows. You have the option of using Cyber Recovery ReadyNodes for vSAN Storage Clusters, which come preconfigured and pretested for optimal storage performance.

To deploy a new vSAN ESA storage cluster that is enabled for cyber recovery using Quick Start:

1. From the vSphere client, navigate to the new cluster.
2. Select the Datacenter node, right-click, and select **New Cluster**.
3. In the New Cluster wizard Basics page, configure the following options:
   a) Enter a name for the cluster.
   b) Select if you want to enable the vSphere Distributed Resource Scheduler (DRS) for the cluster.
   c) Select if you want to enable vSphere High Availability (HA) for the cluster.
   d) Activate the vSAN option to make this cluster a vSAN storage cluster.
   e) Select the vSAN Express Storage Architecture (ESA) option, which optimizes the cluster for high-performance flash storage operations and cyber recovery.
   f) Select the 'Cyber recovery' option to dedicate this cluster for use only as a datastore for cyber recovery and clean room operations.
4. Click **Next**.
5. Select an image for the cluster.
6. Review the cluster details, and then click **Finish**.
7. On the Add hosts page, click **Add** to open the Add hosts wizard. Configure the following settings:
   a) On the Add hosts page, enter information for new hosts, or click **Existing hosts** and select from hosts that are listed in the inventory.
   b) On the Host summary page, verify the host settings.
   c) On the Ready to complete page, click **Finish**.
8. From the vSphere client, select the vSAN storage cluster that you created from the left navigation.
9. On the Configure Cluster card, click **Configure**.
10. On the Cluster type page of the wizard, the 'vSAN Storage Cluster (cyber recovery)' is already selected. Click **Next**.
11. For the remaining vSAN ESA cluster configurations, follow the instructions that are listed in this task, starting at number 5.
12. When you complete the steps, you can now use the vSAN ESA storage cluster for cyber recovery. For more information on configuring a vSAN ESA storage cluster, see Use Quickstart to Configure a vSAN Cluster.

Enable vSAN ESA with Cyber Recovery on an Existing Cluster

If you have already deployed a vSAN ESA storage cluster and you want to use it for cyber recovery, enable the 'Cyber recovery' option for the cluster.

You can also deactivate the cyber recovery option for the cluster.

To enable the cyber recovery option on a vSAN ESA storage cluster:

1. In the vSphere client, select the vSAN ESA storage cluster from the left navigation.
2. Select the **Configure** tab and then select **vSAN** > **Services**.
3. On the Storage card, click **Edit**.
4. From the **vSAN ESA Options** dialog box, you can select or deselect the 'Cyber recovery' option. When enabled, this cluster becomes optimized for recovering workloads from ransomware attacks. This option enables you to use the vSAN ESA storage cluster as a data vault for orchestrated workload recovery.
5. Click **Apply**. You can now see that the storage cluster is enabled for cyber recovery.

Cyber Recovery ReadyNodes for vSAN Storage Clusters

For the best user experience, you can purchase vSAN Cyber Recovery ReadyNodes to use for your vSAN ESA storage clusters.

vSAN ReadyNodes are x86 servers, available from all the leading server vendors, that have been pre-configured, tested, and certified for VMware Hyper-Converged Infrastructure Software. Each ReadyNode™ is optimally configured for vSAN with the required amount of CPU, memory, network, I/O controllers, and storage (SSDs, HDDs, or flash devices). Using ReadyNodes reduces deployment risk with tested and certified configurations.

Cyber Recovery ReadyNodes only work when you Enable vSAN ESA with Cyber Recovery on an Existing Cluster on an existing vSAN vSAN ESA storage cluster, or when you enable Cyber Recovery when deploying a new vSAN ESA storage cluster.

For information about vSAN ReadyNodes for Cyber Recovery, visit vSAN ReadyNodes on the Broadcom compatibility guide site on the Broadcom. To purchase a vSAN ReadyNode, contact your Broadcom sales representative.

Cyber Recovery Health Checks for vSAN Storage Clusters

When you enable a vSAN ESA storage cluster for cyber recovery, vCenter runs periodic health checks for the cluster and its nodes.

Cyber recovery health checks ensure that your vSAN ESA storage cluster is compliant with your organization standards and that the cluster is in good health. Health checks display as events in the vSAN UI.

Health check	Description	Health check conditions	Resolution
Cyber Recovery ReadyNodes	After you deploy a vSAN ESA storage cluster with 'Cyber recovery' enabled, this checks your cluster for Cyber Recovery ReadyNodes.	**Information**: Cyber Recovery ReadyNodes are being used. **Warning**: For cyber recovery, select Cyber Recovery ReadyNodes. Some nodes in the cluster are Cyber Recovery ReadyNodes, and some are not. **Critical**: None of the nodes in your vSAN ESA storage cluster are Cyber Recovery ReadyNodes.	Enable 'Cyber recovery' for your vSAN ESA storage cluster.
Cyber Recovery storage cluster deployment	If you deploy a vSAN ESA storage cluster with 'Cyber recovery' enabled, but the vCenter where you are creating the cluster does not have the protection and recovery appliance deployed on it.	**Warning**: For cyber recovery, you must deploy the protection and recovery appliance.	Click the link to deploy the protection and recovery appliance.
Workload check	Checks to see if you have chosen to run recovered VMs on the vSAN ESA storage cluster. To run recovered VMs on the ESA cluster, you must also have a vSAN compute cluster to run the VMs after recovery.	**Critical**: You need a compute cluster/HCI to run your VMs after recovery.	Create a compute cluster.
Endurance check	Checks the cluster flash drive endurance, which is how efficiently data is being written to the cluster's flash/NVMe drives. For example, how much data can be written before the drive wears out?	**Warning**: Endurance is less than 90%. **Critical**: Endurance is less than 100% and you might encounter sub optimal performance.	

Administering VMware vSAN

Administering VMware vSAN describes how to configure and manage a VMware® vSAN™ cluster in a VMware vSphere® environment. This guide explains how to manage the local physical storage resources that serve as storage capacity devices in a vSAN cluster, and how to define storage policies for virtual machines (VMs) deployed to vSAN datastores.

Intended Audience

This information is for experienced virtualization administrators who are familiar with virtualization technology, day-to-day data center operations, and vSAN concepts. The information in this guide is written for experienced system administrators who are familiar with virtual machine technology and virtual datacenter operations. This manual assumes familiarity with VMware vSphere, including VMware ESX, vCenter and the vSphere Client.

- For more information about network requirements and network design, see the Designing vSAN Network guide.
- For more information about creating vSAN clusters, see the Planning and Configuring vSAN guide.
- For more information about monitoring a vSAN cluster and fixing problems, see the Monitoring and Troubleshooting vSAN guide.

Managing a vSAN Cluster

You can manage a vSAN cluster by using the vSphere Client, esxcli commands, vSphere PowerCLI, and other tools.

Upload Files or Folders to vSAN Datastores

You can upload vmdk files to a vSAN datastore.

You can also upload folders to a vSAN datastore. For more information about datastores, see the vSphere guide. When you upload a vmdk file to a vSAN datastore, the following considerations apply:

- You can upload only stream-optimized vmdk files to a vSAN datastore. VMware stream-optimized file format is a monolithic sparse format compressed for streaming. If you want to upload a vmdk file that is not in stream-optimized format, then before uploading, convert it to stream-optimized format using the vmware-vdiskmanager command-line utility. For more information, see Virtual Disk Development Kit (VDDK) Programming Guide.
- When you upload a vmdk file to a vSAN datastore, the vmdk file inherits the default policy of that datastore. The vmdk does not inherit the policy of the virtual machine from which it was downloaded. vSAN creates the objects by applying the vsanDatastore default policy. You can change the default policy of the datastore. See Change the Default Storage Policy for vSAN Datastores .
- You must upload a vmdk file to virtual machine home folder.

1. In the vSphere Client, navigate to the vSAN datastore.
2. Click the **Files** tab.

Option	Description
Upload Files	1. Select the target folder and click **Upload Files**. You see a message informing that you can upload vmdk files only in VMware stream-optimized format. If you try uploading a vmdk file in a different format, you see an internal server error message. 2. Click **Upload**. 3. Locate the item to upload on the local computer and click **Open**.
Upload Folder	1. Select the target folder and click **Upload Folder**. You see a message informing that you can upload vmdk files only in VMware stream optimized format. 2. Click **Upload**. 3. Locate the item to upload on the local computer and click **Upload**.

Download Files or Folders from vSAN Datastores

You can download files and folders from a vSAN datastore.

The vmdk files are downloaded as stream-optimized files with the filename `<vmdkName>_stream.vmdk`. VMware stream-optimized file format is a monolithic sparse format compressed for streaming.

You can convert a VMware stream-optimized vmdk file to other vmdk file formats using the vmware-vdiskmanager command-line utility. For more information, see Virtual Disk Development Kit (VDDK) Programming Guide.

1. In the vSphere Client, navigate to the vSAN datastore.
2. Click the **Files** tab, select the file and then click **Download**. Ensure that the popups are enabled before you download the file.

   You see a message alerting you that vmdk files are downloaded from the vSAN datastores in VMware stream-optimized format with the filename extension `.stream.vmdk`.

Using vSAN Policies

When you use vSAN, you can define virtual machine storage requirements, such as performance and availability, in a policy.

In most cases, vSAN ensures that each virtual machine deployed to vSAN datastores is assigned at least one storage policy. After they are assigned, the storage policy requirements are pushed to the vSAN layer when a virtual machine is created. The virtual device is distributed across the vSAN datastore to meet the performance and availability requirements.

vSAN uses the VASA storage provider to supply information about underlying storage to the vCenter. This information helps you to make appropriate decisions about virtual machine placement, and to monitor your storage environment.

What are vSAN Policies

vSAN storage policies define storage requirements for your VMs.

vSAN storage policies determine how virtual machine storage objects are provisioned and allocated within the datastore to guarantee the required level of service. When you enable vSAN on a ESX host cluster, a single vSAN datastore is created and a default storage policy is assigned to the datastore.

When you know the storage requirements of your VMs, you can create a storage policy referencing the capabilities that the datastore advertises. You can create several policies to capture different types or classes of requirements.

Each virtual machine is assigned at least one virtual machine storage policy when deployed to a vSAN datastore. You can assign storage policies when you create or edit VMs.

> **Note:**
>
> If you do not assign a storage policy to a virtual machine, vSAN uses the datastore default policy. vSAN ESA clusters have the vSAN ESA Auto RAID policy as the default and uses the vSAN ESA Auto RAID rule. Auto RAID policy automatically determines the optimal Site Disaster Tolerance (SDT) and Failures to Tolerate (FTT).

If vSAN ESA is enabled, every snapshot is not a new object. A base VMDK and its snapshots are contained in one vSAN object. A data digest disk is a file that stores data digests for a specific VM or virtual disk. This digest file is used by the CBRC (Content Based Read Cache) mechanism to improve performance, especially during read-intensive operations like boot storms or antivirus scans. The digest file effectively acts as an index of the data on the virtual disk, which allows vSAN to quickly identify and access specific data blocks. In vSAN ESA, the data digest disk is backed by separate vSAN object.

> **Note:**
>
> You cannot use the vSAN ESA Auto-RAID policy when creating, cloning, or migrating virtual machines on clusters running VCF/ESX 9.0 or earlier versions, but that is managed by a newer version of vCenter that supports Auto-RAID.
>
> This limitation applies to:
>
> - Compute-only clients running ESX 9.0 or earlier.
> - vSAN clusters running VCF 9.0 or earlier versions of the software.
>
> If the remote mounted datastore is VCF 9.1 or later, and you want to use Auto-RAID in this scenario, you must enable "Assume auto RAID" on the remote cluster.

In vSAN OSA, snapshots are separate vSAN objects. In vSAN OSA, the data digest disks are stored in the VM home directory.

Table 1010: Storage Policy - Availability

Capability	Description
Site disaster tolerance **Note:** Starting in VCF 9.1, before you can create and configure a new VM Storage Policy, you must first disable vSAN ESA Auto-RAID.	The site disaster tolerance rule defines whether to use a standard, stretched, or 2-node cluster. If you use a vSAN stretched cluster, you can define whether data is mirrored at both sites or only at one site. For a vSAN stretched cluster, you can choose to keep data on the preferred or secondary site for affinity with a site or location. • **None - standard cluster** is the default value. This means that there is no site disaster tolerance. • **Host mirroring - 2 node cluster** defines that the objects must tolerate host failure. The object data is replicated to both the hosts. The failures to tolerate policy rule specify the number of failures to tolerate within each host. vSAN performs object mirroring at the disk group level (vSAN OSA) or disk level (vSAN ESA) on each host, based on the FTT policy rule. Each host must have sufficient disk groups or disks to use the FTT policy rule. • **Site mirroring - stretched cluster** defines that the objects must tolerate site disaster. The object data is replicated to both the sites. The failures to tolerate policy rule specify the number of failures to tolerate within each site. • **None - keep data on Preferred (stretched cluster)**. Use this option if you do **not** want the objects in a vSAN stretched cluster to have site failure tolerance and you want to make the objects accessible only on the site that is configured as Preferred. These objects are not affected by the Inter-Switch Link (ISL) or witness host failures. They remain accessible if the site chosen by the policy is accessible. • **None - keep data on Secondary (stretched cluster)**. Use this option if you do not want the objects in a vSAN stretched cluster to have site failure tolerance and you want to make the objects accessible only on the secondary site. These objects are not affected by the Inter-Switch Link (ISL) or witness host failures. They remain accessible if the site chosen by the policy is accessible.
Failures to tolerate (FTT)	Defines the number of ESX host and device failures that a virtual machine object can tolerate. For `n` failures tolerated, each piece of data written is stored in `n+1` places, including parity copies if using RAID-5 or RAID-6. If fault domains are configured, `2n+1` fault domains with ESX hosts contributing capacity are required. An ESX host which does not belong to a fault domain is considered its own single-host fault domain. You can select a data replication method that optimizes for performance or capacity: • RAID-1 (Mirroring) uses more disk space to place the components of objects but provides better performance for accessing the objects. • RAID-5/6 (Erasure Coding) uses less disk space, but performance is reduced. You can select one of the following options: • **No data redundancy**: Specify this option if you do **not** want vSAN to protect a single mirror copy of virtual machine objects. Selecting this option means that your data is unprotected, and you might lose data when the vSAN cluster encounters a device failure. The ESX host might experience unusual delays when entering maintenance mode. The delays occur because vSAN must evacuate the object from the host for the maintenance operation to complete successfully. • **1 failure - RAID-1 (Mirroring)**: Specify this option if your virtual machine object can tolerate one ESX host or device failure. To protect a 100 GB virtual machine object by using RAID-1 (Mirroring) with an FTT of 1, you consume 200 GB. • **1 failure - RAID-5 (Erasure Coding)**: Specify this option if your virtual machine object can tolerate one ESX host or device failure. – For vSAN OSA, to protect a 100 GB virtual machine object by using RAID-5 (Erasure Coding) with an FTT of 1, you consume 133.33 GB. **Note:** • If you use vSAN ESA, vSAN creates an optimized RAID-5 format based on the cluster size. • If the number of hosts in the cluster is less than 6, vSAN creates a RAID-5 (2+1) format. • If the number of ESX hosts is equal or greater than 6, vSAN creates a RAID-5 (4+1) format. • When the cluster size eventually expands or shrinks, vSAN

Capability	Description
vSAN ESA Auto-RAID	When you enable vSAN ESA Auto-RAID policy, the site disaster tolerance and failures to tolerate are automatically set based on the cluster configuration, which is the default setting when creating the new storage policy. See the following table for more details. A policy with this rule is supported only on the cluster running vSAN 9.1 and later release. The following policy rules can only be specified along with the vSAN ESA Auto-RAID policy rule: • Object space reservation • Encryption services • Keep data on Primary/Secondary • IOPS Limit

The following table displays the site disaster tolerance, failures to tolerate, and the RAID factor provided by the objects with vSAN ESA Auto-RAID policy rule, or for all objects if you enable Apply Auto-RAID to all objects:

Cluster Configuration	Site Disaster Tolerance	Failures to Tolerate	RAID Factor
Two-node cluster	None	Host FTT=1, RAID-1 (Mirroring)	2
Standard cluster with greater than or equal to six hosts	None	Host FTT=2, RAID -6 (Erasure Coding)	1.5
Standard cluster with greater than or equal to three hosts to less than or equal to five hosts	None	Host FTT=1, RAID -5 (Erasure Coding)	1.5
Standard cluster with fewer than three hosts	None	Host FTT=0, RAID-0	1
Stretched cluster with greater than or equal to six hosts	Site mirroring	FTT per site=1, RAID-6 (Erasure Coding)	3
Stretched cluster with greater than or equal to six hosts with Keep data on Preferred/ Secondary	None	FTT per site=1, RAID-6 (Erasure Coding)	1.5
Stretched cluster with greater than and equal to three hosts and fewer than six hosts	Site mirroring	FTT per site=1, RAID-5 (Erasure Coding)	3
Stretched cluster with greater than and equal to three hosts and fewer than six hosts with Keep data on Preferred/ Secondary	None	FTT per site=1, RAID-5 (Erasure Coding)	1.5
Stretched cluster with less than three hosts	Site mirroring	FTT per site=0, RAID-0	2
Stretched cluster with less than three hosts with Keep data on Preferred/Secondary	None	FTT per site=0, RAID-0	1

After an object is created, the cluster configuration can change. If the cluster configuration changes, the objects are automatically reconfigured to the new optimal site disaster tolerance and failure to tolerate supported by the cluster. Cluster configuration changes include:

- If you evacuate a host, vSAN reconfigures the objects to a lower FTT if required.
- If a host fails and vSAN cannot repair the objects with the same FTT, vSAN reconfigures the objects to a lower FTT after the repair delay.
- If you add a host, vSAN reconfigures the objects to a higher FTT if required.
- If the cluster configuration changes from stretched cluster to standard cluster or vice versa, vSAN reconfigures the objects to meet the new requirements.

For more information on vSAN ESA effective capacity and RAID factor, see Monitor vSAN ESA Effective Capacity.

Table 1011: Storage Policy - Storage rules

Capability	Description
Encryption services	Defines the encryption options ensuring that only datastores that match the selected options remain compliant. Choose one of the following options: • **Data-at-rest encryption**: Specify this option if the VM must be provisioned only on a datastore that has encryption enabled. • **No encryption**: Specify this option if the VM must be provisioned only on a datastore that has encryption disabled. • **No preference**: This is the default option. Specify this option if you do not want to explicitly apply any encryption rules. By selecting this option, you can provision the VM on a datastore with encryption enabled or disabled.
Data reduction	Defines the space efficiency options ensuring that only datastores that matches the selected options remain compliant. Choose one of the following options: • **Deduplication and compression**: Specify this option if the VM must be provisioned only on a datastore that supports deduplication and compression. For vSAN OSA, deduplication and compression are cluster-level settings. For vSAN ESA prior to the 9.1 release, deduplication is not supported, and this policy rule is not compatible. For vSAN ESA 9.1 and later, this policy rule is ignored and the VM data is compressed and deduplicated based on the cluster level setting. • **Compression only**: Specify this cluster-wide option if a VM must be provisioned only on a datastore that supports compression but not deduplication. For vSAN ESA 9.1 and later, this policy rule is ignored. **Note:** For vSAN OSA, compression is a cluster-level setting. For vSAN ESA prior to the 9.1 release, compression is performed at the object level and is enabled if the space efficiency rule specifies **Compression only** or **No preference**. For vSAN ESA 9.1 and later, this policy rule is ignored and the VM data is compressed and deduplicated based on the cluster level setting. Data compression cannot be disabled. • **No space efficiency**: Specify this option if the VM must be provisioned only on a datastore that does not support deduplication or compression. For vSAN OSA, **No space efficiency** is a cluster-level setting. For vSAN ESA prior to the 9.1 release, compression is performed at the object level and is disabled if the space efficiency rule specifies **No space efficiency**. For vSAN ESA 9.1 and later, this policy rule is ignored and the VM data is compressed and deduplicated based on the cluster level setting. • **No preference**: This is the default option. Specify this option if you do not want to explicitly apply any space efficiency rules. By selecting this option, vSAN applies all space efficiency features supported by the datastore to your VMs. **Note:** The space efficiency storage policy rule is not supported with vSAN ESA Auto-RAID policy rule. The default rule of no preference is applied, and vSAN applies all space efficiency features supported by the datastore to your VMs.

Capability	Description
Storage tier	Specify the storage tier for all VMs with the defined storage policy. Choose one of the following options: • **All flash**: Specify this option if you want to make your VMs compatible with all-flash environments. • **Hybrid**: Specify this option if you want to make your VMs compatible with only hybrid environments. • **No preference**: This is the default option. Specify this option if you do not want to explicitly apply any storage tier rules. By selecting this option, vSAN makes the VMs compatible with both hybrid and all flash environments. **Note:** The storage tier policy rule is not supported with vSAN ESA Auto-RAID policy rule.

Table 1012: Storage Policy - Advanced Policy Rules

Capability	Description
Number of disk stripes per object	The minimum number of capacity devices across which each replica of a virtual machine object is striped. A value higher than 1 might result in better performance, but also result in higher use of system resources. The default value is 1. The maximum value is 12. Do not change the default striping value. In a hybrid environment, the disk stripes are spread across magnetic disks. For an all-flash configuration, the striping is across flash devices that make up the capacity layer. Make sure that your vSAN environment has sufficient capacity devices present to accommodate the request. **Note:** The number of disk stripes per object policy rule is not supported with vSAN ESA Auto-RAID policy rule.
IOPS limit for object	Defines the IOPS limit for an object, such as a VMDK. IOPS is calculated as the number of I/O operations, using a weighted size. If the system uses the default base size of 32 KB, a 64-KB I/O represents two I/O operations. When calculating IOPS, read and write are considered equivalent, but cache hit ratio and sequentiality are not considered. If a disk's IOPS exceeds the limit, I/O operations are throttled. If the **IOPS limit for object** is set to 0, IOPS limits are not enforced.
Object space reservation	Percentage of the logical size of the virtual machine disk (vmdk) object that must be reserved, or thick provisioned when deploying VMs. The following options are available: • Thin provisioning (default) • Thick provisioning **Note:** If you select thick provisioning, the deduplication and compression savings become ineffective.

Capability	Description
Disable Object Checksum	If the option is set to **No**, the object calculates checksum information to ensure the integrity of its data. If this option is set to **Yes**, the object does not calculate checksum information. vSAN uses end-to-end checksum to ensure the integrity of data by confirming that each copy of a file is the same as the source file. The system checks the validity of the data during read/write operations, and if an error is detected, vSAN repairs the data or reports the error. If a checksum mismatch is detected, vSAN automatically repairs the data by overwriting the incorrect data with the correct data. Checksum calculation and error-correction are performed as background operations. The default setting for all objects in the cluster is **No**, which means that checksum is enabled. **Note:** For vSAN ESA, object checksum is always on and cannot be deactivated. For vSAN OSA, object checksum is enabled by default.
Force provisioning	If the option is set to **Yes**, the object is provisioned even if the **Failures to tolerate**, **Number of disk stripes per object**, and **Flash read cache reservation** policies that are specified in the storage policy cannot be satisfied by the datastore. Use this parameter in bootstrapping scenarios and during an outage when standard provisioning is no longer possible. The default **No** is acceptable for most production environments. vSAN fails to provision a virtual machine when the policy requirements are not met.

When working with virtual machine storage policies, you must understand how the storage capabilities affect the consumption of storage capacity in the vSAN cluster. For more information about designing and sizing considerations of storage policies, refer to Designing and Sizing a vSAN Cluster.

How vSAN Manages Policy Changes

Transient capacity is generated when vSAN reconfigures objects for a policy change. The transient storage activities include vSAN disk rebalancing and object conversions.

When you modify a policy, the change is accepted but not applied immediately. vSAN batches the policy change requests and performs them asynchronously, to maintain a fixed amount of transient space.

Policy changes are rejected immediately due to policy compliance and the underlying capabilities of the storage, such as changing a RAID-5 policy to RAID-6 on a five-host cluster. vSAN storage policy compares the rules associated with the policy and evaluates the underlying storage capabilities to determine whether the vSAN storage is compatible.

You can view transient capacity usage in the vSAN Capacity monitor (**Cluster** > **Monitor** > **vSAN** > **Capacity** > **Capacity Usage** > **Usage breakdown** > **System usage**). To verify the status of a policy change on an object, use the vSAN health service to check the vSAN object health.

View vSAN Storage Providers

Enabling vSAN automatically configures and registers a vSAN storage provider in the vSAN cluster.

vSAN storage providers are built-in software components that communicate datastore capabilities to vCenter. A storage capability typically is represented by a key-value pair, where the key is a specific property offered by the datastore. The value is a number or range that the datastore can provide for a provisioned object, such as a virtual machine home namespace object or a virtual disk. You can also use tags to create user-defined storage capabilities and reference them when defining a storage policy for a virtual machine.

1. In the vSphere Client, navigate to vCenter.
2. Click the **Configure** tab, and click **Storage Providers**.

The storage provider for vSAN appears on the list.

Note:

You cannot manually unregister storage providers used by vSAN. To remove or unregister the vSAN storage providers, remove corresponding ESX hosts from the vSAN cluster and then add the ESX hosts back. Make sure that at least one storage provider is active.

What are vSAN Default Storage Policies

vSAN requires that the virtual machines deployed on the vSAN datastores are assigned at least one storage policy.

When provisioning a virtual machine, if you do not explicitly assign a storage policy, vSAN assigns a default storage policy to the virtual machine. Each default policy contains vSAN rule sets and a set of basic storage capabilities, typically used for the placement of virtual machines deployed on vSAN datastores. If vSAN does not assign a storage policy to the virtual machine, vSAN health check prompts a warning to reapply the VM storage policy. For more information, see Broadcom knowledge base article 326534.

If you use a vSAN ESA cluster, depending on your cluster size, you can use one of the vSAN ESA policies listed here.

Note:

The objects in a vSAN ESA cluster with RAID 0 or RAID 1 configuration have three disk stripes, though the default storage policy defines only one disk stripe.

Table 1013: vSAN ESA Default Storage Policy - RAID-5 Specifications

Specification	Setting
Failures to tolerate	1
Number of disk stripes per object	1
Object space reservation	Thin provisioning
Force provisioning	No

Note:

RAID-5 in vSAN ESA supports three ESX host clusters. If you enable auto-policy management, the cluster must have four ESX hosts to use RAID-5.

Table 1014: vSAN ESA Default Storage Policy - RAID-6 Specifications

Specification	Setting
Failures to tolerate	2
Number of disk stripes per object	1
Object space reservation	Thin provisioning
Force provisioning	No

Note:

If you enable auto-policy management, the cluster must have six ESX hosts to use RAID-6.

Table 1015: vSAN Default Storage Policy Specifications

Specification	Setting
Failures to tolerate	1
Number of disk stripes per object **Note:** In some cases, vSAN also applies a stripe.	1
Object space reservation	0 **Note:** Setting the Object space reservation to zero means that the virtual disk is thin provisioned, by default.
Force provisioning	No

You can review the configuration settings for the default virtual machine storage policy when you navigate to the **VM Storage Policies** > name of the default storage policy > **Rule-Set 1: VSAN**.

For best results, consider creating and using your own virtual machine storage policies, even if the requirements of the policy are same as those defined in the default storage policy.

When you assign a user-defined storage policy to a datastore, vSAN applies the settings for the user-defined policy on the specified datastore. Only one storage policy can be the default policy for the vSAN datastore.

Auto-Policy Management

Clusters with vSAN ESA can use auto-policy management to generate an optimal default storage policy, based on the cluster type (standard or stretched) and the number of ESX hosts. vSAN clusters prior to the 9.1 release configures the Site disaster tolerance and Failures to tolerate to optimal settings for the cluster. vSAN ESA clusters with the 9.1 or later release enables the vSAN ESA Auto-RAID policy rule.

The name of the auto-generated policy is based on the cluster name, as follows: *ClusterName - Optimal Default Datastore Policy*

When you enable auto-policy management, vSAN assigns a new optimal policy to the vSAN datastore, and that policy becomes the datastore default policy for the cluster.

To enable auto-policy management, use the slide control on vSAN**> Services > Storage > Edit**.

vSAN Default Storage Policy Characteristics

The following characteristics apply to the vSAN datastore default storage policies.

- The vSAN Default Storage Policy is assigned to all virtual machine objects if you do not assign any other vSAN policy when you provision a virtual machine. The **VM Storage Policy** text box is set to **Datastore default** on the **Select Storage** page. For more information about using storage policies, see the *vSphere Storage* guide.

  Note:

  Virtual machine swap and virtual machine memory objects receive a vSAN default storage policy with **Force provisioning** set to **Yes**. With forced provisioning, if there are not enough hosts to satisfy a storage policy, the object is deployed with the bare minimum (No Data Redundancy).

  If you deploy RAID-6, which requires six hosts, but you only have four hosts, vSAN sets No Data Redundancy to the policy until you add the two extra hosts. (Adding these hosts makes it compliant with the policy). When Forced provisioning is set to Yes, it deploys with the minimum. When set to No, it generates an error.
- A vSAN default policy only applies to vSAN datastores. You cannot apply a default storage policy to non- vSAN datastores, such as NFS or a VMFS datastore.
- Objects in a vSAN ESA cluster with RAID 0 or RAID 1 configuration will have 3 disk stripes, even if the default policy defines only 1 disk stripe.

- Because the **vSAN Default Storage Policy** is compatible with any vSAN datastore in the vCenter, you can move your virtual machine objects provisioned with the default policy to any vSAN datastore in the vCenter.
- You can clone the vSAN Default Storage Policy and use it as a template to create a user-defined storage policy.
- You can edit the vSAN Default Storage Policy, if you have the *StorageProfile.View* privilege. You must have at least one vSAN-enabled cluster that contains at least one ESX host. Typically you do not edit the settings of the vSAN Default Storage Policy.
- You cannot edit the name and description of the vSAN Default Storage Policy, or the vSAN storage provider specification. All other parameters including the policy rules are editable.
- You cannot delete the vSAN Default Storage Policy.
- A default storage policy is assigned when the policy that you assign during virtual machine provisioning does not include rules specific to vSAN.

Change the Default Storage Policy for vSAN Datastores

You can change the default storage policy for a selected vSAN datastore.

Verify that the virtual machine storage policy you want to assign as the default policy to the vSAN datastore meets the requirements of virtual machines in the vSAN cluster.

1. In the vSphere Client, navigate to the vSAN datastore.
2. Click the **Configure** tab.
3. Under **General**, click the **Default Storage Policy Edit** button, and select the storage policy that you want to assign as the default policy to the vSAN datastore.

   Note:

   You can also edit the Improved Virtual Disk Home Storage Policy. Click **Edit** and select the home storage policy that you want to assign as the storage policy for the home object.

   You can choose from a list of storage policies that are compatible with the vSAN datastore, such as the vSAN Default Storage Policy and user-defined storage policies that have vSAN rule sets defined.
4. Click **OK**. The storage policy is applied as the default policy when you provision new virtual machines without explicitly specifying a storage policy for a datastore.

You can define a new storage policy for virtual machines. See Define a Storage Policy for vSAN Using vSphere Client.

Define a Storage Policy for vSAN Using vSphere Client

You can create a storage policy that defines storage requirements for a virtual machine and its virtual disks.

- Verify that the vSAN storage provider is available. Refer to View vSAN Storage Providers.
- Required privileges: **Profile-driven storage.Profile-driven storage view** and **Profile-driven storage.Profile-driven storage update**

In this policy, you reference the storage capabilities supported by the vSAN datastore.

1. Navigate to **Policies and Profiles**, then click **VM Storage Policies** in the vSphere Client.
2. Click **Create**.
3. On the **Name and description** page, select a vCenter.
4. Type a name and a description for the storage policy and click **Next**.
5. On the **Policy structure** page, select **Enable rules for vSAN storage**, and click **Next**.
6. On the vSAN page, define the policy rule set, and click **Next**.
   a) On the **Availability** tab, define the **Site disaster tolerance** and **Failures to tolerate**.

      Availability options define the rules for failures to tolerate, data locality, and failure tolerance method.

      - **Site disaster tolerance** defines the type of site failure tolerance used for virtual machine objects.

- **Failures to tolerate** defines the number of ESX host and device failures that a virtual machine object can tolerate, and the data replication method.

For example, if you choose **Dual site mirroring** and **2 failures - RAID-6 (Erasure Coding)**, vSAN configures the following policy rules:

- Failures to tolerate: 1
- Secondary level of failures to tolerate: 2
- Data locality: None
- Failure tolerance method: RAID-5/6 (Erasure Coding) - Capacity

If you enable vSAN ESA Auto-RAID, the site disaster tolerance and failures to tolerate by default displays vSAN managed (based on cluster configuration).

b) On the **Storage Rules** tab, define the encryption, space efficiency, and storage tier rules that can be used along with the vSAN HCI Datastore Sharing to distinguish the remote datastores. With vSAN 9.1 release, vSAN ESA Global Deduplication is a cluster setting that is available only after you configure a cluster.

- **Encryption services**: Defines the encryption rules for virtual machines that you deploy with this policy. You can choose one of the following options:
  - **Data-at-rest encryption**: Encryption is enabled on the virtual machines.
  - **No encryption**: Encryption is not enabled on the virtual machines.
  - **No preference**: Makes the virtual machines compatible with both data-at-rest encryption and No encryption options.
- **Space Efficiency**: Defines the space saving rules for the virtual machines that you deploy with this policy. You can choose one of the following options:
  - **Deduplication and compression**: Enables both deduplication and compression on the virtual machines. Deduplication and compression are available only on all-flash disk groups. For more information, see Deduplication and Compression Design Considerations in a vSAN Cluster.
  - **Compression only**: Enables only compression on the virtual machines. Compression is available only on all-flash disk groups. For vSAN ESA compression is defined in the vSAN policy.
  - **No space efficiency**: Space efficiency features are not enabled on the virtual machines. Choosing this option requires datastores without any space efficiency options to be turned on.
  - **No preference**: Makes the virtual machines compatible with all the options.

Storage tier: Specifies the storage tier for the virtual machines that you deploy with this policy. You can choose one of the following options. Choosing the **No preference** option makes the virtual machines compatible with both hybrid and all flash environments.

- **All flash**
- **Hybrid**
- **No preference**

c) On the **Advanced Policy Rules** tab, define advanced policy rules, such as number of disk stripes per object and IOPS limits.

d) On the **Tags** tab, click **Add Tag Rule**, and define the options for your tag rule.

Make sure that the values you provide are within the range of values advertised by storage capabilities of the vSAN datastore.

7. On the **Storage compatibility** page, review the list of datastores under the **COMPATIBLE** and **INCOMPATIBLE** tabs and click **Next**.

   To be eligible, a datastore does not need to satisfy all the rule sets within the policy. The datastore must satisfy at least one rule set and all rules within this set. Verify that the vSAN datastore meets the requirements set in the storage policy and that it appears on the list of compatible datastores.

8. On the **Review and finish** page, review the policy settings, and click **Finish**.

The new policy is added to the list.

Assign this policy to a virtual machine and its virtual disks. vSAN places the virtual machine objects according to the requirements specified in the policy. For information about applying the storage policies to virtual machine objects, see the vSphere Storage guide.

Expanding and Managing a vSAN Cluster

After you have set up your vSAN cluster, you can add ESX hosts and capacity devices, remove ESX hosts and devices, and manage failure scenarios.

Expanding a vSAN Cluster

You can expand an existing vSAN cluster by adding ESX hosts or adding devices to existing ESX hosts, without disrupting any ongoing operations.

Use one of the following methods to expand your vSAN cluster.

- Add new ESX hosts to the cluster that are configured using the supported cache and capacity devices. See Add an ESX Host to the vSAN Cluster.
- Move existing ESX hosts to the vSAN cluster and configure them by using a host profile. See Configuring ESX Hosts in the vSAN Cluster Using Host Profile.
- Add new capacity devices to ESX hosts that are cluster members. See Add Devices to the Disk Group in vSAN Cluster.

  Note:

  When a vSAN cluster shrinks, objects cannot be reconfigured unless the cluster fails to recover within an hour. When a vSAN cluster grows, the reconfiguration window is set to five minutes.

Expanding vSAN Cluster Capacity and Performance

If your vSAN cluster is out of storage capacity or when you notice reduced performance, you can expand the vSAN cluster for capacity and performance.

- For vSAN ESA, expand the storage capacity of your vSAN cluster by adding flash devices to the storage pools of the existing ESX hosts or by adding one or more new ESX hosts with supported flash devices.
- For vSAN OSA, expand the storage capacity of your vSAN cluster either by adding storage devices to existing disk groups or by adding disk groups. New disk groups require flash devices for the cache. For information about adding devices to disk groups, see Add Devices to the Disk Group in vSAN Cluster. Adding capacity devices without increasing the cache might reduce your cache-to-capacity ratio to an unsupported level. For hybrid configurations, which combine SSDs for caching and HDD for capacity, a 10% cache-to-capacity ratio is recommended. For all-flash there is no fixed cache-to-capacity ratio.

  Improve the vSAN cluster performance by adding at least one cache device (flash) and one capacity device (flash or magnetic disk) to an existing storage I/O controller or to a new ESX host. Or you can add one or more ESX hosts with disk groups to produce the same performance impact after vSAN completes automatic rebalance in the vSAN cluster.

Although compute-only ESX hosts can exist in a vSAN cluster, and consume capacity from other ESX hosts in the vSAN cluster, add uniformly configured ESX hosts for efficient operation. Although it is best to use the same or similar devices in your disk groups or storage pools, any device listed on the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/ is supported. Try to distribute capacity evenly across ESX hosts. For information about adding devices to disk groups or storage pools, see Create a Disk Group or Storage Pool in vSAN Cluster.

After you expand the vSAN cluster capacity, enable automatic rebalance to distribute resources evenly across the vSAN cluster. For more information, see About vSAN Cluster Rebalancing.

Use Quickstart to Add ESX Hosts to a vSAN Cluster

If you configured your vSAN cluster through Quickstart, you can use the Quickstart workflow to add ESX hosts and storage devices to the vSAN cluster.

- The Quickstart workflow must be available for your vSAN cluster.

- Verify that the resources, including drivers, firmware, and storage I/O controllers, are listed in the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
- VMware recommends creating uniformly configured ESX hosts in the vSAN cluster, so you have an even distribution of components and objects across devices in the cluster. However, there might be situations where the vSAN cluster becomes unevenly balanced, particularly during maintenance or if you overcommit the capacity of the vSAN datastore with excessive virtual machine deployments.
- No network configuration performed through the Quickstart workflow has been modified from outside of the Quickstart workflow.

  Networking settings configured while creating the vSAN cluster with Quickstart have not been modified.

When you add new ESX hosts to the vSAN cluster, you can use the Cluster configuration wizard to complete the ESX host configuration. For more information about Quickstart, see Using Quickstart to Configure and Expand a vSAN Cluster.

> **Note:**
>
> If you are running vCenter on an ESX host, the ESX host cannot be placed into maintenance mode as you add it to a vSphere cluster using the Quickstart workflow. All other virtual machines on the ESX host must be powered off.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under **Configuration**, click **Quickstart**.
4. On the **Add hosts** card, click **Add** to open the Add hosts wizard.
   a) On the **Add hosts** page, enter information for new hosts, or click Existing hosts and select from hosts listed in the inventory.
   b) On the **Host Summary** page, verify the host settings.
   c) On the Import Image page, import an image from the ESX host to set as the new image for the cluster. You can either select an existing image or import an image from one of the ESX hosts.
   d) On the **Review** page, click **Finish**.
5. On the Configure Cluster card, click **Configure** to open the Cluster configuration wizard.
   a) On the **Configure the distributed switches** page, enter networking settings for the new hosts.
   b) (optional) On the **Claim disks** page, select disks on each new host. If you enable Auto clam at the cluster level for vSAN ESA, all the compatible disks are automatically claimed.
   c) (optional) On the **Create fault domains** page, move the new hosts into their corresponding fault domains.

      For more information about fault domains, see Managing Fault Domains in vSAN Clusters.
   d) On the **Ready to complete** page, verify the cluster settings, and click **Finish**.

Verify the vSAN Health status.

Add an ESX Host to the vSAN Cluster

You can add ESX hosts to a running vSAN cluster without disrupting any ongoing operations.

- Verify that the resources, including drivers, firmware, and storage I/O controllers, are listed in the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
- VMware recommends creating uniformly configured ESX hosts in the vSAN cluster, so you have an even distribution of components and objects across devices in the cluster. However, there might be situations where the vSAN cluster becomes unevenly balanced, particularly during maintenance or if you overcommit the capacity of the vSAN datastore with excessive virtual machine deployments.
- Verify that you configured the ESX hosts for the vSAN network.

The new ESX host's resources become associated with the cluster.

1. In the vSphere Client, navigate to the cluster.
2. Right-click the vSphere cluster and select **Add New or Existing Hosts**. The Add hosts wizard appears.

Option	Description
New hosts	1. Enter the IP address or FQDN. 2. Enter the username and password associated with the ESX host.
Existing hosts	1. Select ESX hosts that you previously added to vCenter.

3. Click **Next**.
4. View the summary information and click **Next**.
5. Import an image from the ESX host to set as the new image for the cluster. You can either select an existing image or import an image from one of the ESX hosts.
6. Review the settings and click **Finish**.

   The ESX host enters maintenance mode before the ESX host is added to the vSphere cluster.

Verify the vSAN Health status.

Verify that the vSAN Disk Balance health check is green.

For more information about vSAN cluster configuration and fixing problems, see vSAN Cluster Configuration Issues.

Configuring ESX Hosts in the vSAN Cluster Using Host Profile

When you have multiple ESX hosts in the vSAN cluster, you can use a host profile of an existing vSAN host to configure the ESX hosts in the vSAN cluster.

- Verify that the ESX host is in maintenance mode.
- Verify that the hardware components, drivers, firmware, and storage I/O controllers are listed in the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.

The host profile includes information about storage configuration, network configuration, and other characteristics of the ESX host. If you are planning to create a vSphere cluster with many ESX hosts, use the host profile feature. Host profiles enable you to add more than one ESX host at a time to the vSAN cluster.

1. Create an ESX host profile.
   a) Navigate to **Policies and Profiles**, then click **Host Profiles** in the vSphere Client.
   b) Click the **Extract Host Profile** icon.
   c) On the **Select the host** dialog, select the ESX host you intend to use as the reference ESX host and click **Next**.

      The selected ESX host must be an active host.
   d) On the **Name and description** dialog, enter a name for the new profile and click **Finish**.
   e) Review the summary information for the new host profile and click **Finish**.
2. Attach the ESX host to the intended host profile.
   a) From the Profile list in the Host Profiles view, select the host profile to be applied to the ESX host.
   b) Click the **Attach/Detach Hosts and clusters to a host profile** icon
      ( ).
   c) Select the host from the expanded list and click **Attach** to attach the host to the profile.

      The host is added to the Attached Entities list.
   d) Click **Next**.
   e) Click **Finish** to complete the attachment of the host to the profile.
3. Detach the referenced vSAN host from the host profile.

   When a host profile is attached to a cluster, the host or hosts within that cluster are also attached to the host profile. However, when the host profile is detached from the cluster, the association between the host or hosts in the cluster and that of the host profile remains intact.

a) From the Profile List in the Host Profiles view, select the host profile to be detached from a host or cluster.
b) Click the **Attach/Detach Hosts and clusters to a host profile** icon
( ).
c) Select the host or cluster from the expanded list and click **Detach**.
d) Click **Detach All** to detach all the listed hosts and clusters from the profile.
e) Click **Next**.
f) Click **Finish** to complete the detachment of the host from the host profile.

4. Verify the compliance of the vSAN host to its attached host profile and determine if any configuration parameters on the host are different from those specified in the host profile.
   a) Navigate to a host profile.

      The **Objects** tab lists all host profiles, the number of hosts attached to that host profile, and the summarized results of the last compliance check.
   b) Click the **Check Host Profile Compliance** icon
   ( ).

      To view specific details about which parameters differ between the host that failed compliance and the host profile, click the **Monitor** tab and select the Compliance view. Expand the object hierarchy and select the non-compliant host. The parameters that differ are displayed in the Compliance window, below the hierarchy.

      If compliance fails, use the Remediate action to apply the host profile settings to the host. This action changes all host profile-managed parameters to the values that are contained in the host profile attached to the host.
   c) To view specific details about which parameters differ between the host that failed compliance and the host profile, click the **Monitor** tab and select the Compliance view.
   d) Expand the object hierarchy and select the failing host.

      The parameters that differ are displayed in the Compliance window, below the hierarchy.

5. Remediate the host to fix compliance errors.
   a) Select the **Monitor** tab and click **Compliance**.
   b) Right-click the host or hosts to remediate and select **All vCenter Actions** > **Host Profiles** > **Remediate**.

      You can update or change the user input parameters for the host profiles policies by customizing the host.
   c) Click **Next**.
   d) Review the tasks that are necessary to remediate the host profile and click **Finish**.

   The host is part of the vSAN cluster and its resources are accessible to the vSAN cluster. The host can also access all existing vSAN storage I/O policies in the vSAN cluster.

Using vSphere Configuration Profiles (VCP) to Manage vSAN Clusters

You can now use vSphere Configuration Profiles (VCP) to define, validate, and enforce desired vSAN clusters configurations at the cluster level.

Managing configuration at cluster level ensures operational consistency, reduces configuration drift, and improves compliance across your environment. Configuration profiles also ensure that all hosts in the cluster have the same settings, while eliminating the need to configure hosts manually.

> **Important:**
>
> To enable the vSphere Configuration Profiles (VCP) feature for a given vSAN cluster, the target ESX image in VMware vSphere Lifecycle Manager must be at least version 9.1.

Configuration profiles for vSAN use a 'desired state' model, where a user-configurable, cluster-level document serves as a single source of truth for cluster and host configuration. vSAN-specific settings—such as deployments, fault domain assignments, disk claiming strategies, and other data services—can be managed alongside other vSphere configurations. The exportable cluster configuration document is stored in the JSON format that is backed by a JSON schema.

Configuration profiles detect inconsistencies between the desired configurations and the actual vSAN cluster runtime state, and report any drift through the VCP compliance engine. You can also run on-demand compliance checks to

determine the current state of your vSAN cluster. When a configuration is incorrect, you can remediate the setting back to the desired state.

Supported VCP Configurations for vSAN Clusters

VSphere Configuration Profiles (VCP) are supported for the following vSAN cluster types:

- vSAN ESA cluster, non-stretched
- vSAN Max cluster, non-stretched
- vSAN compute cluster, single and stretched

vSAN Cluster Type	Available VCP vSAN Cluster Configurations
vSAN ESA cluster non-stretched	Services • Encryption • DIT • HCI Mesh Configurations • Enable VCP administration. • Deploy a non-stretched vSAN ESA cluster using VCP. • Configure vSAN host fault domain. • Claim storage pool disks in a non-stretched vSAN ESA cluster by specifying candidate disks' canonical names. • Enable or disable vSAN encryption. • Enable or disable DIT. • Mount/unmount vSAN remote datastores. • View and configure all vSAN public advanced options on 9.1 ESX host by default. • Enable the view of hidden configurations to manage all vSAN internal advanced options on 9.1 ESX host
vSAN Max cluster non-stretched	Services • De-converged network • Auto-policy • Encryption • DIT Configurations • Enable VCP administration. • Deploy a non-stretched vSAN Max cluster using VPC. • Enable or disable a de-converged network together with non-stretched vSAN Max cluster deployment. • Configure vSAN host fault domains. • Claim storage pool disks by specifying candidate disks' canonical names. • Enable or disable vSAN auto-policy. • Enable or disable vSAN encryption. • Enable or disable DIT.
vSAN compute-only cluster, single site and stretched	Configurations • Enable VCP administration. • Deploy a vSAN compute-only cluster using VCP. • Configure or unconfigure site fault domains. • Mount or unmount vSAN remote datastores. • Configure site coupling in a compute-only cluster for a remote datastore from a stretched server cluster.

Note:

Configuration profiles for vSAN clusters replace host profiles. Once you enable a configuration profile for an entire cluster, host profiles cannot be attached to the cluster or to any of its hosts. However, certain hosts can have setting overrides.

Create a Configuration Profile During vSAN Cluster Deployment

One of the easiest ways to create a configuration profile is when you deploy a new vSAN cluster.

To deploy a vSAN cluster using a draft configuration profile, import the configuration from one of the hosts in the cluster. Then, you can apply that configuration to all other hosts in the cluster.

Prerequistes

- VMware Cloud Foundation (VCF) 9.1.
- Hosts running ESX 8.0 or above.

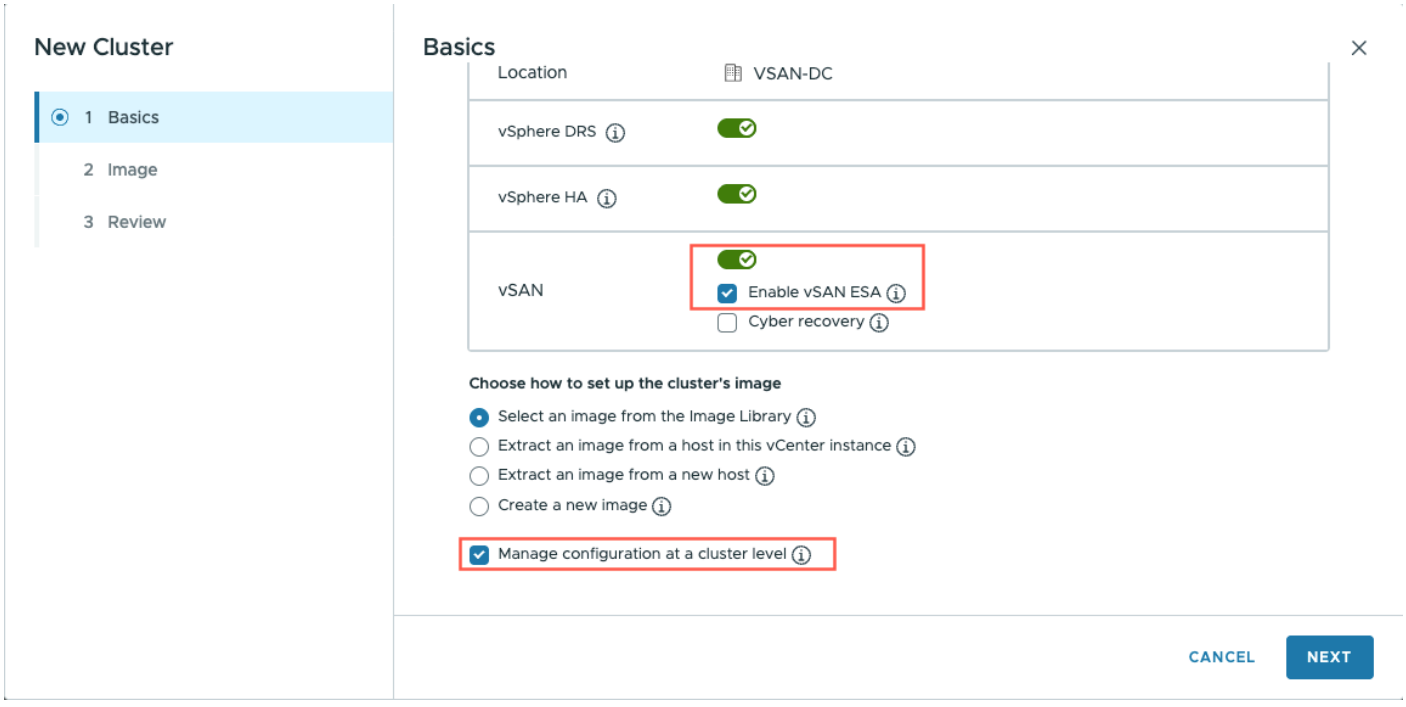
This workflow consists of the following tasks:

1. Deploy a vSAN Cluster with a Configuration Profile
2. Add Hosts to the Cluster
3. Create Configuration Profile by Importing From Host
4. Apply Changes and Remediate Configuration

Deploy a vSAN Cluster with a Configuration Profile

To deploy a vSAN cluster and enable a configuration profile:

1. In the vSphere Client, navigate to the datacenter, right-click, and choose **New Cluster**.
2. In the New Cluster wizard, Basics card, type a name for the cluster in the Name text box.
3. Activate DRS, vSphere HA, and vSAN for the cluster.
4. Under 'Choose how to set up the cluster's image, select 'Create a new image' and also select 'Manage configuration at the cluster level'. When you select to manage the configuration at the cluster level, the configuration profile is enabled for the cluster.



5. Click **Next**.
6. In the Image card of the New Cluster Wizard, select an image from the Image Library of Lifecycle Manager.
7. Click **Next** and review the cluster details.

8. Click **Finish**. The cluster appears in the inventory.

Add Hosts to the Cluster

Next, add hosts to the cluster. You will use one of the hosts to set the baseline configuration for the cluster.

To add hosts to a cluster:

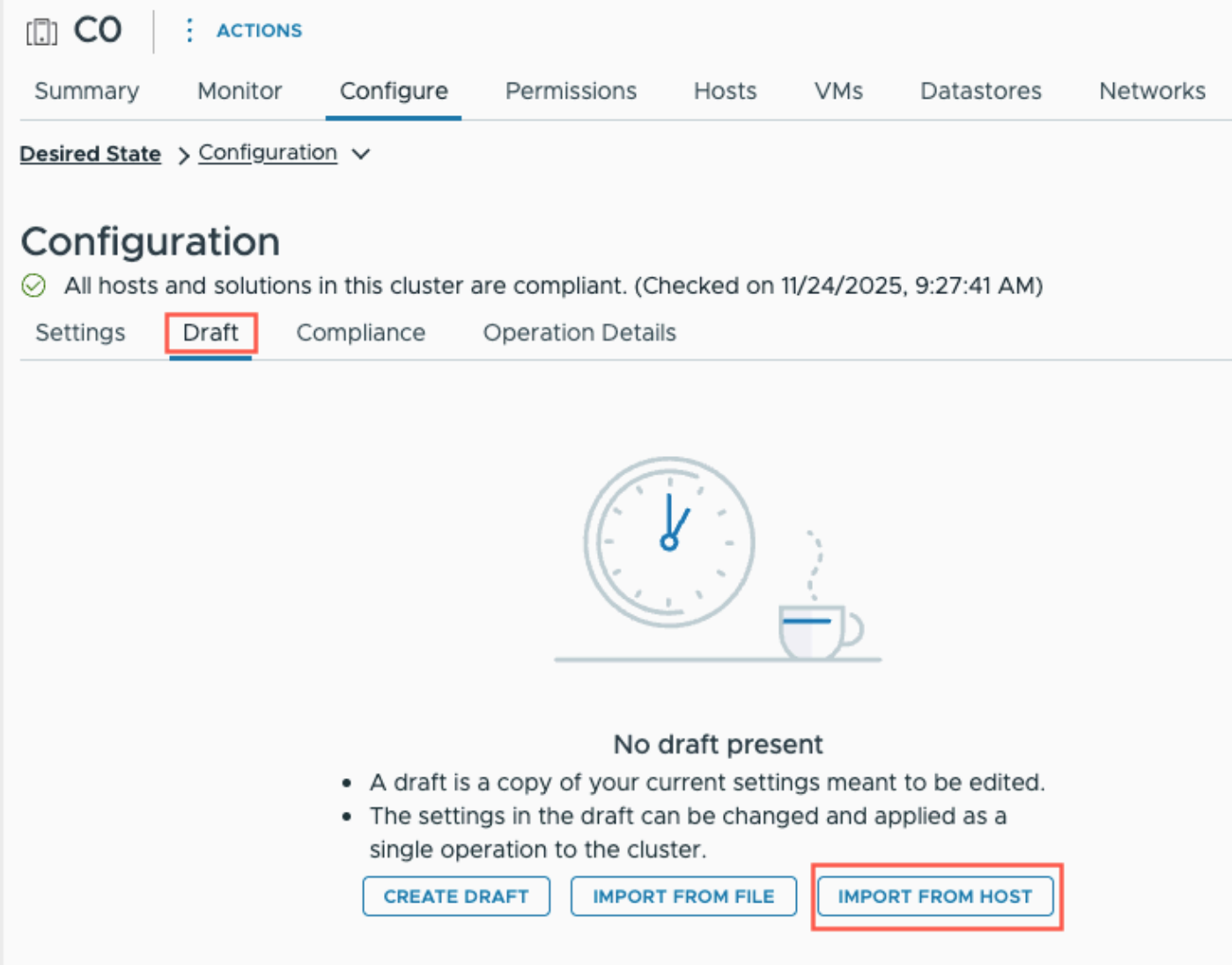
1. In the vSphere Client, navigate to the vSAN cluster node, right-click, and choose **Add hosts**.
2. On the Add hosts page, enter information for new hosts, or click **Existing hosts** and select from hosts that are listed in the inventory.
3. On the Host summary page, verify the host settings.
4. On the Ready to complete page, click **Finish**.

Create Configuration Profile by Importing From Host

Next, create and configure a profile for the cluster by importing the configuration of one host. After you import the host configuration, you can apply (remediate) that configuration to all other hosts in the cluster.

To create a configuration profile by importing from a host:

1. After you have added hosts, select the cluster and then click the **Configure** tab.
2. From the embedded navigation window, select **Desired State** > **Configuration**.
   The desired state configuration is where the cluster-level settings are defined and stored in a JSON file that you can edit and export. You import the configuration of one host in the cluster, and then you use it as the desired state configuration for all hosts in the cluster. If any of the hosts deviate from the cluster-level settings, those changes are saved as overrides.
3. Click the **Draft** tab, and then click **Import from Host**.
   A draft configuration is a copy of your current configuration that can be edited and later applied to the cluster. You import the configuration of one host in the cluster and then use that host's configuration as the draft desired state configuration for all hosts in the cluster. If any of the hosts deviate from the cluster-level settings, those changes are saved as overrides.



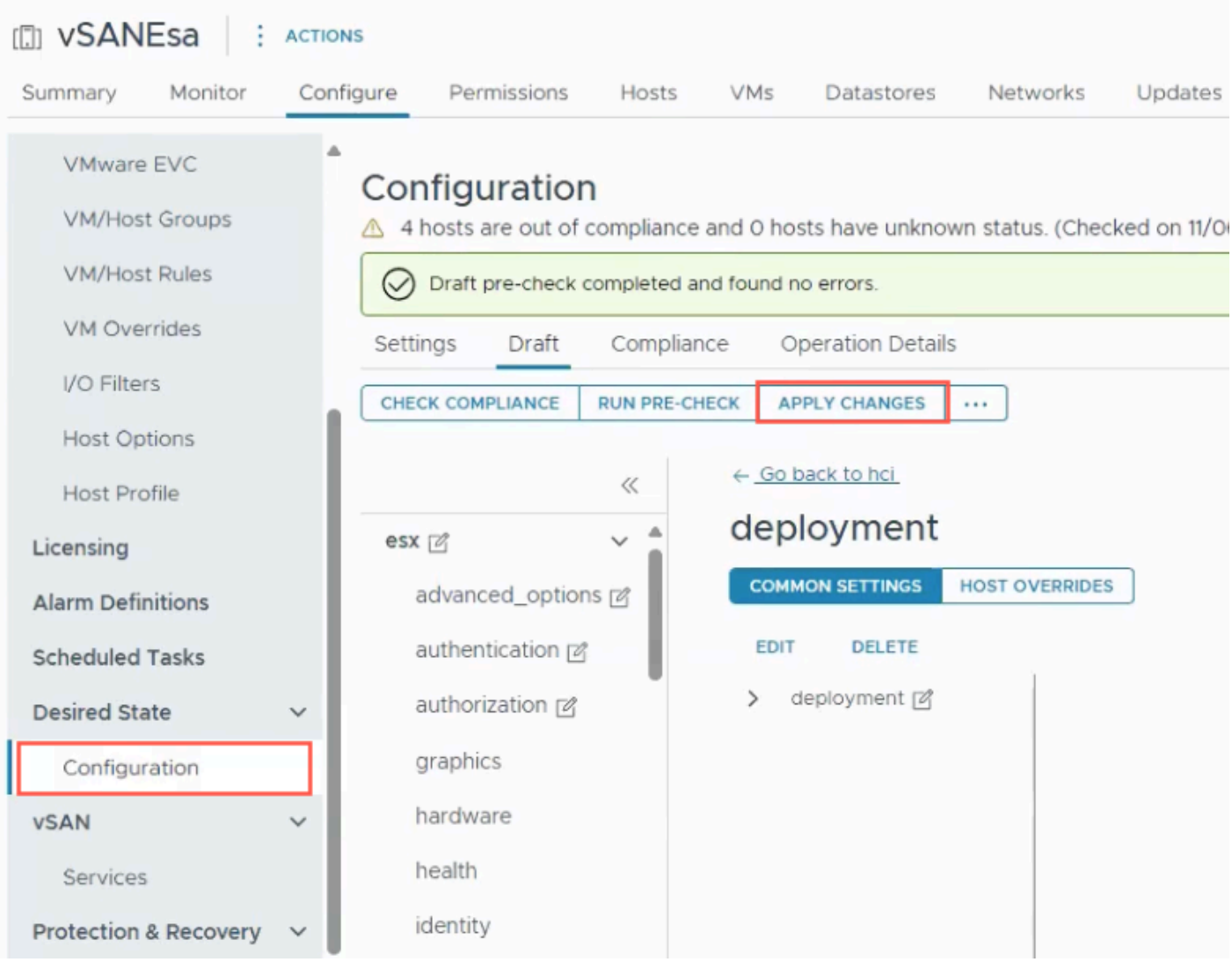
4. In the **Import from host** dialog box, select one of the hosts.
5. Click **Import**. When the host configuration has been imported, click **Close**.
6. After you have imported the configurations of one host into the vSAN cluster's desired state configuration, you can configure your cluster any way you like. Configuration changes apply to all hosts in the cluster.

Apply Changes and Remediate Configuration

You can apply changes to a cluster configuration profile after running a compliance check or a pre-check, which applies the configuration changes to all hosts in the cluster.

To apply the configuration profile:

1. Under **Desired State** > **Configuration**, click the **Apply Changes**.



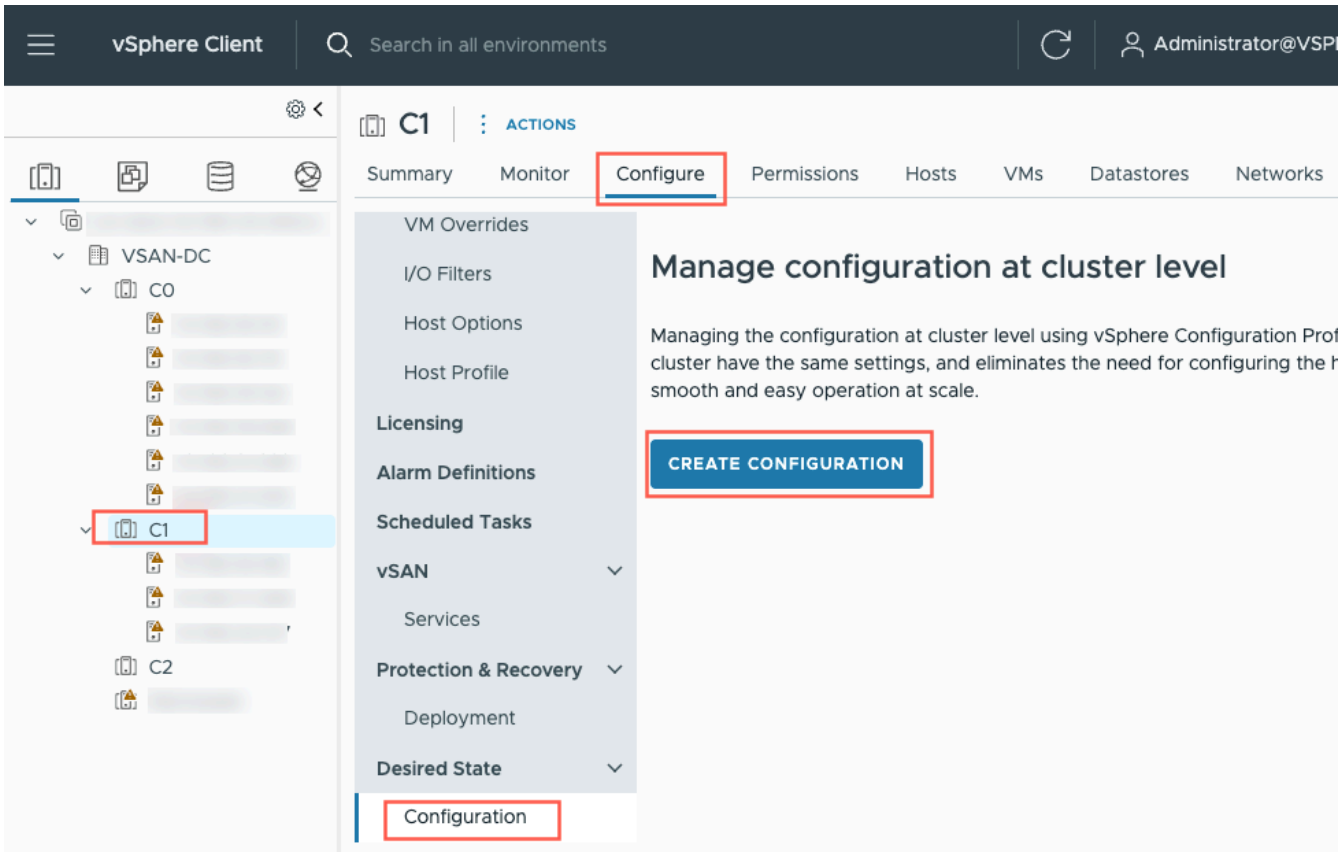
2. In the Remediate wizard pre-check, if no issues are found, click **Next**.
3. Review the remediation settings, and then click **Next**.
4. On the Review Impact page, review the changes that occur when the changes are applied. If any of the changes require approval, acknowledge and approve the changes.
5. Click **Remediate**.
6. After you have applied the change, you can verify the configurations.

Create a Configuration Profile and Apply to an Existing Cluster

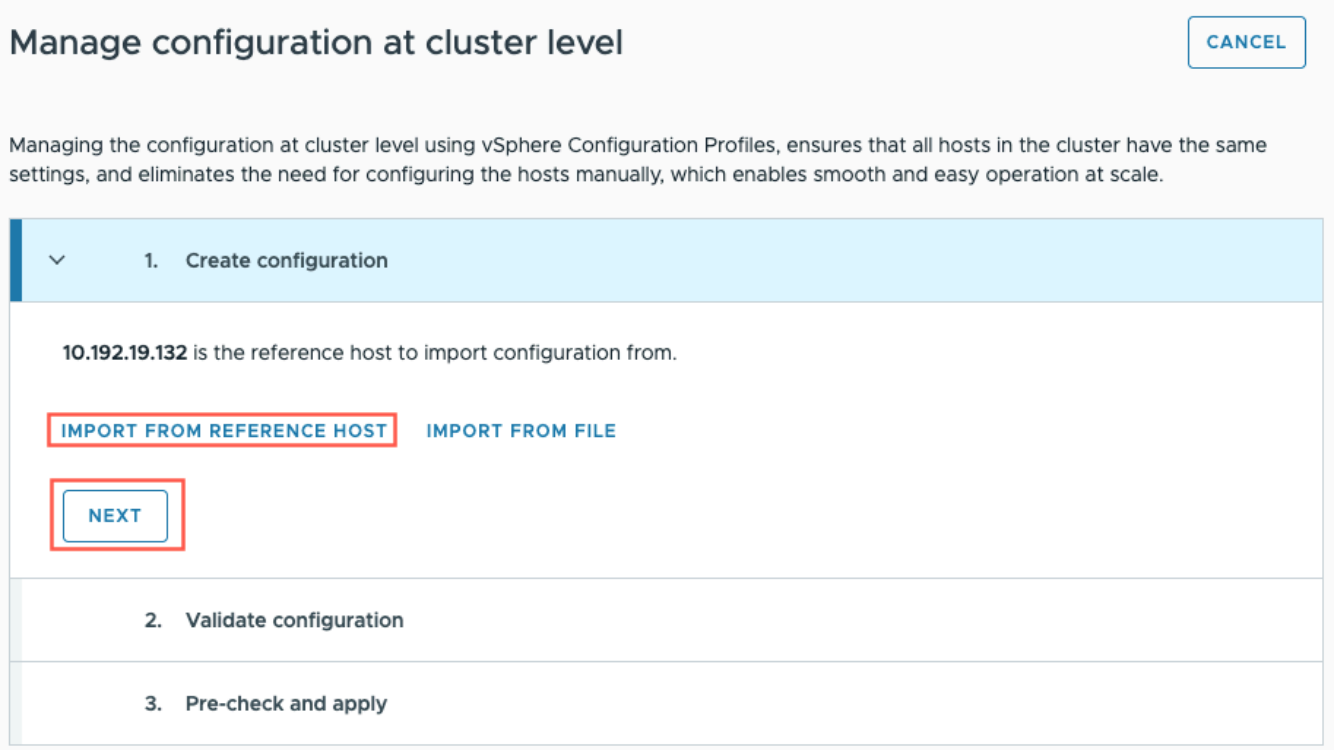
If you have an existing vSAN cluster, you can create a configuration profile for the cluster so you can manage cluster configurations more effectively.

To create a vSAN configuration profile for an existing cluster:

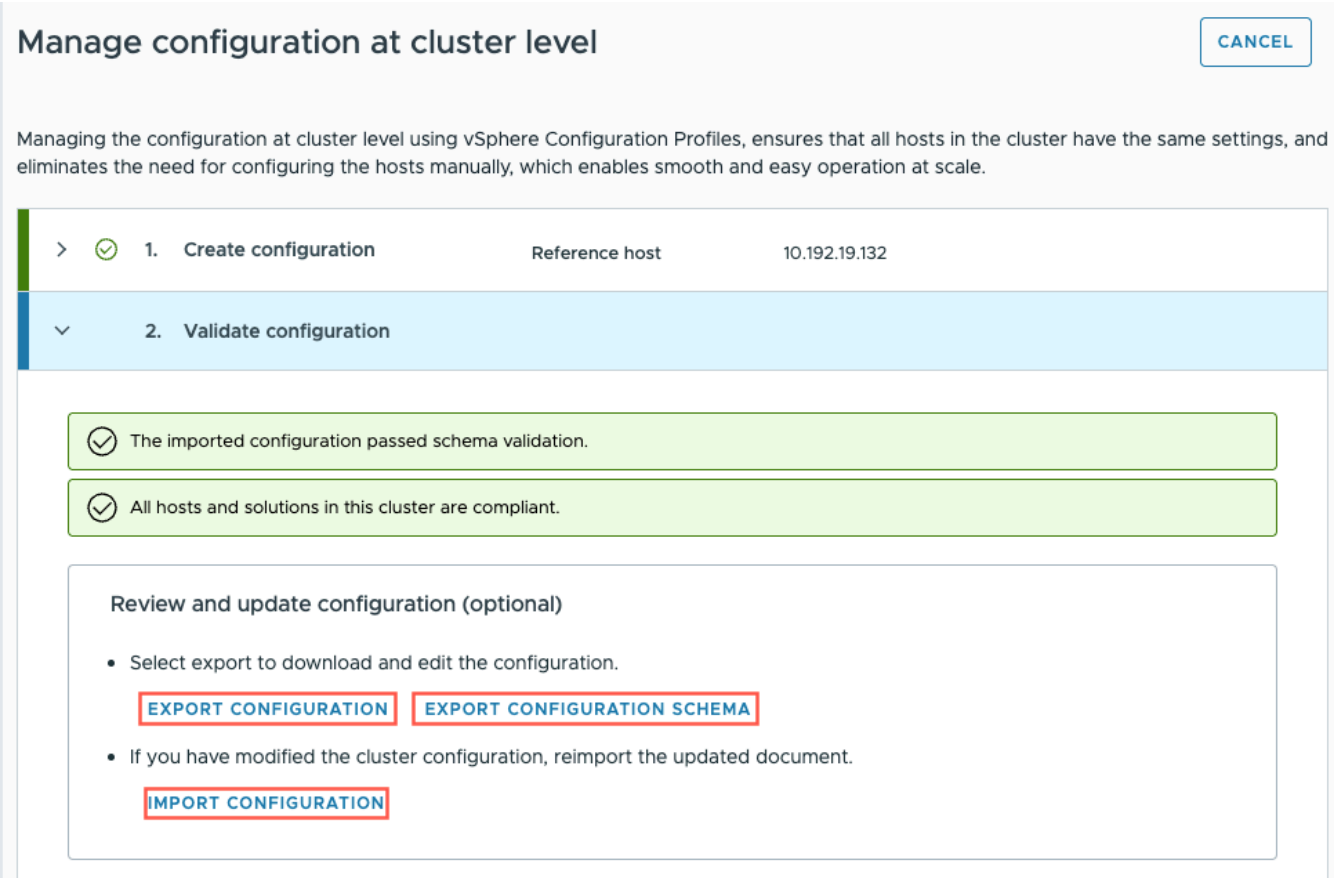
1. In the vSphere Client, navigate to the cluster.
2. Select the **Configure** tab, and then select **Desired State** > **Configuration**.
3. Click **Create Configuration**.



4. You have the option of importing the configuration of an existing host, or from an existing vSAN cluster configuration file in the JSON format.
5. If you selected to import from host, select one of the hosts from the list, and then click **Import**.



6. If you are selecting to import from file, browse to the JSON configuration file and click **Import**. Use this option if you already have a cluster configuration profile you want to use with this cluster.
7. After the host configuration or configuration file has been imported, click **Next** to validate the configuration.
8. After the configuration validates successfully, you can export the configuration as a JSON file. Or, you can export the JSON schema that is used for the configuration file. If you modify the exported JSON configuration file, you can reimport that configuration and can use it for the cluster.



9. Click **Next**. The pre-check assess the impact of applying the configuration to the entire cluster.
10. If the pre-check passes with no errors, you can click **Finish and Apply**. You can also view host level details from the pre-check.
11. In the **Finish and Apply** dialog box, click **Continue**, which apply the configuration to all hosts in the cluster.

    If the changes you are about to apply are destructive in any way, such as a disk remval, you will be asked to confirm the changes.
12. After the configuration has been applied, you can click **Go To Configuration** to view the configuration settings.
13. You can configure as many vSAN cluster settings as you want, then reapply the changes to the remediate the cluster with the new configuration. When making cluster configuration changes with configuration profiles, you always work in a draft state until you are ready to apply the changes.

Edit Cluster Configuration Profile and Apply Changes

Anytime that you change a cluster configuration, you must apply those changes to remediate the configuration.

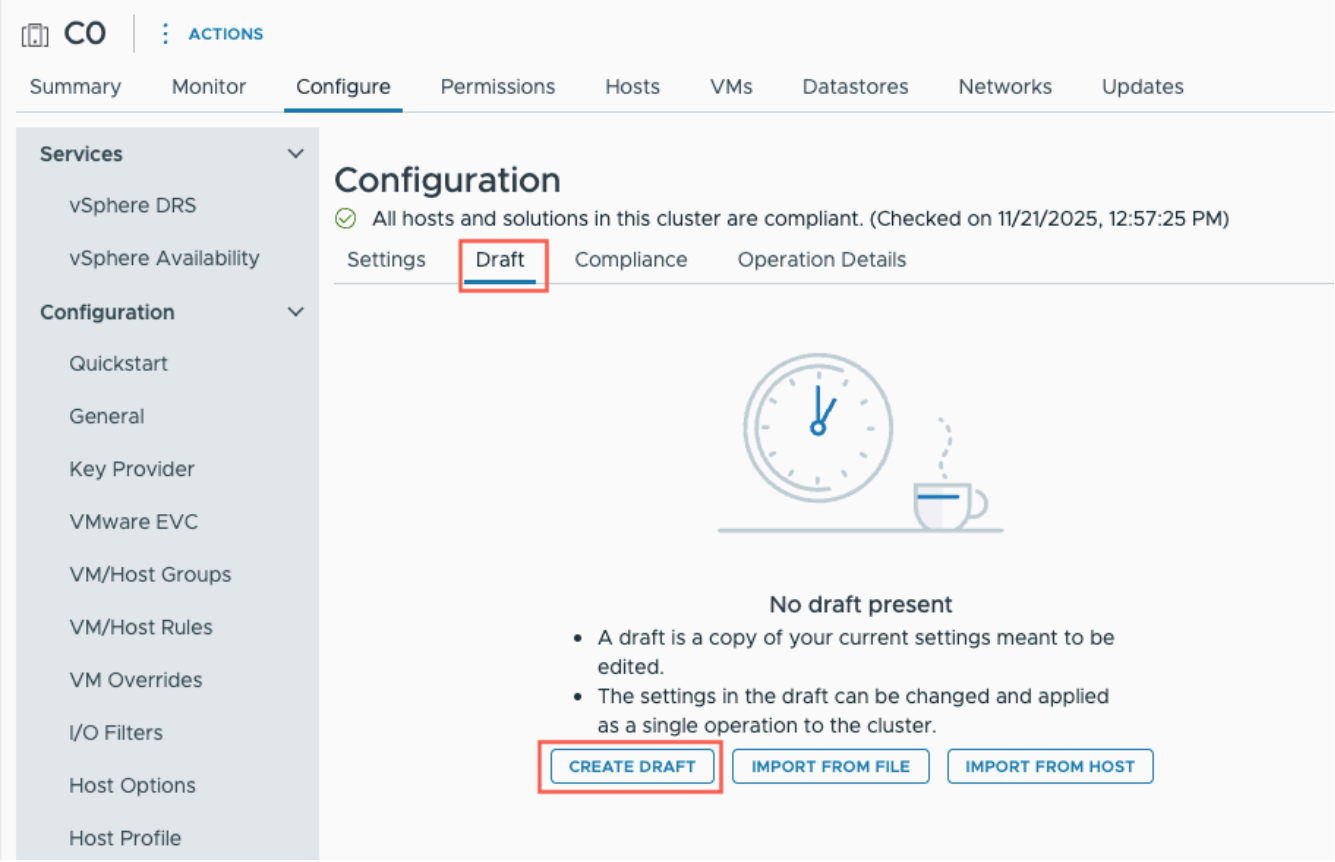
When you change a vSAN cluster configuration profile, you work first in a draft state. The draft acts as a copy of the current configuration that you can edit without affecting the cluster settings. When you apply the changes and then remediate the configuration profile, the draft configuration becomes the new desired configuration and source of truth for the cluster.

Some vSAN cluster configurations can make disruptive changes to disk layouts and other important configurations. You must acknowledge these types of changes before you can proceed with remediation.

To see the supported vSAN cluster configurations that you can manage with VCP, see Support VCP Configurations for vSAN.

To update a vSAN cluster configuration and then remediate it:

1. In the vSphere Client, navigate to the cluster.
2. Select the **Configure** tab, and then from the inner navigation panel select **Desired State** > **Configuration**.
3. On the Configuration panel, select the **Draft** tab.
4. Click **Create Draft**. Creating a draft configuration serves as a copy of the current configuration that you can modify without affecting the cluster.



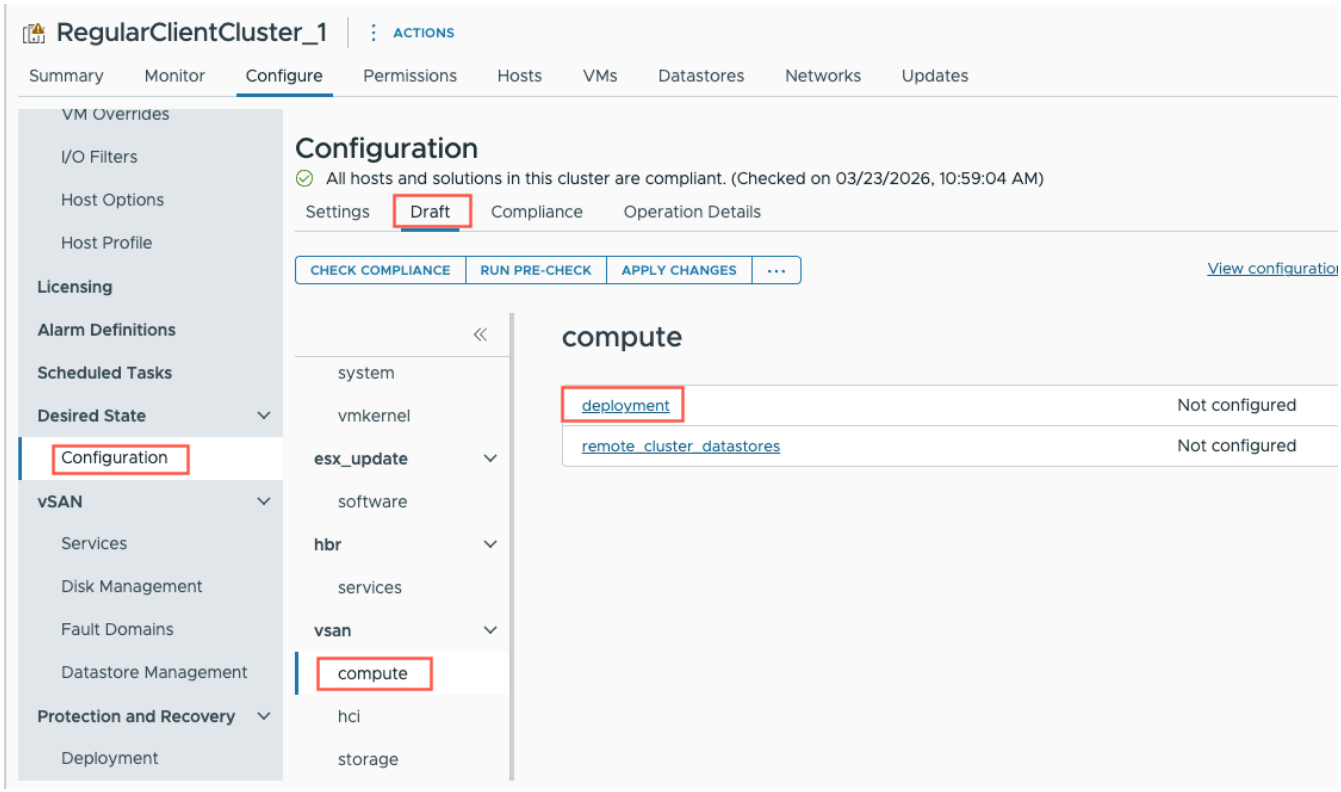
5. From the navigation pane, select a configuration to modify. If no configuration has been created yet, click **Configure Settings** to start configuring the cluster**.**

   You configure vSAN cluster configurations under the 'vsan' namespace for these three configuration categories:

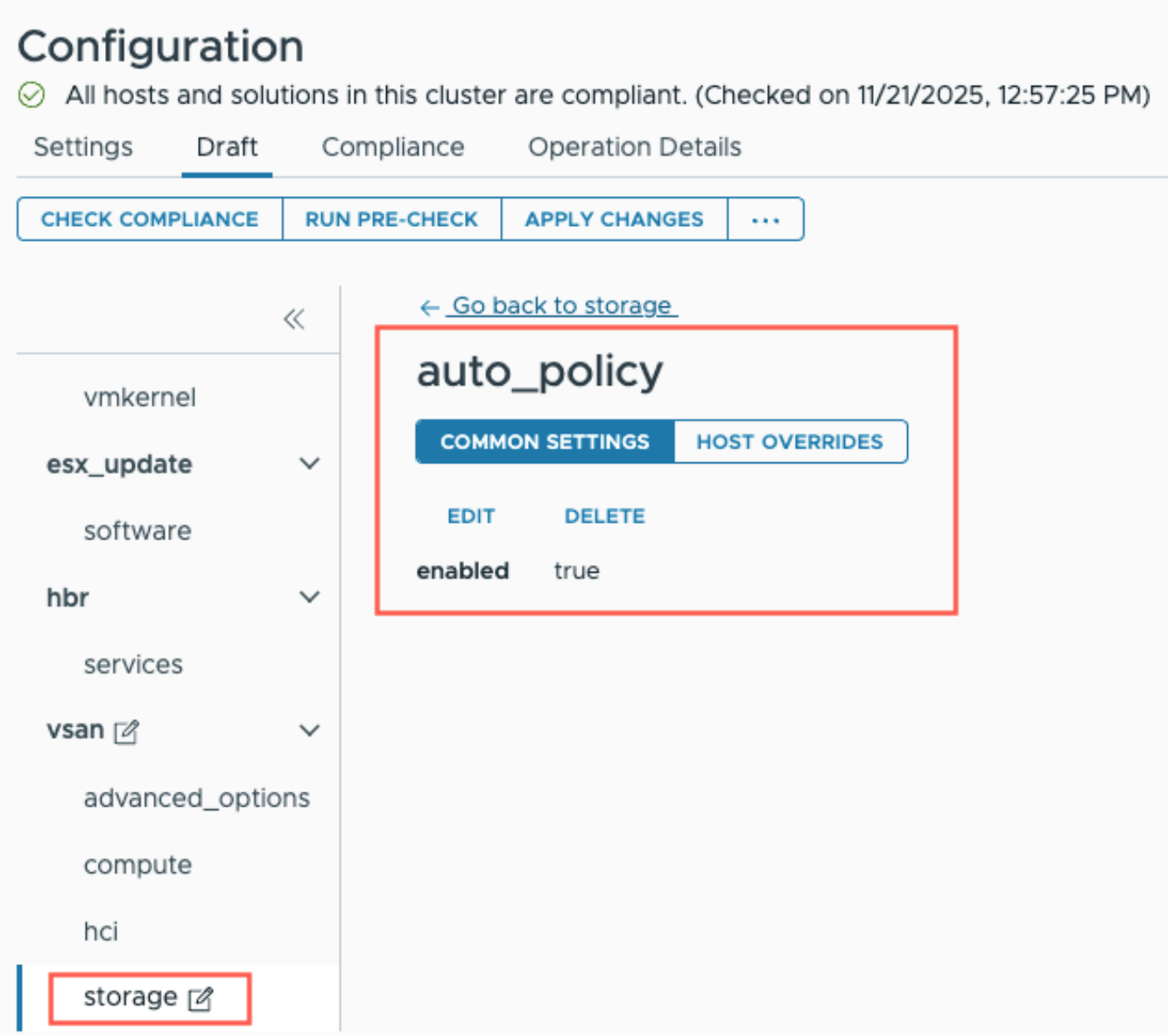
Table 1016: VCP Configurations for vSAN

vSAN Configuration	Description
Compute	• **Deployment**. Configure how vSAN ensures data availability and how it can survive hardware failures. If you choose Compute cluster, you must configure this setting. • **Remote data cluster datastores**. Used for disaggregated storage deployments, where you configure remote data clusters without storage.
HCI	• **Deployment**. Defines the architecture of the vSAN cluster: – vSAN OSA (Original Storage Architecture) – vSAN ESA (Express Storage Architecture) – vSAN Max • **Encryption**. Allows you to control encryption for data at rest. • In transit encryption. Allows you to control data as it moves over the network by encrypting traffic between ESXi hosts as they replicate data to one another. • **Remote cluster datastores**. In a disaggregated storage architecture, this setting allows you to list specific remote vSAN datastores that this cluster is allowed to mount to.
Storage	• **Auto policy**. When enabled, vSAN automatically creates and maintains a Default Storage Policy, based on the current size of your cluster. This setting is for vSAN ESA clusters only. • **Deployment**. Defines the underlying architecture (ESA or OSA) and role (HCI or storage-only) of the cluster. • **Encryption**. You can encrypt all data that is written to the physical NVMe or SSD drives in the cluster (data at rest). • **In transit encryption**. You can encrypt all data as it moves across your network.

For example:


RegularClientCluster_1
ACTIONS
Summary
Monitor
Configure
Permissions
Hosts
VMs
Datastores
Networks
Updates
VM Overrides
I/O Filters
Host Options
Host Profile
Licensing
Alarm Definitions
Scheduled Tasks
Desired State
Configuration
vSAN
Services
Disk Management
Fault Domains
Datastore Management
Protection and Recovery
Deployment
Configuration
All hosts and solutions in this cluster are compliant. (Checked on 03/23/2026, 10:59:04 AM)
Settings
Draft
Compliance
Operation Details
CHECK COMPLIANCE
RUN PRE-CHECK
APPLY CHANGES
View configuratio
system
vmkernel
esx_update
software
hbr
services
vsan
compute
hci
storage
compute
deployment
Not configured
remote_cluster_datastores
Not configured

6. Click **Save**. You see the new configuration.



7. Now, click **Run Pre-Check**. If the pre-check fails, you are notified and given the ability to edit the configuration to fix it.

   If the pre-check fails, banner messages indicate the errors and where to fix them.

8. If the pre–check is successful, the banner message indicates if the pre-check was successful. Click **Apply Changes** to remediate the configuration.
9. In the **Remediate** dialog box, after the pre-check runs, click **Next**.
10. On the Remediation Settings page, view the changes that will be made when you run the remediation and then click **Next**.
11. On the Review Impact page, you can review the impact remediation will have on your cluster and its hosts when you start the remediation process. If you have changed disks for a host, on the Review Impact page you can select the

data migration mode for moving data from the old disk to the new disk. You can also select decommission mode for the old disk.

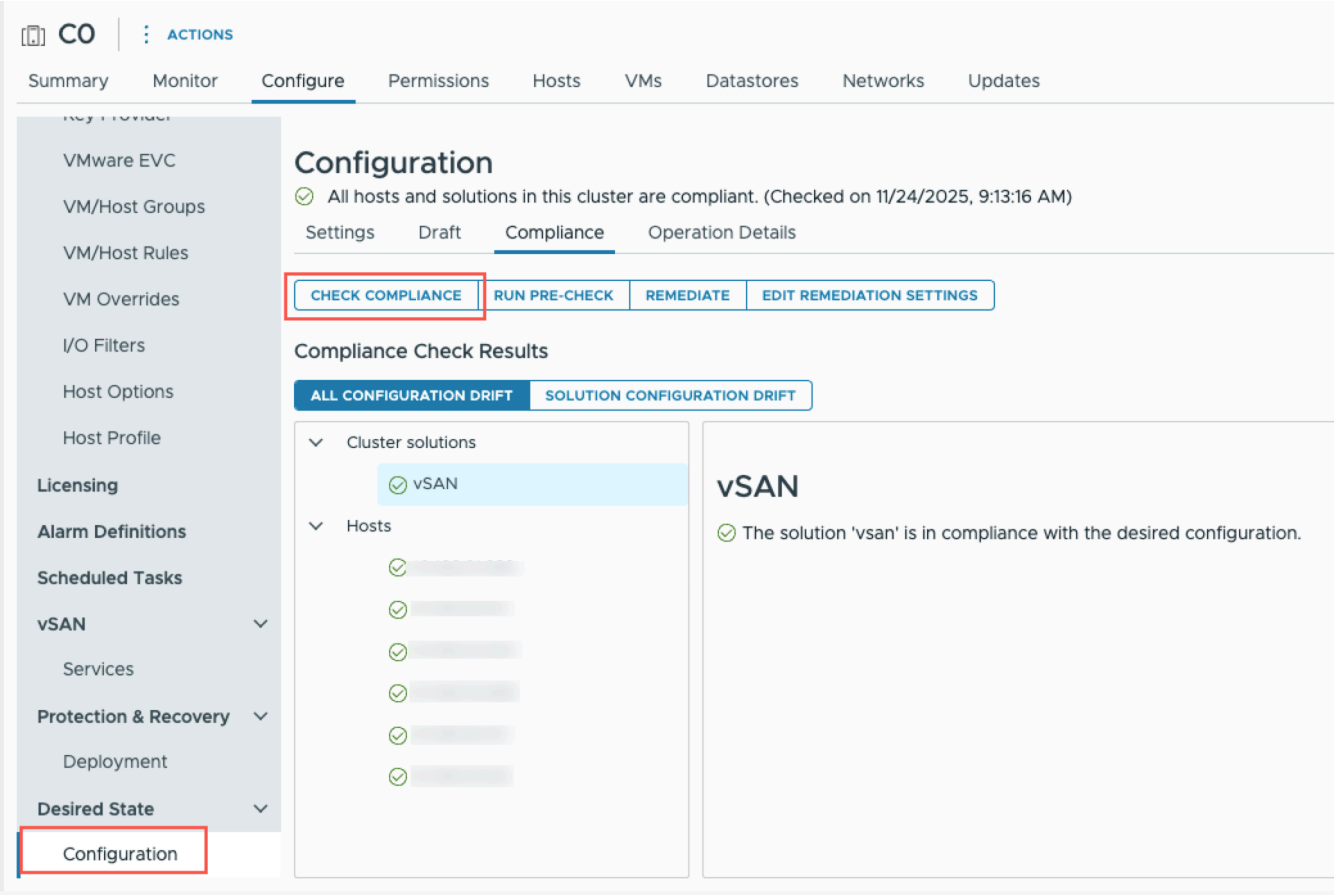
12. Click **Remediate**. If the remediation is successful, you see a banner message that indicates success.

Check for Configuration Drift

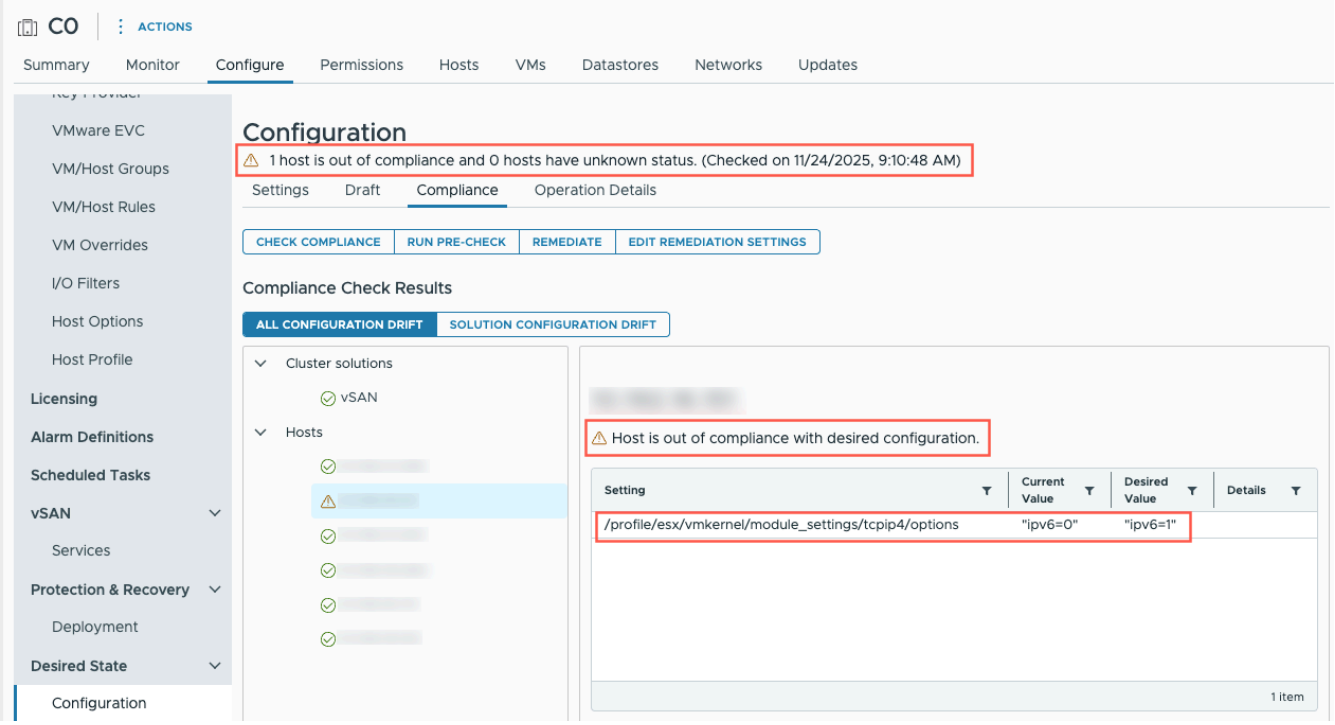
You can check for configuration drift at the solution level only, or you can check both solution and individual host configurations for drift.

To check cluster compliance for configuration drift:

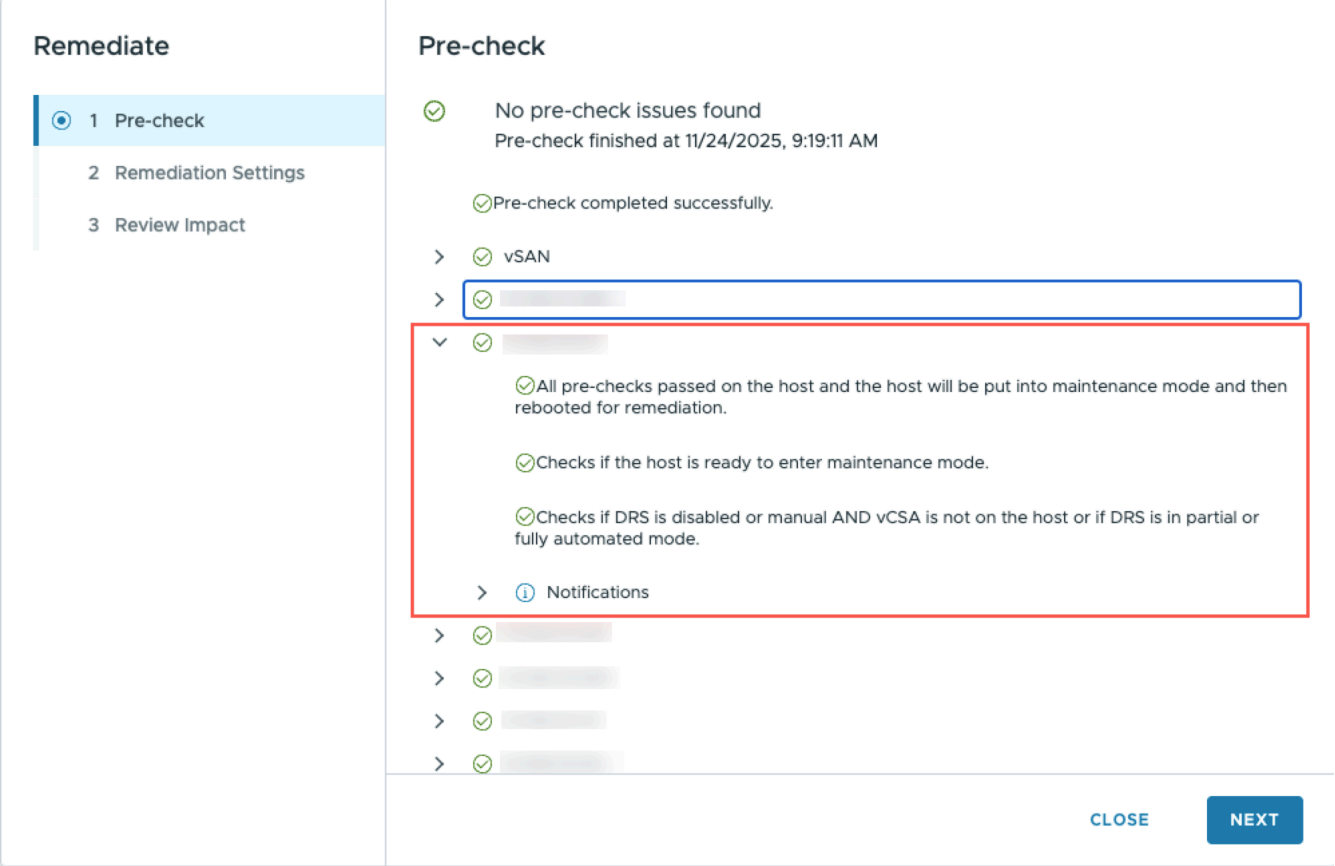
1. Under **Desired State** > **Configuration**, click **Check Compliance** to check for misconfigurations in the cluster.



2. If any configurations are out of compliance, select either the **All Configuration Drift** or **Solution Configuration Drift** tabs to see the misconfigurations. For example, in this image one of the hosts has the IPv6 setting disabled.



3. To reset the proper configuration, run the **Pre-Check** to prepare for remediation.
4. In the Remediate wizard pre-check page, you can expand the cluster nodes to view the details of pre-checks that were run on each host in the cluster.



5. Click **Next**.
6. Review the remediation settings, and then click **Next**.
7. On the Review Impact page, you can review the impact of what happens when you remediate. Click **Remediate**.
8. After you have remediated the misconfigurations, you can verify that the configuration state is correct.

Export or Import a Cluster Configuration Profile
You can export a cluster configuration to the JSON file format, which you can modify and reimport.

You can also import a cluster configuration to apply that configuration to another cluster.

To export a cluster configuration profile:

1. In the vSphere Client, select the vSAN cluster node from the left navigation.
2. On the **Configure** tab, in the navigation select **Desired State** > **Configuration**
3. Click **Export** > **Cluster configuration**.
4. Now, you can either apply this configuration to an existing cluster, or apply it to a new cluster.

To import a cluster configuration profile:

1. In the vSphere Client, select the vSAN cluster node from the left navigation.
2. On the **Configure** tab, in the navigation select **Desired State** > **Configuration**.
3. Click the **Draft** tab, and then click Import from file.
4. Select the file and click **Import**.
5. Next, click **Run Pre-Check** to make sure that the configuration file is valid.
6. Click **Apply changes** to apply the configuration profile to the cluster.

Edit Remediation Settings
You can change any of the default remediation settings for your vSAN cluster configuration profile.

Any changes that you make here will override cluster settings for the cluster and will apply to all future remediations. Changes that you make here do not apply to in-progress remediation operations.

1. In the vSphere Client, select the vSAN cluster node from the left navigation.
2. On the **Configure** tab, in the navigation select **Desired State** > **Configuration**.
3. Click the **Compliance** tab.
4. Click **Edit Remediation Settings**.
5. From the **Edit Cluster Remediation Settings** dialog box, you can modify the following settings:

Setting	Description
Common Settings	
Enable Quick Boot	Quick Boot significantly reduces the time that is taken for a host to reboot, but is supported only on select platforms. For more information on quick boot, see Understanding ESX Quick Boot Compatibility and Configuring vSphere Lifecycle Manager for Fast Upgrades.
Suspend to disk	Saves the virtual machines current memory and CPU state to a file on the host while the cluster is being remediated. When remediation is finished, this option uses the file to restore the VM to the state it was before remediation.
Suspend to memory	Suspending virtual machines to memory and using the Quick Boot functionality can reduce the time that it takes to remediate a configuration. For more information, see the Suspend Virtual Machines to Memory on Configuring vSphere Lifecycle Manager for Fast Upgrades.
Migrate powered off and suspended VMs to other hosts in the cluster, if a host must enter maintenance mode.	Use this option to control whether powered-off and suspended VMs are migrated to other hosts when a host enters maintenance mode. This setting does not apply to running VMs.

Setting	Description
Retry policy	Control how frequently remediation retry operations if remediation fails (in minutes), and how many retries to attempt before ignoring failed operations.
Disable DPM on the cluster	Control if you want the VMware Distributed Power Management (DPM) feature to be disabled during remediation. For more information, see Activating vSphere DPM for a DRS Cluster.
Disable HA admission control	Disabling HA Admission control can help remediation to proceed in clusters with higher resource contention, at the cost of reduced HA availability guarantees. For more information, see vSphere HA Admission Control.
Image specific settings	
Enable Live Patch	Choose to apply security patches and urgent bug fixes to all hosts in a cluster to manage with an image, ensuring minimal disruption to your workloads. If Live Patch is disabled, then you must later perform traditional Maintenance Mode/Reboot remediations to update the latest security patches.
Prevent remediation if hardware compatibility issues are found.	Choose this option if you want to prevent remediation if a hardware compatibility is encountered. For more information, see Cluster-Level Hardware Compatibility Checks.
Parallel remediation	Allows you to remediate all ESX hosts in a cluster against the desired configuration sequentially.
Configuration Profile specific settings	
Auto-remediate newly added hosts	When selected, any new hosts that have been added since the last remediation are remediated with the cluster-wide configuration.
Serial remediation	Allows you to remediate all ESX hosts in a cluster against the desired configuration sequentially.

6. Click **Save** to save the changes.

Unset vSAN Configurations from a VCP

If you want to completely unset all vSAN configurations from a cluster VCP, you must delete all configuration in the profile.

> **Warning:**
>
> Unsetting vSAN configurations from a VCP completely disables vSAN from the cluster. Before you perform this operation, ensure that unsetting all vSAN configurations from a cluster does not affect any other users or processes in your vSphere environment.

To unset vSAN configurations from a VCP:

1. In the vSphere Client, select the vSAN cluster node from the left navigation.
2. On the **Configure** tab, in the navigation select **Desired State** > **Configuration**.
3. On the Configuration panel, select the **Draft** tab.
4. Next, from the inner left navigation, select the 'vsan' node.
5. To completely remove all vSAN configurations from the VCP, you must delete **ALL** settings for the profile:
   - compute
     - deployment
     - clusters
   - HCI
     - deployment
     - encryption
     - in_transit_encryption

  - remote_cluster_datastores
- storage
  - auto_policy
  - deployment
  - encryption
  - in_transit_encryption

For more information about these settings, see Table 1016: VCP Configurations for vSAN .

6. Select a vSAN configuration node on the left (compute, HCI, and storage), and then select one of the settings. For example, select **vsan** > **compute** > **deployment**.
7. For each settings, select **Delete**.
8. Repeat the action by deleting all configurations for vSAN. After you have deleted all vSAN configurations, click **Run Pre-Check** to make sure that the configuration file is valid.
9. Click **Apply changes** to apply the configuration removal to the cluster.

Sharing Remote vSAN Datastores

Remote datastore sharing enables vSAN clusters to share their datastores with other clusters.

You can provision virtual machines running on your local cluster to use storage space on a remote datastore. When you provision a new virtual machine, you can select a remote datastore that is mounted to the client cluster. You can assign any compatible storage policy configured for the remote datastore.

Mounting a remote datastore is a cluster-wide configuration. When you mount a remote datastore to a vSAN cluster, it is available to all hosts in the cluster.

When you prepare a vSphere cluster for mounting remote datastore, select any one of the following vSAN cluster types:

vSAN Custer Type	Description
vSAN HCI cluster	Provides compute resources and storage resources. It can share its datastore across data centers and vCenter instances and mount datastores from other vSAN HCI clusters.
Compute Cluster	A vSphere cluster that can mount a remote datastore from a vSAN storage cluster. These clusters are hosts in a vSphere cluster that only complies with the *Broadcom Compatibility Guide* for vSphere, but not for vSAN.
vSAN storage cluster (deployment model based on vSAN ESA)	Provides storage resources, but not compute resources. Its datastore can be mounted by compute clusters or vSAN HCI clusters across data centers and vCenter instances.

vSAN datastore sharing has the following design considerations:

- vSAN single site and vSAN stretched clusters can share datastores across clusters in the same data center, or across clusters managed by remote vCenter instances, as long as the source and the remote vCenter instances are accessible and they pass the prechecks.
- Client hosts that are not part of a cluster are not supported. You can configure a single host compute-only cluster, however VMs running on this node are affected in the event of a host failure.
- A datastore on a vSAN storage cluster or a vSAN HCI server cluster can be shared across up to 10 client clusters.
- A client cluster can mount up to 5 remote datastores from one or more vSAN storage cluster or vSAN server clusters.
- A single datastore can be mounted by up to 128 vSAN hosts, including hosts in the local vSAN server cluster.
- All objects that make up a virtual machine must reside on the same datastore.
- For vSphere HA to work with vSAN datastore sharing, configure the following failure response for Datastore with APD on the client cluster: Power off and restart virtual machines.

The following configurations are not supported with vSAN datastore sharing:

- Remote provisioning of iSCSI volumes, or CNS persistent volumes. You can provision iSCSI volumes on the local vSAN datastore, but not on any remote vSAN datastore. For remote provisioning of CNS persistent volumes, see vSphere Functionality Supported by vSphere Container Storage Plug-in and Using vSphere Container Storage Plug-in for HCI Mesh Deployment in the *vSphere Storage* guide.
- Air-gapped networks or clusters using multiple vSAN VMkernel ports.
- Client cluster mounting datastores from different vSAN architectures. Starting with VCF 9.1, a client cluster can mount remote datastores from vSAN clusters regardless of storage architecture. A cluster running vSAN OSA can mount datastores from a vSAN ESA cluster, and vice-versa. For more information, see this blog post: Greater Flexibility and Security with VMware vSAN Storage Clusters in VCF 9.1.
- Client cluster mounting datastores from different storage providers. Both the client and the server clusters must use the same storage architecture. you cannot mount a vSAN storage server cluster to a Virtual Volumes client cluster.

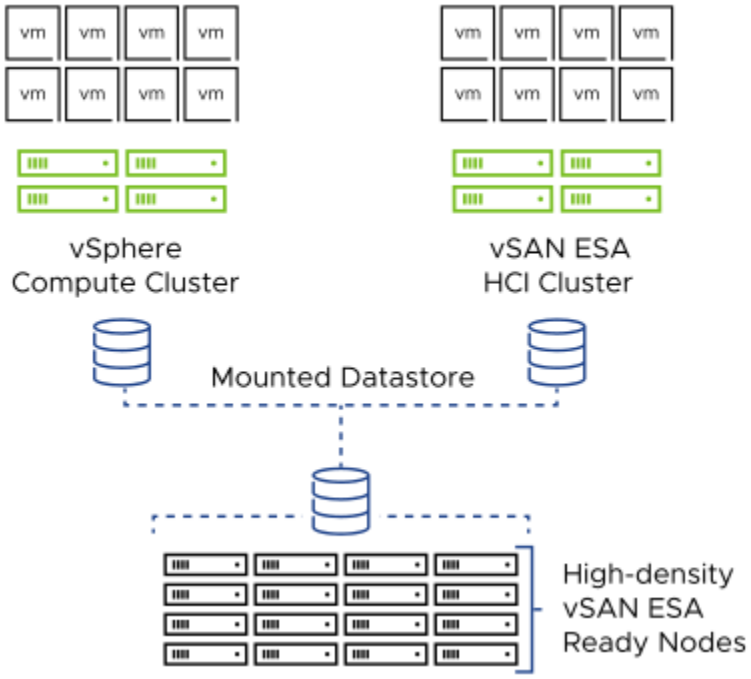
Disaggregated Storage with vSAN Storage Cluster

vSAN storage cluster is a fully distributed, scalable, shared storage solution for vSphere clusters and vSAN clusters. Storage resources are disaggregated from compute resources, so you can scale storage and compute resources independently.

vSAN storage cluster uses vSAN ESA and high-density vSAN Ready Nodes for increased capacity and performance.

> **Note:**
>
> vSAN storage cluster can be deployed by purchasing VMware Cloud Foundation or by acquiring the advanced add-on offer for VMware vSphere Foundation. Licensing for vSAN storage cluster is based on a per-TiB metric, which corresponds to the total amount of raw storage capacity needed for the environments.

A vSAN storage cluster acts as a server cluster that only provides storage. You can mount its datastore to vSphere clusters configured as compute clusters or vSAN HCI client clusters.



vSAN storage clusters have the following design considerations:

- Supported only on vSAN ESA running on vSAN Ready Nodes certified for vSAN storage clusters.
- Acts as a storage server only, not as a client. It is recommended not to run workload virtual machines on vSAN storage clusters.
- Requires a minimum of four ESX hosts, each hosts with a minimum of 20 TiB per host for space efficiency. Broadcom recommends a minimum of six ESX hosts. To optimize performance, use a uniform configuration of storage devices across all hosts.

- Supports the use of 25 GbE network connections between ESX hosts in the vSAN storage cluster, though Broadcom recommends the use of 100 GbE connections. Broadcom also recommends using traffic separation for storage clusters. For more information, see Storage Separation Distributed Switch Model.
- Supports the use of 10 GbE network connections from compute clients to the vSAN storage clusters, though Broadcom recommends the use of 25 GbE connections. For best performance, enable support for jumbo frames (MTU = 9000) and ensure you have sufficient resources at the network spine. For smaller vSAN storage cluster configurations, Broadcom support the use of 10 GbE network connections between ESX hosts. For more information, see vSAN ESA ReadyNode Hardware Guidance.
- Requires a latency below five milliseconds between the client and the server ESX hosts.
- Enable **Auto-Policy management** (Configure > vSAN > Services > Storage > Edit) to ensure optimal levels of resilience and space efficiency.
- Enable **Automatic rebalance** (Configure > vSAN > Services > Advanced Options > Edit) to ensure an evenly balanced, distributed storage system.

Note:

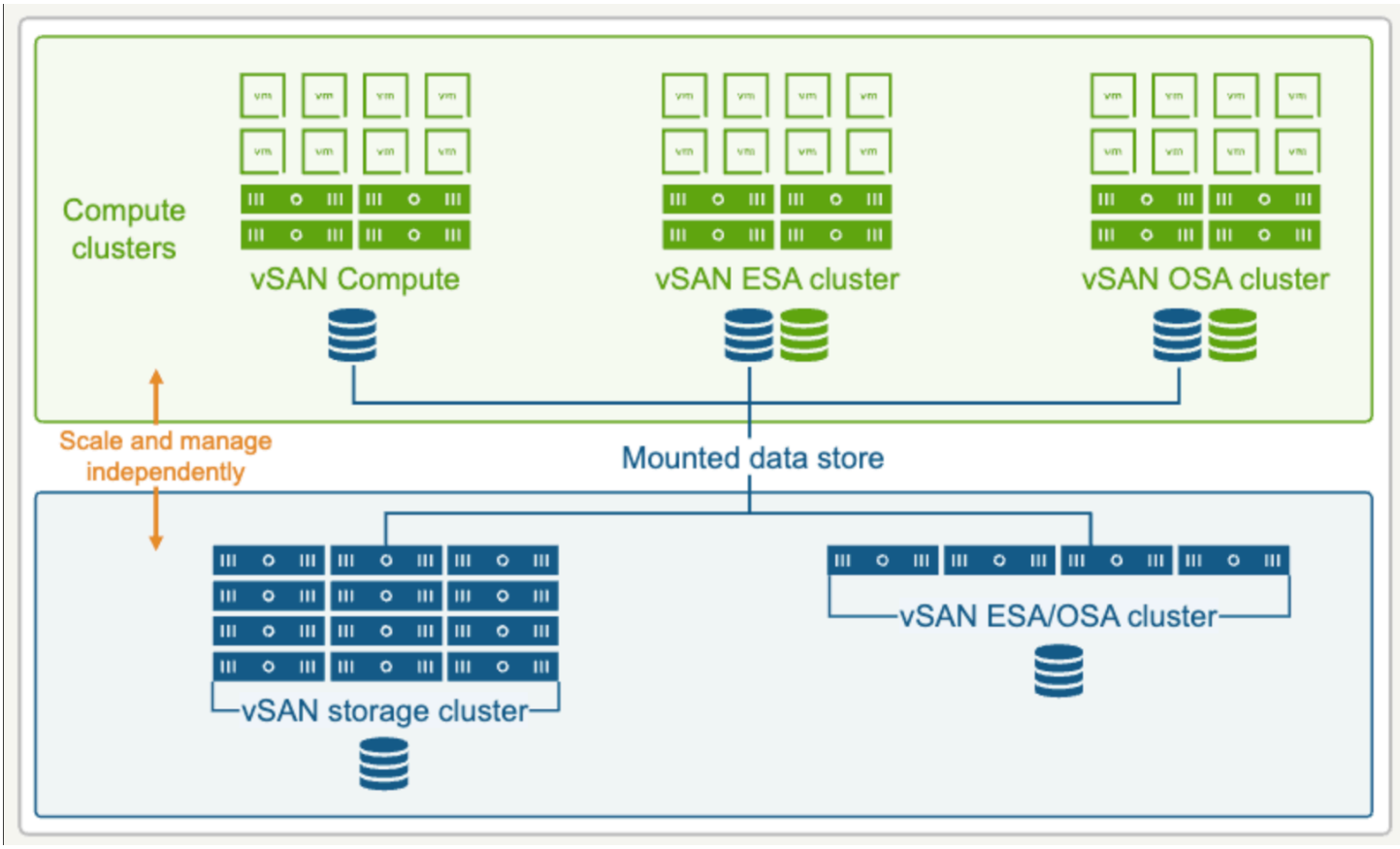
You can configure vSAN storage cluster only during cluster creation. You cannot convert an existing vSAN HCI cluster to a vSAN storage cluster, and you cannot convert a vSAN storage cluster to a vSAN HCI cluster. You must deactivate vSAN and reconfigure the cluster to make the conversion.

Compute cluster	A compute cluster is a vSphere cluster with a small vSAN element that enables it to mount a remote datastore hosted on a vSAN storage cluster. The ESX hosts in a compute cluster do not have local storage. You can monitor the capacity, health, and performance of the remote datastore. **Note:** A compute cluster does not require vSAN Ready Nodes. Compute clusters have the following design considerations: • vSAN networking must be configured on ESX hosts in the compute cluster. • No vSAN storage eligible devices can be present on ESX hosts in a compute cluster. • No data management features can be configured on the compute cluster.
vSAN storage cluster	A vSAN storage cluster can separate the external VM traffic from the internal vSAN storage traffic by using the dedicated VMkernel ports for different traffic types. With the vSAN storage cluster, you have the option to use the Storage cluster client network.

Cross-Cluster Capacity Sharing

vSAN storage clusters or vSAN HCI clusters can share their datastores with other vSAN ESA HCI clusters, vSAN OSA clusters, and vSAN compute clusters.

A vSAN storage cluster can act as a server to provide data storage, while a vSAN HCI cluster or compute cluster can also act as a client that consumes storage from a remote datastore. Adding remote data-in-transit encryption with at rest encryption and local data-in-transit encryption provides full end-to-end encryption for workloads.



Use the Datastore Management view to monitor and manage remote datastores mounted on the local vSAN cluster. Each client vSAN cluster or compute-only client cluster can mount remote datastores from server vSAN clusters. Each compatible vSAN cluster also can act as a server, and allow other vSAN clusters to mount its local datastore.

Monitor views for capacity, performance, health, encryption, and placement of virtual objects show the status of remote objects and datastores.

Using Remote vCenter Instances as Datastore Sources

vSAN HCI and vSAN storage clusters can share remote datastores across vCenter instances. You can add a remote vCenter as a datastore source for clusters on the local vCenter. Client clusters on the local vCenter can mount datastores that reside on the remote vCenter.

Use the Remote Datastore Management page in vCenter to manage remote datastore sources (**Configure** > **vSAN** > **> Remote Datastore Management**). Click the tabs to access information about shared datastores across vCenter instances, add vCenter instances as datastores sources, and mount datastores to local clusters.

Datastore Sources	View and manage datastore sources residing in remote vCenter instances. You can add or remove remote datastore sources for the local vCenter.
Clusters	View and manage clusters residing in the local vCenter. You can mount or unmount datastores from remote vCenter instances to the selected client cluster on the local vCenter.
Datastores	View all datastores available under this vCenter.

vCenter to vCenter datastore sharing has the following design considerations:

- Each vCenter can serve up to 10 client vCenter instances.
- Each client vCenter can add up to 5 remote vCenter datastore sources.
- When a virtual machine on a client cluster managed by one vCenter uses storage from a remote server cluster managed by another vCenter, the storage policy on the client's vCenter takes precedence.

View Remote vSAN Datastores

Use the Datastore Management page to view remote datastores mounted to the local vSAN cluster, and client clusters sharing the local datastore.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Datastore Management**.

This view lists information about each datastore mounted to the local cluster.

- vSAN OSA or vSAN ESA datastore
- Server cluster that hosts the datastore
- vSAN compute cluster
- vCenter of the server cluster (if applicable)
- Capacity usage of the datastore
- Free capacity available
- Number of virtual machines using the datastore (number of virtual machines using the compute resources of the local cluster, but the storage resources of the server cluster)
- Client clusters that have mounted the datastore
- Remote data-in-transit encryption status

You can mount or unmount remote datastores from this page.

Mount Remote vSAN Datastore

You can mount one or more datastores to a local vSAN HCI cluster, compute cluster, or the stretched compute cluster. You can use vSAN storage cluster only as a server cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Datastore Management**.
4. Click **Mount Remote Datastore** to open the wizard.
5. (Optional) Select a remote vCenter as the datastore source.

   Note:

   vSAN 9.1 supports stretched storage clusters to be mounted by HCI enabled or vSAN compute clusters through remote datastores across vCenter instances.
6. Select a datastore.
7. (Optional) If the server cluster is a vSAN stretched cluster, configure Site Coupling to choose the optimal data path between the server and the client cluster.

   A vSAN stretched cluster might have an asymmetrical network, where links within each availability zone have higher bandwidth and lower latency than links between availability zones. A symmetrical network has similar links within each availability zone and across availability zones.
8. Use the toggle button to enable remote data-in-transit encryption and specify a rekey interval. Click **Next**.

   If no value is set for the rekey interval, then the default of 24 hours is used.
9. Check the datastore compatibility, and click **Finish**.

The remote datastore is mounted to the local vSAN cluster.

When you provision a virtual machine, you can select the remote datastore as the storage resource. Assign a storage policy that is supported by the remote datastore.

Edit Remote vSAN Datastore

Use the Datastore Management page to edit remote datastores mounted to the local vSAN cluster, and client clusters sharing the local datastore.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Datastore Management**.
4. Select a datastore and click **Edit** to open the wizard.
5. (Optional) If the server cluster is a vSAN stretched cluster, configure Site Coupling to choose the optimal data path between the server and the client cluster.
6. Use the toggle button to enable or disable remote data-in-transit encryption and click **Next**.

   If no value is set for the rekey interval, then the default of 24 hours is used.
7. Check the datastore compatibility, and click **Finish**.

Unmount Remote vSAN Datastore

You can unmount a remote datastore from a vSAN cluster.

If no virtual machines on the local cluster are using the remote vSAN datastore, you can unmount the datastore from your local vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Datastore Management**.
4. Select a remote datastore, and click **Unmount**.
5. Click **Unmount** to confirm.

The selected datastore is unmounted from the local cluster.

Monitor Datastore Sharing with vSphere Client

You can use the vSphere Client to monitor the status of vSAN datastore sharing operations.

vSAN capacity monitor notifies you when remote datastores are mounted to the cluster. You can select the remote datastore to view its capacity information **(Monitor > vSAN > Capacity**). vSAN displays a banner informing you in case there are existing clusters mounting the vSAN datastore from this cluster.

- The Virtual Objects view (**Monitor** > **> vSAN** > **Virtual Objects**) shows the virtual objects used by the remote cluster VMs .
- The Physical disk placement view (**VM** > **Monitor** > **Physical disk placement**) for a VM located on a remote datastore shows information about its remote location.

vSAN health (**Monitor** > **vSAN** > **Health** > **Health Findings** > **All** > **Filter by Category**) checks report on the status of HCI functions.

- Data > vSAN Object health check shows accessibility information of remote objects.
- Network > Server cluster partition check reports about network partitions between hosts in the client cluster and the server cluster.
- Network > Latency checks the latency between hosts in the client cluster and the server cluster.

vSAN cluster performance views include virtual machine performance charts that display the virtual machine level performance of the client cluster from the perspective of the remote cluster. You can select a remote datastore to view the performance.

You can run pro-active tests on remote datastores to verify virtual machine creation and network performance. The virtual machine creation test creates a virtual machine on the remote datastore. The Network performance test checks the

network performance between all hosts in the client cluster and all hosts the server clusters. For more information, see Proactive Tests on vSAN Cluster.

Add Remote vCenter as Datastore Source

You can add a remote vCenter as a remote datastore source for client clusters on the local vCenter.

1. In the vSphere Client, navigate to the cluster.
2. Select **Configure**.
3. Under **vSAN**, click **Datastore Management**.
4. On the **Datastore Sources** tab, click **Add Remote Datastore Source** to open the wizard.
5. Enter information to specify the remote vCenter.
6. Check the compatibility, review the configuration, and click **Finish**.

The remote vCenter is added as a datastore source. vSAN clusters on the local vCenter can mount remote datastores that reside on the remote vCenter.

Working with Members of the vSAN Cluster in Maintenance Mode

Before you shut down, reboot, or disconnect an ESX host that is a member of a vSAN cluster, you must put the ESX host in maintenance mode.

When working with maintenance mode, consider the following guidelines:

- When you place an ESX host in maintenance mode, you must select a data evacuation mode, such as **Ensure accessibilityFull data migration** or **No data migration**. Ensure accessibility is the default option and moves data, if necessary.
- When any member ESX host of a vSAN cluster enters maintenance mode, the cluster capacity automatically reduces as the member ESX host no longer contributes storage to the vSAN cluster.
- A virtual machine's compute resources might not reside on the ESX host that is being placed in maintenance mode, and the storage resources for virtual machines might be located anywhere in the vSAN cluster.
- The **Ensure accessibility** mode is faster than the **Full data migration** mode because the **Ensure accessibility** migrates only the components from the hosts that are essential for running the virtual machines. When in this mode, if you encounter a failure, the availability of your virtual machine is affected. Selecting the **Ensure accessibility** mode does not reprotect your data during failure and you might experience unexpected data loss.
- The Enhanced durability mode helps to store the incremental data writes in case an unexpected host failure occurs within your cluster while a host is in the maintenance mode. The ESX host can enter maintenance mode when the objects in the cluster use failure to tolerate (FTT) of 1 and Ensure accessibility evacuation mode.
- When you select the **Full data migration** mode, your data is automatically reprotected against a failure, if the resources are available and the **Failures to tolerate** set to 1 or more. When in this mode, all components from the ESX host are migrated and, depending on the amount of data you have on the ESX host, the migration might take longer. With **Full data migration** mode, your virtual machines can tolerate failures, even during planned maintenance.
- When working with a three-host cluster, you cannot place a server in maintenance mode with **Full data migration**. Consider designing a vSAN cluster with four or more ESX hosts for maximum availability.

Before you place an ESX host in maintenance mode, you must verify the following:

- If you are using **Full data migration** mode, verify that the cluster has enough ESX hosts and capacity available to meet the **Failures to tolerate** policy requirements.
- Verify that enough flash capacity exists on the remaining ESX hosts to handle any flash read cache reservations. To analyze the current capacity use per ESX host, and whether a single ESX host failure might cause the cluster to run out of space and impact the vSAN cluster capacity, cache reservation, and cluster components, use the data migration pre-check tests. For more information, see Check the Data Migration Capabilities of an ESX Host in the vSAN Cluster.
- Verify that you have enough capacity devices in the remaining ESX hosts to handle stripe width policy requirements, if selected.

- Make sure that you have enough free capacity on the remaining ESX hosts to handle the amount of data that must be migrated from the ESX host entering maintenance mode.

The Confirm Maintenance Mode dialog box provides information to guide your maintenance activities. You can view the impact of each data evacuation option.

- Whether or not sufficient capacity is available to perform the operation.
- How much data will be moved.
- How many objects will become non-compliant.
- How many objects will become inaccessible.

Check the Data Migration Capabilities of an ESX Host in the vSAN Cluster

Use data migration pre-check to identify the impact of migration options when placing an ESX host into maintenance mode or removing it from the cluster.

Before you place an ESX host into maintenance mode, run the data migration pre-check. The test results provide information to help you determine the impact to cluster capacity, predicted health checks, and any objects that will go out of compliance. If the operation will not succeed, pre-check provides information about what resources are needed.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Data Migration Pre-check**.
4. Select an ESX host, a vSAN data migration option, and click **Pre-check**.

   vSAN runs the data migration precheck tests.
5. View the test results.

   The pre-check results show whether the ESX host can safely enter maintenance mode.

   - The **Object state** tab displays objects that might have issues after the data migration.
   - The **Cluster Capacity** tab displays the impact of data migration on the vSAN cluster before and after you perform the operation.
   - The **Predicted Health** tab displays the health checks that might be affected by the data migration.

If the pre-check indicates that you can place the ESX host into maintenance mode, you can click **Enter Maintenance Mode** to migrate the data and place the ESX host into maintenance mode.

Place a Member of vSAN Cluster in Maintenance Mode

Before you shut down, reboot, or disconnect an ESX host that is a member of a vSAN cluster, you must place the ESX host in maintenance mode.

Verify that your environment has the capabilities required for the option you select.

When you place an ESX host in maintenance mode, you must select a data evacuation mode, such as **Ensure accessibility**, **Full data migration**, or **No data migration**. When any member ESX host of a vSAN cluster enters maintenance mode, the cluster capacity is automatically reduced, because the member ESX host no longer contributes capacity to the cluster. This also results in the vCLS VMs getting powered off and unregistered.

> **Note:**
>
> The vSAN file service virtual machines (FSVM) running on an ESX host are automatically powered off when an ESX host in the vSAN cluster enters maintenance mode.

Any vSAN iSCSI targets served by this ESX host are transferred to other ESX hosts in the vSphere cluster, and thus the iSCSI initiator are redirected to the new target owner.

1. In the vSphere Client, navigate to the cluster.
2. Expand the cluster, right-click the ESX host and select **Maintenance Mode > Enter Maintenance Mode**.
3. Select a data evacuation mode and click **OK**.

Option	Description
Ensure accessibility	This is the default option. When you power off or remove the ESX host from the cluster, vSAN migrates just enough data to ensure every object is accessible after the ESX host goes into maintenance mode. Select this option if you want to take the ESX host out of the cluster temporarily, for example, to install upgrades, and plan to have the ESX host back in the vSphere cluster. This option is not appropriate if you want to remove the ESX host from the vSphere cluster permanently. Typically, only partial data evacuation is required. However, the virtual machine might no longer be fully compliant to a virtual machine storage policy during evacuation. That means, it might not have access to all its replicas. If a failure occurs while the ESX host is in maintenance mode and the **Failures to tolerate** is set to 1, you might experience data loss in the cluster. **Note:** This is the only evacuation mode available if you are working with a three-host cluster or a vSAN cluster configured with three fault domains.
Full data migration	vSAN evacuates all data to other ESX hosts in the vSphere cluster and maintains the current object compliance state. Select this option if you plan to migrate the ESX host permanently. When evacuating data from the last ESX host in the vSphere cluster, make sure that you migrate the virtual machines to another datastore and then place the ESX host in maintenance mode. This evacuation mode results in the largest amount of data transfer and consumes the most time and resources. All the components on the local storage of the selected ESX host are migrated elsewhere in the vSphere cluster. When the ESX host enters maintenance mode, all virtual machines have access to their storage components and are still compliant with their assigned storage policies. **Note:** If there are objects in reduced availability state, this mode maintains this compliance state and does not guarantee that the objects will become compliant. If a virtual machine object that has data on the host is not accessible and is not fully evacuated, the host cannot enter maintenance mode.
No data migration	vSAN does not evacuate any data from this ESX host. If you power off or remove the ESX host from the vSphere cluster, some virtual machines might become inaccessible.

A vSphere cluster with three fault domains has the same restrictions that a three-host vSphere cluster has, such as the inability to use **Full data migration** mode or to reprotect data after a failure.

Alternatively, you can place an ESX host in the maintenance mode by using ESXCLI. Before placing an ESX host in this mode, ensure that you powered off the virtual machines that running on the ESX host.

To perform an action before entering maintenance mode, run the following command on the ESX host:

```
esxcli system maintenanceMode set --enable 1 --vsanmode=<str>
```

Following are the string values allowed for vsanmode:

- ensureObjectAccessibility - Evacuate data from the disk to ensure object accessibility in the vSAN cluster, before entering maintenance mode.

  Note:
  The default value is ensureObjectAccessibility. This value will be used if you do not specify any value for the vsanmode.
- evacuateAllData - Evacuate all data from the disk before entering maintenance mode.
- noAction - Do not move vSAN data out of the disk before entering maintenance mode.

To verify the status of the ESX host, run the following command:

```
esxcli system maintenanceMode get
```

To exit maintenance mode, run the following command:

```
esxcli system maintenanceMode set --enable 0
```

You can track the progress of data migration in the cluster.

Managing Fault Domains in vSAN Clusters

Fault domains enable you to protect against rack or chassis failure if your vSAN cluster spans across multiple racks or blade server chassis.

You can create fault domains and add one or more hosts to each fault domain. A fault domain consists of one or more vSAN hosts grouped according to their physical location in the data center. When configured, fault domains enable vSAN to tolerate failures of entire physical racks as well as failures of a single host, capacity device, network link, or a network switch dedicated to a fault domain.

The **Failures to tolerate** policy for the cluster depends on the number of failures a virtual machine is provisioned to tolerate. When a virtual machine is configured with the **Failures to tolerate** set to 1 `(FTT=1)`, vSAN can tolerate a single failure of any kind and of any component in a fault domain, including the failure of an entire rack.

When you configure fault domains on a rack and provision a new virtual machine, vSAN ensures that protection objects, such as replicas and witnesses, are placed in different fault domains. For example, if a virtual machine's storage policy has the **Failures to tolerate** set to N `(FTT=n)`, vSAN requires a minimum of `2*n+1` fault domains in the cluster. When VMs are provisioned in a cluster with fault domains using this policy, the copies of the associated virtual machine objects are stored across separate racks.

A minimum of three fault domains are required to support FTT=1. For best results, configure four or more fault domains in the cluster. A cluster with three fault domains has the same restrictions that a three host cluster has, such as the inability to reprotect data after a failure and the inability to use the **Full data migration** mode. For information about designing and sizing fault domains, see Designing and Sizing vSAN Fault Domains .

Consider a scenario where you have a vSAN cluster with 16 hosts. The hosts are spread across four racks, that is, four hosts per rack. To tolerate an entire rack failure, create a fault domain for each rack. You can configure a cluster of such capacity with the **Failures to tolerate** set to 1. If you want the **Failures to tolerate** set to 2, configure five fault domains in the cluster.

When a rack fails, all resources including the CPU, memory in the rack become unavailable to the cluster. To reduce the impact of a potential rack failure, configure fault domains of smaller sizes. Increasing the number of fault domains increases the total amount of resource availability in the cluster after a rack failure.

When working with fault domains, follow these best practices.

- Configure a minimum of three fault domains in the vSAN cluster. For best results, configure four or more fault domains.
- A host not included in any fault domain is considered to reside in its own single-host fault domain.
- You do not need to assign every vSAN host to a fault domain. If you decide to use fault domains to protect the vSAN environment, consider creating equal sized fault domains.
- When moved to another cluster, vSAN hosts retain their fault domain assignments.
- When designing a fault domain, place a uniform number of hosts in each fault domain.
- You can add any number of hosts to a fault domain. Each fault domain must contain at least one host.

Create a New Fault Domain in vSAN Cluster

To ensure that the virtual machine objects continue to run smoothly during a rack failure, you can group hosts in different fault domains.

- Choose a unique fault domain name. vSAN does not support duplicate fault domain names in a cluster.
- Verify the version of your ESX hosts.
- Verify that your vSAN hosts are online. You cannot assign hosts to a fault domain that is offline or unavailable due to hardware configuration issue.

When you provision a virtual machine on the vSphere cluster with fault domains, vSAN distributes protection components, such as witnesses and replicas of the virtual machine objects across different fault domains. As a result, the vSAN environment becomes capable of tolerating entire rack failures in addition to a single host, storage disk, or network failure.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click the plus icon.

   The New Fault Domain wizard opens.
5. Enter the fault domain name.
6. Select one or more hosts to add to the fault domain.

   A fault domain cannot be empty. You must select at least one host to include in the fault domain.
7. Click **Create**.

   The selected hosts appear in the fault domain. Each fault domain displays the used and reserved capacity information. This enables you to view the capacity distribution across the fault domain.

Move ESX Host into Selected Fault Domain in vSAN Cluster

You can move an ESX host into a selected fault domain in the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click and drag the ESX host that you want to add onto an existing fault domain.

   The **Add Hosts To Domain** dialog appears.
5. Click **Move**.

   The selected ESX host appears in the fault domain.

Move ESX Hosts out of a Fault Domain in vSAN Cluster

Depending on your requirement, you can move ESX hosts out of a fault domain.

Verify that the ESX host is online. You cannot move ESX hosts that are offline or unavailable from a fault domain.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
   a) Click and drag the ESX host from the fault domain to the Standalone Hosts area.
   b) Click **Move** to confirm.

The selected ESX host is no longer part of the fault domain. Any ESX host that is not part of a fault domain is considered to reside in its own single-host fault domain.

You can add ESX hosts to fault domains. See Move ESX Host into Selected Fault Domain in vSAN Cluster.

Rename a Fault Domain in vSAN Cluster

You can change the name of an existing fault domain in your vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
   a) Click the Actions icon on the right side of the fault domain, and choose **Edit**.
   b) Enter a new fault domain name.
4. Click **Apply** or **OK**.
   The new name appears in the list of fault domains.

Remove Selected Fault Domains from vSAN Cluster

When you no longer need a fault domain, you can remove it from the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Click the Actions icon on the right side of the fault domain, and select **Delete**.
5. Click **Delete** to confirm.

All ESX hosts in the fault domain are removed and the selected fault domain is deleted from the vSAN cluster. Each ESX host that is not part of a fault domain is considered to reside in its own single-host fault domain.

Tolerate Additional Failures with Fault Domain in vSAN Cluster

Fault domains in a vSAN cluster provides resilience and assures that the data is available even with failures based on policy.

With failures to tolerate (FTT) set to 1, the object can tolerate a failure. However, a temporary failure followed by a permanent failure in a cluster can result in data loss. An additional fault domain provides vSAN the ability to create a durability component without having additional FTTs for the object. vSAN triggers this extra component during planned and unplanned failures. Unplanned failures include network disconnect, disk failures, and host failures. Planned failures include Entering Maintenance Mode (EMM). For example, a 6 host cluster with RAID 6 object cannot create a durability component if there is a host failure.

vSAN ensures the data availability of the objects when the components go offline and comes back online unexpectedly based on the FTTs specified in the storage policy. During a failure, the writes of the failed component is redirected to the durability component. When the component recovers from the transient failure and comes back online, the durability component disappears and results in the resynchronization of the component.

Without the durability component in place, if there is a second permanent failure in the cluster and the mirror object is affected, the object data gets permanently lost even if the failure is resolved.

Using vSAN Snapshots and Replication

vSAN snapshots and replication allows for local virtual machine protection and remote virtual machine replication.

vSAN snapshots and replication supports vSAN ESA enabled HCI clusters and compute-only clusters that mount a datastore from a vSAN storage cluster. It allows you to quickly recover VMs from operational failures or ransomware attacks, using native snapshots stored locally on the cluster. The replication of VMs enables you to protect VMs by ensuring data back up, minimal data loss, and quick data recovery in case of a site failure. vSAN snapshots and replication supports VMFS and NFS on the client-side, but only for replication. For more information, see the vSAN snapshots and replication guide.

Deploy and Configure the Protection and Recovery Appliance

To activate vSAN snapshots and replication, you must deploy the protection and recovery appliance from the vSphere Client.

After deploying, you must configure the protection and recovery appliance to connect to a vCenter instance on both the protected and the recovery sites. For more information on deploying and configuring the protection and recovery appliance, see Deploy the Protection and Recovery Appliance and Configure the Protection and Recovery Appliance to Connect to a vCenter instance.

Clustering Applications on vSAN

vSAN supports two main clustering options such as shared VMDKs using SCSI-3 persistent reservations for Windows Server Failover Cluster (WSFC) and multi-writer VMDKs for Oracle RAC.

For example, applications such as Linux-based clustering solutions require shared disk access control, and can use shared VMDKs with SCSI-3 persistent reservations. Solutions such as IBM Db2 pureScale or specific clustered file systems can be configured to use multi-writer VMDKs for concurrent shared access.

WSFC ensures high availability for SQL Server Failover Cluster Instances (FCI). It requires shared storage with SCSI-3 persistent reservations to allow hosts safe shared disk access. vSAN support multiple WSFC deployment methods, including native vSAN shared VMDKs and vSAN iSCSI.

Oracle RAC enables multiple Oracle instances to access a single datastore providing high availability and scalability. vSAN aggregates the local disks of ESX hosts into a shared datastore. The virtual disks that are configured in multi-writer mode allow multiple VMs (RAC nodes) to access the same disk.

Virtual Clustering Solutions with Shared Virtual Disks on vSAN

vSAN supports high availability (HA) clustering applications running inside virtual machines that leverage shared virtual disks. This capability enables multiple virtual machines, residing on the same or different ESXi hosts, to simultaneously access a shared virtual disk on a vSAN datastore.

The following shared-disk configuration models are supported:

Shared Disk Configuration	Description
Shared VMDKs with Physical Bus Sharing	This configuration uses a shared virtual SCSI bus and supports SCSI-3 Persistent Reservations (SCSI-3 PR). On vSAN, SCSI-3 PR support is implemented specifically for use with Windows Server Failover Clustering (WSFC). Only the SCSI-3 PR primitives required for WSFC are supported. Use of SCSI-3 PR–based clustering solutions other than WSFC is not currently supported on vSAN. Applications that depend on additional or different SCSI-3 PR reservation mechanisms may not function as expected.
Shared VMDKs with Multi-Writer Mode	Multi-writer mode enables a virtual disk to be accessed concurrently by multiple virtual machines. Unlike Physical Bus Sharing, however, multi-writer mode does not provide SCSI-3 persistent reservations or any storage-level locking mechanism. In this setup, the clustering software running within the guest operating systems is fully responsible for coordinating disk access, enforcing locking semantics, and ensuring data integrity. Broadcom does not perform application-level validation of the various clustering mechanisms implemented within guest operating systems. Oracle RAC has been used as the reference application for qualifying multi-writer mode. Customers should refer to their respective vendor documentation for supported configurations and recommended best practices.

Online Capacity Expansion

Support for hot extension (online expansion) of Shared VMDKs on vSAN will be available starting with the VMware vSphere 9.0 update 1 release. Online expansion will be supported for both clustering configurations that use Physical Bus Sharing (for example, Windows Server Failover Clustering – WSFC) and those that use multi-writer mode (for example, Oracle RAC).

vSAN Support for WSFC

ESX supports vSAN framework and up to five node WSFC clusters.

- Supports vSAN for Windows Server 2012 and later.
- WSFC on vSAN can work with any type of disk, "Thin" and "Thick"-provisioned disks.
- Enables customers to move away from using pRDM.
- WSFC on vSAN supports vSphere HA, DRS, vMotion, and hot extension of a shared vSAN disk.
- The underlying transport protocol can be FC, iSCSI, FCoE, or NVMe Fabric storage (FC, TCP). For more information, see Setup for Windows Server Failover Clustering on VMware vSphere.
- Virtual adapter NVMe support is added as part of vSphere 9.1 along with PVSCSI.

vSAN Support for Oracle RAC

ESX supports vSAN framework and up to five node Oracle RAC 19c clusters.

- Supports vSAN for Oracle RAC 19c and later.
- Oracle RAC on vSAN can work with any type of disk, "Thin" and "Thick"-provisioned disks.
- Enables to move away from using pRDM.
- Oracle RAC on vSAN supports vSphere HA, DRS, vMotion, and hot extension of a shared vSAN disk.
- Virtual adapter NVMe support is added as part of vSphere 9.1 along with PVSCSI.

vSAN Support for Red Hat and SUSE Enterprise Linux High Availability Clusters

Red Hat Enterprise Linux High Availability Add-On and SUSE Linux Enterprise High Availability Extension clusters are supported on vSAN when configured using multi-writer mode only.

Important to consider:

- Cluster fencing mechanisms that rely on SCSI-3 Persistent Reservations (such as fence_scsi or fence_mpath) are not currently supported on vSAN.
- Customers should consult the respective vendor documentation for guidance on supported configurations and best practices.

Add Hard Disks to the First Virtual Machine of a WSFC or Oracle RAC Cluster Using vSAN Shared VMDKs

In a WSFC cluster, storage disks are shared between nodes. In an Oracle RAC cluster with vSAN shared VMDKs, you must add shared disks to the first VM across physical hosts.

- Use Hardware Version 13 or later. See Broadcom Hardware Compatibility Guide.
- Prepare vSAN deployment.

1. In the vSphere Client, select the newly created virtual machine, right-click, and select **Edit Settings**.
2. From the **Add New Device** drop-down menu, select **SCSI Controller**.
3. Perform the following:
   - For WSFC, in New SCSI controller, select **VMware Paravirtual** as the Change Type and set SCSI Bus Sharing to **Physical**.

- For Oracle RAC, in New SCSI controller, select **VMware Paravirtual** as the Change Type and set SCSI Bus Sharing to **None**.

  From the **Sharing** drop-down list, select **Multi-writer**.

4. From the **Add New Device** drop-down menu, select **Hard Disk**.
5. Expand the **New Hard disk** and select the required disk size.
6. In the Virtual Device Node, select **New SCSI controller** and **SCSI (1:0) New Hard disk** from the drop-down lists.
7. Click **OK** to create a new hard disk.

Add Hard Disks to the Additional Virtual Machines of the WSFC or Oracle RAC Cluster Using vSAN Shared VMDKs

To allow shared access to disk resources, point to existing disks on the VM, the first node of a WSFC cluster or an Oracle RAC cluster. Use the same SCSI IDs while assigning disks to all additional nodes.

- Obtain SCSI IDs for all virtual disks to be shared.
- Obtain disk file path on datastore for all shared disks.

1. In the vSphere Client, select the newly created virtual machine, right-click, and select **Edit Settings**.
2. Click the **New device** drop-down menu, select **SCSI Controller**.
3. Perform one of the following:
   - For WSFC, in new SCSI Controller, select **VMware Paravirtual** and set **SCSI Bus Sharing** to Physical. Click **OK**.
   - For Oracle RAC, in New SCSI controller, select **VMware Paravirtual** as the Change Type and set SCSI Bus Sharing to **None**.

     From the **Sharing** drop-down list, select **Multi-writer**.

     Note:

     You must select the same SCSI controller type, for example, VMware Paravirtual, that you selected for the first virtual machine's shared storage disks.
4. Select the newly created virtual machine in step 1, right-click, and select **Edit Settings**.
5. Click the **New device** drop-down menu, select **Existing Hard Disk**.
6. In **Disk File Path**, browse to the location of the disk to be shared specified for the first node.
7. Expand **New Hard disk**.
8. Select the same SCSI ID you chose for the first virtual machine's shared storage disks, (for example, select **SCSI (1:0)**).

   The disk SCSI ID for this virtual machine's shared storage must match the corresponding SCSI ID for the first virtual machine.
9. Click **OK**.

   For more information on HA and DRS, see Use WSFC in an vSphere HA and vSphere DRS Environment.

Hot Extend a Shared VMDK (WSFC) or Multi-Writer Disk (Oracle RAC)

The hot extension of a shared vSAN disk allows you to increase the size of a shared disk without shutting down the Windows Server Failover Clustering (WSFC) or Oracle RAC cluster.

Extending the size of shared vSAN disk is a two-step process. First you must increase the size of shared vSAN disk using vCenter. Then extend the partition size of the disk from the guest OS.

Following are the best practices for hot extending shared vSAN disk:

- You can only extend one shared vSAN disk at a time from a single VM. You cannot extend multiple shared vSAN disks simultaneously.

- Do not reconfigure any other VM parameter while you are changing the size of a shared vSAN disk. Any other change that you make in VM configuration will not be performed during this operation.
- All VMs sharing the disk must be accessible and registered with vCenter. VMs cannot be in a suspended or powered off state. VMs must be in a healthy state.
- All hosts must be vSAN 9.1 and later and upgrades cannot be performed during this operation.
- Hot extend of a shared vSAN disk is not supported for disks that have IOFilters attached.

1. Select any one VM that is sharing the vSAN disk that you want to extend and click **Edit Settings**.
2. Select the disk that you want to extend and enter the new size.
3. Select **OK**.

   All VM operations on all VMs that are sharing the vSAN disk that is being extended are delayed until the extended operation is complete. After the operation is complete, each VM that is sharing the vSAN disk shows the new disk size. Ensure that the hot extension of a shared vSAN disk does not cause I/O failures.

   If an operation fails, vCenter displays the following error messages:

Error Message	Action
The operation can not be performed because host {arg} is not active	Review the current configuration and state of the VM, host, and datastore.
The operation can not be performed because vm {arg} is in suspended state	Review the current configuration and state of the VM, host, and datastore.
The operation can not be performed because datastore is inaccessible	Review the current configuration and state of the VM, host, and datastore.
The operation can not be performed because disk extend on multiple shared disks is not allowed	Review the current configuration and state of the VM, host, and datastore.
The operation can not be performed because host {arg} version is not supported	Review the current configuration and state of the VM, host, and datastore.
The operation can not be performed because both multiwriter mode and physical bus sharing is enabled	Review the current configuration and state of the VM, host, and datastore.
Reconfigure VM failed for {VM} on shared disk	Hot extend the failed VMs.

4. After the size of the vSAN disk is extended, extend the disk partition size from the guest OS. This must be done from the VM which owns the disk resource. Use the appropriate utility depending on your guest OS. Before you extend the size, verify all VMs that are sharing the disk display the new extended size. For example, on Windows you can verify using the "disk mgmt tool" where the amount of increased size is displayed as unallocated. The Windows Failover clustering tool shows the increased size of the shared vSAN disk.

   For more information, see Broadcom knowledge base article 323136.

Using the vSAN iSCSI Target Service

Use the iSCSI target service to enable ESX hosts and physical workloads that reside outside the vSAN cluster to access the vSAN datastore.

This feature enables an iSCSI initiator on a remote ESX host to transport block-level data to an iSCSI target on a storage device in the vSAN cluster. vSAN support Windows Server Failover Clustering (WSFC), so WSFC nodes can access vSAN iSCSI targets. For more information, see Broadcom knowledge base article 313230.

After you configure the vSAN iSCSI target service, you can discover the vSAN iSCSI targets from a remote host. To discover vSAN iSCSI targets, use the IP address of any host in the vSAN cluster, and the TCP port of the iSCSI target. To ensure high availability of the vSAN iSCSI target, configure multipath support for your iSCSI application. You can use the IP addresses of two or more hosts to configure the multipath. vSAN iSCSI VIP is an IP address that you can use to discovery IP address for initiators to discover vSAN iSCSI target service without having to provide the underlying host vSAN iSCSI vmkernel IP address details.

Note:

vSAN iSCSI target service does not support other vSphere or ESX clients or initiators, third-party hypervisors, or migrations using raw device mapping (RDMs).

vSAN iSCSI target service supports the following CHAP authentication methods:

CHAP	In CHAP authentication, the target authenticates the initiator, but the initiator does not authenticate the target.
Mutual CHAP	In mutual CHAP authentication, an extra level of security enables the initiator to authenticate the target.

For more information about using the vSAN iSCSI target service, see iSCSI Target Usage Guide.

iSCSI Targets

You can add one or more iSCSI targets that provide storage blocks as logical unit numbers (LUNs). vSAN identifies each iSCSI target by a unique iSCSI qualified Name (IQN). You can use the IQN to present the iSCSI target to a remote iSCSI initiator so that the initiator can access the LUN of the target.

Each iSCSI target contains one or more LUNs. You define the size of each LUN, assign a vSAN storage policy to each LUN, and enable the iSCSI target service on a vSAN cluster. You can configure a storage policy to use as the default policy for the home object of the vSAN iSCSI target service.

iSCSI Initiator Groups

You can define a group of iSCSI initiators that have access to a specified iSCSI target. The iSCSI initiator group restricts access to only those initiators that are members of the group. If you do not define an iSCSI initiator or initiator group, then each target is accessible to all iSCSI initiators.

A unique name identifies each iSCSI initiator group. You can add one or more iSCSI initiators as members of the group. Use the IQN of the initiator as the member initiator name.

Enable the vSAN iSCSI Target Service

Before you can create iSCSI targets and LUNs and define iSCSI initiator groups, you must enable the iSCSI target service on the vSAN cluster. You must create a VMkernel port on each host to use the vSAN iSCSI default network or select an existing port.

Ensure that you have defined the default network for vSAN iSCSI target service.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN click **Services**.
4. On the vSAN iSCSI Target Service row, click **Enable**.

   The Enable vSAN iSCSI Target Service wizard opens.
5. In the Basic tab, you can select the default network, TCP port, and Authentication method at this time. You also can select a vSAN storage policy.
6. In the Virtual IP tab, click the Enable vSAN iSCSI Virtual IP slider to turn it on.
7. Select the network device, IP address, subnet mask, and gateway. When you enable virtual IP, you can use this IP for iSCSI connections. You can disable the virtual IP, as required. If you disable virtual IP, it might cause iSCSI traffic interruptions.

   Note:

   vSAN stretched cluster does not support vSAN iSCSI virtual IP.

8. Click **Enable**.

The vSAN iSCSI target service is enabled.
After the iSCSI target service is enabled, you can create iSCSI targets and LUNs, and define iSCSI initiator groups.

Create a vSAN iSCSI Target

You can create or edit an iSCSI target and its associated LUN.

Verify that the vSAN iSCSI target service is enabled.

1. Navigate to the vSAN cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **iSCSI Target Service**.
4. Click the iSCSI Targets tab.
5. Click **Add**. The **New iSCSI Target** dialog box is displayed. If you leave the target IQN field blank, the IQN is generated automatically.
6. Enter a target **Alias**.
7. Select a **Storage policy**, **Network**, **TCP port**, and **Authentication** method.
8. Select the **I/O Owner Location**. This feature is available only if you have configured vSAN cluster as a stretched cluster. It allows you to specify the site location for hosting the iSCSI target service for a target. This helps in avoiding the cross site iSCSI traffic. If you have set the policy as HFT>=1, in the event of a site failure the I/O owner location changes to the alternate site. After the site failure recovery, the I/O owner location automatically changes back to the original I/O owner location as per the configuration.

   Note:

   The I/O Owner Location setting operates on a best-effort basis. During cluster events such as host transitions or witness reconnections, the iSCSI target owner may temporarily move to the opposite site, regardless of the configured site affinity. This transition is typically brief (less than one minute) while vSAN restores the target owner to the configured site. To maintain uninterrupted iSCSI connectivity during these transitions, ensure that iSCSI initiators have network access to all hosts in both sites of the stretched cluster.

   You can select one of the following options to set the site location:

   - **Either**: Hosts the iSCSI target service either on Preferred or Secondary site.
   - **Preferred**: Hosts the iSCSI target service on the Preferred site.
   - **Secondary**: Hosts the iSCSI target service on the Secondary site.
9. Click **OK**.

iSCSI target is created and listed under the vSAN iSCSI Targets section with the information such as IQN, I/O owner host, and so on. You can use the filter icon to filter the group name and initiator count.

Define a list of iSCSI initiators that can access this target.

Add a LUN to a vSAN iSCSI Target

You can add one or more LUNs to a vSAN iSCSI target, or edit an existing LUN.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
   a) Under vSAN, click **iSCSI Targets**.
   b) Click the **iSCSI Targets** tab, and select a target.
   c) In the vSAN iSCSI LUNs section, click **Add**. The **Add LUN to Target** dialog box is displayed.
   d) Enter the size of the LUN. The vSAN Storage Policy configured for the iSCSI target service is assigned automatically. You can assign a different policy to each LUN.
3. Click **Add**.

Resize a LUN on a vSAN iSCSI Target

Depending on your requirement, you can increase the size of an online LUN.

Online resizing of the LUN is enabled only if all hosts in the cluster are upgraded to vSAN 9.1 or later.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **iSCSI Targets**.
4. Click the **iSCSI Targets** tab and select a target.
5. In the vSAN iSCSI LUNs section, select a LUN and click **Edit**. The Edit LUN dialog box is displayed.
6. Increase the size of the LUN depending on your requirement.
7. Click **OK**.

Place a LUN in Offline Mode

You can place a LUN on a vSAN iSCSI target in offline mode.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **iSCSI Targets**.
4. Click the **iSCSI Targets** tab and select a target.
5. In the vSAN iSCSI LUNs section, select a LUN and click **Offline**. The Place iSCSI LUN offline dialog box is displayed.
6. Click **Yes**.

Remove a LUN from a vSAN iSCSI Target

You can delete a LUN from a vSAN iSCSI target.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **iSCSI Targets**.
4. Click the **iSCSI Targets** tab and select a target.
5. In the vSAN iSCSI LUNs section, select a LUN and click **Remove**. The Remove iSCSI LUNs from Target dialog box is displayed.
6. Click **Remove**.

Create a vSAN iSCSI Initiator Group

You can create a vSAN iSCSI initiator group to provide access control for vSAN iSCSI targets.

Only iSCSI initiators that are members of the initiator group can access the vSAN iSCSI targets.

Note:
The initiators outside the initiator group cannot access the target if the initiator group for access control is created on the iSCSI target. The existing connections from these initiators will be lost and cannot be recovered until they are added to the initiator group. You must check the current initiator connections and ensure that all the authorized initiators are added to the initiator group before group creation.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
   a) Under vSAN, click **iSCSI Targets**.
   b) Click the **Initiator Groups** tab, and click **Add**. The **New Initiator Group** dialog box is displayed.
   c) Enter a name for the iSCSI initiator group.
   d) (Optional) To add members to the initiator group, enter the IQN of each member. Use the following format to enter the member IQN:

      iqn.YYYY-MM.domain:name

      Where:

      - YYYY = year, such as 2016
      - MM = month, such as 09
      - domain = domain where the initiator resides
      - name = member name (optional)
3. Click **Create**.

Add members to the iSCSI initiator group.

Add Initiator to a vSAN iSCSI Initiator Group

You can add a member to the vSAN iSCSI initiator group.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
   a) Under vSAN, click **iSCSI Targets**.
   b) Click the **Initiator Groups** tab.
   c) In the Accessible Targets section, click **Add**. The **Add Initiators** dialog box is displayed.
   d) Enter an initiator name in the **Member initiator name** field.
   e) Click **Add** to add a member to the initiator group.
3. Click **Add**.

   The vSAN iSCSI initiator is created and listed.

Assign a Target to a vSAN iSCSI Initiator Group

You can assign a vSAN iSCSI target to an iSCSI initiator group.

Verify that you have an existing iSCSI initiator group.

Only those initiators that are members of the initiator group can access the assigned targets.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
   a) Under vSAN, click **iSCSI Targets**.
   b) Select the **Initiator Groups** tab.
   c) In the Initiators section, click **Add**. The **Add Initiators** dialog box is displayed.
   d) Select a target from the list of available targets.
3. Click **Add**.

Monitor vSAN iSCSI Target Service

You can monitor the iSCSI target service to view the physical placement of iSCSI target components and to check for failed components.

Verify that you have enabled the vSAN iSCSI target service and created targets and LUNs.

You also can monitor the health status of the iSCSI target service.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN click **Virtual Objects**. The iSCSI targets are listed on the page.
4. Select a target and click **View Placement Details**. The Physical Placement shows where the data components of the target are located, the LUNs associated with the target, and its physical location.
5. Click **Group components by host placement** to view the hosts associated with the iSCSI data components.

You can view the vSAN iSCSI target performance. For more information, see View vSAN Host Performance.

Turn Off the vSAN iSCSI Target Service

You can turn off the vSAN iSCSI target service.

Workloads running on iSCSI LUNs are stopped when you turn off the iSCSI target service. Before you turn it off, ensure that there are no workloads running on iSCSI LUNs.

Turning off vSAN iSCSI target service does not delete the LUNs/Targets. If you wish to reclaim the space, delete the LUNs/targets manually before you turn off vSAN iSCSI target service. For more information, see Remove a LUN from a vSAN iSCSI Target.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. On the vSAN**iSCSI Target Service** row, click **EDIT**.

   The Edit vSAN iSCSI Target Service wizard opens.
5. Click the **Enable vSAN iSCSI Target Service** slider to turn it off and click **Apply**.

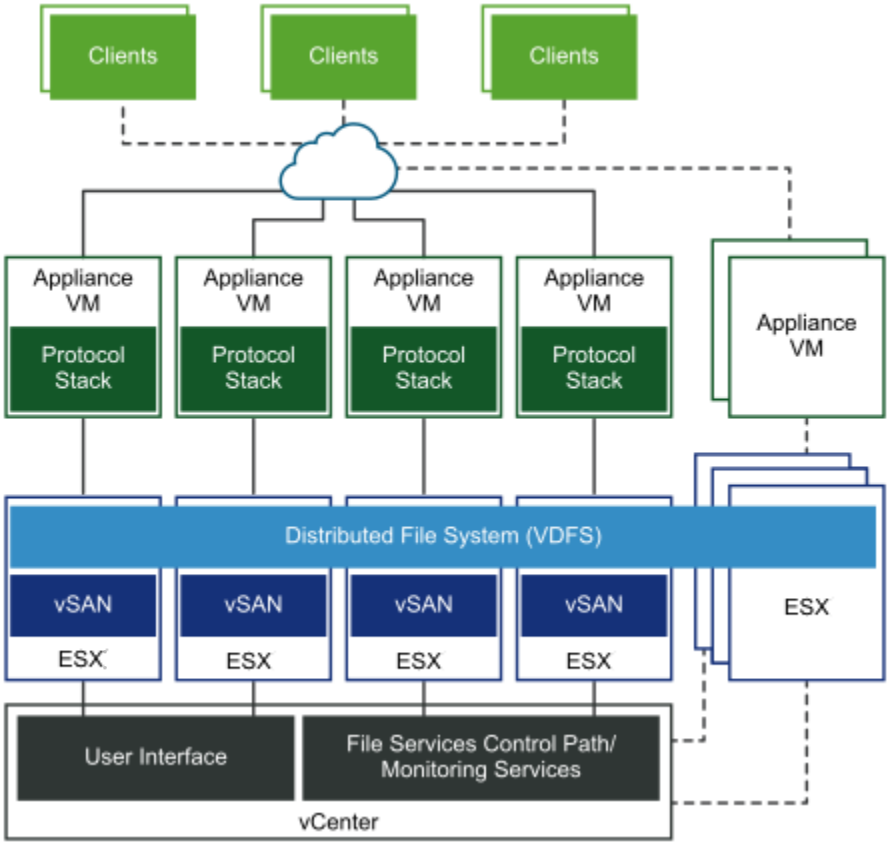
The vSAN iSCSI target service is not enabled.

vSAN File Service

Use the vSAN file service to create file shares in the vSAN datastore that client workstations or virtual machines can access.

The data stored in a file share can be accessed from any device that has access rights. vSAN file service is a layer that sits on top of vSAN to provide file shares. It currently supports SMB, NFSv3, and NFSv4.1 file shares. vSAN file service comprises of vSAN Distributed File System (vDFS) which provides the underlying scalable filesystem by aggregating vSAN objects, a Storage Services Platform which provides resilient file server end points and a control plane for deployment, management, and monitoring. File shares are integrated into the existing vSAN Storage Policy Based

Management, and on a per-share basis. vSAN file service brings in capability to host the file shares directly on the vSAN cluster.



When you configure vSAN file service, vSAN creates a single VDFS distributed file system for the cluster which will be used internally for management purposes. A file service VM (FSVM) is placed on each host. The FSVMs manage file shares in the vSAN datastore. Each FSVM contains a file server that provides both NFS and SMB service.

A static IP address pool should be provided as an input while enabling file service workflow. One of the IP addresses is designated as the primary IP address. The primary IP address can be used for accessing all the shares in the file services cluster with the help of SMB and NFSv4.1 referrals. A file server is started for every IP address provided in the IP pool. A file share is exported by only one file server. However, the file shares are evenly distributed across all the file servers. To provide computing resources that help manage access requests, the number of IP addresses must be equal to the number of hosts in the vSAN cluster.

vSAN file service supports vSAN stretched clusters and two-node vSAN clusters. A two-node vSAN cluster should have two data node servers in the same location or office, and the witness in a remote or shared location.

For more information about Cloud Native Storage (CNS) file volumes, see the VMware vSphere Kubernetes Service Components documentation.

Limitations and Considerations of vSAN File Service

Consider the following when configuring vSAN file service:

- vSAN supports two-node configurations and stretched clusters.
- vSAN supports 64 file servers in a 64 host setup.
- vSAN OSA cluster supports 100 file shares.
- vSAN supports file service on ESA.
- vSAN ESA cluster supports 500 file shares. Out of those 500 file shares, maximum 100 file shares can be SMB. For example, if you create 100 SMB file shares then the cluster can only support additional 400 NFS file shares.
- vSAN file service can connect only to a single network or port group.

- vSAN file services does not support the following:
  - Read-Only Domain Controllers (RODC) for joining domains because the RODC cannot create computer accounts. As a security best practice, a dedicated org unit should be pre-created in the Active Directory and the user name mentioned here should be controlling this organization.
  - Disjoint namespace. When the primary DNS suffix of a server within an Active Directory domain does not match the DNS name of the domain itself, it is referred to as disjoint namespace.
  - Multiple domains and Single Active Directory Forest environments.
  - NFSv3 UDP mounts.
- When a host enters maintenance mode, the file server moves to another FSVM. The FSVM on the host that entered maintenance mode is powered off. After the host exits maintenance mode, the FSVM is powered on.
- Network reconfigurations or migrations affecting the VM network used by vSAN file service can cause file services to not function properly. Avoid these types of changes while using the vSAN file service:
  - Migration from NSX-T N-VDS (NSX Virtual Distributed Switch) to VDS (Virtual Distributed Switch).
  - Replacement of opaque networks with standard VDS networks.
  - Any other network migration or reconfiguration that changes the network backing for File Service VMs.

Enable vSAN File Service

You can enable vSAN file services on a vSAN OSA cluster or a vSAN ESA cluster.

Ensure that the following are configured before enabling the vSAN file services:

- The vSAN cluster must be a regular vSAN cluster, a vSAN stretched cluster, or a vSAN ROBO cluster.
- Every ESX host in the vSAN cluster must have minimal hardware requirements such as:
  - Minimum 16 Core CPU
  - Minimum 128 GB physical memory
  - Minimum 10 GbE network
- You must ensure to prepare the network as vSAN file service network:
  - If using standard switch based network, the Promiscuous Mode and Forged Transmits are enabled as part of the vSAN file services enablement process.
  - If using vSphere Distributed Switch (DVS) based network, vSAN file services are supported on vSphere Distributed Switch (DVS). Create a dedicated distributed port group for vSAN file services in the DVS. MacLearning and Forged Transmits are enabled as part of the vSAN file services enablement process for a provided DVS port group.
  - **Important:**

    If using NSX-based network, ensure that Mac Changes and Mac Learning are enabled for the provided network entity from the NSX admin console, and all the hosts and File Services nodes are connected to the desired NSX network. For more information, see Create an NSX MAC Discovery Segment Profile.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN click **Services**.
4. On the File Service row, click **Enable**.

   The Enable File Service wizard opens.

5. From the **Select** drop-down, select a network.
6. In the File service agent, select one of the following options to download the OVF file.

Option	Description
Automatically load latest OVF	This option lets the system search and download the OVF. **Note:** • Ensure that you have configured the proxy and firewall to allow vCenter outbound access to `depot.vmware.com` and `download3.vmware.com`, so that vCenter can download the appropriate JSON file and OVF. For more information about configuring the vCenter DNS, IP address, and proxy settings, see the vCenter Configuration guide. • **Use current OVF**: Allows you to use the OVF that is already available. • **Automatically load latest OVF**: Allows the system to search and download the latest OVF.
Manually load OVF	This option allows you to browse and select an OVF that is already available on your local system. **Note:** If you select this option, you should upload all the following files: • `VMware-vSAN-File-Services-Appliance-x.x.x.x-x_OVF10.mf` • `VMware-vSAN-File-Services-Appliance-x.x.x.x-x-x_OVF10.cert` • `VMware-vSAN-File-Services-Appliance-x.x.x.x-x-x-system.vmdk` • VMware-vSAN-File-Services-Appliance-x.x.x.x-x-cloud-components.vmdk • VMware-vSAN-File-Services-Appliance-x.x.x.x-x-log.vmdk • VMware-vSAN-File-Services-Appliance-x.x.x.x-x_OVF10.ovf

7. Click **Enable**.

- The OVF is downloaded and deployed.
- The vSAN file services is enabled.
- A file services VM (FSVM) is placed on each host.

Note:
The FSVMs are managed by the vSAN file services. Do not perform any operation on the FSVMs.

Configure vSAN File Service

You can configure the file service, which enable you to create file shares on your vSAN datastore.

Ensure the following before configuring the vSAN file service:

- Enable vSAN file service.
- Allocate static IP addresses as file server IPs from vSAN file service network, each IP is the single point access to vSAN file shares.
  - For best performance, the number of IP addresses must be equal to the number of hosts in the vSAN cluster.

  - All the static IP addresses must be from the same subnet.
  - Every static IP address has a corresponding FQDN, which must be part of the Forward lookup and Reverse lookup zones in the DNS server.
- If you are planning to create a Kerberos based SMB file share or a Kerberos based NFS file share, you need the following:
  - Microsoft Active Directory (AD) domain to provide authentication to create an SMB file share or an NFS file share with the Kerberos security.
  - (Optional) Active Directory Organizational Unit to create all file server computer accounts.
  - A domain user in the directory service with the sufficient privileges to create and delete computer accounts.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. On the File Service row, click **Configure Domain**.

   The File Service Domain wizard opens.
5. In the File Service Domain page, enter the unique namespace and click **Next**. The domain name must have minimum two characters. The first character must be an alphabet or a number. The remaining characters can include an alphabet, a number, an underscore (_), a period (.), a hyphen (-).
6. In the Networking page, enter the following information, and click **Next**:
   - **Protocol**: You can select IPv4 or IPv6. vSAN file service only supports IPv4 or IPv6 stack. The reconfiguration between IPv4 and IPv6 is not supported.
   - **DNS servers**: Enter a valid DNS server to ensure the proper configuration of file service.
   - **DNS suffixes**: Provide the DNS suffix that is used with the file service. All other DNS suffixes from where the clients can access these file servers must also be included. File service does not support DNS domain with single label, such as "app", "wiz", "com" and so on. A domain name given to file service must be of the format thisdomain.registerdrootdnsname. DNS name and suffix must adhere to the best practices detailed in https://docs.microsoft.com/en-us/windows-server/identity/ad-ds/plan/selecting-the-forest-root-domain.
   - **Subnet mask**: Enter a valid subnet mask. This text box appears when you select IPv4.
   - **Prefix length**: Enter a number between 1 and 128. This text box appears when you select IPv6.
   - **Gateway**: Enter a valid gateway.
   - **IP Pool**: Enter IP addresses and the corresponding DNS names.

   vSAN ESA cluster supports 500 file shares. Out of those 500 file shares, maximum 100 file shares can be SMB. For example, if you create 100 SMB file shares then the cluster can only support additional 400 NFS file shares.

   Each file server on a vSAN ESA cluster can support a maximum of 50 file shares and requires at least 10 IPs to have the maximum of 500 shares. With the increase in the file servers or file shares per host, there might be an impact on the performance of vSAN file service. For best performance, the number of IP address must to be equal to the number of hosts in the vSAN cluster.

   Affinity site option is available if you are configuring vSAN file service on a vSAN stretched cluster. This option allows you to configure the placement of the file server on **Preferred** or **Secondary** site. This helps in reducing the cross-site traffic latency. The default value is **Either**, which indicates that no site affinity rule is applied to the file server.

   Note:
   If your cluster is a ROBO cluster, ensure that the Affinity site value is set to **Either**.

   In a site failure event, the file server affiliated to that site fails over to the other site. The file server fails back to the affiliated site when it is recovered. Configure more file servers to one site if more workloads can be expected from a certain site.

   Note:
   If the file server contains SMB file shares, then it does not failback automatically even if the site failure is recovered.

   Consider the following while configuring the IP addresses and DNS names:

- To ensure proper configuration of file service, the IP addresses you enter in the Networking page must be static addresses and the DNS server must have records for those IP addresses. For best performance, the number of IP addresses must be equal to the number of hosts in the vSAN cluster.
- You can have a maximum of 64 hosts in the cluster. If large scale cluster support is configured, you can enter up to 64 IP addresses.
- You can use the following options to automatically fill the IP address and DNS server name text boxes:
  AUTO FILL: This option is displayed after you enter the first IP address in the IP address text box. Click the AUTO FILL option to automatically fill the remaining fields with sequential IP addresses, based on the subnet mask and gateway address of the IP address that you have provided in the first row. You can edit the auto filled IP addresses.
  LOOK UP DNS: This option is displayed after you enter the first IP address in the IP address text box. Click the LOOK UP DNS option to automatically retrieve the FQDN corresponding to the IP addresses in the IP address column.

  Note:

  - All valid rules apply for the FQDNs. For more information, see https://tools.ietf.org/html/rfc953.
  - The first part of the FQDN, also known as NetBIOS Name, must not have more than 15 characters.

  The FQDNs are automatically retrieved only under the following conditions:
  – You must have entered a valid DNS server in the Domain page.
  – The IP addresses entered in the IP Pool page must be static addresses and the DNS server must have records for those IP addresses.

7. In the Directory service page, enter the following information and click **Next**.

Option	Description
Directory service	Configure an Active Directory domain to vSAN file service for authentication. If you are planning to create an SMB file share or an NFSv4.1 file share with Kerberos authentication, then you must configure an AD domain to vSAN file service.
AD domain	Fully qualified domain name joined by the file server.
Preferred AD Server	Enter the IP address of the preferred AD server. In case of multiple IP addresses, ensure that they are separated by comma.
Organizational unit (Optional)	Contains the computer account that the vSAN file service creates. In an organization with complex hierarchies, create the computer account in a specified container by using a forward slash mark to denote hierarchies (for example, organizational_unit/inner_organizational_unit). **Note:** By default, the vSAN file service creates the computer account in the Computers container.
AD username	User name to be used for connecting and configuring the Active Directory service. This user name authenticates the active directory on the domain. A domain user authenticates the domain controller and creates vSAN file service computer accounts, related SPN entries, and DNS entries (when using Microsoft DNS). As a best practice, create a dedicated service account for the file service. A domain user in the directory service with the following sufficient privileges to create and delete computer account: • (Optional) Add/Update DNS entries
Password	Password for the user name of the Active Directory on the domain. vSAN file service use the password to authenticate to AD and to create the vSAN file service computer account.

Note:

- vSAN file service does not support the following:
  - Read-Only Domain Controllers (RODC) for joining domains because the RODC cannot create computer accounts. As a security best practice, a dedicated org unit must be pre-created in the Active Directory and the user name mentioned here must be controlling this organization.
  - Disjoint namespace.
  - Multiple domains and Single Active Directory Forest environments.
- Only English characters are supported for Active Directory user name.
- Only single AD domain configuration is supported. However, the file servers can be put on a valid DNS subdomain. For example, an AD domain with the name `example.com` can have file server FQDN as `name1.eng.example.com`.
- Pre-created computer accounts for file servers are not supported. Make sure that the user provided here have sufficient privilege over the organizational unit.
- vSAN file service also has a Health Check to indicate if the forward and reverse lookups for file servers are working properly.

8. Review the settings and click **Finish**.

The file service domain is configured. File servers are started with the IP addresses that were assigned during the vSAN file service configuration process.

Run vSAN health check and verify the health findings.

Edit vSAN File Service

You can edit and reconfigure the settings of a vSAN File Service.

The vSAN File Service stores domain configuration, including IP addresses, on the VDFS datastore. Disabling and re-enabling File Service does not reset this configuration. The service restarts with the previously configured IP addresses and domain settings. If the file service domain is empty (has no shares), the File Service domain will be removed when disabling file service.

If there are active shares, changing the Active Directory domain is not permitted as this action can disrupt the user permissions on the active shares.

If your Active Directory password has been changed, then you can edit the Active Directory configuration settings and provide the new password.

Note:

This action might cause minor disruption to the inflight I/Os on the file shares.

Before you start, review the following constraints before editing vSAN File Service settings. Some changes require that active file shares do not exist on the cluster.

Setting	Editable when shares exist	Editable when no shares exist
Active Directory password	Yes	Yes
File server IP addresses including primary file server.	Yes.	Yes
File server FQDNs	No — requires an IP address change	Yes
Active Directory domain, organizational unit, username	Only the 'Preferred AD' field of the AD configuration is editable, regardless of whethere there are shares. The DNS name must needs to be reconfigured together with the IP.	

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. On the File Service row, click **Edit** > **Edit domain**.
   The File Service Domain wizard opens.
5. In the File Service Domain page, edit the file service domain name and click **Next**.
6. In the Networking page, make the appropriate configuration changes and click **Next**. You can edit the primary IP addresses, static IP addresses, and DNS names. You can add or remove the primary IP addresses or static IP addresses. You cannot change the DNS name without changing the IP.

   Note:
   Changing domain information is a disruptive action. It might require all clients to use new URLs to reconnect to the file shares.
7. In the Directory service page, make appropriate directory related changes if required, and click **Next**.

   Note:
   You cannot change the AD domain, organizational unit, and username after initially configuring vSAN file services.
8. In the Review page, click **Finish** after making necessary changes.

The changes are applied to the vSAN file service configuration.

Reconfigure Network Settings for the vSAN File Service

If you recently segmented your network and need to reconfigure it, first disable the File Service, then enable the File Service with a different network. Any existing shares will not be deleted when you disable the file service.

After enabling file service network, you can reconfigure File Service IP addresses to different subnet.

To reconfigure network settings after network segmentation:

1. Navigate to the vSAN cluster and click **Configure** > **vSAN** > **Services**.
2. On the File Service row, click **Edit**> **Edit domain**. The File Service Domain wizard opens.
3. In the Networking page, update the IP addresses and DNS names to match the new network configuration and click **Next**.
4. Complete the wizard and click **Finish**.
5. Verify that the File Service restarts successfully with the new network settings.

Disable vSAN File Service

You can disable vSAN file service.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. On the File Service row, click **Edit** > **Disable**.

   The Disable File Service wizard opens.
5. Click **Disable**.
   - If you disable file service with existing file shares, the file service domain will not be deleted. File shares remain stored in the vSAN datastore, and their configuration is preserved for future use.
   - If you disable file service without existing file shares, the empty file service domains without any active file shares gets automatically cleaned up. This ensures that your vSAN environment remains free of unused or orphaned domains, with optimized system performance and resource usage.

Create a vSAN File Share

When the vSAN file service is enabled, you can create one or more file shares on the vSAN datastore.

- If you are creating an SMB file share or a NFSv4.1 file share with Kerberos security, then ensure that you have configured vSAN file service to an Active Directory domain.
- Ensure that you have set `Host.Config.Storage` privilege.

Considerations for Share Name and Usage

- Usernames with non-ascii characters can be used to access share data.
- Share names cannot exceed 80 characters and can contain English characters, numbers, and hyphen character. Every hyphen character must be preceded and followed by a number or alphabet. Consecutive hyphens are not allowed.
- For SMB type shares, file and directories can contain any Unicode compatible strings.
- For pure NFSv4 type shares, the file and directories can contain any UTF-8 compatible strings. See Network File System (NFS) Version 4 Protocol.
- For pure NFSv3 and NFSv3+NFSv4 shares file and directories can contain only ASCII compatible strings.
- Migrating any share data from older NFSv3 to new vSAN file service shares with NFSv4 only requires conversion of all file and directories names to UTF-8 encoding. There are third part tools to achieve the same.

vSAN file service does not support using NFS file shares on ESX .

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Shares**.

   vSAN ESA cluster supports 500 file shares. Out of those 500 file shares, maximum 100 file shares can be SMB. For example, if you create 100 SMB file shares then the cluster can only support additional 400 NFS file shares.

   Each file server on a vSAN ESA cluster can support a maximum of 50 file shares and requires at least 10 IPs to have the maximum of 500 shares. With the increase in the file servers or file shares per host, there might be an impact on the performance of vSAN file service. For best performance, the number of IP addresses must to be equal to the number of hosts in the vSAN cluster.
4. Click **Add**.

   The Create file share wizard opens.
5. In the General page, enter the following information and click **Next**.
   - **Name**: Enter a name for the file share.
   - **Protocol**: Select an appropriate protocol. vSAN file service supports SMB and NFS file system protocols.
     If you select the **SMB** protocol, you can also configure the SMB file share to accept only the encrypted data using the **Protocol encryption** or Access based enumeration option. The access based enumeration displays only the files and folders that you have permissions to access. The files or folders are hidden if you do not have the Read (or equivalent) permissions. You can enable both protocol encryption and access based enumeration in a vSAN cluster.
     If you select the **NFS** protocol, you can configure the file share to support either **NFS 3**, **NFS 4**, or both **NFS 3 and NFS 4** versions. If you select **NFS 4** version, you can set either **AUTH_SYS** or **Kerberos** security.

     Note:

     SMB protocol and Kerberos security for NFS protocol can be configured only if the vSAN file service is configured with Active Directory. For more information, see Configure vSAN File Service.
   - With SMB protocol, you can hide the files and folders that the share client user does not have permission to access using the **Access based enumeration** option.
   - **Storage Policy**: Select an appropriate storage policy.
   - **Affinity site**: This option is available if you are creating a file share on a vSAN stretched cluster. This option helps you place the file share on a file server that belongs to the site of your choice. Use this option when you prefer low

latency while accessing the file share. The default value is **Either**, which indicates that the file share is placed on a site with less traffic on either preferred or secondary site.

- **Storage space quotas**: You can set the following values:
  - **Share warning threshold**: When the share reaches this threshold, a warning message is displayed.
  - **Share hard quota**: When the share reaches this threshold, new block allocation is denied.
- **Labels**: A label is a key-value pair that helps you organize file shares. You can attach labels to each file share and then filter them based on their labels. A label key is a string with 1~250 characters. A label value is a string and the length of the label value should be less than 1k characters. vSAN file service supports up to 5 labels per share.

6. The Net access control page, provides options to define access to the file share. Net access control options are available only for NFS shares. Select one of the following options and click **Next**.
   - **No access**: Select this option to make the file share inaccessible from any IP address.
   - **Allow access from any IP**: Select this option to make the file share accessible from all IP addresses.
   - **Customize net access**: Select this option to define permissions for specific IP addresses. Using this option you can specify whether a particular IP address can access, make changes, or only read the file share. You can also enable **Root squash** for each IP address. Root squash protects the NFS server from unauthorized root-level client access. You can enter the IP addresses in the following formats:
     - A single IP address. For example, 123.23.23.123
     - CIDR (Classless Inter-Domain Routing) notation by using a slash followed by a number, indicating how many bits of the IP address are dedicated to the network portion.
     - A range by specifying a starting IP address and ending IP address separated by a hyphen (-). For example, 123.23.23.123-123.23.23.128
     - Asterisk (*) to imply all the clients.
7. In the Review page, review the settings, and then click **Finish**.

   A new file share is created on the vSAN datastore.

View vSAN File Shares

You can view the list of vSAN file shares.

The vSAN ESA cluster displays the number of existing file shares and the maximum file share limit allowed in a cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Shares**.

A list of vSAN file shares appears. For each file share, you can view information such as storage policy, hard quota, usage over quota, actual usage, and so on.

Access vSAN File Shares

You can access a file share from a host client.

Access NFS File Share

You can access a file share from a host client, using an operating system that communicates with NFS file systems. For RHEL-based Linux distributions, NFS 4.1 support is available in RHEL 7.3 and CentOS 7.3-1611 running kernel 3.10.0-514 or later. For Debian based Linux distributions, NFS 4.1 support is available in Linux kernel version 4.0.0 or later. All NFS clients must have unique hostnames for NFSv4.1 to work. You can use the Linux mount command with the Primary IP to mount a vSAN file share to the client. For example: `mount -t nfs4 -o minorversion=1,sec=sys <primary ip>:/vsanfs/<share name>`. NFSv3 support is available for RHEL-based and Debian based Linux distributions. You can use the Linux mount command to mount a vSAN file share to the client. For example: mount `-t nfs vers=3 <nfsv3_access_point> <localmount_point>`.

Sample v41 commands for verifying the NFS file share from a host client:

```
[root@localhost ~]# mount -t nfs4 -o minorversion=1,sec=sys <primary ip address>:/vsanfs/
TestShare-0 /mnt/TestShare-0
```

```
[root@localhost ~]# cd /mnt/TestShare-0/
[root@localhost TestShare-0]# mkdir bar
[root@localhost TestShare-0]# touch foo
[root@localhost TestShare-0]# ls -l
total 0
drwxr-xr-x. 1 root root 0 Feb 19 18:35 bar
-rw-r--r--. 1 root root 0 Feb 19 18:35 foo
```

Access NFS Kerberos File Share

A Linux client accessing an NFS Kerberos share should have a valid Kerberos ticket.

Sample NFSv4 commands for verifying the NFS Kerberos file share from a host client:

An NFS Kerberos share can be mounted using the following mount command:

```
[root@localhost ~]# mount -t nfs4 -o minorversion=1,sec=krb5/krb5i/krb5p <primary ip address>:/
vsanfs/TestShare-0 /mnt/TestShare-0
[root@localhost ~]# cd /mnt/TestShare-0/
[root@localhost TestShare-0]# mkdir bar
[root@localhost TestShare-0]# touch foo
[root@localhost TestShare-0]# ls -l
total 0
drwxr-xr-x. 1 root root 0 Feb 19 18:35 bar
-rw-r--r--. 1 root root 0 Feb 19 18:35 foo
```

Changing Ownership of a NFS Kerberos share

You must log in using the AD domain user name for changing the ownership of a share. The AD domain user name provided in the file service configuration acts as a sudo user for the Kerberos file share.

```
[root@localhost ~]# mount -t nfs4 -o minorversion=1,sec=sys <primary ip address>:/vsanfs/
TestShare-0 /mnt/TestShare-0
[fsadmin@localhost ~]# chown user1 /mnt/TestShare-0
[user1@localhost ~]# ls -l /mnt/TestShare-0
total 0
drwxr-xr-x. 1 user1 domain  users 0 Feb 19 18:35 bar
-rw-r--r--. 1 user1 domain users 0 Feb 19 18:35 foo
```

Access SMB File Share

You can access an SMB file share from a Windows client.

Ensure that the Windows client is joined to the Active Directory domain that is configured with vSAN file service.

1. Copy the SMB file share path using the following procedure:
    1. In the vSphere Client, navigate to the cluster.
    2. Click the **Configure** tab.
    3. Under vSAN, click **File Service Shares**.
       List of all the vSAN file shares appears.
    4. Select the SMB file share that you want to access from the Windows client.
    5. Click **COPY PATH** > **SMB**.
       The SMB file share path gets copied to your clipboard.
2. Log into the Windows client as a normal Active Directory domain user.
3. Access the SMB file share using path that you have copied. For example: \\vsan-file-server-dns-name\namespace \share-name.

Edit a vSAN File Share

You can edit the settings of a vSAN file share.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Service Shares**.

   List of all the vSAN file shares appears.
4. Select the file share that you want to modify and click **Edit**.
5. In the Edit file share page, make appropriate changes to the file share settings and click **Finish**.

The file share settings are updated.

Note:
vSAN does not allow file share protocol change between SMB and NFS.

Manage SMB File Share on vSAN Cluster

vSAN file service supports the shared folders snap-in for the Microsoft Management Console (MMC) for managing the SMB shares on the vSAN cluster.

You can perform the following tasks on vSAN file system SMB shares using the MMC tool:

- Manage Access Control List (ACL).
- Close open files.
- View active sessions.
- View open files.
- Close client connections.

1. Copy the MMC Command using the following procedure:
   1. In the vSphere Client, navigate to the cluster.
      Click the **Configure** tab.
      Under vSAN, click **File Service Shares**.
      List of all the vSAN file shares appears.
   2. Select the SMB file share that you want to manage from the Windows client using the MMC tool.
   3. Click **COPY MMC COMMAND**.
      The MMC command gets copied to your clipboard. For example: fsmgmt.msc /computer:\\vsan-file-service-dns-name.
2. Log into the Windows client as a file service admin user. The file service admin user is configured when you create the file service domain. A file service admin user has all the privileges on the file server.
3. In the search box on the taskbar, type Run, and then select **Run**.
4. In the Run box, run the MMC command that you have copied to access and manage the SMB share using the MMC tool.

Delete a vSAN File Share

You can delete a file share when you no longer need it.

When you delete a file share, all the snapshots associated with that file share are also deleted.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Service Shares**.

   List of all the vSAN file shares appears.

4. Select the file share that you want to modify and click **Delete**.
5. On the Delete file shares dialogue, click **Delete**.

   The data gets deleted.

vSAN Distributed File System Snapshot

A snapshot provides a space-efficient and time-based archive of the data.

It provides the ability to retrieve data from a file or a set of files in the event of accidental deletion of a file. A file system level snapshot provides you information about the files that have been changed and the changes made to the file. It provides you an automated file recovery service and it is more efficient compared to the traditional tape-based backup method. A snapshot on its own does not provide a full disaster recovery solution but it can be used by the third-party backup vendors to copy the changed files (incremental backup) to a different physical location.

vSAN file services has a built-in feature that allows you to create a point- in-time image of the vSAN file share. When the vSAN file service is enabled, you can create up to 32 snapshots per share. A vSAN file share snapshot is a file system snapshot that provides a point-in-time image of a vSAN file share.

Considerations for File System Snapshot

- Use Default as the snapshot name to retrieve data.
- Snapshot name cannot exceed 100 characters and can contain English characters, numbers, and special characters except the following:
  - " (ASCII 34)
  - $ (ASCII 36)
  - % (ASCII 37)
  - & (ASCII 38)
  - * (ASCII 42)
  - / (ASCII 47)
  - : (ASCII 58)
  - < (ASCII 60)
  - > (ASCII 62)
  - ? (ASCII 63)
  - \ (ASCII 92)
  - ^ (ASCII 94)
  - | (ASCII 124)
  - ~ (ASCII 126)

Create a Snapshot

When the vSAN file service is enabled, you can create one or more snapshots that provide a point-in-time image of the vSAN file share. You can create a maximum of 32 snapshots per file share.

You should have created a vSAN file share.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Service Shares**.

   A list of vSAN file shares appears.
4. Select the file share for which you want to create a snapshot and then click **Snapshots** > **New Snapshot**.

   Create new snapshot dialogue appears.
5. On the Create new snapshot dialogue, provide a name for the snapshot, and click **Create**.

A point-in-time snapshot for the selected file share is created.

View a Snapshot
You can view the list of snapshots along with the information such as date and time of the snapshot creation, and its size.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Service Shares**.

   A list of vSAN file shares appears.
4. Select a file share and click **Snapshots**.

A list of snapshots for that file share appears. You can view information such as date and time of the snapshot creation, and its size.

Delete a Snapshot
You can delete a snapshot when you no longer need it.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **File Service Shares**.

   A list of vSAN file shares appears.
4. Select a file share and click **Snapshots**.
   A list of snapshots of that belongs to the file share you have selected appears.
5. Select the snapshot that you want to delete and click **Delete**.

   You cannot delete the last snapshot available.

Rebalance Workload on vSAN File Service Hosts

Health displays the workload balance health status for all the hosts that are part of the vSAN file service Infrastructure.

If there is an imbalance in the workload of a host, you can correct it by rebalancing the workload.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Health**.
4. Under Health, expand **File Service** and then click **Infrastructure Health**.

   The Infrastructure Health tab displays a list of all the hosts that are part of the vSAN file service infrastructure. For each host, the status of workload balance is displayed. If there is an imbalance in the workload of a host, an alert is displayed in the **Workload Balance** column.
5. Click **Remidiate Imbalance** and then **Rebalance** to fix the imbalance.

   Before proceeding with rebalancing, consider the following:

   - During rebalancing, containers in the hosts with an imbalanced workload might be moved to other hosts. The rebalancing activity might also impact the other hosts in the cluster.
   - During the rebalance process, the workloads running on NFS shares are not disrupted. However, the I/O to SMB shares located in the containers that have moved are disrupted.

The host workload is balanced and the workload balance status turns green.

Monitor Performance of vSAN File Service

You can monitor the performance of NFS and SMB file shares.

Ensure that vSAN Performance Service is enabled. If you are using the vSAN Performance Service for the first time, you see a message alerting you to enable it. For more information about vSAN Performance Service, see Monitoring vSAN Performance.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Performance**.
4. Click the **File Share** tab.
5. Select one of the following options:

Option	Action
File share	Select the file share for which you want to generate and view the performance report.
Time Range	• Select **Last** to select the number of hours for which you want to view the performance report. • Select **Custom** to select the date and time for which you want to view the performance report. • Select **Save** to add the current setting as an option to the Time Range list.

6. Click **Show Results**.

The throughput, IOPS, and latency metrics of the vSAN file service for the selected period are displayed.

For more information on vSAN Performance Graphs, see the Broadcom knowledge base article 214493.

Monitor vSAN File Share Capacity

You can monitor the capacity for both native file shares and CNS-managed file shares.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Capacity**.
4. Click **Capacity Usage** tab.
5. In the Usage breakdown section, expand **User data** > **File Shares**.

The file share capacity information is displayed.
For more information about monitoring vSAN capacity, see Monitor vSAN Capacity.

Monitor vSAN File Service and File Share Health

You can monitor the health of both vSAN file service and file share objects.

View vSAN File Service Health

You can monitor the vSAN file service health.

Ensure that vSAN Performance Service is enabled.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Health**.
4. In the Health section, expand **File Service**.
5. Click the following file service health parameters to view the status.

Option	Action
Infrastructure health	Displays the file service infrastructure health status per ESX host. For more information, click the **Info** tab.
File Server Health	Displays the file server health status. For more information, click the **Info** tab.
Share health	Displays the file service share health. For more information, click the **Info** tab.

Monitor vSAN File Share Objects Health
You can monitor the health of file share objects.

To view the file share object health, navigate to the vSAN cluster and then click **Monitor** > **vSAN** > **Virtual Objects**.

The device information such as name, identifier or UUID, number of devices used for each virtual machine, and how they are mirrored across hosts is displayed in **View Placement Details**.

Migrate a Hybrid vSAN Cluster to an All-Flash Cluster

You can migrate the disk groups in a hybrid vSAN cluster to all-flash disk groups. This is applicable only for vSAN OSA all-flash architecture.

- Ensure that all the vSAN policies that the cluster uses specify **No preference** for encryption services, space efficiency, and storage tier.
- You must use RAID-1 (Mirroring) for **Failures to tolerate** until all the disk groups are converted to all-flash.

The vSAN hybrid cluster uses magnetic disks for the capacity layer and flash devices for the cache layer. You can change the configuration of the disk groups in the cluster so that it uses flash devices on the cache layer and the capacity layer.

Note:
Follow the steps to migrate a hybrid vSAN cluster to Solid State Drive (SSD), hybrid vSAN cluster to NVMe, or SSD to NVMe.

1. Remove the hybrid disk groups on the host.
   a) In the vSphere Client, navigate to the cluster.
   b) Click the **Configure** tab.
   c) Under vSAN, click **Disk Management**.
   d) Under Disk Groups, select the disk group to remove, click **…**, and then click **Remove**.
   Select **Full data migration** as a migration mode and click **Yes**.

   Note:
   Migrate the disk groups on each host in the vSAN cluster.
2. Remove the physical HDD disks from the host.
3. Add the flash devices to the host.
   Verify that no partitions exist on the flash devices.

4. Create the all-flash disk groups on the host.
5. Repeat the steps from 1 through 4 on each host until all the hybrid disk groups are converted to the all-flash disk groups.

   Note:

   If you cannot hot-plug disks on the host, place the host in maintenance mode before removing disks in the vSphere Client. Shut down the host to replace the disks with flash devices. Then power on the host, exit maintenance mode, and create new disk groups.

Manage Site Maintenance for a vSAN Stretched Cluster

vSAN stretched clusters extend a vSAN cluster from a single data site to two sites, to provide higher availability and intersite load balancing. vSAN ESA and vSAN OSA support placing an entire site into maintenance mode.

- Ensure that you have configured a vSAN stretched cluster.
- vCenter and ESX hosts must have vSAN 9.1 and later.
- Ensure that you have assigned your hosts to the preferred and secondary fault domains.

For maintenance of hosts, you must place only one host into maintenance mode at a time, complete the maintenance operation, and then exit the host from maintenance. You must then repeat this process for each host in the cluster. Placing hosts into maintenance mode one at a time ensures that if a site fails at any point in time, all the objects remain accessible on the active site without any data loss.

If you require site infrastructure maintenance and need to place all the hosts in a site into maintenance mode, doing this for each host one at a time until all the hosts are in maintenance can cause inaccessibility of objects, if the active site fails.

The objects become accessible once the active site that is not in maintenance recovers. However, if that site incurs a failure that prevents the hosts on that site from becoming available, you must restore the objects from the backup.

With the vSAN 9.1 release, the site maintenance operation ensures point-in-time consistency across all object components within the maintenance site. If the active site fails while this site is in maintenance mode, you can recover objects on the maintenance site. You should wait for the site to become available so that the objects become accessible to the most recent state. If the active site fails and is not recoverable, you can recover the objects to an older point in time using the site takeover operation. This is similar to restoring the objects to an older point in time. However, you can do it quickly because you do not need to copy all the data from the backup.

Note:

vSAN storage clusters support site maintenance, but they do not support site takeover.

When you initiate the site maintenance operation, vSAN ensures that all hosts in a site are placed into maintenance mode and the data remains consistent. All components on the hosts of the site entering maintenance are synchronized to the same point in time. The VMs having data locality to this site are powered off. Ensure that you perform site maintenance precheck before initiating site maintenance operation. The precheck result shows all the details, including the VMs to be migrated, VMs to be powered off, and so on.

After the site enters maintenance mode, all new writes occur on the hosts in the active data site. vSAN also migrates workloads from the maintenance site to the active site. You can then return the site to normal operation by exiting site maintenance mode.

When the hosts on the site exit maintenance mode, vSAN starts a resync process to update data modified during maintenance. The resync occurs across different objects or specific object components. Until vSAN fully synchronizes the delta data, the site remains in an inconsistent state. If the site holding the consistent copy fails before the resync completes, vSAN may not recover the objects on the site that exited maintenance as their data is stale.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Select the site that you want to move to site maintenance.
5. Select
   
   > **Enter Maintenance Mode**.
6. From the Enter Maintenance Mode wizard, click **Go To Pre-Check**. vSAN runs the data migration precheck tests.
7. If the pre-check indicates that you can place the site into maintenance mode, click **Enter Maintenance Mode** to migrate the data and place the site into site maintenance mode.

Exit the Site Maintenance for a Stretched Cluster

When the site infrastructure maintenance operations are complete, you can exit the site maintenance for a stretched cluster.

Once you start the operation, all stale writes from the active site starts resyncing to this site. Each object component is resynced independently and if the resync is in progress, the components of an object or different objects could be at a different point in time. During this time, if the active site fails and cannot be recovered, it may not be possible to use the site takeover operation to recover the object back to an older point in time.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Fault Domains**.
4. Select the site that you want to move out of site maintenance mode.
5. Select
   
   > **Exit Maintenance Mode**.
6. In the Exit maintenance mode dialog, click **Exit Maintenance Mode**.

Perform the Site Takeover for a vSAN Stretched Cluster

vSAN site takeover recovers the data availability of the underlying vSAN objects in a stretched cluster, when a data site and witness are down.

Ensure the following before initiating a vSAN site takeover:

- All hosts in the failed site and the witness host are unavailable and cannot be recovered quickly.
- vCenter, ESX host operations, and VM operations are disabled during the vSAN site takeover.

Site takeover allows you to handle a situation where one data site is in maintenance and the other two sites in a cluster fail at the same time. The site in maintenance can be recovered to an available state so that VM instances can start and use the storage.

If a site takeover runs on a data site in maintenance mode, all the objects recover to the same older point in time. In other scenarios, some objects may not recover, or those objects that may recover to different points in time. vSAN cannot guarantee that all applications can handle recovery to older data, when their data depends on other objects or application states in the same or different clusters.

A data site and the witness host can fail with the vCenter VM when the vCenter VM is deployed on the same vSAN datastore. You can still perform site takeover and recover the vCenter VM after the site takeover of the vSAN datastore.

vSAN site takeover is not supported with storage cluster or compute-only cluster. The site takeover does not recover the vSAN objects for vSAN file service and iSCSI target objects.

1. Log in to one of the ESX hosts in the surviving site using the command-line interface as the root user.
2. Run the following site takeover command on the ESX host in the surviving site and verify the precheck results:

   ```
   /usr/lib/vmware/vsan/bin/site-takeover
   ```

   During site takeover, hosts are disconnected from vCenter. A reboot is required to access the recovered vSAN objects from the hosts that did not participate in the site takeover. vSAN health displays the list of hosts that require a reboot after the site takeover.
3. Verify the vSAN health if you want to bring some of the hosts in the failed site back to the cluster.

- The command performs validation checks and displays the expected recovery state of vSAN objects.
- All inaccessible objects are removed after the site takeover.
- A summary is displayed after the site takeover is complete. If the other site recovers, its hosts require a reboot. All the existing components in the site are discarded and a full resync is required as a result of site takeover. If the site is not recoverable, its hosts must be replaced or removed to make the cluster a standard cluster.

Additional Information About Site Force Takeovers

When performing a vSAN site force takeover, keep in mind the following topics which might apply to your environment:

- vSAN Data Protection Workload Recovery After Site Force Takeover
- Site Maintenance and Force Takeover Availability During Cluster Upgrades
- VADP Workload Recovery After Site Force Takeover

vSAN Data Protection Workload Recovery After Site Force Takeover

Ideally, you should recover VMs from the backup repository. However, if the backup does not contain the most recent critical data, you can perform a site force takeover on the surviving site to recover the workloads from that site's data. This is possible because the Recovery Point Objective (RPO) window was exceeded or the surviving site had already entered site maintenance mode before the failure occurred

After site force takeover completes successfully, vSAN initiates a full data synchronization. The data on the surviving site is treated as the authoritative source. The backup repository is updated to reflect the recovered data after synchronization completes.

> **Note:**
>
> If the site force takeover is performed in a cluster where vSAN data protection uses the surviving site as a replication target, the same behavior applies. Although the backup data is recoverable after the takeover, a full data synchronization is still triggered. The backup data is synchronized to match the source data.

No additional action is required for the synchronization to occur. vSAN initiates and manages the full resync automatically after the takeover completes.

Site Maintenance and Force Takeover Availability During Cluster Upgrades

Site maintenance and site force takeover operations require all ESX hosts in the cluster to run vSAN 9.1 or later. If the cluster contains hosts running different versions ("mixed mode"), the site maintenance and force takeover options are unavailable until every host completes the upgrade.

VMware software determines whether these operations are available by checking the version of each host at runtime. If any host has not been upgraded to vSAN 9.1 or a later release, the operations are disabled for the entire cluster.

This constraint also applies during an in-progress upgrade. If site maintenance was initiated in vSAN 9.0 and the cluster is partway through an upgrade to vSAN 9.1, site maintenance and force takeover options remain disabled until all hosts have completed the upgrade and the cluster exits mixed mode.

> **Tip:**
>
> Complete the cluster upgrade to vSAN 9.1 or later on all hosts before initiating site maintenance or force takeover operations. You can monitor upgrade progress in the vSphere Client.

VADP Workload Recovery After Site Force Takeover

When VADP (VMware vStorage APIs for Data Protection) is configured to back up VMs stored on a vSAN datastore in a stretched cluster, and those VMs' vSAN objects have site failure tolerance, losing one data site and the witness host simultaneously causes the VADP objects to lose quorum. Those objects become inaccessible on the surviving site.

Broadcom recommends recovering VM data from the backup repository. If the backup does not contain the most recent critical data , because the Recovery Point Objective (RPO) window was exceeded or the surviving site had already entered site maintenance mode before the failure occurred, you can perform a site force takeover on any host in the surviving site to recover the data from that site.

After site force takeover completes successfully, vSAN triggers a full data synchronization. The data on the surviving site is treated as the authoritative source. The backup repository data is synchronized to match the recovered data after synchronization completes.

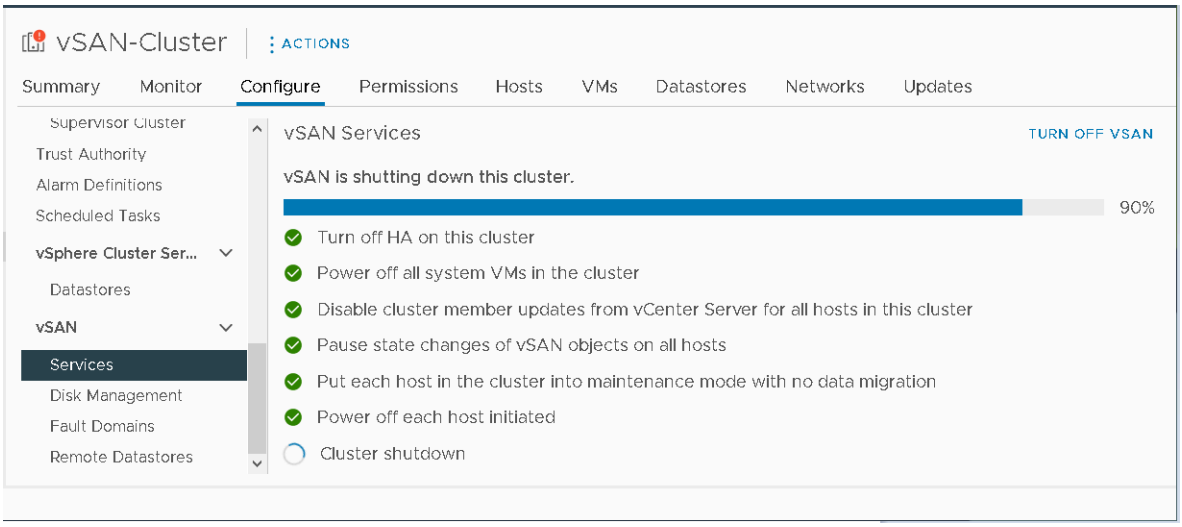
No additional action is required. vSAN initiates and manages the full resync automatically after the takeover completes.

Shutting Down and Restarting the vSAN Cluster

You can shut down the entire vSAN cluster to perform maintenance or troubleshooting.

Use the Shutdown Cluster wizard to shutdown the vSAN cluster. The wizard performs the necessary steps and alerts you when it requires user action. You also can manually shut down the cluster, if necessary.

> **Note:**
> When you shut down a vSAN stretched cluster, the witness host remains active.



Shut Down the vSAN Cluster Using the Shutdown Cluster Wizard

Use the Shutdown cluster wizard to gracefully shut down the vSAN cluster for maintenance or troubleshooting.

Ensure vSAN client clusters have unmounted any vSAN datastores or vSAN storage clusters that have been shared.

The Shutdown Cluster Wizard is available with vSAN 9.1 and later releases.

1. Prepare the vSAN cluster for shutdown.
   a) Check the vSAN Health to confirm that the cluster is healthy.
   b) Power off all virtual VMs stored in the vSAN cluster, except for vCenter VMs and file service VMs. If vCenter is hosted on the vSAN cluster, do not power off the vCenter VM or virtual machine service VMs (such as DNS, Active Directory) used by vCenter.
   c) If this is an vSAN HCI Datastore Sharing server cluster, power off all client VMs stored on the cluster. If the client cluster's vCenter virtual machine is stored on this cluster, either migrate or power off the virtual machine. Once this server cluster is shutdown, its shared datastore is inaccessible to clients.
   d) Verify that all resynchronization tasks are complete.

      Click the **Monitor** tab and select **vSAN** > **Resyncing Objects**.

      Note:

      If any member ESX hosts are in lockdown mode, add the host's root account to the security profile Exception User list. For more information, see the vSphere Security guide.
2. Right-click the vSAN cluster in the vSphere Client, and select menu **Shutdown cluster**.

   You also can click **Shutdown Cluster** on the vSAN Services page.
3. On the Shutdown cluster wizard, verify that the Shutdown pre-checks are green checks. Resolve any issues that are red exclamations. Click **Next**.

   If vCenter appliance is deployed on the vSAN cluster, the Shutdown cluster wizard displays the vCenter notice. Note the IP address of the orchestration host, in case you need it during the cluster restart. If vCenter uses service VMs such as DNS or Active Directory, note them as exceptional VMs in the Shutdown cluster wizard.
4. Select a reason for performing the shutdown, and click **Shutdown**.

   The vSAN Services page changes to display information about the shutdown process.
5. Monitor the shutdown process.

   vSAN performs the steps to shutdown the cluster, powers off the system VMs, and powers off the ESX hosts.

Restart the vSAN cluster. See Restart the vSAN Cluster.

Restart the vSAN Cluster

You can restart a vSAN cluster that is shut down for maintenance or troubleshooting.

Ensure that the ESX hosts are in maintenance mode.

1. Power on the cluster ESX hosts.

   If the vCenter is hosted on the vSAN cluster, wait for vCenter to restart.
2. Right-click the vSAN cluster in the vSphere Client, and select menu **Restart cluster**.

   You also can click **Restart Cluster** on the vSAN Services page.
3. On the Restart Cluster dialog, click **Restart**.

   The vSAN Services page changes to display information about the restart process.
4. After the cluster has restarted, check the vSAN Health and resolve any outstanding issues.

Manually Shut Down and Restart the vSAN Cluster

You can manually shut down the entire vSAN cluster to perform maintenance or troubleshooting.

Use the Shutdown Cluster wizard unless your workflow requires a manual shut down. When you manually shut down the vSAN cluster, do not deactivate vSAN on the cluster.

1. Shut down the vSAN cluster.
   a) Check the vSAN Health to confirm that the cluster is healthy.
   b) Power off all VMs running in the vSAN cluster, if vCenter is not hosted on the cluster. If vCenter is hosted in the vSAN cluster, do not power off the vCenter VM or service VMs (such as DNS, Active Directory) used by vCenter.
   c) If vSAN file service is enabled in the vSAN cluster, you must deactivate the file service. Deactivating the vSAN file service removes an empty file service domain. If you want to retain the empty file service domain after restarting the vSAN cluster, you must create an NFS or SMB file share before deactivating the vSAN file service.
   d) Click the **Configure** tab and turn off HA. As a result, the cluster does not register host shutdowns as failures.

      For vSphere 9.1 and later, enable vCLS retreat mode is deprecated. For more information, see the Broadcom knowledge base article 316514.
   e) Verify that all resynchronization tasks are complete.

      Click the **Monitor** tab and select **vSAN** > **Resyncing Objects**.
   f) If vCenter is hosted on the vSAN cluster, power off the vCenter virtual machine.

      Make a note of the host that runs the vCenter VM. It is the host where you must restart the vCenter VM.
   g) Deactivate cluster member updates from vCenter by running the following command on the ESX hosts in the cluster. Ensure that you run the following command on all the ESX hosts.

      ```
      esxcfg-advcfg -s 1 /VSAN/IgnoreClusterMemberListUpdates
      ```
   h) Log in to any host in the cluster other than the witness host.
   i) Run the following command only on that host. If you run the command on multiple ESX hosts concurrently, it may cause a race condition causing unexpected results.

      ```
      python /usr/lib/vmware/vsan/bin/reboot_helper.py prepare
      ```

      The command returns and prints the following:

      ```
      Cluster preparation is done.
      ```

      Note:

      - The cluster is fully partitioned after the successful completion of the command.
      - If you encounter an error, resolve the issue based on the error message and try enabling vCLS retreat mode again.
      - If there are unhealthy or disconnected ESX hosts in the cluster, remove the ESX hosts and retry the command.
   j) Place all the ESX hosts into maintenance mode with **No Action**. If the vCenter is powered off, use the following command to place the ESX hosts into maintenance mode with **No Action**.

      ```
      esxcli system maintenanceMode set -e true -m noAction
      ```

      Perform this step on all the hosts.

      To avoid the risk of data unavailability while using **No Action** at the same time on multiple hosts, followed by a reboot of multiple hosts, see the Broadcom knowledge base article 60424. To perform simultaneous reboot of all hosts in the cluster using a built-in tool, see the Broadcom knowledge base article 70650.
   k) After all ESX hosts have successfully entered maintenance mode, perform any necessary maintenance tasks and power off the ESX hosts.
2. Restart the vSAN cluster.
   a) Power on the ESX hosts.

      Power on the physical box where ESX is installed. The ESX host starts, locates the VMs, and functions normally.

      If any ESX hosts fail to restart, you must manually recover the ESXhosts or move the bad hosts out of the vSAN cluster.
   b) When all the ESX hosts are back after powering on, exit all ESX hosts from maintenance mode. If the vCenter is powered off, use the following command on the ESX hosts to exit maintenance mode.

      ```
      esxcli system maintenanceMode set -e false
      ```

      Perform this step on all the ESX hosts.

c) Log in to one of the ESX hosts in the cluster other than the witness ESX host.
d) Run the following command only on that ESX host. If you run the command on multiple ESX hosts concurrently, it may cause a race condition causing unexpected results.

```
python /usr/lib/vmware/vsan/bin/reboot_helper.py recover
```

The command returns and prints the following:

```
Cluster reboot/power-on is completed successfully!
```

e) Verify that all the ESX hosts are available in the cluster by running the following command on each ESX host.

```
esxcli vsan cluster get
```

f) Enable cluster member updates from vCenter by running the following command on the ESX hosts in the cluster. Ensure that you run the following command on all the ESX hosts.

```
esxcfg-advcfg -s 0 /VSAN/IgnoreClusterMemberListUpdates
```

g) Restart the vCenter virtual machine if it is powered off. Wait for the vCenter virtual machine to be powered up and running. To deactivate vCLS retreat mode, see the Broadcom knowledge base article 80472.
h) Verify again that all the ESX hosts are participating in the vSAN cluster by running the following command on each ESX host.

```
esxcli vsan cluster get
```

i) Restart the remaining VMs through vCenter.
j) Check the vSAN Health and resolve any outstanding issues.
k) (Optional) Enable vSAN file service.
l) (Optional) If the vSAN cluster has vSphere Availability enabled, you must manually restart vSphere Availability to avoid the following error: `Cannot find vSphere HA master agent`.

To manually restart vSphere Availability, select the vSAN cluster and navigate to:

1. **Configure** > **Services** > **vSphere Availability** > **EDIT** > **Disable vSphere HA**
2. **Configure** > **Services** > **vSphere Availability** > **EDIT** > **Enable vSphere HA**

3. If there are unhealthy or disconnected ESX hosts in the cluster, recover or remove the ESX hosts from the vSAN cluster. If vCenter uses service VMs such as DNS or Active Directory, note them as exceptional VMs in the Shutdown cluster wizard.

   Retry the above commands only after the vSAN Health shows all available ESX hosts in the green state.

   If you have a three-node vSAN cluster, the command `reboot_helper.py recover` cannot work in a one ESX host failure situation. As an administrator, do the following:

   1. Temporarily remove the failure ESX host information from the unicast agent list.
   2. Add the ESX host after running the following command.

      ```
      reboot_helper.py recover
      ```

   Following are the commands to remove and add the ESX host to a vSAN cluster:

   ```
   #esxcli vsan cluster unicastagent remove -a <IP Address> -t node -u <NodeUuid>
   #esxcli vsan cluster unicastagent add -t node -u <NodeUuid> -U true -a <IP Address> -p 12321
   ```

Restart the vSAN cluster. See Restart the vSAN Cluster.

Device Management in a vSAN Cluster

You can perform various device management tasks in a vSAN cluster.

You can create hybrid or all-flash disk groups, enable vSAN to claim devices for capacity and cache, turn LED indicators on or off, mark devices as flash, mark remote devices as local, and so on.

> **Note:**
> Marking devices as flash and marking remote devices as local are not supported in a vSAN ESA cluster.

Managing Storage Devices in vSAN Cluster

When you configure vSAN on a cluster, you claim storage devices on each host to create the vSAN datastore.

Each vSAN cluster contains a single vSAN datastore. As you claim disks for disk groups (vSAN OSA) or storage pool (vSAN ESA) on each host, the size of the datastore increases according to the amount of physical capacity added by those devices.

vSAN has a uniform workflow for claiming disks across all scenarios. You can list all available disks by model and size, or by host.

Add a Storage Pool (vSAN ESA)	Each ESX host that contributes storage contains a single storage pool of flash devices. Each flash device provides caching and capacity to the cluster. You can add a storage pool using any compatible devices. vSAN creates only one storage pool per host, irrespective of the number of storage disks the ESX host is attached to.
Add a Disk Group (vSAN OSA)	When you add a disk group, you must specify the ESX host and the devices to claim. Each disk group contains one flash cache device and one or more capacity devices. You can create multiple disk groups on each ESX host, and claim a cache device for each disk group. When adding a disk group, consider the ratio of flash cache to consumed capacity. The ratio depends on the requirements and workload of the cluster. For a hybrid cluster, consider using at least 10 percent of flash cache to consumed capacity ratio (not including replicas such as mirrors). **Note:** If a new ESX host is added to the vSAN cluster, the local storage from that ESX host is not added to the vSAN datastore automatically. You must add a disk group to use the storage from the new ESX host.
Claim Disks for vSAN Direct	Use vSAN Direct to enable stateful services to access raw, non-vSAN local storage through a direct path. You can claim host-local devices for vSAN Direct, and use vSAN to manage and monitor those devices. On each local device, vSAN Direct creates and independent VMFS datastore and makes it available to your stateful application. Each local vSAN Direct datastore appears as a vSAN-D datastore. **Note:** If vSAN ESA is enabled for the cluster, you cannot claim disks for vSAN Direct.

Create a Disk Group or Storage Pool in vSAN Cluster

Depending on the storage architecture you use in your cluster, you can decide to create a disk group or storage pool.

Create a Storage Pool on a Host (vSAN ESA)
You can claim disks to define a storage pool on a vSAN host. Each host that contributes storage contains a single storage pool of flash devices. Each flash device provides caching and capacity to the cluster. You can create a storage pool with any devices that are compatible for ESA. vSAN creates only one storage pool per host. vSAN ESA uses a single tier storage architecture where all devices contribute to capacity.

Use vSAN Managed Disk Claim to automatically claim all compatible disks on the cluster hosts. When you add new hosts, vSAN will also claim compatible disks on those hosts. Any disks added manually are not affected by this setting. You can manually add such disks to the storage pool.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Click **Claim Unused Disks**.

   You can change the disk claim mode to use **vSAN Managed Disk Claim**. vSAN will automatically claim all compatible devices on cluster hosts.

5. Group by host.
6. Select compatible disks to claim.
7. Click **Create** to confirm your selections.

   The disk management page appears with the hosts listed. There will be an indication that disks are claimed on the hosts in the **Disks in use** column reflecting the updated number of disks per host. To see the claimed disks for the host, click **View disks**.

Create a Disk Group on a Host (vSAN OSA)
You can claim cache and capacity devices to define disk groups on a vSAN host. Select one cache device and one or more capacity devices to create the disk group. vSAN OSA uses a tiered storage architecture comprised of disk group constructs.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a host from the list, and click **View Disks**.
5. Click **Create Disk Group**.
6. Select disks to claim.
   1. Select one flash device to use for the cache tier.
   2. Select at least one disk for the capacity tier.
7. Click **Create** to confirm your selections.

   The new disk group appears in the list.

Claim Storage Devices for vSAN Express Storage Architecture Cluster

You can select a group of devices from an ESX host, and vSAN organizes them into a storage pool.

After vSAN ESA is enabled, you can claim disks either manually or automatically. In the manual method, you can select a group of storage devices to be claimed.

In automatic disk claim, vSAN automatically selects all compatible disks from the hosts. When new hosts are added to the cluster, vSAN automatically claims the compatible disks available in those hosts and adds the storage to the vSAN datastore.

You can choose devices that are not reported as certified for vSAN ESA and those devices will be considered in the storage pool, but such configuration is not recommended and can impact performance.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. To manually claim disks, click **Claim Unused Disks**.
   a) Select the devices you want to claim
   b) Click **Create**.
5. To automatically claim disks, click **Change Disk Claim Mode** and click the **vSAN managed disk claim** toggle button.

   Note:

   If you chose to use vSAN managed disk claiming when configuring the cluster, the toggle button would be already enabled.

   vSAN claims the devices that you selected and organizes them into storage pools that support the vSAN datastore. By default, vSAN creates one storage pool for each ESX host that contributes storage to the cluster. If the selected devices are not certified for vSAN ESA, those devices are not considered for creating storage pools.

Claim Storage Devices for vSAN Original Storage Architecture Cluster

You can select a group of cache and capacity devices, and vSAN organizes them into default disk groups.

In this method, you select devices to create disk groups for the vSAN cluster. You need one cache device and at least one capacity device for each disk group.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Click **Claim Unused Disks**.
5. Select devices to add to disk groups.
   - For hybrid disk groups, each host that contributes storage must contribute one flash cache device and one or more HDD capacity devices. You can add only one cache device per disk group.
     - Select a flash devices to be used as cache and click **Claim for cache tier**.
     - Select one or more HDD device to be used as capacity and click **Claim for capacity tier** for each of them.
     - Click **Create** or **OK**.
   - For all-flash disk groups, each host that contributes storage must contribute one flash cache device and one or more flash capacity devices. You can add only one cache device per disk group.
     - Select one or more flash devices to be used as cache and click **Claim for cache tier** for reach of them.
     - Select a flash device to be used for capacity and click **Claim for capacity tier**.
     - Click **Create** or **OK**.

   vSAN claims the devices that you selected and organizes them into default disk groups that contribute the vSAN datastore.

   To verify the role of each device added to the all-flash disk group, navigate to the "Claimed as" column for a given host on the Disk Management page. The table shows the list of devices and their purpose in a disk group. For all-flash and hybrid disk groups, the cache disk is always shown first in the disk group grid.

Claim Disks for vSAN Direct

You can claim local storage devices as vSAN Direct for use with the vSAN Data Persistence Platform.

> **Note:**
>
> Only the vSAN Data Persistence platform can consume vSAN Direct storage. The vSAN Data Persistence platform provides a framework for software technology partners to integrate with Broadcom infrastructure. Each partner must develop their own plug-in for Broadcom customers to receive the benefits of the vSAN Data Persistence platform. The platform is not operational until the partner solution running on top is operational. For more information, see the *vSphere Supervisor Concepts* guide.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Click **Claim Unused Disks**.
5. On the Claim Unused Disks dialog, select the vSAN Direct tab.
6. Select a device to claim by selecting the checkbox in the **Claim for vSAN Direct** column.

   > **Note:**
   >
   > Devices claimed for your vSAN cluster do not appear in the vSAN Direct tab.

7. Click **Create**.

For each device you claim, vSAN creates a new vSAN Direct datastore.

You can click the Datastores tab to display the vSAN Direct datastores in your cluster.

Working with Individual Devices in vSAN Cluster

You can perform various device management tasks in the vSAN cluster.

You can add devices to a disk group, remove devices from a disk group, enable or disable locator LEDs, and mark devices. You can also add or remove disks that are claimed using the vSAN Direct.

Add Devices to the Disk Group in vSAN Cluster

When you configure vSAN to claim disks in manual mode, you can add additional local devices to existing disk groups.

The devices must be the same type as the existing devices in the disk groups, such as SSD or magnetic disks.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select the disk group, and click the **Add Disks**.
5. Select the device that you want to add and click **Add**.

   If you add a used device that contains residual data or partition information, you must first clean the device. For information about removing partition information from devices, see Remove Partition From Devices.

Verify that the vSAN Disk Balance health check is green. If the Disk Balance health check issues a warning, perform automatic rebalance operation during off-peak hours. For more information, see Configure Automatic Rebalance in vSAN Cluster.

Check a Disk or Disk Group's Data Migration Capabilities from vSAN Cluster

Use the data migration pre-check to find the impact of migration options when unmounting a disk or disk group, or removing it from the vSAN cluster.

Run the data migration pre-check before you unmount or remove a disk or disk group from the vSAN cluster. The test results provide information to help you determine the impact to cluster capacity, predicted health checks, and any objects that will go out of compliance. If the operation will not succeed, pre-check provides information about what resources are needed.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Data Migration Pre-check**.
4. Select a disk or disk group, choose a data migration option, and click **Pre-check**.

   vSAN runs the data migration precheck tests.
5. View the test results.

   The pre-check results show whether you can safely unmount or remove the disk or disk group.

   - The Object Compliance and Accessibility tab displays objects that might have issues after the data migration.
   - The Cluster Capacity tab displays the impact of data migration on the vSAN cluster before and after you perform the operation.
   - The Predicted Health tab displays the health checks that might be affected by the data migration.

If the pre-check indicates that you can unmount or remove the device, click the option to continue the operation.

Remove Disk Groups or Devices from vSAN

You can remove selected devices from a disk group, or you can remove an entire disk group from a vSAN OSA cluster.

Run data migration pre-check on the device or disk group before you remove it from the cluster.

Because removing unprotected devices might be disruptive for the vSAN datastore and virtual machines in the datastore, avoid removing devices or disk groups.

Typically, you delete devices or disk groups from vSAN when you are upgrading a device or replacing a failed device, or when you must remove a cache device. Other vSphere storage features can use any flash-based device that you remove from the vSAN cluster.

Deleting a disk group permanently deletes the disk membership and the data stored on the devices.

> **Note:**
> Removing one flash cache device or all capacity devices from a disk group removes the entire disk group.
>
> **Note:**
> If the cluster uses deduplication and compression, you cannot remove a single disk from the disk group. You must remove the entire disk group.

Evacuating data from devices or disk groups might result in the temporary noncompliance of virtual machine storage policies.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Remove a disk group or selected devices.

Option	Description
Remove the Disk Group	1. Under Disk Groups, select the disk group to remove, and click **…**, then **Remove**. 2. Select a data evacuation mode.
Remove the Selected Device	1. Under Disk Groups, select the disk group that contains the device that you are removing. 2. Under Disks, select the device to remove, and click the **Remove Disk(s)**. 3. Select a data evacuation mode.

5. Click **Yes** or **Remove** to confirm.

   The data is evacuated from the selected devices or disk group. You can use locator LEDs to identify the location of storage devices. For more information, see Using Locator LEDs in vSAN.

Recreate a Disk Group in vSAN Cluster

When you recreate a disk group in the vSAN cluster, the existing disks are removed from the disk group, and the disk group is deleted.

vSAN recreates the disk group with the same disks. When you recreate a disk group on a vSAN cluster, vSAN manages the process for you. vSAN evacuates data from all disks in the disk group, removes the disk group, and creates the disk group with the same disks.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Under Disk Groups, select the disk group to recreate.
5. Click **…**, then click the **Recreate**.

   The Recreate Disk Group dialog box appears.

6. Select a data migration mode, and click **Recreate**.

All data residing on the disks is evacuated. The disk group is removed from the cluster, and recreated.

Using Locator LEDs in vSAN

You can use locator LEDs to identify the location of storage devices.

vSAN can light the locator LED on a failed device so that you can easily identify the device. This is particularly useful when you are working with multiple hot plug and host swap scenarios.

Consider using I/O storage controllers with pass-through mode, because controllers with RAID 0 mode require additional steps to enable the controllers to recognize locator LEDs.

For information about configuring storage controllers with RAID 0 mode, see your vendor documentation.

Locator LEDs

You can turn locator LEDs on vSAN storage devices on or off. When you turn on the locator LED, you can identify the location of a specific storage device.

- Verify that you have installed the supported drivers for storage I/O controllers that enable this feature. For information about the drivers that are certified by Broadcom, see the *Brodcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
- In some cases, you might need to use third-party utilities to configure the Locator LED feature on your storage I/O controllers. For example, when you are using HP you should verify that the HP SSA CLI is installed.

When you no longer need a visual alert on your vSAN devices, you can turn off locator LEDs on the selected devices.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a host to view the list of devices.
5. At the bottom of the page, select one or more storage devices from the list, and perform the desired action for the locator LEDs.

Option	Action
Turn on LED	Turns on locator LED on the selected storage device. You also can use the **Manage** tab and click **Storage** > **Storage Devices**.
Turn off LED	Turns off locator LED on the selected storage device. You also can use the **Manage** tab and click **Storage** > **Storage Devices**.

Mark Devices as Flash in vSAN

When flash devices are not automatically identified as flash by ESX hosts, you can manually mark them as local flash devices.

- Verify that the device is local to your ESX host.
- Verify that the device is not in use.
- Make sure that the virtual machines accessing the device are powered off and the datastore is unmounted.

Flash devices might not be recognized as flash when they are enabled for RAID 0 mode rather than passthrough mode. When devices are not recognized as local flash, they are excluded from the list of devices offered for vSAN and you cannot use them in the vSAN cluster. Marking these devices as local flash makes them available to vSAN.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select the host to view the list of available devices.
5. From the **Show** drop-down menu at the bottom of the page, select **Not in Use**.
6. Select one or more flash devices from the list and click the **Mark as Flash Disk**.
7. Click **Yes** to save your changes.
   The Drive type for the selected devices appears as Flash.

Mark Devices as HDD in vSAN

When local magnetic disks are not automatically identified as HDD devices by ESX hosts, you can manually mark them as local HDD devices.

- Verify that the magnetic disk is local to your ESX host.
- Verify that the magnetic disk is not in use and is empty.
- Verify that the virtual machines accessing the device are powered off.

If you marked a magnetic disk as a flash device, you can change the disk type of the device by marking it as a magnetic disk.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select the host to view the list of available magnetic disks.
5. From the **Show** drop-down menu at the bottom of the page, select **Not in Use**.
6. Select one or more magnetic disks from the list and click **Mark as HDD Disk**.
7. Click **Yes** to save.
   The Drive Type for the selected magnetic disks appears as HDD.

Mark Devices as Local in vSAN

When ESX hosts are using external SAS enclosures, vSAN might recognize certain devices as remote and might be unable to automatically claim them as local.

Make sure that the storage device is not shared.

In such cases, you can mark the devices as local.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a host to view the list of devices.
5. From the **Show** drop-down menu at the bottom of the page, select **Not in Use**.
6. From the list of devices, select one or more remote devices that you want to mark as local and click the **Mark as local disk**.
7. Click **Yes** to save your changes.

Mark Devices as Remote in vSAN

ESX hosts that use external SAS controllers can share devices.

You can manually mark those shared devices as remote, so that vSAN does not claim the devices when it creates disk groups. In vSAN, you cannot add shared devices to a disk group.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a host to view the list of devices.
5. From the **Show** drop-down menu at the bottom of the page, select **Not in Use**.
6. Select one or more devices that you want to mark as remote and click the **Mark as remote**.
7. Click **Yes** to confirm.

Add a Capacity Device to vSAN Disk Group

You can add a capacity device to an existing vSAN disk group.

Verify that the device is formatted and is not in use.

You cannot add a shared device to a disk group.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a disk group.
5. Click the **Add Disks** at the bottom of the page.
6. Select the capacity device that you want to add to the disk group.
7. Click **OK** or **Add**.
   The device is added to the disk group.

Remove Partition From Devices

You can remove partition information from a device so vSAN can claim the device for use.

Verify that the device is not in use by ESX as boot disk, VMFS datastore, or vSAN.

If you have added a device that contains residual data or partition information, you must remove all preexisting partition information from the device before you can claim it for vSAN use. Broadcom recommends adding clean devices to disk groups.

When you remove partition information from a device, vSAN deletes the primary partition that includes disk format information and logical partitions from the device.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management**.
4. Select a host to view the list of available devices.
5. From the **Show** drop-down menu, select **Ineligible**.
6. Select a device from the list and click **Erase partitions**.
7. Click **OK** to confirm.
   The device is clean and does not include any partition information.

Increasing Space Efficiency in a vSAN Cluster

You can use space efficiency techniques to reduce the amount of space for storing data.

These techniques reduce the total storage space required to meet your needs.

vSAN Space Efficiency Features

You can use space efficiency techniques to reduce the amount of space for storing data.

These techniques reduce the total storage capacity required to meet your needs. vSAN supports SCSI unmap commands that enable you to reclaim storage space that is mapped to a deleted vSAN object.

You can use deduplication and compression on a vSAN cluster to eliminate duplicate data and reduce the amount of space required to store data. Or you can use compression-only vSAN to reduce storage requirements without compromising server performance.

You can set the **Failure tolerance method** policy attribute on virtual machines to use RAID 5 or RAID 6 erasure coding. Erasure coding can protect your data while using less storage space than the default RAID 1 mirroring on vSAN OSA. vSAN ESA does not use RAID-1 by default.

You can use deduplication and compression, and RAID 5 or RAID 6 erasure coding to increase storage space savings. RAID 5 or RAID 6 each provide clearly defined space savings over RAID 1. Deduplication and compression can provide additional savings.

Reclaiming Storage Space in vSAN with SCSI Unmap

SCSI UNMAP commands enable you to reclaim storage space that is mapped to deleted files in the file system created by the guest on the vSAN object.

Deleting or removing files frees space within the file system. This free space is mapped to a storage device until the file system releases or unmaps it. vSAN supports reclamation of free space, which is also called the unmap operation. You can free storage space in the vSAN datastore when you delete or migrate a virtual machine, consolidate a snapshot, and so on.

Reclaiming storage space can provide a higher host-to-flash I/O throughput and improve the flash endurance.

In vSAN OSA, unmap capability is not enabled by default. Enable **Guest Trim/Unmap** on the vSAN Services Advanced options tab. When you enable unmap on a vSAN cluster, you must power off and then power on all virtual machines. Virtual machines must use virtual hardware version 13 or above to perform unmap operations.

vSAN also supports the SCSI UNMAP commands issued directly from a guest operating system to reclaim storage space. vSAN supports offline unmaps and inline unmaps. On Linux OS, offline unmaps are performed with the `fstrim(8)` command, and inline unmaps are performed when the `mount -o discard` command is used. On Windows OS, NTFS performs inline unmaps by default.

Using Deduplication and Compression in vSAN OSA Cluster

vSAN can perform block-level deduplication and compression to save storage space.

When you enable deduplication and compression on a vSAN all-flash cluster, duplicate data within each disk group is reduced. Deduplication removes redundant data blocks, whereas compression removes additional redundant data within each data block. These techniques work together to reduce the amount of space required to store the data. vSAN applies deduplication and then compression as it moves data from the cache tier to the capacity tier. Use compression-only vSAN for workloads that do not benefit from deduplication, such as online transactional processing.

Deduplication occurs inline when data is written back from the cache tier to the capacity tier. The deduplication algorithm uses a fixed block size and is applied within each disk group. Redundant copies of a block within the same disk group are deduplicated.

For the vSAN OSA, deduplication and compression are enabled as a cluster-wide setting, but they are applied on a disk group basis. Additionally, you cannot enable compression on specific workloads as the settings cannot be changed through vSAN policies. When you enable deduplication and compression on a vSAN cluster, redundant data within a particular disk group is reduced to a single copy.

Note:

Compression-only vSAN is applied on a per-disk basis.

You can enable deduplication and compression when you create a vSAN all-flash cluster or when you edit an existing vSAN all-flash cluster. For more information, see Enable Deduplication and Compression on an Existing vSAN Cluster.

When you enable or disable deduplication and compression, vSAN performs a rolling reformat of every disk group on every host. Depending on the data stored on the vSAN datastore, this process might take a long time. Do not perform these operations frequently. If you plan to disable deduplication and compression, you must first verify that enough physical capacity is available to place your data.

Note:

Deduplication and compression might not be effective for encrypted virtual machines, because virtual machine encryption encrypts data on the host before it is written out to storage. Consider storage tradeoffs when using virtual machines encryption.

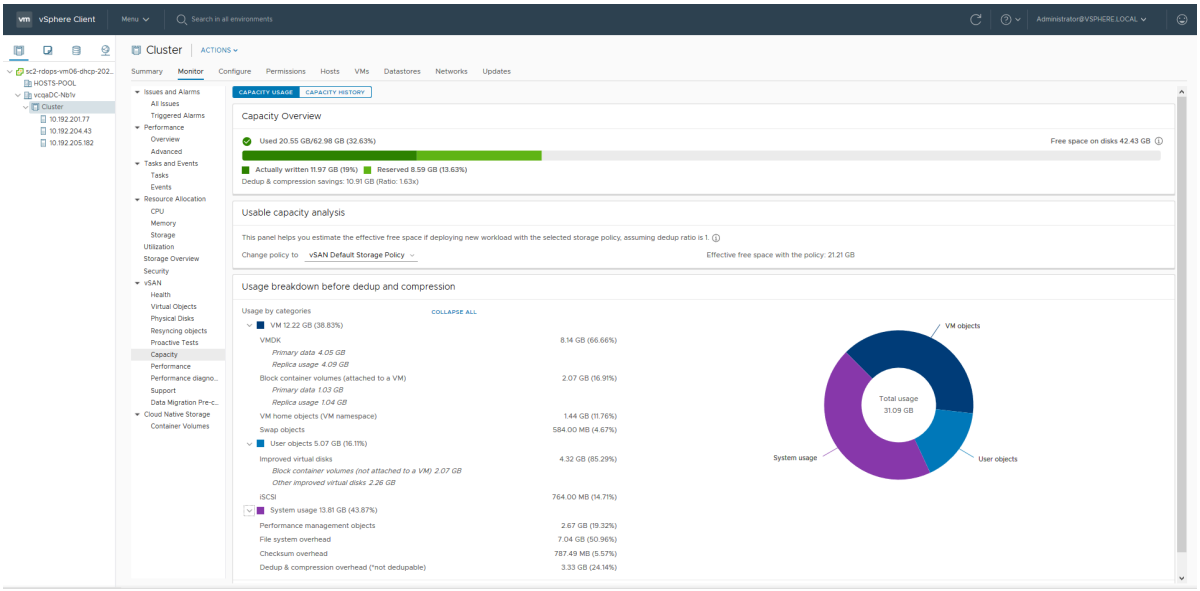
How to Manage Disks in a Cluster with Deduplication and Compression

Consider the following guidelines when managing disks in a cluster with deduplication and compression enabled. These guidelines do not apply to compression-only vSAN.

- Avoid adding disks to a disk group incrementally. For more efficient deduplication and compression, consider adding a disk group to increase the cluster storage capacity.
- When you add a disk group manually, add all the capacity disks at the same time.
- You cannot remove a single disk from a disk group. You must remove the entire disk group to make modifications.
- A single disk failure causes the entire disk group to fail.

Verifying Space Savings from Deduplication and Compression

The amount of storage reduction from deduplication and compression depends on many factors, including the type of data stored and the number of duplicate blocks. Larger disk groups tend to provide a higher deduplication ratio. You can check the results of deduplication and compression by viewing the Usage breakdown before dedup and compression in the vSAN Capacity monitor.



You can view the Usage breakdown before dedup and compression when you monitor vSAN capacity in the vSphere Client. It displays information about the results of deduplication and compression. The Used Before space indicates the logical space required before applying deduplication and compression, while the Used After space indicates the physical space used after applying deduplication and compression. The Used After space also displays an overview of the amount of space saved, and the Deduplication and Compression ratio.

The Deduplication and Compression ratio is based on the logical (Used Before) space required to store data before applying deduplication and compression, in relation to the physical (Used After) space required after applying deduplication and compression. Specifically, the ratio is the Used Before space divided by the Used After space. For example, if the Used Before space is 3 GB, but the physical Used After space is 1 GB, the deduplication and compression ratio is 3x.

When deduplication and compression are enabled on the vSAN cluster, it might take several minutes for capacity updates to be reflected in the Capacity monitor as disk space is reclaimed and reallocated.

Deduplication and Compression Design Considerations in a vSAN Cluster

Consider these guidelines when you configure deduplication and compression in a vSAN cluster.

- Deduplication and compression are available only on all-flash disk groups.
- On-disk format version 3.0 or later is required to support deduplication and compression.
- You must have a valid license to enable deduplication and compression on a cluster.
- When you enable deduplication and compression on a vSAN cluster, all disk groups participate in data reduction through deduplication and compression.
- vSAN can eliminate duplicate data blocks within each disk group, but not across disk groups (applicable only for vSAN OSA).
- Capacity overhead for deduplication and compression is approximately five percent of total raw capacity.
- Policies must have either 0 percent or 100 percent object space reservations. Policies with 100 percent object space reservations are always honored, but can make deduplication and compression less efficient.

Enable Deduplication and Compression on a New vSAN OSA Cluster

You can enable deduplication and compression when you configure a new vSAN all-flash cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Click **Edit** under **Data Services**.
   b) Select a space efficiency option: Deduplication and compression, or Compression only.
   c) Under **Encryption**, enable data-at-rest encryption by using the toggle button.
   d) (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN reduces the protection level of your virtual machines while enabling Deduplication and Compression. For more details, see Reduce Virtual Machine Redundancy for vSAN Cluster.
4. Complete your cluster configuration.

Enable Deduplication and Compression on an Existing vSAN Cluster

You can enable deduplication and compression by editing configuration parameters on an existing all-flash vSAN cluster.

To enable on a vSAN OSA cluster:

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**
   a. Click to edit Space Efficiency.
   b. Select a space efficiency option: Deduplication and compression, or Compression only.
   c. (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN reduces the protection level of your virtual machines while enabling Deduplication and Compression. For more details, see Reduce Virtual Machine Redundancy for vSAN Cluster.
4. Click **Apply** to save your configuration changes.

While enabling deduplication and compression on a vSAN OSA cluster, vSAN updates the on-disk format of each disk group of the cluster.

While enabling deduplication and compression, vSAN performs a rolling reformat of each disk group of the cluster. For more information, see this KB:How to enable/disable "Deduplication and Compression" in vSAN and the caveats to consider.

To accomplish this change, vSAN evacuates data from the disk group, removes the disk group, and recreates it with a new format that supports deduplication and compression.

The enablement operation does not require virtual machine migration or DRS. The time required for this operation depends on the number of hosts in the cluster and the amount of data. You can monitor the progress on the **Tasks and Events** tab.

Disable Deduplication and Compression on vSAN Cluster

You can disable deduplication and compression on your vSAN cluster.

When deduplication and compression are disabled on the vSAN cluster, the size of the used capacity in the cluster can expand (based on the deduplication ratio). Before you disable deduplication and compression, verify that the cluster has enough capacity to handle the size of the expanded data.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
   a) Under vSAN, select **Services**.
   b) Click **Edit**.
   c) Disable Deduplication and Compression.
   d) (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN reduces the protection level of your virtual machines, while disabling Deduplication and Compression. See Reduce Virtual Machine Redundancy for vSAN Cluster.
3. Click **Apply** or **OK** to save your configuration changes.

While disabling deduplication and compression, vSAN changes the disk format on each disk group of the cluster. It evacuates data from the disk group, removes the disk group, and recreates it with a format that does not support deduplication and compression.

The time required for this operation depends on the number of hosts in the cluster and amount of data. You can monitor the progress on the **Tasks and Events** tab.

Reduce Virtual Machine Redundancy for vSAN Cluster

When you enable deduplication and compression, in certain cases, you might need to reduce the level of protection for your virtual machines.

Enabling deduplication and compression requires a format change for disk groups. To accomplish this change, vSAN evacuates data from the disk group, removes the disk group, and recreates it with a new format that supports deduplication and compression.

In certain environments, your vSAN cluster might not have enough resources for the disk group to be fully evacuated. Examples for such deployments include a three-node cluster with no resources to evacuate the replica or witness while maintaining full protection. Or a four-node cluster with RAID-5 objects already deployed. In the latter case, you have no place to move part of the RAID-5 stripe, since RAID-5 objects require a minimum of four nodes.

You can still enable deduplication and compression and use the Allow Reduced Redundancy option. This option keeps the virtual machines running, but the virtual machines might be unable to tolerate the full level of failures defined in the virtual machine storage policy. As a result, temporarily during the format change for deduplication and compression, your virtual machines might be at risk of experiencing data loss. vSAN restores full compliance and redundancy after the format conversion is completed.

Add or Remove Disks with Deduplication and Compression Enabled

When you add disks to a vSAN cluster with enabled deduplication and compression, specific considerations apply.

- For vSAN OSA, you can add a capacity disk to a disk group with enabled deduplication and compression. However, for more efficient deduplication and compression, instead of adding capacity disks to an existing disk group, create a new disk group to increase cluster storage capacity. For vSAN ESA, with global deduplication you can add a storage devices to an existing host's storage pool to expand a cluster.
- When you remove a disk from a cache tier, the entire disk group is removed. Removing a cache tier disk when deduplication and compression are enabled triggers data evacuation.
- For vSAN OSA, deduplication and compression are implemented at a disk group level. You cannot remove a capacity disk from the cluster with enabled deduplication and compression. You must remove the entire disk group. For vSAN ESA, deduplication is implemented at the vSAN object level, making it transparent to storage devices in the storage pool.
- If a capacity disk fails, the entire disk group becomes unavailable. To resolve this issue, identify and replace the failing component immediately. When removing the failed disk group, use the No Data Migration option.

Using Global Deduplication and Compression in vSAN ESA Cluster

vSAN can perform cluster-level global deduplication and compression to save storage space.

After configuring a vSAN ESA cluster, you can enable global deduplication. Compression in a vSAN ESA cluster is enabled by default. You can enable global deduplication on a vSAN ESA cluster, at the vSAN object level. Global deduplication removes redundant data blocks, whereas compression removes additional redundant data within each data block. These techniques work together to reduce the amount of space required to store the data.

In vSAN ESA, global deduplication is active for all VMs except those using thick provisioning or HFTT-3 policies. vSAN stops deduplicating a VM upon switching to thick provisioning or HFTT-3. If you change a VM's policy from HFTT less than 3 to HFTT=3, vSAN stops global deduplication and marks the VM as non-compliant. Deduplication automatically resumes if the VM is reverted to thin provisioning or an HFTT-2 or less.

After you enable global deduplication, you can pause data deduplication as required. The new data written to the vSAN cluster is not deduplicated. The deduplicated data cannot be reverted when you pause deduplication.

> **Note:**
>
> If you enable vSAN ESA Global Deduplication, you cannot convert a single site vSAN cluster to a vSAN stretched cluster.

Global Deduplication Impact on vSAN Clusters

As a new feature, global deduplication impacts the behavior of a vSAN cluster in the following ways:

- **VM Storage policy**. After global deduplication is enabled on a vSAN cluster, deduplication is effective on almost all VMs (except thick provisioned, HFTT-3 VMs), even if the data reduction setting in a VM storage policy is applied to the VM.
- **Thick provisioned objects**. After global deduplication is enabled on a vSAN cluster, the thick provisioned VMs hosted on a vSAN cluster will not be deduplicated. After the VM is reconfigured from thin to thick provision, vSAN will stop further deduplication.
- **Storage policy with HFFT-3**. When global deduplication is enabled on a vSAN cluster, the VMs configured with HFTT-3 and above storage policy will not be deduplicated. After the VMs is reconfigured with storage policy from HFTT < 3 to HFTT-3, vSAN will stop further deduplication and the VM health status would become in-compliant.
- **Stretch clusters**. When global deduplication is enabled on a standard vSAN cluster, it cannot be converted into a stretched cluster.

Enable Global Deduplication on a New vSAN ESA Cluster

You can enable global deduplication on a vSAN ESA cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Click **Edit** under **Data Reduction and Encryption**.
   b) Use the toggle button to enable global deduplication. After you enable global deduplication, you can pause deduplication. The deduplicated data cannot be reverted when you pause global deduplication. For vSAN ESA, compression is enabled by default. On a vSAN ESA cluster, global deduplication is applied to most VMs except thick-provisioned VMs and those configured with an Host Failures to Tolerate of 3 (HFTT-3) or higher storage policy, regardless of the data reduction settings in the applied VM storage policy.

      > **Note:**
      >
      > If you enable vSAN ESA Global Deduplication, you cannot convert a single site vSAN cluster to a vSAN stretched cluster.

c) Under Encryption, enable data-at-rest encryption and data-in-transit encryption by using the toggle button.
d) (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN reduces the protection level of your virtual machines, while disabling deduplication. The Allow Reduced Redundancy option applies to data at rest encryption only. For more information, see Reduce Virtual Machine Redundancy for vSAN Cluster.
e) Complete your cluster configuration.

Enable Global Deduplication on an Existing vSAN ESA Cluster

You can enable global deduplication by editing configuration parameters on an existing vSAN ESA cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Click **Edit** under **Data Reduction and Encryption**.
   b) Use the toggle button to enable deduplication. In a vSAN ESA cluster, compression is enabled by default. Once global deduplication is enabled on a standard vSAN cluster, you cannot convert the cluster into a stretched cluster.
   c) (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN OSA reduces the protection level of your virtual machines while enabling deduplication. For more details, see Reduce Virtual Machine Redundancy for vSAN Cluster.
4. Click **Apply** to save your configuration changes.

   In a vSAN ESA cluster, vSAN enables compression by default.

Pause Global Deduplication on vSAN ESA Cluster

You can pause global deduplication on a vSAN ESA cluster.

The new data written to the vSAN cluster is not deduplicated. The deduplicated data cannot be reverted when you pause global deduplication.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
   a) Click **Edit** under **Data Reduction and Encryption**.
   b) Use the toggle button to pause global deduplication.

      Note:

      You cannot disable global deduplication.
   c) (Optional) Select **Allow Reduced Redundancy**. If needed, vSAN OSA reduces the protection level of your virtual machines while pausing global deduplication. For more details, see Reduce Virtual Machine Redundancy for vSAN Cluster.
4. Click **Apply** to save your configuration changes.

Selecting the Best RAID Configuration for a vSAN Storage Cluster

You can use RAID 5 or RAID 6 erasure coding to protect against data loss and increase storage efficiency.

Erasure coding can provide the same level of data protection as mirroring (RAID 1), while using less storage capacity. RAID 5 or RAID 6 erasure coding enables vSAN to tolerate the failure of up to two capacity devices in the datastore. You can configure RAID 5 on all-flash clusters with four or more fault domains. You can configure RAID 5 or RAID 6 on all-flash clusters with six or more fault domains.

RAID 5 or RAID 6 erasure coding requires less additional capacity to protect your data than RAID 1 mirroring. For example, a virtual machine protected by a **Failures to tolerate** value of 1 with RAID 1 requires twice the virtual disk size, but with RAID 5 it requires 1.33 times the virtual disk size.

Note:

You cannot use the vSAN ESA Auto-RAID policy when creating, cloning, or migrating virtual machines on clusters running VCF/ESX 9.0 or earlier versions, but that is managed by a newer version of vCenter that supports Auto-RAID.

This limitation applies to:

- Compute-only clients running ESX 9.0 or earlier.
- vSAN clusters running VCF 9.0 or earlier versions of the software.

If the remote mounted datastore is VCF 9.1 or later, and you want to use Auto-RAID in this scenario, you must enable "Assume auto RAID" on the remote cluster

The following tables show a general comparison between RAID 1 and RAID 5 or RAID 6 for both vSAN OSA and vSAN ESA architectures.

Table 1017: vSAN ESA (Express Storage Architecture) RAID Options

Note:

Protection overhead includes both site mirroring (PFTT) and local raid protection.

FTT	RAID Type	Protection Overhead	Layout	Min Hosts
0	RAID-0	1x	Single copy of data	1
1	RAID-1	2x	2 mirrors	3
2	RAID-1	3x	3 mirrors	5
3	RAID-1	4x	4 mirrors	7
1	RAID-5 (2+1)	1.5x	2 data + 1 parity	3
1	RAID-5 (4+1)	1.25x	4 data + 1 parity	6
2	RAID-6 (4+2)	1.5x	4 data + 2 parity	6

Table 1018: vSAN ESA (Express Storage Architecture) - Stretched Cluster RAID Options

Site Mirroring (PFTT)	FTT	Local RAID Type	Protection Overhead	Layout	Min Hosts
1	0	RAID-0	2x	2-node stretched: 1 host per site + witness	3 (1+1+1)
1	1	RAID-1	2x	2 host per site + witness	5 (2+2+1)
1	1	RAID-1	4x	Mirror per site + witness	7 (3+3+1)
1	1	RAID-5 (2+1)	3x	RAID-5 per site + witness	7 (3+3+1)
1	1	RAID-5 (4+1)	2.5x	RAID-5 per site + witness	13 (6+6+1)
1	2	RAID-6 (4+2)	3x	RAID-6 per site + witness	13 (6+6+1)

Table 1019: vSAN ESA (Express Storage Architecture) - Auto Raid

FTT	RAID Type	Protection Overhead	Layout	Min Hosts
0	RAID-0	1.0x	Single copy of data	< 3 hosts
1	RAID-1	2.0x	2 mirrors	2 hosts
1	RAID-5 (2+1)	1.5x	2 data + 1 parity	3-5
2	RAID-6 (4+2)	1.5x	4 data + 2 parity	6+

Table 1020: vSAN ESA (Express Storage Architecture) - Auto Raid - Stretched Cluster

Site Mirroring (PFTT)	FTT	Local RAID Type	Protection Overhead	Layout	Min Hosts
1	1	RAID-5 (2+1)	3x	RAID-5 per site + witness	7 (3+3+1)
1	2	RAID-6 (4+2)	3x	RAID-6 per site + witness	13 (6+6+1)

RAID-6, RAID-5, and RAID-1 erasure coding is a policy attribute that you can apply to virtual machine components.

- To use RAID-6, set **Failure tolerance method** to **RAID-5/6 (Erasure Coding)** and **Failures to tolerate** to 2. RAID 5 or RAID 6 erasure coding does not support a **Failures to tolerate** value of 3.
- To use RAID-5, set **Failure tolerance method** to **RAID-5/6 (Erasure Coding)** and **Failures to tolerate** to 1.
- To use RAID-1, set **Failure tolerance method** to **RAID-1 (Mirroring)**. RAID 1 mirroring requires fewer I/O operations to the storage devices, so it can provide better performance. For example, a cluster resynchronization takes less time to complete with RAID 1.

Note:
In a vSAN stretched cluster, the **Failure tolerance method** of **RAID-5/6 (Erasure Coding)** applies only to the **Site disaster tolerance** setting.

Note:

For a vSAN ESA cluster, depending on the number of fault domains that you use, the number of components listed under **RAID 5** (**Monitor** > **vSAN** > **Virtual Objects** > testVM > **View Placement Details**) will vary. If six or more fault domains are available in the cluster, then five components will be listed under **RAID 5**. If five or fewer fault domains are available, then three components will be listed.

For more information about configuring policies, see Using vSAN Policies.

RAID 5 or RAID 6 Design Considerations in vSAN Cluster

Consider these guidelines when you configure RAID 5 or RAID 6 erasure coding in a vSAN cluster.

- RAID 5 or RAID 6 erasure coding is available only on all-flash disk groups for vSAN OSA clusters.
- On-disk format version 3.0 or later is required to support RAID 5 or RAID 6.
- You must have a valid license to enable RAID 5/6 on a cluster.
- You can achieve additional space savings by enabling deduplication and compression on the vSAN cluster.

Using Encryption in a vSAN Cluster

You can encrypt data-in-transit in your vSAN cluster, and encrypt data-at-rest in your vSAN datastore.

vSAN can encrypt data-in-transit across hosts in the vSAN cluster. Data-in-transit encryption protects data as it moves around the vSAN cluster.

vSAN can encrypt data-at-rest in the vSAN datastore. Data-at-rest encryption protects data on storage devices, in case a device is removed from the cluster.

vSAN Data-In-Transit Encryption

vSAN data-in-transit encryption enables you to encrypt data as it moves across hosts in your vSAN cluster.

vSAN provides data-in-transit encryption across hosts in the cluster. When you enable data-in-transit encryption, vSAN encrypts all data and metadata traffic between hosts.

Notice:

In VCF 9.1, vSAN also supports remote data-in-transit for HCI-enabled cluster with remote datastores.

vSAN data-in-transit encryption has the following characteristics:

- vSAN uses AES-256 bit encryption on data-in-transit.
- vSAN data-in-transit encryption is not related to data-at-rest-encryption. You can enable or disable each one separately.
- Forward secrecy is enforced for vSAN data-in-transit encryption.
- Traffic between data hosts and witness hosts is encrypted.
- File service data traffic between the VDFS proxy and VDFS server is encrypted.
- vSAN file service inter-host connections are encrypted.

vSAN uses symmetric keys that are generated dynamically and shared between hosts. Hosts dynamically generate an encryption key when they establish a connection, and they use the key to encrypt all traffic between the hosts. You do not need a key management server to perform data-in-transit encryption.

Each host is authenticated when it joins the cluster, ensuring that connections only to trusted hosts are allowed. When a host is removed from the cluster, its authentication certificate is removed.

vSAN data-in-transit encryption is a cluster-wide setting. When enabled, all data and metadata traffic is encrypted as it transits across hosts.

Enable Data-In-Transit Encryption on a vSAN Cluster

You can enable data-in-transit encryption by editing the configuration parameters of a vSAN cluster.

To enable data-in-transit encryption on a vSAN cluster:

1. In the vSphere Client, navigate to an existing cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Under Data-In-Transit Encryption, click **Edit** .
5. Click to enable **Data-In-Transit encryption**.
6. Next, select the rekey interval, which determines how frequently vSAN automatically rotates (rekeys) the encryption keys that are used for data-in-transit encryption. Choose a default interval (6 hours, 12 hours, 1 day, 3 days, or 7 days), or set your own custom interval (in minutes). If no rekey interval is set, then the default of 24 hours is used.
7. Click **Apply**.

Encryption of data-in-transit is enabled on the vSAN cluster. vSAN encrypts all data moving across hosts and file service inter-host connections in the cluster.

Enable Data-In-Transit for a vSAN HCI Cluster

You can enable data-in-transit encryption by editing the configuration parameters of a vSAN HCI cluster.

To enable data-in-transit encryption on a vSAN HCI cluster:

1. In the vSphere Client, navigate to an existing cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Under Data Reduction and Encryption, select **Edit**.
5. In the Data Reduction and Encryption dialog box, select Data-In-Transit encryption.
6. Next, select the rekey interval, which determines how frequently vSAN automatically rotates (rekeys) the encryption keys that are used for data-in-transit encryption. Choose a default interval (6 hours, 12 hours, 1 day, 3 days, or 7 days), or set your own custom interval (in minutes). If no rekey interval is set, then the default of 24 hours is used.
7. Click **Apply**.

vSAN Data-At-Rest Encryption

vSAN can encrypt data-at-rest in your vSAN datastore.

Data-at-rest encryption integrates with other vSAN data services in the following ways:

- **vSAN Original Storage Architecture (OSA):** vSAN performs inline processing, such as deduplication and compression, before applying data-at-rest encryption.
- **vSAN Express Storage Architecture (ESA):** vSAN performs inline compression and applies data-at-rest encryption on the write path. Deduplication in ESA runs as an offline background process.

When you enable data-at-rest encryption, vSAN encrypts data after all other processing, such as deduplication, is performed. Data-at-rest encryption protects data on storage devices, in case a device is removed from the cluster.

Using encryption on your vSAN datastore requires some preparation. After your environment is set up, you can enable data-at-rest encryption on your vSAN cluster.

Data-at-rest encryption requires an external Key Management Server (KMS) or a vSphere Native Key Provider. For more information about vSphere encryption, see the vSphere Security guide.

You can use an external Key Management Server (KMS), the vCenter system, and your ESX hosts to encrypt data in your vSAN cluster. vCenter requests encryption keys from an external KMS. The KMS generates and stores the keys, and vCenter obtains the key IDs from the KMS and distributes them to the ESX hosts.

vCenter does not store the KMS keys, but keeps a list of key IDs.

How vSAN Data-At-Rest Encryption Works with the Key Management Server (KMS)

When you enable data-at-rest encryption, vSAN encrypts everything in the vSAN datastore.

All files are encrypted, so all virtual machines and their corresponding data are protected. Only administrators with encryption privileges can perform encryption and decryption tasks. vSAN uses encryption keys as follows:

- vCenter requests an AES-256 Key Encryption Key (KEK) from the KMS. vCenter stores only the ID of the KEK, but not the key itself.
- The ESX host encrypts disk data using the industry standard AES-256 XTS mode. In vSAN OSA, each disk has a different randomly generated Data Encryption Key (DEK). In vSAN ESA, all the disks in the cluster use the same DEK to encrypt object data.
- Each ESX host uses the KEK to encrypt its DEKs, and stores the encrypted DEKs on disk. The host does not store the KEK on disk for security purposes. If a host reboots, it requests the KEK with the corresponding ID from the KMS. The host can then decrypt its DEKs as needed. However, if the host has a Trusted Platform Module (TPM) hardware security chip, then it can safely store the KEK locally.
- A host key is used to encrypt core dumps, not data. All hosts in the same cluster use the same host key. When collecting support bundles, a random key is generated to re-encrypt the core dumps. You can specify a password to encrypt the random key.

When a host reboots, it does not mount its disk groups, or storage devices(for vSAN ESA), until it receives the KEK. This process can take several minutes or longer to complete. You can monitor the status of the disk groups in the vSAN health service, under **Physical disks > Software state health**.

Encryption Key Persistence

Data-at-rest encryption can continue to function even when the key server is temporarily offline or unavailable. With key persistence enabled, the ESX hosts can persist the encryption keys even after a reboot.

Each ESX host obtains the encryption keys initially and retains them in its key cache. If the ESX host has a Trusted Platform Module (TPM), the encryption keys are persisted in the TPM across reboots. The host does not need to request encryption keys. Encryption operations can continue when the key server is unavailable, because the keys have persisted in the TPM.

Use the following commands to enable key persistence on a cluster host.

```
esxcli system settings encryption set --mode=TPM

esxcli system security keypersistence enable
```

For more information about encryption key persistence, see "Key Persistence Overview" in the vSphere Security guide.

Using vSphere Native Key Provider

vSAN supports vSphere Native Key Provider. If your environment is set up for vSphere Native Key Provider, you can use it to encrypt virtual machines in your vSAN cluster. For more information, see "Configuring and Managing vSphere Native Key Provider" in the vSphere Security guide.

vSphere Native Key Provider does not require an external Key Management Server (KMS). vCenter generates the Key Encryption Key and pushes it to the ESX hosts. The ESX hosts then generate Data Encryption Keys.

> **Note:**
> If you use vSphere Native Key Provider, you must backup the Native Key Provider to ensure reconfiguration tasks run smoothly.

vSphere Native Key Provider can coexist with an existing key server infrastructure.

Design Considerations for vSAN Data-At-Rest Encryption

Consider these guidelines when working with data-at-rest encryption.

- Do not deploy your KMS server on the same vSAN datastore that you plan to encrypt.
- Encryption is CPU intensive. AES-NI significantly improves encryption performance. Enable AES-NI in your BIOS.
- The witness host in a vSAN stretched cluster does not participate in vSAN encryption. The witness host does not store customer data, only metadata, such as the size and UUID of vSAN object and components.

  > **Note:**
  > If the witness host is an appliance running on another cluster, you can encrypt the metadata stored on it. Enable data-at-rest encryption on the cluster that contains the witness host.
- Establish a policy regarding core dumps. Core dumps are encrypted because they can contain sensitive information. If you decrypt a core dump, carefully handle its sensitive information. ESX core dumps might contain keys for the ESX host and for the data on it.
  - Always use a password when you collect a `vm-support` bundle. You can specify the password when you generate the support bundle from the vSphere Client or using the `vm-support` command.
    The password recrypts core dumps that use internal keys to use keys that are based on the password. You can later use the password to decrypt any encrypted core dumps that might be included in the support bundle. Unencrypted core dumps or logs are not affected.
  - The password that you specify during `vm-support` bundle creation is not persisted in vSphere components. You are responsible for keeping track of passwords for support bundles.
- Ensure that vSAN supports KMS. For more information, see the Supported KMS Vendors.
- Verify KMS is healthy before enabling encryption.

- Ensure that you use highly available KMS clusters.

Set Up the Standard Key Provider

Use a standard key provider to distribute the keys that encrypt the vSAN datastore.

Before you can encrypt the vSAN datastore, you must set up a standard key provider to support encryption. That task includes adding the KMS to vCenter and establishing trust with the KMS. vCenter provisions encryption keys from the key provider.

The KMS must support the Key Management Interoperability Protocol (KMIP) 1.1 standard. See the vSphere Compatibility Matrices for details.

Add a KMS to vCenter

You add a Key Management Server (KMS) to your vCenter system from the vSphere Client.

- Verify that the key server is in the *vSphere Compatibility Matrices* and is KMIP 1.1 compliant.
  - Verify that you have the required privileges: **Cryptographer**.**ManageKeyServers**
- Connecting to a KMS by using only an IPv6 address is not supported.
- Connecting to a KMS through a proxy server that requires user name or password is not supported.

vCenter creates a KMS cluster when you add the first KMS instance. If you configure the KMS cluster on two or more vCenters, make sure you use the same KMS cluster name.

> **Note:**
>
> Do not deploy your KMS servers on the vSAN cluster you plan to encrypt. If a failure occurs, hosts in the vSAN cluster must communicate with the KMS.

- When you add the KMS, you are prompted to set this cluster as a default. You can later change the default cluster explicitly.
- After vCenter creates the first cluster, you can add KMS instances from the same vendor to the cluster.
- You can set up the cluster with only one KMS instance.
- If your environment supports KMS solutions from different vendors, you can add multiple KMS clusters.

1. Navigate to vCenter.
2. Click **Configure** tab.
3. Browse the inventory list and select the vCenter instance.
4. Under Security, click **Key Providers**.
5. Click **Add KMS**, specify the KMS information in the wizard, and click **OK**.

Option	Value
KMS cluster	Select **Create new cluster** for a new cluster. If a cluster exists, you can select that cluster.
Cluster name	Name for the KMS cluster. You can use this name to connect to the KMS if your vCenter instance becomes unavailable.
Server alias	Alias for the KMS. You can use this alias to connect to the KMS if your vCenter instance becomes unavailable.
Server address	IP address or FQDN of the KMS.
Server port	Port on which vCenter connects to the KMS.
Proxy address	Optional proxy address for connecting to the KMS.
Proxy port	Optional proxy port for connecting to the KMS.
User name	Some KMS vendors allow users to isolate encryption keys that are used by different users or groups by specifying a user name and password. Specify a user name only if your KMS supports this functionality, and if you intend to use it.

Option	Value
Password	Some KMS vendors allow users to isolate encryption keys that are used by different users or groups by specifying a user name and password. Specify a password only if your KMS supports this functionality, and if you intend to use it.

Establish a Standard Key Provider Trusted Connection by Exchanging Certificates

After you add the standard key provider to the vCenter system, you can establish a trusted connection.

Add the standard key provider.

The exact process depends on the certificates that the key provider accepts, and on your company policy.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider.
   The KMS for the key provider is displayed.
5. Select the KMS.
6. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
7. Select the option appropriate for your server and follow the steps.

Option	See
vCenter Root CA certificate	Use the Root CA Certificate Option to Establish a Standard Key Provider Trusted Connection.
vCenter Certificate	Use the Certificate Option to Establish a Standard Key Provider Trusted Connection.
Upload certificate and private key	Use the Upload Certificate and Private Key Option to Establish a Standard Key Provider Trusted Connection.
New Certificate Signing Request	Use the New Certificate Signing Request Option to Establish a Standard Key Provider Trusted Connection.

Use the Root CA Certificate Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload your root CA certificate to the KMS.

All certificates that are signed by your root CA are then trusted by this KMS. The root CA certificate that vSphere Virtual Machine Encryption uses is a self-signed certificate that is stored in a separate store in the VMware Endpoint Certificate Store (VECS) on the vCenter system.

> **Note:**
> Generate a root CA certificate only if you want to replace existing certificates. If you do, other certificates that are signed by that root CA become invalid. You can generate a new root CA certificate as part of this workflow.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider with which you want to establish a trusted connection.

   The key server (KMS) for the key provider is displayed.
5. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
6. Select **vCenter Root CA Certificate** and click **Next**.

   The Download Root CA Certificate dialog box is populated with the root certificate that vCenter uses for encryption. This certificate is stored in VECS.

7. Copy the certificate to the clipboard or download the certificate as a file.
8. Follow the instructions from your KMS vendor to upload the certificate to their system.

   Note:
   Some KMS vendors require that the KMS vendor restarts the KMS to pick up the root certificate that you upload.

Finalize the certificate exchange. See Finish the Trust Setup for a Standard Key Provider.

Use the Certificate Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload the vCenter certificate to the KMS.

After the upload, the KMS accepts traffic that comes from a system with that certificate. vCenter generates a certificate to protect connections with the KMS. The certificate is stored in a separate key store in the VMware Endpoint Certificate Store (VECS) on the vCenter system.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider with which you want to establish a trusted connection.

   The key server (KMS) for the key provider is displayed.
5. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
6. Select **vCenter Certificate** and click **Next**.

   The Download Certificate dialog box is populated with the root certificate that vCenter uses for encryption. This certificate is stored in VECS.

   Note:
   Do not generate a new certificate unless you want to replace existing certificates.
7. Copy the certificate to the clipboard or download it as a file.
8. Follow the instructions from your KMS vendor to upload the certificate to the KMS.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Use the New Certificate Signing Request Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that vCenter generate a Certificate Signing Request (CSR) and send that CSR to the KMS.

The KMS signs the CSR and returns the signed certificate. You can upload the signed certificate to vCenter. Using the **New Certificate Signing Request** option is a two-step process. First you generate the CSR and send it to the KMS vendor. Then you upload the signed certificate that you receive from the KMS vendor to vCenter.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider with which you want to establish a trusted connection.

   The key server (KMS) for the key provider is displayed.
5. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
6. Select **New Certificate Signing Request (CSR)** and click **Next**.
7. In the dialog box, copy the full certificate in the text box to the clipboard or download it as a file.

   Use the **Generate new CSR** button in the dialog box only if you explicitly want to generate a CSR.

8. Follow the instructions from your KMS vendor to submit the CSR.
9. When you receive the signed certificate from the KMS vendor, click **Key Providers** again, select the key provider, and from the **Establish Trust** drop-down menu, select **Upload Signed CSR Certificate**.
10. Paste the signed certificate into the bottom text box or click **Upload File** and upload the file, and click **Upload**.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Use the Upload Certificate and Private Key Option to Establish a Standard Key Provider Trusted Connection

Some Key Management Server (KMS) vendors require that you upload the KMS server certificate and private key to the vCenter system.

- Request a certificate and private key from the KMS vendor. The files are X509 files in PEM format.

Some KMS vendors generate a certificate and private key for the connection and make them available to you. After you upload the files, the KMS trusts your vCenter instance.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider with which you want to establish a trusted connection.

   The key server (KMS) for the key provider is displayed.
5. From the **Establish Trust** drop-down menu, select **Make KMS trust vCenter**.
6. Select **KMS certificate and private key** and click **Next**.
7. Paste the certificate that you received from the KMS vendor into the top text box or click **Upload a File** to upload the certificate file.
8. Paste the key file into the bottom text box or click **Upload a File** to upload the key file.
9. Click **Establish Trust**.

Finalize the trust relationship. See Finish the Trust Setup for a Standard Key Provider.

Set the Default Key Provider Using the vSphere Client

You can use the vSphere Client to set the default key provider at the vCenter level.

As a best practice, verify that the Connection Status in the Key Providers tab shows Active and a green check mark.

You must set the default key provider if you do not make the first key provider the default, or if your environment uses multiple key providers and you remove the default one.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider.
5. Click **Set as Default**.

   A confirmation dialog box appears.
6. Click **Set as Default**.

   The key provider displays as the current default.

Finish the Trust Setup for a Standard Key Provider

Unless the **Add Standard Key Provider** dialog prompted you to trust the KMS, you must explicitly establish trust after certificate exchange is complete.

You can complete the trust setup, that is, make vCenter trust the KMS, either by trusting the KMS or by uploading a KMS certificate. You have two options:

- Trust the certificate explicitly by using the **Upload KMS certificate** option.

- Upload a KMS leaf certificate or the KMS CA certificate to vCenter by using the **Make vCenter Trust KMS** option.

  Note:
  If you upload the root CA certificate or the intermediate CA certificate, vCenter trusts all certificates that are signed by that CA. For strong security, upload a leaf certificate or an intermediate CA certificate that the KMS vendor controls.

1. Navigate to vCenter.
2. Click the **Configure** tab.
3. Under **Security**, select **Key Providers**.
4. Select the key provider with which you want to establish a trusted connection.

   The key server (KMS) for the key provider is displayed.
5. Select the KMS.
6. Select one of the following options from the **Establish Trust** drop-down menu.

Option	Action
Make vCenter Trust KMS	In the dialog box that appears, click **Trust**.
Upload KMS certificate	1. In the dialog box that appears, either paste in the certificate, or click **Upload a file** and browse to the certificate file. 2. Click **Upload**.

Enable Encryption on a New vSAN OSA Cluster

You can enable encryption when you configure a new vSAN cluster.

- Required privileges:
  - **Host**.**Inventory**.**EditCluster**
  - **Cryptographer**.**ManageEncryptionPolicy**
  - **Cryptographer**.**ManageKeyServers**
  - **Cryptographer**.**ManageKeys**
- You must have set up a KMS cluster and established a trusted connection between vCenter and the KMS.

1. In the vSphere Client, navigate to an existing cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services.**
4. Click **Data Services** and click **Edit**.
5. Click the **Configure vSAN** button.
6. On the **vSAN capabilites** page, select the **Encryption** check box, and select a KMS cluster.

   Note:
   Make sure the **Erase disks before use** check box is deselected, unless you want to wipe existing data from the storage devices as they are encrypted.
7. On the **Claim disks** page, specify which disks to claim for the vSAN cluster.
   a) Select a flash device to be used for capacity and click the **Claim for capacity tier** icon
   ( ).
   b) Select a flash device to be used as cache and click the **Claim for cache tier** icon
   ( ).
8. Complete your cluster configuration.

Encryption of data-at-rest is enabled on the vSAN cluster. vSAN encrypts all data added to the vSAN datastore.

Generate New Encryption Keys

You can generate new encryption keys, in case a key expires or becomes compromised.

- Required privileges:
  - **Host**.**Inventory**.**EditCluster**
  - **Cryptographer**.**ManageKeys**
- You must have set up a KMS cluster and established a trusted connection between vCenter and the KMS.

The following options are available when you generate new encryption keys for your vSAN cluster.

- If you generate a new KEK, all hosts in the vSAN cluster receive the new KEK from the KMS. Each host's DEK is re-encrypted with the new KEK.
- If you choose to re-encrypt all data using new keys, a new KEK and new DEKs are generated. A rolling disk re-format is required to re-encrypt data.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Click **Data Services** and click **Edit**.
5. In the vSAN is turned ON pane, click the **Generate new encryption keys** button.
6. To generate a new KEK, click **OK**. The DEKs will be re-encrypted with the new KEK.
   - To generate a new KEK and new DEKs, and re-encrypt all data in the vSAN cluster, select the following check box: **Also re-encrypt all data on the storage using new keys**.
   - If your vSAN cluster has limited resources, select the **Allow Reduced Redundancy** check box. If you allow reduced redundancy, your data might be at risk during the disk reformat operation.

Enable vSAN Encryption on Existing vSAN Cluster

You can enable encryption by editing the configuration parameters of an existing vSAN cluster.

- Required privileges:
  - **Host**. **Inventory**. **EditCluster**
  - **Cryptographer**. **ManageEncryptionPolicy**
  - **Cryptographer.ManageKeyServers**
  - **Cryptographer** . **ManageKeys**
- You must have set up a KMS cluster and established a trusted connection between vCenter and the KMS.
- The cluster's disk-claiming mode must be set to manual.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Click **Data Services** and click **Edit**.
5. On the vSAN Services dialog, click **Data-At-Rest encryption**.
6. Select **Wipe residual data** check box to wipe existing data from the storage devices before the data is encrypted.
7. Select a key provider from the drop-down.
8. Click **Data-In-Transit encryption** to enable fault tolerance traffic encryption.
9. Select Default or Custom as the **Rekey interval**. You can specify the interval based on your selection.
10. Select **Allow reduced redundancy** check box to reduce the protection level of your VMs.
11. Click **Apply**.

A rolling reformat of all disks in the storage pool in a vSAN ESA cluster or disk groups in an OSA cluster takes place as vSAN encrypts all data in the vSAN datastore.

vSAN Encryption and Core Dumps

If your vSAN cluster uses data-at-rest encryption, and if an error occurs on the ESX host, the resulting core dump is encrypted to protect data.

Core dumps that are included in the `vm-support` package are also encrypted.

> **Note:**
> Core dumps can contain sensitive information. Follow your organization's data security and privacy policy when handling core dumps.

Core Dumps on ESX Hosts

When an ESX host crashes, an encrypted core dump is generated and the host reboots. The core dump is encrypted with the host key that is in the ESX key cache. What you can do next depends on several factors.

- In most cases, vCenter retrieves the key for the ESX host from the KMS and attempts to push the key to the ESX host after reboot. If the operation is successful, you can generate the `vm-support` package and you can decrypt or re-encrypt the core dump.
- If vCenter cannot connect to the ESX host, you might be able to retrieve the key from the KMS.
- If the host used a custom key, and that key differs from the key that vCenter pushes to the ESX host, you cannot manipulate the core dump. Avoid using custom keys.

Core Dumps and vm-support Packages

When you contact Broadcom Technical Support because of a serious error, your support representative usually asks you to generate a `vm-support` package. The package includes log files and other information, including core dumps. If support representatives cannot resolve the issues by looking at log files and other information, you can decrypt the core dumps to make relevant information available. Follow your organization's security and privacy policy to protect sensitive information, such as host keys.

Core Dumps on vCenter Systems

A core dump on a vCenter system is not encrypted. vCenter already contains potentially sensitive information. At the minimum, ensure that the vCenter is protected. You also might consider turning off core dumps for the vCenter system. Other information in log files can help determine the problem.

Collect a vm-support Package for an ESX Host in an Encrypted vSAN Datastore

If data-at-rest encryption is enabled on a vSAN cluster, any core dumps in the `vm-support` package are encrypted.

Inform your support representative that data-at-rest encryption is enabled for the vSAN datastore. Your support representative might ask you to decrypt core dumps to extract relevant information.

> **Note:**
> Core dumps can contain sensitive information. Follow your organization's security and privacy policy to protect sensitive information such as host keys.

You can collect the package, and you can specify a password if you expect to decrypt the core dump later. The `vm-support` package includes log files, core dump files, and more.

1. Log in to vCenter using the vSphere Client.
2. Click **Hosts and Clusters**, and right-click the ESX host.
3. Select **Export System Logs**.
4. In the dialog box, select **Password for encrypted core dumps**, and specify and confirm a password.
5. Leave the defaults for other options or make changes if requested by Broadcom Technical Support, and click **Finish**.
6. Specify a location for the file.
7. If your support representative asked you to decrypt the core dump in the `vm-support` package, log in to any ESX host and follow these steps.
   a) Log in to the ESX and connect to the directory where the `vm-support` package is located.
      The filename follows the pattern `esx.date_and_time.tgz`.
   b) Make sure that the directory has enough space for the package, the uncompressed package, and the recompressed package, or move the package.
   c) Extract the package to the local directory.
      ```
      vm-support -x *.tgz .
      ```
      The resulting file hierarchy might contain core dump files for the ESX host, usually in `/var/core`, and might contain multiple core dump files for virtual machines.
   d) Decrypt each encrypted core dump file separately.
      ```
      crypto-util envelope extract --offset 4096 --keyfile vm-support-incident-key-file
      --password encryptedZdumpdecryptedZdump
      ```
      `vm-support-incident-key-file` is the incident key file that you find at the top level in the directory.

      `encryptedZdump` is the name of the encrypted core dump file.

      `decryptedZdump` is the name for the file that the command generates. Make the name similar to the *`encryptedZdump`* name.
   e) Provide the password that you specified when you created the `vm-support` package.
   f) Remove the encrypted core dumps, and compress the package again.
      ```
      vm-support --reconstruct
      ```
8. Remove any files that contain confidential information.

Decrypt or Re-Encrypt an Encrypted Core Dump on ESX Host

You can decrypt or re-encrypt an encrypted core dump on your ESX host by using the `crypto-util` CLI.

The ESX host key that was used to encrypt the core dump must be available on the ESX host that generated the core dump.

You can decrypt and examine the core dumps in the `vm-support` package yourself. Core dumps might contain sensitive information. Follow your organization's security and privacy policy to protect sensitive information, such as host keys.

For details about re-encrypting a core dump and other features of `crypto-util`, see the command-line help.

> **Note:**
> `crypto-util` is for advanced users.

1. Log directly in to the ESX host on which the core dump occurred.
   If the ESX host is in lockdown mode, or if SSH access is not enabled, you might have to enable access first.

2. Determine whether the core dump is encrypted.

Option	Description
Monitor core dump	`crypto-util envelope describe vmmcores.ve`
zdump file	`crypto-util envelope describe --offset 4096 zdumpFile`

3. Decrypt the core dump, depending on its type.

Option	Description
Monitor core dump	`crypto-util envelope extract vmmcores.ve vmmcores`
zdump file	`crypto-util envelope extract --offset 4096 zdumpEncrypted zdumpUnencrypted`

Monitoring and Troubleshooting vSAN

Monitoring and Troubleshooting vSAN describes how to monitor and troubleshoot VMware® vSAN™ by using the vSphere Client.

In addition, *Monitoring and Troubleshooting vSAN* explains how to monitor and troubleshoot a vSAN cluster using esxcli and other tools.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we create content using inclusive language.

Intended Audience

This guide is intended for anyone who wants to monitor vSAN operation and performance, or troubleshoot problems with a vSAN cluster. The information in this guide is written for experienced system administrators who are familiar with virtual machine technology and virtual datacenter operations. This manual assumes familiarity with VMware vSphere, including VMware ESX , vCenter, and the vSphere Client.

- For more information about network requirements and network design, see the Designing vSAN Network guide.
- For more information about creating vSAN clusters, see the Planning and Configuring vSAN guide.
- For more information about vSAN features and how to configure a vSAN cluster, see the Administering VMware vSAN guide.

Monitoring the vSAN Cluster

You can monitor the vSAN cluster and all the objects related to it.

You can monitor all of the objects in a vSAN environment, including ESX hosts that participate in a vSAN cluster and the vSAN datastore. For more information about monitoring objects and storage resources in a vSAN cluster, see the vSphere Monitoring and Performance guide.

Monitor vSAN ESA Capacity

You can use the vSAN ESA Capacity page to monitor vSAN ESA effective capacity and vSAN ESA legacy capacity.

Monitor vSAN ESA Effective Capacity

The capacity page displays the effective capacity of the vSAN datastore that includes the total usable capacity, used capacity, and the free usable capacity.

To view effective capacity for a vSAN datastore, enable Auto-RAID using one of these methods:

- Enable vSAN ESA Auto-RAID in the vSAN datastore default storage policy.

- Select the 'Apply Auto-RAID' option to all objects in the vSAN ESA cluster configuration.

For clusters that have data reduction techniques such as vSAN ESA Global Deduplication and compression enabled, you can view the deduplication and compression savings and the deduplication and compression ratio. For more information on the Legacy Capacity View, see Monitor vSAN Capacity.

Note:

The compute client cluster displays vSAN ESA effective capacity for the remote vSAN datastore. When you enable effective capacity on the remote vSAN datastore, the compute client cluster displays total usable capacity, used capacity, and free effective capacity. If you do not enable the effective capacity, the client cluster displays the legacy capacity usage of the remote datastore that includes total capacity, used capacity, and actually written capacity.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Capacity** to view the vSAN effective capacity information.

Term	Description
Total usable capacity	Usable physical storage space in the vSAN datastore, excluding system reservations and RAID redundancy overhead. For more information on vSAN ESA Auto-RAID, see What are vSAN Policies.
Used capacity	Used physical storage space in the vSAN datastore after global deduplication and compression. It also includes the capacity of the system objects such as performance stats objects, native trace objects, file service root objects, iSCSI home objects, data protection namespace objects, FCD namespace objects, and shared user data.
Actually written	Actually used capacity. The amount of data physically written in the vSAN datastore, excluding system reservations, RAID overhead, and space reserved by thick provisioning.
Object reserved	Over reservation for objects created with thick provisioning in the policy. This capacity is not actually used by the objects.
Free usable capacity	Remaining usable capacity available in the vSAN datastore, excluding all system reservations. Free usable capacity = total usable capacity - used capacity.
Total provisioned capacity	Sum of the effective provisioning capacity. This excludes RAID overhead for all the namespace objects.
Data reduction ratio	Ratio of the original data size to the actual space used after applying compression and vSAN ESA global deduplication.
Deduplication ratio	Ratio of the original size of the deduplication-eligible objects to the actual space used after vSAN ESA global deduplication.
Compression ratio	Ratio of the original size of the compressible data to the actual space used after compression.
Thin provisioning saving ratio	Ratio of the total provisioned capacity to the actual used capacity.
Overprovisioning ratio	Measures how much the total provisioned storage exceeds the total usable capacity available in the vSAN datastore.
Snapshot saving ratio	Measures the data reduction of snapshots. It is the ratio of the original snapshot size that would be consumed if every snapshot was fully inflated to the actual physical capacity currently used by all snapshots.
Fully inflated snapshot capacity	Total storage that all snapshots can use at full logical size without data reduction techniques such as local snapshots or vSAN ESA global deduplication.

Term	Description
vSAN snapshot used capacity	Total storage used by all snapshots in the cluster including the shared data blocks.

Use the following formula to calculate the total usable capacity:

Total Usable Capacity = (Total Raw Capacity - System Usage) / RAID Factor.

Term	Description
Total Raw Capacity	Total claimed disk volumes on the vSAN ESA datastore.
System Usage	Includes System Overhead and System Reservation: • System Overhead includes the following: – Global Metadata overhead - The capacity of the performance tier. – File system overhead - vSAN on-disk format overhead that may take up the capacity drives. – vSAN ESA objects overhead - vSAN ESA uses the capacity to store object metadata and to provide high performance. – Dedup overhead - Overhead to get the benefits of global deduplication in vSAN ESA cluster. This data is visible only if you enable vSAN ESA global deduplication. • System Reservations is the minimum space required for vSAN operations.
RAID Factor	Determined based on the cluster configuration. The RAID factor can be 1.5 for a standard cluster with auto managed RAID policy. For more information on RAID factor, see What are vSAN Policies.

For example, if the total raw capacity is 40 TB, the system overhead (that includes system overhead and system reservations) is 14.5 TB, and the RAID factor is 1.5, the total usable capacity available is 17 TB (40-14.5)/1.5.

Monitor vSAN ESA Legacy Capacity

You can monitor the capacity of the vSAN ESA datastore, vSAN Direct storage, and Persistent Memory (PMem) storage.

If you do not enable vSAN ESA Auto-RAID in the default storage policy and do not select Apply Auto-RAID to all objects, the Legacy Capacity View becomes the default. This view displays the physical capacity information.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Capacity** to view the vSAN capacity information.

   If vSAN ESA effective capacity is the default view, click **Legacy Capacity View** to view the capacity information.

Using vSAN ESA legacy capacity page, you can analyze and view the What if analysis and capacity breakdown at the vSAN ESA cluster level.

Note:

The effective free space calculation in What if analysis is based on the free usable capacity available in Effective Capacity if:

- Apply Auto-RAID to all objects is selected in the vSAN ESA cluster configuration.
- vSAN ESA Auto-RAID is enabled in the selected storage policy.

For more details on the vSAN ESA legacy capacity view, see Monitor vSAN Capacity.

Monitor vSAN Capacity

You can monitor the capacity of the vSAN datastore, vSAN Direct storage, and Persistent Memory (PMem) storage.

You can analyze usage and view the capacity breakdown at the cluster level. The vSAN OSA capacity view is applicable for vSAN ESA 9.1 or the earlier verions.

The cluster Summary page includes a summary of vSAN capacity. You also can view more detailed information in the Capacity monitor.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Capacity** to view the vSAN capacity information.

The Capacity Overview displays the storage capacity of the vSAN datastore, including free space available, used space, and the space that is actually written and physically consumed on the vSAN disks. For clusters that have deduplication and compression enabled, you can view the deduplication and compression savings and the deduplication and compression ratio.

Terms	Description
Free space on disks	Raw free space on the physical disks, without considering the storage policy and overhead.
Used	Total written physical space
Actually written	Actually used capacity. This capacity is displayed when deduplication or compression are not enabled.
Reservations and Alerts	Reserved capacity for a vSAN cluster for internal operations. Customize the alert thresholds to receive warning and error alerts.

What if analysis enables you to estimate the effective free space while keeping the deduplication ratio as 1. **Effective free space** is an estimate of free space available based on the selected storage policy. The effective free space typically is smaller than the free space available on the disks. This can be due to the cluster topology or the distribution of space across fault domains. For example, consider a cluster with 100 GB free space available on the disks. However, 100 GB cannot be provisioned as a single 100 GB object due to the distribution of free space across fault domains. If there are three fault domains and each fault domain has 33 GB free space, then the largest object that you can create with FTT 1 is 33 GB.

Oversubscription reports the vSAN capacity required if all the thin provisioned VMs and user objects are used at full capacity. It shows a ratio of the required usage compared with the total available capacity. While calculating the oversubscription, vSAN includes all the available VMs, user objects, and storage policy overhead, and does not consider the vSAN namespace and swap objects.

Note:

Persistent Memory (PMem) storage does not support What if analysis and oversubscription.

- The Usage breakdown before deduplication and compression displays the amount of storage space used by VMs, user objects, and the system. You can view a pie chart that represents the different usage categories. Click the pie chart to view the details of the selected category.

Following are the different usage categories available:

Category	Description
VM (user objects) usage	Displays the following: • VM home objects - Usage of VM namespace object. • Swap objects - Usage of VM swap files. • VMDK - Capacity consumed by VMDK objects that reside on the vSAN datastore that can be categorized as primary data and replica usage. Primary data includes the actual user data written into the physical disk which does not include any overhead. Replica usage displays the RAID overhead for the virtual disk. • Shared user data - Usage of shared data after deduplication for all the deduplicated objects. • VM memory snapshots - Usage of memory snapshot file for VMs. • Block container volumes (attached to a VM) - Capacity consumed by the container objects that are attached to a VM. • vSphere replication persistent state file - vSAN object used to store the persistent state file (PSF) at source site.
Non-VM (user objects) usage	Displays iSCSI objects, block container volumes that are not attached to VM, user-created files, ISO files, VM templates, files shares, file container volumes, and vSAN objects used by the vSphere replication service at the target site.
System usage	Displays the following: • Performance management objects - Capacity consumed by objects created for storing performance metrics when you enable the performance service. • File system overhead - vSAN on-disk format overhead that may take up on the capacity drives. • ESA object overhead - vSAN ESA uses the capacity to store object metadata and to provide high performance. • Checksum overhead - Overhead to store all the checksums in vSAN OSA. • Dedup & compression overhead - Overhead to get the benefits of deduplication and compression in vSAN OSA. This data is visible only if you enable deduplication and compression. • Operations usage - Temporary space usage in a cluster. The temporary space usage includes temporary capacity used for rebalance operations or moving objects due to FTT changes. • Native trace objects - Capacity consumed by objects created for storing vSAN traces.

Note:
PMEM only supports VMDK, Non-Volatile Dual In-line Memory Module (NVDIMM), and file system overhead.

When you enable deduplication and compression, it might take several minutes to hours for capacity updates to be reflected in the Capacity monitor, as disk space is reclaimed and reallocated. For more information about deduplication and compression, see Using Deduplication and Compression in vSAN OSA Cluster.

In vSAN ESA, Usage by Snapshots displays the snapshot usage by the vSAN datastore. You can delete one or more snapshots and free the used space, thus managing space consumption. To delete a snapshot, right-click the virtual machine > **Snapshots** > **Manage Snapshots**. Click **Delete** to delete a snapshot. Click **Delete All Snapshots** to delete all the snapshots of the selected VM.

The following are the different usage snapshots available:

Snapshot	Description
Container volume snapshots	Displays the container volume snapshot usage in the vSAN datastore.
VMDK snapshots	Displays the VMDK snapshot usage in the vSAN datastore.
vSAN file share snapshots	Displays the file share snapshot usage in the vSAN datastore.
Current data	Displays the usage data that is not included in the snapshot usage data. You can calculate the current data by subtracting the total snapshot usage from the total used space.

You can check the history of capacity usage in the vSAN datastore. Click **Capacity History**, specify a date or custom time and date range, and click **Show Results**.

The Capacity monitor displays two thresholds represented as vertical markers in the bar chart:

- Operations threshold - Displays the space vSAN requires to perform internal operations in the cluster. If the used space reaches beyond that threshold, vSAN might not be able to operate properly.
- Host rebuild threshold - Displays the space vSAN requires to tolerate one ESX host failure. If the used space reaches beyond the host rebuild threshold and the host fails, vSAN might not successfully restore all data from the failed host.

If you enable reserved capacity, the Capacity monitor displays the following:

- Operations reserve - Reserved space in the cluster for internal operations.
- Host rebuild reserve - Reserved space for vSAN to be able to repair objects in case of single ESX host failure. The Capacity monitor displays the host rebuild threshold only when the host rebuild reserve is enabled. For more information, see .

If the resynchronization of objects is in progress in a cluster, vSAN displays the capacity used in the capacity chart as operations usage. In case there is enough free space in the cluster, vSAN might use more space than the operations threshold for the resyncing operations to complete faster.

Click **Configure** tab to enable the capacity reserve. You can also click **Configure** > **vSAN** > **Services** to enable the capacity reserve. For more information on configuring the reserved capacity, see Configure Reserved Capacity for vSAN Cluster.

In a cluster, if there is more utilization than the host rebuild threshold and the reserved capacity is not enabled, the capacity chart turns yellow as a warning. If the most consumed host fails, vSAN cannot recover the data. If you enable the host rebuild reserve, the capacity chart turns yellow at 80% of the host rebuild threshold. If the used space reaches beyond the operations threshold and the reserved capacity is not enabled, vSAN cannot perform or complete operations such as rebalance, resync object components due to policy changes, and so on. In that case, the capacity chart turns red to indicate that the disk usage exceeds the operations threshold. For more information about capacity reserve, see About Reserved Capacity in vSAN Cluster.

Monitor Physical Devices in vSAN Cluster

You can monitor ESX hosts, cache devices, and capacity devices used in the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Disk Management** to review all ESX hosts, cache devices, and capacity devices in the cluster. The physical location is based on the hardware location of cache and capacity devices on ESX hosts.

   The cache and capacity devices are available only in the vSAN OSA cluster.

Monitor Devices that Participate in vSAN Datastores

Verify the status of the devices that back up the vSAN datastore. You can check whether the devices experience any problems.

1. In the vSphere Client, navigate to the vSAN datastore.
2. Select the vSAN datastore.
3. Click the **Configure** tab.

   You can view general information about the vSAN datastore, including capacity, capabilities, and the default storage policy.

Monitor Virtual Objects in vSAN Cluster

You can view the status of virtual objects in the vSAN cluster.

When one or more ESX hosts are unable to communicate with the vSAN datastore, the information about virtual objects might not be displayed.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under **vSAN**, select **Virtual Objects** to view the corresponding virtual objects in the vSAN cluster.
4. Click

   to filter the virtual objects based on name, type, storage policy, and UUID.
   a) Select the check box on one of the virtual objects and click **View Placement Details** to open the Physical Placement dialog box. You can view the device information, such as name, identifier or UUID, number of devices used for each virtual machine, and how they are distributed across ESX hosts.
   b) On the Physical Placement dialog box, select the **Group components by host placement** check box to organize the objects by ESX host and by disk.

      Note:

      At the cluster level, the Container Volumes filter displays detached container volumes. To view attached volumes, expand the VM to which the container is attached.
5. Select the check box of the attached block type or file volumes and click **View Performance**. You can use the vSAN cluster performance charts to monitor the workload in your cluster. For more information on the vSAN cluster performance charts, see View vSAN Cluster Performance.
6. Select the check box on one of the container volumes and click **View Container Volume**. For more information about monitoring container volumes, see Monitor Container Volumes in vSAN Cluster.
7. Select the check box on one of the file volumes and click **View File Share**. For more information about file volume, see View vSAN File Shares.

Monitor Container Volumes in vSAN Cluster

You can view the status of the container volumes in the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under **Cloud Native Storage**, select **Container Volumes** to view the container volumes in the vSAN cluster. You can view information about the volume name, label, datastore, compliance status, health status, and capacity quota.
4. Click

   to view the following:
   - Click the **Basics** tab to view the volume details such as volume type, ID, datastore, storage policy, compliance, and health status.

- Click the **Kubernetes objects** tab to view Kubernetes related data such as Kubernetes cluster, namespace, pod, persistent volume claim, labels, and so on.
- Click the **Physical Placement** tab to view the type, ESX host, cache, and capacity disk of the virtual object components.
- Click the **Performance** tab to view the performance of the container volumes.

5. Select the check box for the volumes that have an out-of-date policy status that you can view under **Compliance Status** column. Click **Reapply Policy** to reapply the policy on the selected volumes.
6. Select the check box for the container volume you want to delete and click **Delete**.
7. Click **Migrate** to migrate a container to a different datastore.
8. Use the **Add Filter** option to add filters to the container volumes.

Migrate Container Volumes in vSAN Cluster

You can migrate container volumes to a destination datastore in the vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under **Cloud Native Storage**, select **Container Volumes** to view the container volumes in the vSAN cluster.
4. Click **Migrate** to migrate a container to a different datastore.
5. In the Migrate volume dialog box, you can view the volume storage policy and actual size of the volume.
6. Filter and select the destination datastore.
7. Select **I acknowledge that** check box as volume migration is an advanced operation.
8. Click **Migrate**.

About Reserved Capacity in vSAN Cluster

vSAN requires capacity for its internal operations.

For a cluster to be able to tolerate a single ESX host failure, vSAN requires free space to restore the data of the failed host. You can reserve the amount of the largest host to ensure that that the data can be recreated. These values are represented as thresholds in the Capacity Monitor page:

- Operations threshold - Displays the space vSAN requires to run its internal operations such as rebalancing, host rebuild, and so on in the cluster. If the used space exceeds the operations threshold, vSAN might not operate properly.
- Host rebuild threshold - Displays the space vSAN requires to tolerate one host failure. If the used space exceeds the host rebuild threshold and the host fails, vSAN might not successfully restore all data from the failed host.

For more information on the capacity thresholds, see Monitor vSAN Capacity.

> **Note:**
>
> The reserved capacity is not supported on a vSAN stretched cluster, cluster with fault domains and nested fault domains, two-node cluster, or if the number of ESX hosts in the cluster is less than four.

vSAN provides you the option to reserve the capacity in advance so that it has enough free space available to perform internal operations and to repair data back to compliance following a single host failure. By enabling reserve capacity in advance, vSAN prevents you from using the space to create workloads and intends to save the capacity available in a cluster. By default, the reserved capacity is not enabled.

If there is enough free space in the vSAN cluster, you can enable the operations reserve and/or the host rebuild reserve.

- Operations reserve - Reserved space in the cluster for vSAN internal operations.
- Host rebuild reserve - Reserved space for vSAN to be able to repair in case of a single host failure.

These soft reservations prevent the creation of new VMs or powering on VMs if such operations consume the reserved space. Once the reserved capacity is enabled, vSAN does not prevent powered on VM operations, such as I/O from the

guest operating system or applications from consuming the space even after the threshold limits are reached. After you enable the reserved capacity, you must monitor the disk space health alerts and capacity usage in the cluster and take appropriate actions to keep the capacity usage below the threshold limits.

To enable reserved capacity for the host rebuild, you must first enable the operations reserve. When you enable operations reserve, vSAN reserves 5% additional capacity in the operations reserve as a buffer to ensure you have time to react to the capacity fullness before the actual threshold is reached.

vSAN indicates when the capacity usage is high in a cluster. The indications can be in the form of health alerts, capacity chart turning yellow or red, and so on. Due to the reservation, vSAN might not have enough free space left. This results in the inability to create VMs or VM snapshots, creating or extending virtual disks, and so on.

Note:
You cannot enable reserved capacity, if the cluster is at a capacity higher than the specified threshold.

Capacity Reservation Considerations

Following are the considerations if you enable reserved capacity:

- When you enable reserved capacity with the host rebuild reserve and place a host into maintenance mode, the host might not come back online. In this case, vSAN continues to reserve capacity for another host failure, on top of the one already in maintenance mode. If the capacity usage exceeds the host rebuild threshold, this can cause operations to fail.
- When you enable reserved capacity with the host rebuild reserve and a host fails, vSAN might not start repairing the affected objects until the repair timer expires. During this time, vSAN continues to reserve capacity for another host failure. This can cause failure of operations if the capacity usage is above the current host rebuild threshold, after the first host failure. After the repairs are complete, you can deactivate the reserved capacity for the host rebuild reserve if the cluster does not have the capacity for another host failure.

Configure Reserved Capacity for vSAN Cluster

You can configure reserved capacity for a vSAN cluster to reserve capacity for internal operations.

You can also configure reserve capacity to reserve capacity for data repair following a single ESX host failure. Ensure that you have the following required privileges: **Host.Inventory.EditCluster** and **Host.Config.Storage**.

Verify that the vSAN cluster:

- Is not configured as a vSAN stretched cluster or two-node cluster.
- Has no fault domains and nested fault domains created.
- Has a minimum of four ESX hosts.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Click to edit the Reservations and Alerts.
5. Click to activate or deactivate the operations reserve. On enabling the operations reserve, vSAN ensures that the cluster has enough space to complete the internal operations such as rebalancing, host rebuild, and so on.
6. Click to enable or deactivate the ESX host rebuild reserve. On enabling the host rebuild reserve, vSAN provides the reservation of space to repair data back to compliance following a single host failure. You can enable the host rebuild

reserve only after you enable the operations reserve. After enabling, if you deactivate the operations reserve, the host rebuild reserve gets automatically deactivated.

7. Select **Customize alerts**. You can set a customized threshold to receive warning and error alerts. The threshold percentage is calculated based on the available capacity, which is the difference between the total capacity and the reserved capacity. If you do not set a customized value, vSAN uses the default thresholds to generate alerts.
8. Click **Apply**.

About vSAN Cluster Resynchronization

You can monitor the status of virtual machine objects that are being resynchronized in the vSAN cluster.

When a hardware device, ESX host, or network fails, or if a ESX host is placed into maintenance mode, vSAN initiates resynchronization in the vSAN cluster. However, vSAN waits 60 minutes for the failed components to come back online before initiating resynchronization tasks.

The following events trigger resynchronization in the cluster:

- Editing a virtual machine (VM) storage policy. When you change VM storage policy settings, vSAN might initiate object recreation and subsequent resynchronization of the objects.
  Certain policy changes such as changing the stripe width might cause vSAN to create another version of an object and synchronize it with the previous version. When the synchronization is complete, the original object is discarded.
  vSAN ensures that VMs continue to run and are not interrupted by this process. This process might require additional temporary capacity.
- Restarting an ESX host after a failure.
- Recovering ESX hosts from a permanent or long-term failure. If an ESX host is unavailable for more than 60 minutes (by default), vSAN creates copies of data to recover the full policy compliance.
- Evacuating data by using the full data migration mode before you place a ESX host in maintenance mode.
- Exceeding the utilization threshold of a capacity device. Resynchronization is triggered when capacity device utilization in the vSAN cluster approaches or exceeds the threshold level of 80 percent.

If a VM is not responding due to latency caused by resynchronization, you can throttle the IOPS used for resynchronization. For more information, see the Broadcom knowledge base article 326830.

Monitor the Resynchronization Tasks in vSAN Cluster

To evaluate the status of objects that are being resynchronized, you can monitor the resynchronization tasks that are currently in progress.

Verify that ESX hosts in your vSAN cluster are running ESX 9.1 or later.

1. In the vSphere Client, navigate to the cluster.
2. Select the **Monitor** tab.
3. Under **vSAN** select **Resyncing Objects**.
4. Track the progress of resynchronization of virtual machine objects.

   The object repair time defines the time vSAN waits before repairing a non-compliant object after placing an ESX host in a failed state or maintenance mode. The default setting is 60 minutes. To change the setting, edit the Object Repair Timer (**Configure > vSAN > Services > Advanced Options**).

   You can also view the following information about the objects that are resynchronized:

Objects	Description
Total resyncing objects	Total number of objects to be resynchronized in the vSAN cluster.
Bytes left to resync	Data (in bytes) that is remaining before the resynchronization is complete.

Objects	Description
Total resyncing ETA	Estimated time left for the resynchronization to complete. The objects to be resynchronized are categorized as active, queued, and suspended. The objects that are actively synchronizing fall in the active category. The objects that are in the queue for resynchronization are the queued objects. The objects that were actively synchronizing but are now in the suspended state falls in the suspended category.
Scheduled resyncing	Remaining number of objects to be resynchronized. You can classify scheduled resyncing into two categories: scheduled and pending. The scheduled category displays the objects that are not resyncing because the delay timer has not expired. Resynchronization of objects starts once the timer expires. The pending category displays the objects with the expired delay timer that cannot be resynchronized. This can be due to insufficient resources in the current cluster or the vSAN FTT policy set on the cluster not being met.

You can also view the resynchronization objects based on various filters such as **Intent** and **Status**. Using **Show first**, you can modify the view to display the number of objects.

About vSAN Cluster Rebalancing

When any capacity device in your cluster reaches 80 percent full, vSAN automatically rebalances the data among the other devices in the cluster.

The vSAN cluster rebalancing continues until the space and components available on all capacity devices is below the threshold. Cluster rebalancing evenly distributes resources across the cluster to maintain consistent performance and availability.

The following operations can cause disk capacity to reach 80% and initiate cluster rebalancing:

- Hardware failures occur on the cluster.
- ESX hosts are placed in maintenance mode with the **Evacuate all data** option.
- ESX hosts are placed in maintenance mode with **Ensure data accessibility** when objects assigned FTT=0 reside on the host.

> **Note:**
> To provide enough space for maintenance and reprotection, and to minimize automatic rebalancing events in the vSAN cluster, consider enabling Operations Reserve or keeping 30-percent capacity available at all times.

Configure Automatic Rebalance in vSAN Cluster

vSAN automatically rebalances data on the disks by default. You can configure settings for automatic rebalancing.

Your vSAN cluster can become unbalanced based on the space or component usage for many reasons such as when you create objects of different sizes, when you add new ESX hosts or capacity devices, or when objects write different amounts of data to the disks. If the cluster becomes unbalanced, vSAN automatically rebalances the disks. Based on the space or component usage, this operation moves components from over-utilized disks to under-utilized disks.

You can enable or deactivate automatic rebalance, and configure the variance threshold for triggering an automatic rebalance. If any two disks in the cluster have a variance in capacity or component usage that exceeds the rebalancing threshold, vSAN begins rebalancing the cluster.

Disk rebalancing can impact the I/O performance of your vSAN cluster. By default the rebalance threshold is set at 30 percentage and ensures that the cluster remains relatively balanced without significantly impacting the performance. If the cluster becomes severely imbalanced, such as after adding one or more hosts or disks, temporarily using a lower threshold of 10 or 20 percentage makes the cluster evenly balanced. This must be done during off-peak periods to minimize the performance impact during the rebalancing activity. Once the rebalancing is complete, you can change the threshold back to the default 30 percentage.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, select **Services**.
4. Under **Advanced Options** > click **Edit**.
5. Click to enable or deactivate Automatic Rebalance.
   You can change the variance threshold to any percentage from 10 to 75.

You can use the vSAN Health to check the disk balance. Expand the Cluster category, and select vSAN **Disk Balance**.

Using the vSAN Default Alarms

You can use the default vSAN alarms to monitor the cluster, ESX hosts, and existing vSAN licenses.

The default alarms are automatically triggered when the events corresponding to the alarms are activated or if one or all the conditions specified in the alarms are met. You cannot edit the conditions or delete the default alarms. To configure alarms that are specific to your requirements, create custom alarms for vSAN. See Creating a vCenter Alarm for a vSAN Event.

For information about monitoring alarms, events, and editing existing alarm settings, see the vSphere Monitoring and Performance guide.

View vSAN Default Alarms

Use the default vSAN alarms to monitor your cluster, ESX hosts, analyze any new events, and assess the overall cluster health.

1. In the vSphere Client, navigate to the cluster.
2. Click **Configure** and then click **Alarm Definitions**.
3. Click

   next to the **Alarm Name** and type **vSAN** in the search box to display the alarms that are specific to vSAN.

   Type vSANHealth Service Alarm to search for vSAN health service alarms.

   The default vSAN alarms are displayed.
4. From the list of alarms, click each alarm to view the alarm definition.

View vSAN Network Alarms

vSAN network diagnostics queries the latest network metrics and compares the metrics statistics with the defined threshold values.

The vSAN performance service must be turned on.

If the value reaches above the threshold that you have set, vSAN network diagnostics raises an alarm. You must acknowledge and manually reset the triggered alarms to green after fixing the network issues.

1. In the vSphere Client, navigate to a host in the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Performance**.
4. Select **Physical Adapters**, and select a NIC. Select a time range for your query. vSAN displays performance charts for the physical NIC (pNIC), including throughput, packets per second, and packets loss rate.
5. Select
   
   In the Threshold settings dialog box, enter a threshold value to receive warning and error alert.
6. Click **Save**.

vSAN displays the performance statistics of all the network I/Os in use. vSAN network diagnostics result appears in the vCenter alerts. The redirection to the related performance charts is available in the vSAN network alerts generated by the network diagnostics service.

Using the VMkernel Observations for Creating vSAN Alarms

VMkernel Observations (VOBs) are system events that you can use to set up vSAN alarms.

vSAN alarms are used for monitoring and troubleshooting performance and networking issues in the vSAN cluster. In vSAN, these events are known as observations.

VMware ESX Observation IDs for vSAN

Each VOB event is associated with an identifier (ID). Before you create a vSAN alarm in the vCenter, you must identify an appropriate VOB ID for the vSAN event for which you want to create an alert. You can create alerts in the VMware ESX Observation Log file (`vobd.log`). For example, use the following VOB IDs to create alerts for any device failures in the cluster.

- `esx.problem.vob.vsan.lsom.diskerror`
- `esx.problem.vob.vsan.pdl.offline`

To review the list of VOB IDs for vSAN, open `/usr/lib/vmware/hostd/extensions/hostdiag/locale/en/event.vmsg` file located on your ESX host in the `/var/log` directory. The log file contains the following VOB IDs that you can use for creating vSAN alarms.

Table 1021: VOB IDs for vSAN

VOB ID	Description
esx.audit.vsan.clustering.enabled	The vSAN clustering service is enabled.
esx.clear.vob.vsan.pdl.online	The vSAN device has come online.
esx.clear.vsan.clustering.enabled	The vSAN clustering service is enabled.
esx.clear.vsan.vsan.network.available	vSAN has one active network configuration.
esx.clear.vsan.vsan.vmknic.ready	A previously reported vmknic has acquired a valid IP.
esx.problem.vob.vsan.lsom.componentthreshold	vSAN reaches the near node component count limit.
esx.problem.vob.vsan.lsom.diskerror	A vSAN device is in a permanent error state.
esx.problem.vob.vsan.lsom.diskgrouplimit	vSAN fails to create a disk group.
esx.problem.vob.vsan.lsom.disklimit	vSAN fails to add devices to a disk group.
esx.problem.vob.vsan.lsom.diskunhealthy	vSAN disk is unhealthy.
esx.problem.vob.vsan.pdl.offline	A vSAN device is offline.
esx.problem.vsan.clustering.disabled	vSAN clustering services are not enabled.
esx.problem.vsan.lsom.congestionthreshold	vSAN device memory or SSD congestion has been updated.

VOB ID	Description
esx.problem.vsan.net.not.ready	A vmknic is added to vSAN network configuration without a valid IP address. This happens when the vSAN network is not ready.
esx.problem.vsan.net.redundancy.lost	The vSAN network configuration does not have the required redundancy.
esx.problem.vsan.no.network.connectivity	vSAN does not have existing networking configuration, which is in use.
esx.problem.vsan.vmknic.not.ready	A vmknic is added to the vSAN network configuration without a valid IP address.
esx.problem.vob.vsan.lsom.devicerepair	The vSAN device is offline and in a repaired state because of I/O failures.
esx.problem.vsan.health.ssd.endurance	One or more vSAN disks exceed the warning usage of estimated endurance threshold.
esx.problem.vsan.health.ssd.endurance.error	A vSAN disk exceeds the estimated endurance threshold.
esx.problem.vsan.health.ssd.endurance.warning	A vSAN disk exceeds 90% of its estimated endurance threshold.

Creating a vCenter Alarm for a vSAN Event

You can create alarms to monitor events on the selected vSAN object, including the cluster, ESX hosts, datastores, networks, and virtual machines.

You must have the required privilege level of `Alarms.Create Alarm` or `Alarm.Modify Alarm`

1. In the vSphere Client, navigate to the cluster.
2. On the **Configure** tab, select **Alarm Definitions** and click **Add**.
3. In the Name and Targets page, enter a name and description for the new alarm.
4. From the **Target type** drop-down menu, select the type of inventory object that you want this alarm to monitor and click **Next**.

   Depending on the type of target that you choose to monitor, the summary that follows the **Targets**, change.
5. In the Alarm Rule page, select a trigger from the drop-down menu.

   The combined event triggers are displayed. You can set the rule for a single event only. You must create multiple rules for multiple events.

6. Click **Add Argument** to select an argument from the drop-down menu.
   a) Select an operator from the drop-down menu.
   b) Select an option from the drop-down menu to set the threshold for triggering an alarm.
   c) Select severity of the alarm from the drop-down menu. You can set the condition to either **Show as Warning** or **Show as Critical**, but not for both. You must create a separate alarm definition for warning and critical status.
7. Select **Send email notifications**, to send email notifications when alarms are triggered.
   a) Select the **Repeat** check box if you want to repeat the alarm in minutes at the specified interval.
   b) In the **Subject** text box, enter the alarm name and target name.
   c) In the **Email to** text box, enter recipient addresses. Use commas to separate multiple addresses.
8. Select **Send SNMP traps** to send traps when alarms are triggered on a vCenter instance.
9. Select the **Repeat** check box if you want to repeat the alarm in minutes at the specified interval.
10. Select **Run script** to run scripts when alarms are triggered.
11. Select the **Repeat** check box if you want to repeat the alarm in minutes at the specified interval.
12. Select an advanced action from the drop-down menu. You can define the advanced actions for virtual machine and ESX hosts. You can add multiple advanced actions for an alarm.
13. Click **Next** to set the Reset Rule.
14. Select **Reset the alarm to green** and select a trigger and a condition.
15. Click **Next** to review the alarm definition.
16. Click the **Repeat actions every** drop-down if you want to repeat the alarm in minutes at the specified interval.
17. Select **Enable this alarm** to enable the alarm and click **Create**.

The alarm is configured.

Monitoring vSAN Health

You can check the overall health of the vSAN cluster, including hardware compatibility and networking configuration and operations.

You can also check the advanced vSAN configuration options, storage device health, and virtual machine object health.

About the vSAN Health

Use the vSAN health to monitor the health of your vSAN cluster.

You can use the vSAN health to monitor the status of cluster components, diagnose issues, and troubleshoot problems. The health findings cover hardware compatibility, network configuration and operation, advanced vSAN configuration options, storage device health, and virtual machine objects.

You can use Overview to monitor the core health issues of your vSAN cluster. You can also view the following:

- Cluster health score based on the health findings
- View the health score trend for 24 hours
- View the health score trend for a particular period

Ensure that the **Historical Health Service** is enabled to view details of the Health score trend. Click **View Details** in the Health score trend chart to examine the health state of the cluster for a selected time point within 24 hours. Use **Custom** to customize the time range as per your requirement. If you disable **Historical Health Service**, the vSAN Health score trend and the past health checks data gets deleted.

You can use the vSAN Health findings to diagnose issues, troubleshoot problems, and remediate the problems.

The health findings are classified as follows:

- **Unhealthy** – Critical or important issue(s) being detected that needs attention.

- **Healthy** – There are no issues found that needs attention.
- **Info** – Health findings which may not impact the cluster running state but important for awareness.
- **Silenced** – Health findings have been silenced without triggering vSAN health alarm by intention.

To troubleshoot an issue, you can sort the findings by root cause to resolve the primary issues initially and then verify if the impacted issues can also be resolved.

vSAN periodically retests each health finding and updates the results. To run the health findings and update the results immediately, click the **Retest** button.

If you participate in the Customer Experience Improvement Program (CEIP), you can run health findings and send the data to VMware for advanced analysis. Click **Retest with Online health** and then click **OK**. Online notifications is enabled by default if the vCenter can connect to VMware Analytics Cloud without enrolling CEIP. If you do not want to participate in CEIP, you can still receive vSAN health notifications for software and hardware issues using Online notifications.

Viewing vSAN Health History

The vSAN health history helps you examine health issues by querying the historical health records. You can only view the historical health data of a cluster. By default, the health history is enabled. To deactivate the health history, select the cluster and navigate to the **Configure** > **vSAN** > **Services** > **Historical Health Service** > **Edit**. Use the toggle **Enable vSAN Historical Health Service** and click **Apply** to deactivate the health history. If you deactivate the health history, all the health data collected on the vCenter database gets purged. The database stores the health data for up to 30 days depending on the available capacity.

Using the Health view, you can view the health history for a selected time range. The start date of the time range must not be earlier than 30 days from the current date. The end date must not be later than the current date. Based on your selection, you can view the historical health findings. Click **View History Details** to view the history of a health finding within a selected time period. The historical data is displayed as a graphical representation with green circles, yellow triangles, and red squares showing success, warning, and failure respectively. The detailed information about each health finding result is displayed in a table.

Using vSAN Support Insight

vSAN support insight is a platform that helps you maintain a reliable and consistent compute, storage, and network environment. VMware support uses the vSAN support insight to monitor the vSAN performance diagnostics and resolve performance issues. vSAN uses Customer Experience Improvement Program (CEIP) to send data to VMware for analysis on a regular basis. To deactivate CEIP, click the icon next to **vSphere Client** > **Administration** > Deployment > **Customer Experience Improvement Program** > **Leave Program**.

Check vSAN Health

You can view the status of vSAN health findings to verify the configuration and operation of your vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under **vSAN**, select **Health** to review the vSAN health finding.
4. Under Health findings, perform the following:
   - Click **Unhealthy** to view the issues and the details. Click **Troubleshoot** to troubleshoot and fix an issue. You can sort the findings by root cause to resolve the primary issues and then verify if the impacted issues can be resolved. Click the **Ask VMware** button to open a knowledge base article that describes the health finding and provides information about how to resolve the issue. You can also view the status history of the health finding for a given period using **History Details** tab.
   - Click **View History Details** to identify the status history of the health finding for a particular time period. The default time period is 24 hours. You can also customize the time period as per your requirement. The status of an unhealthy finding is displayed in yellow or red.
   - You can click **Silence alert** on a health finding, so it does not display any warnings or failures.

- Click **All** to view health findings that are healthy. Click **View Current Result** to view the current status of the health finding. Click **View History Details** to identify the status history of the health finding for a particular time period. The status is displayed in green. You can also view the status history of the health finding for a given period using **History Details** tab.

Monitor vSAN from ESX Host Client

You can monitor vSAN health and basic configuration through the ESX host client.

The ESX host client is a browser-based interface for managing a single ESX host. It enables you to manage the host when vCenter is not available. The host client provides tabs for managing and monitoring vSAN at the host level.

- The **vSAN** tab displays basic vSAN configuration.
- The **Hosts** tab displays the ESX hosts participating in the vSAN cluster.
- The **Health** tab displays host-level health findings.

1. Open a browser and enter the IP address of the host.
   The browser redirects to the login page for the host client.
2. Enter the username and password for the host, and click **Login**.
3. In the host client navigator, click **Storage**.
4. In the main page, click the vSAN datastore to display the Monitor link in the navigator.
5. Click the tabs to view vSAN information for the host.
   a) Click the **Events** tab to display the ESX host events.
   b) Click the **vSAN** tab to display basic vSAN configuration.
   c) Click the **Hosts** tab to display the ESX hosts participating in the vSAN cluster.
   d) Click the **Health** tab to display host-level health findings.
6. (Optional) On the **vSAN** tab, click **Edit Settings** to correct configuration issues at the host level.
   Select the values that match the configuration of your vSAN cluster, and click **Save**.

Proactive Tests on vSAN Cluster

You can initiate a health test on your vSAN cluster to verify that the cluster components are working as expected.

> **Note:**
> You must not conduct the proactive test in a production environment as it creates network traffic and impacts the vSAN workload.

Run the VM creation test to verify the vSAN cluster health. Running the test creates a virtual machine on each host in the cluster. The test creates a VM per each ESX host in the vSAN cluster. The VM gets deleted if the test is successful. VM creates and and deletes tasks that you can monitor on the task console. If the VM creation and deletion tasks are successful, assume that the cluster components are working as expected and the cluster is functional. The test results shows the last run date and time with their status. You can also view the list of all the ESX hosts where the test was run.

Run the Network performance test to detect and diagnose connectivity issues, and to make sure the network bandwidth between the ESX hosts supports the requirements of vSAN.It also allows to select Enable Network diagnostic mode which creates a ramdisk on a host to collect and save network metrics generated during the test. The test is performed between the ESX hosts in the cluster. It verifies that the network bandwidth between ESX hosts, and reports a warning if the bandwidth is less than 850Mbps. You can run the proactive test at a maximum speed limit of 10Gbps. In vSAN ESA, the proactive test reports error when the result is zero bps and the Health Status displays the test results as info when the result is a non-zero number.

To access a proactive test, select your vSAN cluster in the vSphere Client, and click the **Monitor** tab. Click **vSAN** > **Proactive Tests**.

Managing Proactive Hardware

vSAN Proactive Hardware Management (PHM) informs you of any dying device based on disk predictive failure events generated by the Original Equipment Manufacturer (OEM) vendor.

Ensure that you configured a supported Hardware Support Manager (HSM) and registered to vCenter to prevent predicative failures using PHM.

Based on this information provided by the vendor, you can take the necessary remediation. PHM resides within the vSAN management service on the vCenter. The HSM is registered with the vCenter. PHM collects vendor hardware information from HSM and sends it to vSAN.

About Hardware Support Managers

The deployment method and the management of a hardware support manager are determined by the respective OEM vendor.

Several of the major OEM vendors develop and supply hardware support managers. For example:

- Dell - The hardware support manager that Dell provides is part of their host management solution, OpenManage Integration for VMware vCenter (OMIVV), which you deploy as an appliance.
- HPE - The hardware support managers that HPE provides are part of their management tools, iLO Amplifier and OneView, which you deploy as appliances.
- Lenovo - The hardware support manager that Lenovo provides is part of their server management solution, Lenovo XClarity Integrator for VMware vCenter, which you deploy as an appliance.

You can find the full list of all VMware-certified hardware support managers in the *Broadcom Compatibility Guide* under Platform & Compute section at https://compatibilityguide.broadcom.com/.

Deploying and Configuring Hardware Support Managers

Regardless of the hardware vendor, you must deploy the hardware support manager appliance on a host with sufficient memory, storage, and processing resources.

Typically, hardware support manager appliances are distributed as OVF or OVA templates. You can deploy them on any host in any vCenter instance.

After you deploy the appliance, you must power on the appliance virtual machine and register the appliance as a vCenter extension. You might need to log in to the appliance as an administrator. Each hardware support manager might register with only one or multiple vCenter systems.

For detailed information about deploying, configuring, and managing hardware support managers, refer to the respective OEM-provided documentation and see Deploying Hardware Support Managers in the *Managing Host and Cluster Lifecycle* guide.

Registering Hardware Support Manager

You must register HSM with PHM that resides within the vSAN management service on the vCenter using the vendor management service.

For detailed information about registering hardware support managers, refer to the respective OEM-provided documentation and see Deploying Hardware Support Managers in the *Managing Host and Cluster Lifecycle* guide.

Associating and Dissociating ESX Hosts

After registering HSM with PHM, you need to associate appropriate ESX hosts available in the vCenter with the HSM.

This enables PHM on each host. HSM informs PHM on any change in the managed host list. PHM associates the managed ESX hosts available in a vSAN cluster. When a host is associated or dissociated with PHM, vCenter event gets generated. For detailed information about associating and dissociating ESX hosts, refer to the respective OEM-provided documentation.

Processing Hardware Failures

PHM checks for HSM generated hardware failure events every 10 minutes (600 seconds).

You can customize the time interval using the vSAN configuration file.

1. Log in to vCenter console as root.
2. Open the `/usr/lib/vmware-vsan/VsanVcMgmtConfig.xml` file.
3. Set the interval value using `healthUpdatePollIntervalInSeconds` xml tag in seconds.
4. Restart the vSAN Health service using the command `service-control --restart vmware-vsan-health`.

PHM uses these events to generate alarms, which appears in the vSAN Health. For more information on the vSAN Health events, see the Broadcom knowledge base article 367770.

Monitoring vSAN Performance

You can monitor the performance of your vSAN cluster.

Performance charts are available for clusters, ESX hosts, physical disks, virtual machines, and virtual disks.

About the vSAN Performance Service

You can use vSAN performance service to monitor the performance of your vSAN environment, and investigate potential problems.

The performance service collects and analyzes performance statistics and displays the data in a graphical format. You can use the performance charts to manage your workload and determine the root cause of problems.

When the vSAN performance service is turned on, the cluster summary displays an overview of vSAN performance statistics, including IOPS, throughput, and latency. You can view detailed performance statistics for the cluster, and for each host, disk group, and disk in the vSAN cluster. You also can view performance charts for virtual machines and virtual disks.

Configure vSAN Performance Service

Use the vSAN Performance Service to monitor the performance of vSAN clusters, ESX hosts, disks, and VMs.

- All ESX hosts in the vSAN cluster must be running ESX 9.1 or later.
- Before you configure the vSAN Performance Service, make sure that the cluster is properly configured and has no unresolved health problems.

  Note:

  When you create vSAN OSA, you can optionally enable or deactivate the Performance Service. You can enable and configure the Performance Service. When you create vSAN ESA, the Performance Service is enabled by default. You can then configure the Performance Service.

To support the Performance Service, vSAN uses a Stats database object to collect statistical data. The Stats database is a namespace object in the cluster's vSAN datastore.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. (Optional for vSAN OSA cluster.) Click the Performance Service **Enable** button.
5. (Optional for vSAN OSA cluster.) In vSAN Performance Service Settings, select a storage policy for the stats database object.
6. (Optional for vSAN OSA cluster.) Click **Enable** to enable vSAN Performance Service.
7. Click **Edit** if you want to select a different storage policy in the vSAN Performance Service Settings. You can use the default policy of the vSAN datastore, or the same policy that the VM used.
8. (Optional) Click to enable the verbose mode. This check box appears only after enabling vSAN Performance Service. When enabled, vSAN collects and saves the additional performance metrics to a Stats DB object. If you enable the verbose mode for more than 5 days, a warning message appears indicating that the verbose mode can be resource-intensive. Ensure that you do not enable it for a longer duration.
9. (Optional) Click to enable the network diagnostic mode. This check box appears only after enabling vSAN Performance Service. When enabled, vSAN collects and saves the additional network performance metrics to a RAM disk stats object. If you enable the network diagnostic mode for more than a day, a warning message appears indicating that the network diagnostic mode can be resource-intensive. Ensure that you do not enable it for a longer duration.
10. Click **Apply**.

Use Saved Time Range in vSAN Cluster

You can select saved time ranges from the time range picker in performance views.

- The vSAN performance service must be turned on.
- All ESX hosts in the vSAN cluster must be running ESX 9.1 or later.

You can manually save a time range with a customized name. When you run a storage performance test, the selected time range is saved automatically. You can save a time range for any of the performance views.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, click **Performance**.
4. Select any tab, such as **Backend**.
5. Select a time range and click **Show Results**.
6. Click **Save**. The **Save Time Range** dialog opens.
7. Enter a name for the selected time range.
8. Click **Create**.

   You can save the selected time range at the VM and the host level.

View vSAN Cluster Performance

You can use the vSAN cluster performance charts to monitor the workload in your cluster and determine the root cause of problems.

The vSAN performance service must be turned on before you can view performance charts.

When the performance service is turned on, the cluster summary displays an overview of vSAN performance statistics, including vSAN IOPS, throughput, and latency. At the cluster level, you can view detailed statistical charts for virtual machine consumption and the vSAN back end.

vSAN storage cluster provides shared storage to vSphere clusters. The vSAN compute clients and HCI clients can access storage from the vSAN storage cluster. For virtual machines running on these clients and using the remote vSAN storage, you can view performance metrics such as IOPS, latency, and throughtput.

Note:

- To view iSCSI performance charts, all ESX hosts in the vSAN cluster must be running ESX 9.1 or later.
- To view file service performance charts, you must enable vSAN File Service.
- To view vSAN Direct performance charts, you must claim disks for vSAN Direct.
- To view PMem performance charts, you must have PMem storage attached to the ESX hosts in the cluster.

1. In the vSphere Client, navigate to a standard cluster, storage cluster, or a compute client cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Performance**.
4. Select **Top Contributors**.

   Perform one of the following:

   - Select a time range to view the hotspot entities in the charts. You can view the top 10 hotspot entities as aggregated metrics for the selected time range. You can view the hotspots of VMs, disk groups (vSAN OSA) or disks (vSAN ESA), host (frontend), or host (backend), host (middle layer), vSAN network, or clusters. You have the option to enable separate charts. Based on your selection of VM, host (frontend), or clusters from the drop-down, the HCI cluster displays the local and the remote VMs, host (frontend), or clusters.

     Note:

     With the vSAN 9.1 release, the vSAN storage cluster allows you to view the metrics of the compute-only clients and HCI clients. You can view the hotspots of VMs, disk groups (vSAN OSA) or disks (vSAN ESA), host (frontend), or host (backend), host (middle layer), vSAN network, or clusters.

   - Select a single timestamp to identify the VMs, disk groups (vSAN OSA) or disks (vSAN ESA), host (backend), or host (frontend) that consume the most IOPS, have the highest I/O throughput, or I/O latency. For example, based on the I/O latency graph of the cluster, you can select a timestamp and get the top contributors with latency statistics. You can also select a single contributor and view the latency graph. You have the option to switch between the combined view and table view. If you select a point in time, you can correlate the metrics between different metric types.

5. Select **VM**.

   Perform one of the following:

   - Select **Cluster level metrics** to display the aggregated performance metrics for the cluster that you selected.
   - Select **Show specific VMs** to display metrics for all the VMs selected. If you enable **Show separate chart by VMs**, vSAN displays separate metrics for all the VMs selected.

   Select a time range for your query. vSAN displays performance charts for clients running on the cluster, including IOPS, throughput, latency, congestions, and outstanding I/Os. The statistics on these charts are aggregated from the ESX hosts within the cluster. You can also select **Real-Time** as the time range that displays real-time data that is automatically refreshed. vSAN compute client clusters do not support.

   Note:

   vSAN compute client cluster uses Remote VM to monitor the workloads.

6. (Optional) If you have a compute cluster or HCI cluster, select **Remote VM** to view the performance metrics of the VM hosted in a remote datastore, such as vSAN storage cluster.

   Perform one of the following:

   - Select a time range for your query. vSAN displays performance charts for clients running on the cluster, including IOPS, throughput, latency, congestions, and outstanding I/Os. The statistics on these charts are aggregated from the ESX hosts within the client cluster. You can also select **Real-Time** as the time range that displays real-time data that is automatically refreshed.

- Select a server cluster such as a storage cluster where the remote VMs are available.

7. In vSAN ESA, select **Middle Layer**. Select a time range for your query. vSAN displays the performance charts for the middle layer operations of the host, including IOPS, throughput, and latency. The statistics on these charts are aggregated from the ESX hosts within the cluster. The middle layer is not available in vSAN compute client clusters.
8. Select **Backend**. Select a time range for your query. vSAN displays performance charts for the cluster backend operations, including IOPS, throughput, latency, congestions, and outstanding I/Os. The statistics on these charts are aggregated from the ESX hosts within the cluster. Backend is not available in vSAN compute client clusters.
9. Select **File Share** and choose a file. Select a time range for your query. Select **NFS performance** or **File system performance** based on the protocol layer performance or file system layer performance that you want to display. vSAN displays performance charts for vSAN file services, including IOPS, throughput, and latency.
10. Select **iSCSI** and select an iSCSI target or LUN. Select a time range for your query. vSAN displays performance charts for iSCSI targets or LUNs, including IOPS, bandwidth, latency, and outstanding I/O.
11. (Optional) Select **I/O Insight**. For more information on I/O Insight, see Use vSAN I/O Insight.
12. Select **vSAN Direct** to display the performance data of the vSAN direct disks. Select a time range for your query. vSAN displays performance charts for vSAN direct, including IOPS, bandwidth, latency, and outstanding I/O.
13. Select **PMEM** to display the performance data of all VMs placed on the PMem storage. Select a time range for your query. You can also select **Real-time** as the time range that displays real time data that is automatically refreshed every 30 seconds. PMem displays performance charts including IOPS, bandwidth, and latency. For more information about PMem metrics collection settings, see Broadcom knowledge base article 89100.
14. Click **Refresh** or **Show Results** to update the display.

View vSAN Host Performance

You can use the vSAN host performance charts to monitor the workload on your ESX hosts and determine the root cause of problems.

The vSAN performance service must be turned on before you can view performance charts.

To view the following performance charts, ESX hosts in the vSAN cluster must be running ESX 9.1 or later: Physical Adapters, VMkernal Adapters, VMkernal Adapters Aggregation, iSCSI, vSAN - Backend resync I/Os, resync IOPS, resync throughput, Disk Group resync latency.

- You can view vSAN performance charts for ESX hosts, disk groups, and individual storage devices. When the performance service is turned on, the host summary displays performance statistics for each host and its attached disks.
- At the host level, you can view detailed statistical charts for virtual machine consumption and the vSAN back end, including IOPS, throughput, latency, and congestion.
- Additional charts are available to view the local client cache read IOPS and hit rate. At the disk group level, you can view statistics for the disk group. At the disk level, you can view statistics for an individual storage device.

vSAN storage cluster provides shared storage to vSphere clusters. The vSAN compute clients and HCI clients can access storage from the vSAN storage cluster. For virtual machines running on these clients and using the remote vSAN storage, you can view vSAN host performance metrics such as IOPS, latency, and throughput.

1. In the vSphere Client, navigate to the cluster and select a host.
2. Click the **Monitor** tab.
3. Under vSAN, select **Performance**.
4. Select **VM**.
   - Select **Host level metrics** to display the aggregated performance metrics for the host that you selected.
   - Select **Show specific VMs** to display metrics for all the VMs selected on the host. If you enable **Show separate chart by VMs**, vSAN

Note:

vSAN compute client clusters does not support VM.

Select a time range for your query. vSAN displays performance charts for clients running on the host, including IOPS, throughput, latency, congestions, and outstanding I/Os. You can also select **Real-Time** as the time range that displays real-time data that is automatically refreshed every 30 seconds.

5. (Optional) If you have a compute client cluster or HCI cluster, select **Remote VM** to view the performance metrics of the remote vSAN VM.

   Perform one of the following:

   - Select a time range for your query. vSAN displays performance charts for clients running on the cluster, including IOPS, throughput, latency, congestions, and outstanding I/Os. The statistics on these charts are aggregated from the ESX hosts within the cluster. You can also select **Real-Time** as the time range that displays real-time data that is automatically refreshed.
   - Select a server cluster such as a storage cluster where the remote VMs are available.

6. In vSAN ESA, select **Middle Layer**. Select a time range for your query. vSAN displays the performance charts for the middle layer operations of the host, including the overall middle layer statistics, the overall cache miss by the different types, cache miss by types for the different transactions, and the catch latency for the different transactions. vSAN compute cluster does not support the Middle layer.
7. Select **Backend**. Select a time range for your query. vSAN displays performance charts for the host back-end operations, including IOPS, throughput, latency, congestions, outstanding I/Os, and resync I/Os.
8. Perform one of the following:

   - Select **Disks**, and select a disk group. Select a time range for your query. vSAN displays performance charts for the disk group, including front end (Guest) IOPS, throughput, and latency, as well as overhead IOPS and latency. It also displays the read-cached hit rate, evictions, write-buffer free percentage, capacity and usage, cache disk destage rate, congestions, outstanding I/O, outstanding I/O size, delayed I/O percentage, delayed I/O average latency, internal queue IOPS, internal queue throughput, resync IOPS, resync throughput, and resync latency.
   - In vSAN ESA, select **Disks**, and then select a disk. Select a time range for your query. vSAN displays performance charts for the disk, including vSAN layer IOPS, throughput, and latency. It also displays the physical or firmware layer IOPS, throughput, and latency.

9. Select **Physical Adapters**, and select a NIC. Select a time range for your query. vSAN displays performance charts for the physical NIC (pNIC), including throughput, packets per second, and packets loss rate. vSAN compute client clusters does not support Physical Adapters.
10. Select **Host Network**, and select a VMkernel adapter, such as vmk1. Select a time range for your query. vSAN displays performance charts for all network I/Os processed in the network adapters used by vSAN, including throughput, packets per second, and packets loss rate. vSAN compute client clusters does not support Host Network.
11. Select **iSCSI**. Select a time range for your query. vSAN displays performance charts for all the iSCSI services on the host, including IOPS, bandwidth, latency, and outstanding I/Os.
12. (Optional) Select **I/O Insight**. For more information on I/O Insight, see Use vSAN I/O Insight.
13. Select **vSAN Direct** to display the performance data of the vSAN direct disks. Select a time range for your query. vSAN displays performance charts for vSAN direct, including IOPS, bandwidth, latency, and outstanding I/O.
14. Select **PMEM** to display the performance data of all VMs placed on the PMem storage. Select a time range for your query. You can also select **Real-time** as the time range that displays real time data that is automatically refreshed every 30 seconds. PMem displays the performance charts including IOPS, bandwidth, and latency. For more information about PMem metrics collection settings, see Broadcom knowledge base article 89100.
15. Click **Refresh** or **Show Results** to update the display.

View vSAN VM Performance

You can use the vSAN VM performance charts to monitor the workload on your virtual machines and virtual disks.

The vSAN performance service must be turned on before you can view performance charts.

When the performance service is turned on, you can view detailed statistical charts for virtual machine performance and virtual disk performance. VM performance statistics cannot be collected during migration between ESX hosts, so you might notice a gap of several minutes in the VM performance chart.

vSAN storage cluster provides shared storage to vSphere clusters. The vSAN compute clients and HCI clients can access storage from the vSAN storage cluster.. For virtual machines running on these clients and using the remote vSAN storage, you can view VM performance metrics such as IOPS, latency, and throughput.

> **Note:**
> The performance service supports only virtual SCSI controllers for virtual disks. Virtual disks using other controllers, such as IDE, are not supported.

1. In the vSphere Client, navigate to the cluster and select a VM.
2. Click the **Monitor** tab.
3. Under vSAN, select **Performance**.
4. Select **VM**. Select a time range for your query. vSAN displays performance charts for the VM, including IOPS, throughput, and latency. The VM in the compute client cluster displays the performance charts.
5. Select **Virtual Disk**. Select a time range for your query. vSAN displays performance charts for the virtual disks, including IOPS, delayed normalized IOPS, virtual SCSI IOPS, virtual SCSI throughput, and virtual SCSI latency. The virtual SCSI latency performance charts display a highlighted area due to the IOPS limit enforcement.
6. Click **I/O Insight**. For more information on I/O Insight, see Use vSAN I/O Insight.
7. Click **Refresh** or **Show Results** to update the display.

Use vSAN I/O Insight

I/O Insight allows you to select and view I/O performance metrics of virtual machines in a vSAN cluster.

By understanding the I/O characteristics of VMs, you can ensure better capacity planning and performance tuning. The vSAN performance might be impacted when you enable I/O Insight.

1. In the vSphere Client, navigate to the cluster or host.

   You can also access I/O Insight from the VM. Select the VM and navigate to **Monitor** > **vSAN** > **Performance** > **IO Insight**.

   > **Note:**
   >
   > vSAN compute client clusters does not support vSAN I/O Insight.
2. Click the **Monitor** tab.
3. Under **vSAN**, select **Performance**.
4. Select the **I/O Insight** tab and click **New Instance**.
5. Select the required ESX hosts or VMs that you want to monitor. You can also search for VMs.
6. Click **Next**. Based on the host or VM selected, the name is automatically populated.
7. Select a duration in minutes or hours.
8. Click **Next** and review the instance information.
9. Click **Finish**.

   I/O Insight instance monitors the selected VMs for the specified duration. However, you can stop an instance before completion of the duration that you specified.

   > **Note:**
   >
   > VMs monitored by I/O Insight must not be vMotioned. vMotion stops the VMs from being monitored and will result in an unsuccessful trace preventing vSAN I/O insight monitoring from successful completion. DRS analyzes

resource usage and availability in a cluster and decides when and where to move VMs to balance the load and optimize resource allocation.

vSAN displays performance charts for the VMs in the cluster, including IOPS, throughput, I/O size distribution, I/O latency distribution, and so on.

You can view metrics for the I/O Insight instance that you created.

View vSAN I/O Insight Metrics

I/O Insight performance metrics chart displays the metrics at the virtual disk level.

When I/O Insight is running, vSAN collects and displays the metrics for selected VMs, for a set duration. You can view the performance metrics for up to 90 days. The I/O Insight instances are automatically deleted after this period.

1. In the vSphere Client, navigate to the cluster or host.
   You can also access I/O Insight from the VM. Select the VM and navigate to **Monitor** > **vSAN** > **Performance** > **Virtual Disks**.
2. Click the **Monitor** tab.
3. Under **vSAN**, select **Performance**.
4. Select the **I/O Insight** tab. You can organize the instances based on time or ESX hosts.
5. To view the metrics of an instance, click
   
   and click **View Metrics**. You can optionally stop a running instance before completing the specified duration. For more information on the metrics, see the Broadcom knowledge base article 326953.
   You can rerun an instance, and rename or delete the existing instances.

Use vSAN I/O Trip Analyzer

You can use vSAN I/O trip analyzer to diagnose the virtual machine I/O latency issues.

The vSAN performance service must be enabled before you can run the I/O trip analyzer and view the test results.

vSAN latency issues can be caused by outstanding I/Os, network hardware issues, network congestions, or disk slowness. The I/O trip analyzer allows you to get the breakdown of the latencies at each layer of the vSAN stack. The topology diagram shows only the ESX hosts with VM I/O traffic.

Note:

All the ESX hosts and vCenter in the vSAN cluster must be running 9.1 or later.

Using the I/O trip analyzer scheduler, you can set the recurrence for I/O trip analyzer diagnostic operations. You can either set a one time occurrence or set the recurrence to later. On reaching the recurrence time, the scheduler automatically collects the results. You can view the results collected within 30 days.

Note:
The I/O trip analyzer supports stretched clusters and multiple VMs (maximum 8 VMs and 64 VMDKs) in one diagnostic run for a single cluster.

1. In the vSphere Client, navigate to the cluster.
2. Select a VM.
3. Click the **Monitor** tab.
4. Under vSAN, select **I/O Trip Analyzer**.
5. Click **Run New Test**.
6. In the Run VM I/O Trip Analyzer Test, select the duration of the test.
7. (Optional) Select **Scheduling** to schedule the test for a later time. You can either select **Start now** or enter a time based on your requirement in the **Custom time** field. Select the repeat options and click **Schedule**.

   Note:

   You can schedule only a single I/O trip analyzer per cluster. You can schedule another I/O trip analyzer after deleting the current scheduler. To delete a scheduler, click **Schedules** > **Delete**. You can also modify a schedule that you created. Click **Schedules** > **Edit**. You can repeat a schedule that you have defined.
8. Click **Finish**. The trip analyzer test data is persisted and is available only for 30 days.

   Note:
   vSAN does not support I/O trip analyzer for virtual disks in a remote vSAN datastore.
9. Click **View Result** to view the visualized I/O topology.
10. From the Virtual Disks drop-down, select the disk for which you want to view the I/O topology. You can also view the performance details of the network and the disk groups. Click the edge points of the topology to view the latency details.

    Click the edge points of the topology to view the latency details. If there is a latency issue, click the red icon to focus on that area.
11. Click **Export Data** to generate a zip file that includes the results graph PNG images and raw data in CSV file format.

View vSAN Performance Metrics for Support Cases

Use the vSAN cluster performance metrics to monitor the performance of your cluster and determine the root cause of the performance issues.

The vSAN performance service must be turned on before you can view performance charts.

You can use the vSAN Obfuscation Map to identify the obfuscated data sent to VMware. For more information on obfuscation map, see View vSAN Obfuscation Map.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Support** > **Performance For Support**.
4. Select a performance dashboard from the drop-down menu.
5. Select ESX hosts, disks, or NICs from the drop-down menu.
6. Select a time range for your query.

   The default time range is the most recent hour. You can increase the range to include the last 24 hours, or define a custom time range within the last 90 days. If you used the HCIbench tool to run performance benchmark tests on the vSAN cluster, the time ranges of those tests appear in the drop-down menu.
7. Click **Show Results**.

   vSAN displays performance charts for selected entities, such as IOPS, throughput, latency, congestions, and outstanding I/Os.

Using vSAN Performance Diagnostics

You can use vSAN performance diagnostics to improve the performance of your vSAN OSA cluster, and resolve performance issues.

- The vSAN performance service must be turned on.
- vCenter requires Internet access to download ISO images and patches and to send data to VMware to analyze vSAN performance data.
- You must participate in the Customer Experience Improvement Program (CEIP).

The vSAN performance diagnostics tool analyzes previously run benchmarks gathered from the vSAN performance service. It can detect issues, suggest remediation steps, and provide supporting performance graphs for further insight.

The vSAN performance service provides the data used to analyze vSAN performance diagnostics. vSAN uses CEIP to send data to VMware for analysis.

> **Note:**
>
> vSAN ESA cluster does not support vSAN performance diagnostics. Do not use vSAN performance diagnostics for general evaluation of performance on a production vSAN cluster.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Performance Diagnostics**.
4. Select a benchmark goal from the drop-down menu.

   You can select a goal based on the performance improvement that you want to achieve, such as maximum IOPS, maximum throughput, or minimum latency.
5. Select a time range for your query.

   The default time range is the most recent hour. You can increase the range to include the last 24 hours, or define a custom time range within the last 90 days. If you used the HCIbench tool to run performance benchmark tests on the vSAN cluster, the time ranges of those tests appear in the drop-down menu.
6. Click **Show Results**.

When you click **Show Results**, vSAN transmits performance data to the vSphere backend analytics server. After analyzing the data, the vSAN performance diagnostics tool displays a list of issues that might have affected the benchmark performance for the chosen goal.

You can click to expand each issue to view more details about each issue, such as a list of affected items. You also can click **See More** or **Ask VMware** to display a Knowledge Base article that describes recommendations to address the issue and achieve your performance goal.

View vSAN Obfuscation Map

You can use the vSAN Obfuscation Map to identify the obfuscated data sent to VMware.

vSAN Obfuscation Map provides mapping of the obfuscated data sent to VMware as part of Customer Experience Improvement Program (CEIP) to facilitate communication during the Support Request process between vSAN user and VMware Global Support. Use notepad or any text editor to view the obfuscation map. For more information on obfuscation map, see the Broadcom knowledge base article 51120.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Monitor** tab.
3. Under vSAN, select **Support**.
4. Click **Obfuscation**.
5. Click **View Online** or **Download Obfuscation Map** to view or download the vSphere entities.

Handling Failures and Troubleshooting vSAN

If you encounter problems when using vSAN, you can use troubleshooting topics.

The topics help you understand the problem and offer you a workaround, when it is available.

Uploading a vSAN Support Bundle

You can upload a vSAN support bundle so VMware service personnel can analyze the diagnostic information.

Broadcom Technical Support routinely requests diagnostic information from your vSAN cluster when a support request is addressed. The support bundle is an archive that contains diagnostic information related to the environment, such as product specific logs, configuration files, and so on.

The log files, collected and packaged into a zip file, include the following:

- vCenter support bundle
- Host support bundle

The host support bundle in the cluster includes the following:

```
["Userworld:HostAgent", "Userworld:FDM",
 "System:VMKernel", "System:ntp", "Storage:base", "Network:tcpip",
 "Network:dvs", "Network:base", "Logs:System", "Storage:VSANMinimal",
 "Storage:VSANHealth", "System:BaseMinmal", "Storage:VSANTraces"]
```

vSAN performs an automated upload of the support bundle, and does not allow you to review, obfuscate, or otherwise edit the contents of your support data prior to it being sent to VMware. vSAN connects to the FTP port 21 or HTTPS port 443 of the target server with the domain name *vmware.com*, to automatically upload the support bundle.

> **Note:**
> Data collected in the support bundle may be considered sensitive. If your support data contains regulated data, such as personal, health care, or financial data, you may want to avoid uploading the support bundle.

1. Right-click the vSAN cluster in the vSphere Client.
2. Choose menu **vSAN > Upload support bundle...**
3. Enter your service request ID and a description of your issue.
4. Click **Upload**.

Using Esxcli Commands with vSAN

Use Esxcli commands to obtain information about vSAN OSA or vSAN ESA and to troubleshoot your vSAN environment.

The following commands are available:

Command	Description
`esxcli vsan network list`	Verify which VMkernel adapters are used for vSAN communication.
`esxcli vsan storage list`	List storage disks claimed by vSAN.
`esxcli vsan storagepool list`	List storage pool claimed by vSAN ESA. This command is applicable only for vSAN ESA cluster.
`esxcli vsan cluster get`	Get vSAN cluster information.
`esxcli vsan health`	Get vSAN cluster health status.
`esxcli vsan debug`	Get vSAN cluster debug information.

The `esxcli vsan debug` commands can help you debug and troubleshoot the vSAN cluster, especially when vCenter is not available.

Use: `esxcli vsan debug {cmd} [cmd options]`

Debug commands:

Command	Description
`esxcli vsan debug disk`	Debug vSAN physical disks.
`esxcli vsan debug object`	Debug vSAN objects.
`esxcli vsan debug resync`	Debug vSAN resyncing objects.
`esxcli vsan debug controller`	Debug vSAN disk controllers.
`esxcli vsan debug limit`	Debug vSAN limits.
`esxcli vsan debug vmdk`	Debug vSAN VMDKs.

Example `esxcli vsan debug` commands:

```
esxcli vsan debug disk summary get
   Overall Health: green
   Component Metadata Health: green
   Memory Pools (heaps): green
   Memory Pools (slabs): green
esxcli vsan debug disk list
UUID: 52e1d1fa-af0e-0c6c-f219-e5e1d224b469
   Name: mpx.vmhba1:C0:T1:L0
   SSD: False
   Overall Health: green
   Congestion Health:
         State: green
         Congestion Value: 0
         Congestion Area: none
   In Cmmds: true
   In Vsi: true
   Metadata Health: green
   Operational Health: green
   Space Health:
         State: green
         Capacity: 107365793792 bytes
         Used: 1434451968 bytes
         Reserved: 150994944 bytes
esxcli vsan debug object health summary get
    Health Status                                 Number Of Objects
    --------------------------------------------  -----------------
    reduced-availability-with-no-rebuild-delay-timer                0
    reduced-availability-with-active-rebuild                        0
    inaccessible                                                    0
    data-move                                                       0
    healthy                                                         1
    nonavailability-related-incompliance                            0
    nonavailability-related-reconfig                                0
    reduced-availability-with-no-rebuild                            0
esxcli vsan debug object list
    Object UUID: 47cbdc58-e01c-9e33-dada-020010d5dfa3
       Version: 5
       Health: healthy
       Owner:
       Policy:
          stripeWidth: 1
          CSN: 1
```

```
          spbmProfileName: vSAN Default Storage Policy
          spbmProfileId: aa6d5a82-1c88-45da-85d3-3d74b91a5bad
          forceProvisioning: 0
          cacheReservation: 0
          proportionalCapacity: [0, 100]
          spbmProfileGenerationNumber: 0
          hostFailuresToTolerate: 1

       Configuration:
          RAID_1
             Component: 47cbdc58-6928-333f-0c51-020010d5dfa3
               Component State: ACTIVE,  Address Space(B): 273804165120 (255.00GB),
               Disk UUID: 52e95956-42cf-4d30-9cbe-763c616614d5, Disk Name: mpx.vmhba1..
               Votes: 1,  Capacity Used(B): 373293056 (0.35GB),
               Physical Capacity Used(B): 369098752 (0.34GB),  Host Name: sc-rdops...
             Component: 47cbdc58-eebf-363f-cf2b-020010d5dfa3
               Component State: ACTIVE,  Address Space(B): 273804165120 (255.00GB),
               Disk UUID: 52d11301-1720-9901-eb0a-157d68b3e4fc,  Disk Name: mpx.vmh...
               Votes: 1,  Capacity Used(B): 373293056 (0.35GB),
               Physical Capacity Used(B): 369098752 (0.34GB),  Host Name: sc-rdops-vm..
          Witness: 47cbdc58-21d2-383f-e45a-020010d5dfa3
            Component State: ACTIVE,  Address Space(B): 0 (0.00GB),
            Disk UUID: 52bfd405-160b-96ba-cf42-09da8c2d7023,  Disk Name: mpx.vmh...
            Votes: 1,  Capacity Used(B): 12582912 (0.01GB),
            Physical Capacity Used(B): 4194304 (0.00GB),  Host Name: sc-rdops-vm...

       Type: vmnamespace
       Path: /vmfs/volumes/vsan:52134fafd48ad6d6-bf03cb6af0f21b8d/New Virtual Machine
       Group UUID: 00000000-0000-0000-0000-000000000000
       Directory Name: New Virtual Machine
esxcli vsan debug controller list
    Device Name: vmhba1
       Device Display Name: LSI Logic/Symbios Logic 53c1030 PCI-X Fusion-MPT Dual Ult..
       Used By VSAN: true
       PCI ID: 1000/0030/15ad/1976
       Driver Name: mptspi
       Driver Version: 4.23.01.00-10vmw
       Max Supported Queue Depth: 127
esxcli vsan debug limit get
       Component Limit Health: green
       Max Components: 750
       Free Components: 748
       Disk Free Space Health: green
       Lowest Free Disk Space: 99 %
       Used Disk Space: 1807745024 bytes
       Used Disk Space (GB): 1.68 GB
       Total Disk Space: 107365793792 bytes
       Total Disk Space (GB): 99.99 GB
       Read Cache Free Reservation Health: green
       Reserved Read Cache Size: 0 bytes
       Reserved Read Cache Size (GB): 0.00 GB
       Total Read Cache Size: 0 bytes
       Total Read Cache Size (GB): 0.00 GB
esxcli vsan debug vmdk list
    Object: 50cbdc58-506f-c4c2-0bde-020010d5dfa3
       Health: healthy
       Type: vdisk
```

```
       Path: /vmfs/volumes/vsan:52134fafd48ad6d6-bf03cb6af0f21b8d/47cbdc58-e01c-9e33-
             dada-020010d5dfa3/New Virtual Machine.vmdk
       Directory Name: N/A
esxcli vsan debug resync list
   Object            Component              Bytes Left To Resync  GB Left To Resync
   ----------------  ---------------------  --------------------  -----------------
   31cfdc58-e68d...  Component:23d1dc58...             536870912  0.50
   31cfdc58-e68d...  Component:23d1dc58...            1073741824  1.00
   31cfdc58-e68d...  Component:23d1dc58...            1073741824  1.00
```

Using vsantop Command-Line Tool

Use the command-line tool - vsantop - that runs on ESX hosts to view the real time vSAN performance metrics.

You can use this tool to monitor vSAN performance. To display the different performance views and metrics in vsantop, enter the following commands:

Command	Description
`^L`	Redraw screen
`Space`	Update display
`h` or `?`	Help; show this text
`q`	Quit
`f/F`	Add or remove fields
`o/O`	Change the order of displayed fields
`s`	Set the delay in seconds between updates
`#`	Set the number of instances to display
`E`	Change the selected entity type
`L`	Change the length of the field
`l`	Limit display to specific node id
`.`	Sort by column, same number twice to change sort order

vSAN Configuration on an ESX Host Might Fail

In certain circumstances, the task of configuring vSAN on a particular host might fail.

An ESX host that joins a vSAN cluster fails to have vSAN configured.

If a host does not meet hardware requirements or experiences other problems, vSAN might fail to configure the host. For example, insufficient memory on the host might prevent vSAN from being configured.

1. Place the host that causes the failure in Maintenance Mode.
2. Move the host out of the vSAN cluster.
3. Resolve the problem that prevents the host to have vSAN configured.
4. Move the host back into the vSAN cluster.
5. Exit Maintenance Mode.

Not Compliant Virtual Machine Objects Do Not Become Compliant Instantly

When you use the **Check Compliance** button, a virtual machine object does not change its status from Not Compliant to Compliant even though vSAN resources have become available and satisfy the virtual machine profile.

When you use force provisioning, you can provision a virtual machine object even when the policy specified in the virtual machine profile cannot be satisfied with the resources available in the vSAN cluster. The object is created, but remains in the non-compliant status.

vSAN is expected to bring the object into compliance when storage resources in the cluster become available, for example, when you add an ESX host. However, the object's status does not change to compliant immediately after you add resources.

This occurs because vSAN regulates the pace of the reconfiguration to avoid overloading the system. The amount of time it takes for compliance to be achieved depends on the number of objects in the cluster, the I/O load on the cluster and the size of the object in question. In most cases, compliance is achieved within a reasonable time.

vSAN Cluster Configuration Issues

After you change the vSAN configuration, vCenter performs validation checks for vSAN configuration.

Error messages indicate that vCenter has detected a problem with vSAN configuration.

Note:
Validation checks are also performed as a part of a host synchronization process.

If vCenter detects any configuration problems, it displays error messages. Use the following methods to fix vSAN configuration problems.

Table 1022: vSAN Configuration Errors and Solutions

vSAN Configuration Error	Solution
Host with the vSAN service enabled is not in the vCenter cluster	Add the host to the vSAN cluster. 1. Right-click the host, and select **Move To**. 2. Select the vSAN cluster and click **OK**.
Host is in a vSAN enabled cluster but does not have vSAN service enabled	Verify whether vSAN network is properly configured and enabled on the host. See Configure a Cluster for vSAN Using the vSphere Client.
vSAN network is not configured	Configure vSAN network. See Configuring the vSAN Network.
Host cannot communicate with all other nodes in the vSAN enabled cluster	Might be caused by network isolation.
Found another host participating in the vSAN service which is not a member of this host's vCenter cluster.	Make sure that the vSAN cluster configuration is correct and all ESX hosts are in the same vCenter inventory.

Handling Failures in vSAN

vSAN handles failures of the storage devices, ESX hosts and network in the cluster according to the severity of the failure.

You can diagnose problems in vSAN by observing the performance of the vSAN datastore and network.

Failure Handling in vSAN

vSAN implements mechanisms for indicating failures and rebuilding unavailable data for data protection.

Failure States of vSAN Components

In vSAN, components that have failed can be in absent or degraded state.

According to the component state, vSAN uses different approaches for recovering virtual machine data. vSAN also provides alerts about the type of component failure. See Using the VMkernel Observations for Creating vSAN Alarms and Using the vSAN Default Alarms.

vSAN supports two types of failure states for components:

Table 1023: Failure States of Components in vSAN

Component Failure State	Description	Recovery	Cause
Degraded	A component is in degraded state if vSAN detects a permanent component failure and assumes that the component is not going to recover to working state.	vSAN starts rebuilding the affected components immediately if there are adequate resources in the cluster.	• Failure of a flash caching device • Magnetic or flash capacity device failure • Storage controller failure
Absent	A component is in absent state if vSAN detects a temporary component failure where the component might recover and restore its working state.	vSAN starts rebuilding the affected components immediately if there are adequate resources in the cluster if they are not available within a certain time interval. By default, vSAN starts rebuilding absent components after 60 minutes.	• Lost network connectivity • Failure of a physical network adapter • ESX host failure • Unplugged flash caching device • Unplugged magnetic disk or flash capacity device

Examine the Failure State of a Component

You can determine whether a component is in the absent or degraded failure state.

If a failure occurs in the cluster, vSAN marks the components for an object as absent or degraded based on the failure severity.

1. In the vSphere Client, navigate to the cluster.
2. On the **Monitor** tab, click **vSAN** and select **Virtual Objects**.

   The home directories and virtual disks of the virtual machines in the cluster appear.
3. Select the check box on one of the virtual objects and click **View Placement Details** to open the Physical Placement dialog. You can view device information, such as name, identifier or UUID, number of devices used for each virtual machine, and how they are mirrored across ESX hosts.

Object States That Indicate Problems in vSAN

Examine the compliance status and the operational state of a virtual machine object to find how a failure in the cluster affects the virtual machine.

Table 1024: Object State

Object State Type	Description
Compliance Status	The compliance status of a virtual machine object indicates whether it meets the requirements of the assigned VM storage policy.
Operational State	The operational state of an object can be healthy or unhealthy. It indicates the type and number of failures in the cluster. An object is healthy if an intact replica is available and more than 50 percent of the object's votes are still available. An object is unhealthy if an entire replica is not available or less than 50 percent of the object's votes are unavailable. For example, an object might become unhealthy if a network failure occurs in the cluster and a host becomes isolated.

To determine the overall influence of a failure on a virtual machine, examine the compliance status and the operational state. If the operational state remains healthy although the object is noncompliant, the virtual machine can continue using the vSAN datastore. If the operational state is unhealthy, the virtual machine cannot use the datastore.

Examine the Health of an Object in vSAN

Use the vSphere Client to examine whether a virtual object is healthy.

A virtual machine is considered as healthy when a replica of the VM object and more than 50 percent of the votes for an object are available.

1. In the vSphere Client, navigate to the cluster.
2. On the **Monitor** tab, click **vSAN** and select **Virtual Objects**.
   The home directories and virtual disks of the virtual machines in the cluster appear.
3. Select an object type in the **Affected inventory objects** area at the top of the page to display information about each object, such as object state, storage policy, and vSAN UUID.
   If the inventory object is Unhealthy, the vSphere Client indicates the reason for the unhealthy state in brackets.

Examine the Compliance of a Virtual Machine in vSAN

Use the vSphere Client to examine whether a virtual machine object is compliant with the assigned VM storage policy.

1. Examine the compliance status of a virtual machine.
   a) In the vSphere Client, navigate to the virtual machine in the cluster.
   b) On the **Summary** tab, examine the value of the VM Storage Policy Compliance property under VM Storage Policies.
2. Examine the compliance status of the objects of the virtual machine.
   a) In the vSphere Client, navigate to the virtual machine in the cluster.
   b) On the **Monitor** tab, click **vSAN** and select **Virtual Objects**.
   c) Select an object type in the **Affected inventory objects** area at the top of the page to display information about each object, such as object state, storage policy, and vSAN UUID.
   d) Select the check box on one of the virtual objects and click **View Placement Details** to open the Physical Placement dialog. You can view device information, such as name, identifier or UUID, number of devices used for each virtual machine, and how they are mirrored across ESX hosts.
   e) On the Physical Placement dialog, check the **Group components by host placement** check box to organize the objects by host and by disk.

Accessibility of Virtual Machines Upon a Failure in vSAN

If a virtual machine uses vSAN storage, its storage accessibility might change according to the type of failure in the vSAN cluster.

Changes in the accessibility occur when the cluster experiences more failures than the policy for a virtual machine object tolerates.

As a result from a failure in the vSAN cluster, a virtual machine object might become inaccessible. An object is inaccessible if a full replica of the object is not available because the failure affects all replicas, or when less than 50 percent of the object's votes are available.

According to the type of object that is inaccessible, virtual machines behave in the following ways:

Table 1025: Inaccessibility of Virtual Machine Objects

Object Type	Virtual Machine State	Virtual Machine Symptoms
VM Home Namespace	• Inaccessible • Orphaned if vCenter or the ESX host cannot access the `.vmx` file of the virtual machine.	The virtual machine process might crash and the virtual machine might be powered off.
VMDK	Inaccessible	The virtual machine remains powered on but any I/O operations on the VMDK will fail. Depending on the guest operating system, the guest ends the operation and triggers a event indicating a disk I/O timeout.

Virtual machine inaccessibility is not a permanent state. After the underlying issue is resolved, and a full replica and more than 50 percent of the object's votes are restored, the virtual machine automatically becomes accessible again.

Storage Device is Failing in vSAN Cluster

vSAN monitors the performance of each storage device and proactively isolates unhealthy devices.

It detects gradual failure of a storage device and isolates the device before congestion builds up within the affected host and the entire vSAN cluster.

If a disk experiences sustained high latencies or congestion, vSAN considers the device as a dying disk, and evacuates data from the disk. vSAN handles the dying disk by evacuating or rebuilding data. No user action is required, unless the cluster lacks resources or has inaccessible objects.

Component Failure State and Accessibility

The vSAN components that reside on the magnetic disk or flash capacity device are marked as absent.

Behavior of vSAN

vSAN responds to the storage device failure in the following ways.

Parameter	Behavior
Alarms	An alarm is generated from each host whenever an unhealthy device is diagnosed. A warning is issued whenever a disk is suspected of being unhealthy.
Health finding	The **Disk operation** health finding issues a warning for the dying disk.
Health status	On the Disk Management page, the health status of the dying disk is listed as **Unhealthy**. When vSAN completes evacuation of data, the health status is listed as **DyingDiskEmpty**.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded. If resources are available, vSAN automatically reprotects the data.

If vSAN detects a disk with a permanent error, it makes a limited number of attempts to revive the disk by unmounting and mounting it.

Capacity Device Not Accessible in vSAN Cluster

When a magnetic disk or flash capacity device fails, vSAN evaluates the accessibility of the objects on the device.

vSAN rebuilds them on another host if space is available and the **Primary level of failures to tolerate** is set to 1 or more.

Component Failure State and Accessibility

The vSAN components that reside on the magnetic disk or flash capacity device are marked as degraded.

Behavior of vSAN

vSAN responds to the capacity device failure in the following ways.

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the failed capacity device. Restore the virtual machine from a backup.
I/O operations on the capacity device	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded. If resources are available, an automatic reprotect will occur.

Storage Pool Device Is Not Accessible in vSAN ESA Cluster

When a storage pool device fails, vSAN evaluates the accessibility of the objects on the device.

vSAN rebuilds them on another host if space is available and the Primary level of failures to tolerate is set to 1 or more.

Component Failure State and Accessibility

vSAN responds to the storage pool device failure in the following ways.

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the failed capacity device. Restore the virtual machine from a backup.
I/O operations on the capacity device	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded. If resources are available, an automatic reprotect will occur.

A Caching Device Is Not Accessible in a vSAN Cluster

When a flash caching device fails, vSAN evaluates the accessibility of the objects on the disk group that contains the cache device.

vSAN rebuilds them on another host if possible and the **Primary level of failures to tolerate** is set to 1 or more.

Component Failure State and Accessibility

Both cache device and capacity devices that reside in the disk group, for example, magnetic disks, are marked as degraded. vSAN interprets the failure of a single flash caching device as a failure of the entire disk group.

Behavior of vSAN

vSAN responds to the failure of a flash caching device in the following way:

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the failed capacity device.
I/O operations on the disk group	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded. If resources are available, an automatic reprotect will occur.

A Host is Not Responding in vSAN Cluster

If a host stops responding due to failure or reboot of the host, vSAN waits for the host to recover and rebuilds the components elsewhere in the cluster.

Component Failure State and Accessibility

The vSAN components that reside on the host are marked as absent.

Behavior of vSAN

vSAN responds to the host failure in the following way:

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the failed capacity device.
I/O operations on the host	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded. If resources are available, an automatic reprotect will occur.

Network Connectivity is Lost in vSAN Cluster

When the connectivity between the ESX hosts in the cluster is lost, vSAN determines the active partition.

vSAN rebuilds the components from the isolated partition on the active partition if the connectivity is not restored.

Component Failure State and Accessibility

vSAN determines the partition where more than 50 percent of the votes of an object are available. The components on the isolated hosts are marked as absent.

Behavior of vSAN

vSAN responds to a network failure in the following way:

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the failed capacity device.
I/O operations on the isolated hosts	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.

Parameter	Behavior
Rebuilding data	If the host rejoins the cluster with 60 minutes, vSAN synchronizes the components on the host. If the host does not rejoin the cluster within 60 minutes, vSAN examines whether some of the other ESX hosts in the cluster can satisfy the requirements for cache, space, and placement rules for the objects on the inaccessible host. If such a host is available, vSAN starts the recovery process. If the host rejoins the cluster after 60 minutes and recovery has started, vSAN evaluates whether to continue the recovery or stop it and resynchronize the original components.

A Storage Controller Fails in vSAN Cluster

When a storage controller fails, vSAN evaluates the accessibility of the objects on the disk groups that are attached to the controller.

vSAN rebuilds them on another host.

Symptoms

If a host contains a single storage controller and multiple disk groups, and all devices in all disk groups are failed, then you might assume that a failure in the common storage controller is the root cause. Examine the VMkernel log messages to determine the nature of the fault.

Component Failure State and Accessibility

When a storage controller fails, the components on the flash caching devices and capacity devices in all disk groups that are connected to the controller are marked as degraded.

If a host contains multiple controllers, and only the devices that are attached to an individual controller are inaccessible, then you might assume that this controller has failed.

Behavior of vSAN

vSAN responds to a storage controller failure in the following way:

Parameter	Behavior
Level of failures to tolerate	If the **Level of failures to tolerate** in the VM storage policy is equal to or greater than 1, the virtual machine objects are still accessible from another ESX host in the cluster. If resources are available, vSAN starts an automatic reprotection. If the **Level of failures to tolerate** is set to 0, a virtual machine object is inaccessible if one of the object's components resides on the disk groups that are connected to the storage controller.
I/O operations on the isolated hosts	When a vSAN object experiences a failure or a failed component, I/O operations stop between 5-7 seconds until it revaluates if the object is available. If vSAN determines that the object is available, all running operations are resumed.
Rebuilding data	vSAN examines whether the ESX hosts and the capacity devices can satisfy the requirements for space and placement rules for the objects on the failed device or disk group. If such a host with capacity is available, vSAN starts the recovery process immediately because the components are marked as degraded.

vSAN Stretched Cluster Site Fails or Loses Network Connection

A vSAN stretched cluster manages failures that occur due to the loss of a network connection between sites or the temporary loss of one site.

vSAN Stretched Cluster Failure Handling

In most cases, the vSAN stretched cluster continues to operate during a failure and automatically recovers after the failure is resolved.

Table 1026: How vSAN Stretched Cluster Handles Failures

Type of Failure	Behavior
Network Connection Lost Between Active Sites	If the network connection fails between the two active sites, the witness host and the preferred site continue to service storage operations, and keep data available. When the network connection returns, objects protected by a policy with a site tolerance attribute are resynchronised
Secondary Site Fails or Loses Network Connection	If the secondary site goes offline or becomes isolated from the preferred site and the witness host, the witness host and the preferred site continue to service storage operations, and keep data available, objects protected by a policy with a site tolerance attribute are resynchronised
Preferred Site Fails or Loses Network Connection	If the preferred site goes offline or becomes isolated from the secondary site and the witness host, the secondary site continues storage operations if it remains connected to the witness host, objects protected by a policy with a site tolerance attribute are resynchronised
Witness Host Fails or Loses Network Connection	If the witness host goes offline or becomes isolated from the preferred site or the secondary site, objects become noncompliant but data remains available. VMs that are currently running are not affected.

Examine Inaccessible Objects in a vSAN Stretched Cluster

When a data site and the witness site in a vSAN stretched cluster fail, vSAN objects that are configured with site failure tolerance become inaccessible, and the associated VMs become unavailable.

All hosts in the failed site and the witness host are unavailable and cannot be recovered quickly.

For objects with site failure tolerance, vSAN stores data replicas on both the data sites. After the failure of a witness followed by a data site, the objects become inaccessible as the vSAN objects has lost quorum. A vSAN site takeover promotes the surviving replicas and makes the objects accessible again.

During this process, the data may roll back to an earlier point in time:

- If the surviving site enters site maintenance, all recovered objects roll back to the same point in time.
- If the surviving site is not in site maintenance, objects are not guaranteed to be recoverable or roll back to different points in time.

For more details on the vSAN site takeover, see Perform the Site Takeover for a vSAN Stretched Cluster.

Troubleshooting vSAN

Examine the performance and accessibility of virtual machines to diagnose problems in the vSAN cluster.

Verify Drivers, Firmware, Storage I/O Controllers Against the *VMware Compatibility Guide*

Use the vSAN Health to verify whether your hardware components, drivers, and firmware are compatible with vSAN.

Using hardware components, drivers, and firmware that are not compatible with vSAN might cause problems in the operation of the vSAN cluster and the virtual machines running on it.

The hardware compatibility health findings verify your hardware against the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/. For more information about using the vSAN Health, see Monitoring vSAN Health.

Examining Performance in vSAN Cluster

Monitor the performance of virtual machines, ESX hosts, and the vSAN datastore to identify potential storage problems.

Monitor regularly the following performance indicators to identify faults in vSAN storage, for example, by using the performance charts in the vSphere Client:

- Datastore. Rate of I/O operations on the aggregated datastore.
- Virtual Machine. I/O operations, memory and CPU usage, network throughput and bandwidth.

You can use the vSAN performance service to access detailed performance charts. For information about using the performance service, see Monitoring vSAN Performance.

Network Misconfiguration Status in vSAN Cluster
After you enable vSAN on a cluster, the datastore is not assembled correctly because of a detected network misconfiguration.

After you enable vSAN on a cluster, a vSAN Alarm triggers on the Summary tab indicating a network partition.

One or more members of the cluster cannot communicate because of either of the following reasons:

- A host in the cluster does not have a VMkernel adapter for vSAN.
- The ESX hosts cannot connect each other in the network.

Troubleshoot network partition errors using the vSAN Health and the Broadcom knowledge base article 318839.

Virtual Machine Appears as Noncompliant, Inaccessible, or Orphaned in the vSAN Cluster
The state of a virtual machine that stores data on a vSAN datastore appears as noncompliant, inaccessible, or orphaned due to the vSAN cluster failures.

A virtual machine on a vSAN datastore is in one of the following states that indicate a fault in the vSAN cluster.

- The virtual machine or some of its objects are non-compliant to the configured policy. See Examine the Compliance of a Virtual Machine in vSAN.
- The virtual machine object is inaccessible or orphaned. See Examine the Failure State of a Component.

If an object replica is still available on another host, vSAN forwards the I/O operations of the virtual machine to the replica.

If the object of the virtual machine can no longer satisfy the requirement of the assigned VM storage policy, vSAN considers it noncompliant. For example, a host might temporarily lose connectivity. See Object States That Indicate Problems in vSAN.

If vSAN cannot locate a full replica or more than 50 percent of the votes for the object, the virtual machine becomes inaccessible. If a vSAN detects that the `.vmx` file is not accessible because the VM Home Namespace is maybe inacessbile, the virtual machine may become orphaned. See Accessibility of Virtual Machines Upon a Failure in vSAN.

If the cluster contains enough resources, vSAN automatically recovers the failed components if the failure is permanent.

If the cluster does not have enough resources to rebuild the failed components, extend the space in the cluster.

Attempt to Create a Virtual Machine on vSAN Fails
When you try to deploy a virtual machine in a vSAN cluster, the operation fails with an error that the virtual machine files cannot be created.

The operation for creating a virtual machine fails with an error status: `Cannot complete file creation operation`.

The deployment of a virtual machine on vSAN might fail for several reasons.

- vSAN cannot allocate space for the virtual machine storage policies and virtual machine objects. Such a failure might occur if the datastore does not have enough usable capacity, for example, if a physical disk is temporarily disconnected from the host.
- The virtual machine has very large virtual disks and the ESX hosts in the cluster cannot provide storage for them based on the placement rules in the VM storage policy
  For example, if the **Primary level of failures to tolerate** in the VM storage policy is set to 1, vSAN must store two replicas of a virtual disk in the cluster, each replica on a different host. The datastore might have this space after aggregating the free space on all ESX hosts in the cluster. However, no two ESX hosts can be available in the cluster, each providing enough space to store a separate replica of the virtual disk.
  vSAN does not move components between ESX hosts or disks groups to free space for a new replica, even though the cluster might contain enough space for provisioning the new virtual machine.

Verify the state of the capacity devices in the cluster.

a) In the vSphere Client, navigate to the cluster.
b) On the **Monitor** tab, click **vSAN** and select **Physical Disks**.
c) Examine the capacity and health status of the devices on the ESX hosts in the cluster.

vSAN Stretched Cluster Configuration Error When Adding a Host

Before adding new ESX hosts to a vSAN stretched cluster, all current ESX hosts must be connected. If a current host is disconnected, the configuration of the new host is incomplete.

After you add a host to a vSAN stretched cluster in which some ESX hosts are disconnected, on the Summary tab for the cluster the Configuration Status for vSAN appears as `Unicast agent unset on host`.

When a new host joins a stretched cluster, vSAN must update the configuration on all hosts in the cluster. If one or more hosts are disconnected from the vCenter, the update fails. The new host successfully joins the cluster, but its configuration is incomplete.

Verify that all ESX hosts are connected to vCenter, and click the link provided in the Configuration Status message to update the configuration of the new host.

If you cannot rejoin the disconnected host, remove the disconnected host from the cluster, and click the link provided in the Configuration Status message to update the configuration of the new host.

Cannot Add or Remove the Witness Host in vSAN Stretched Cluster

Before adding or removing the witness host in a vSAN stretched cluster, all current ESX hosts must be connected. If a current host is disconnected, you cannot add or remove the witness host.

When you add or remove a witness host in a vSAN stretched cluster in which some ESX hosts are disconnected, the operation fails with an error status: `The operation is not allowed in the current state. Not all ESX hosts in the cluster are connected to Virtual Center`.

When the witness host joins or leaves a stretched cluster, vSAN must update the configuration on all ESX hosts in the cluster. If one or more ESX hosts are disconnected from the vCenter, the witness host cannot be added or removed.

Verify all ESX hosts are connected to vCenter, and retry the operation. If you cannot rejoin the disconnected host, remove the disconnected host from the cluster, and then you can add or remove the witness host.

Disk Becomes Locked in vSAN Cluster

In an encrypted vSAN cluster, when communication between a host and the KMS is lost, the disks in the storage pool can become locked if the host reboots.

vSAN locks a disk when the host reboots and it cannot get the KEK from the KMS. The disks behave as if they are unmounted. Objects on the disks become inaccessible.

You can view health status of a disk on the Disk Management page in the vSphere Client. An Encryption health finding warning notifies you that a disk is locked.

ESX hosts in an encrypted vSAN cluster do not store the KEK on disk. If a host reboots and cannot get the KEK from the KMS, vSAN locks the host's disks.

To exit the locked state, you must restore communication with the KMS and reestablish the trust relationship. If the KMS is down, you can unlock a storage pool if you have encryption key persistence configured. For more information, see How vSAN Data-At-Rest Encryption Works with the Key Management Server (KMS).

Disk Group Becomes Locked in vSAN Cluster

In an encrypted vSAN cluster, when communication between a host and the KMS is lost, the disk group can become locked if the host reboots.

vSAN locks a host's disk groups when the host reboots and it cannot get the KEK from the KMS. The disks behave as if they are unmounted. Objects on the disks become inaccessible.

You can view a disk group's health status on the Disk Management page in the vSphere Client. An Encryption health finding warning notifies you that a disk is locked.

ESX hosts in an encrypted vSAN cluster do not store the KEK on disk. If a host reboots and cannot get the KEK from the KMS, vSAN locks the host's disk groups.

To exit the locked state, you must restore communication with the KMS and reestablish the trust relationship.

Replacing Existing Hardware Components in vSAN Cluster

Under certain conditions, you must replace hardware components, drivers, firmware, and storage I/O controllers in the vSAN cluster.

In vSAN, you should replace hardware devices when you encounter failures or if you must upgrade your cluster.

vSAN ESA contains a single storage pool of flash devices. Each flash device provides caching and capacity to the cluster.

Replace a Caching Device on a Host in vSAN Cluster

You must replace a flash caching device if you detect a failure or when there is a disk group upgrade.

- Verify that the storage controllers on the hosts are configured in passthrough mode and support the hot-plug feature. If the storage controllers are configured in RAID 0 mode, see the vendor documentation for information about adding and removing devices.
- If you upgrade the caching device, verify the following requirements:
  - If you upgrade the flash caching device, verify that the cluster contains enough space to migrate the data from the disk group that is associated with the flash device.
  - Place the host in maintenance mode. See Place a Member of vSAN Cluster in Maintenance Mode.

Removing the cache device removes the entire disk group from the vSAN cluster. When you replace a flash caching device, the virtual machines on the disk group become inaccessible and the components on the group are marked as degraded. See A Caching Device Is Not Accessible in a vSAN Cluster.

1. In the vSphere Client, navigate to the cluster.
2. On the **Configure** tab, click **Disk Management** under vSAN.
3. Select the entire disk group that contains the flash caching device that you want to remove. vSAN does not allow you to remove the cache disk. To remove the cache disk, you must remove the entire disk group.
4. Click

   and click **Remove**.
5. In the Remove Disk Group dialog box, select any of the following data migration mode to evacuate the data on the disks.
   - **Full data migration** - Transfers all the data available on the host to other ESX hosts in the cluster.
   - **Ensure accessibility** - Transfers data available on the host to the other ESX hosts in the cluster partially. During the data transfer, all virtual machines on the host remains accessible.
   - **No data migration** - There is no data transfer from the host. At this time, some objects might become inaccessible.
6. Click **Go To Pre-Check** to find the impact on the cluster if the object is removed or placed in maintenance mode.
7. Click **Remove** to remove the disk group.

vSAN removes the flash caching device along with the entire disk group from the cluster.

1. Add a new device to the host.
   The host automatically detects the device.
2. If the host is unable to detect the device, perform a device rescan.

For more information on creating a disk group, claiming storage devices, or adding devices to the disk group in the vSAN Cluster, see Device Management in a vSAN Cluster.

Replace a Storage Pool Device in vSAN ESA Cluster

The storage pool represents the amount of capacity provided by the host to the vSAN datastore.

If you upgrade the storage pool device, verify that the cluster contains enough space to migrate the data from the storage pool device.

Each host's storage devices claimed by vSAN form a storage pool. All storage devices claimed by vSAN contribute to capacity and performance.

1. In the vSphere Client, navigate to the cluster.
2. On the **Configure** tab, click **Disk Management** under vSAN.
3. Select the storage pool device, and click **Remove Disk**.
4. In the Remove Disk dialog box, select **Full data migration** to transfer all the data available on the host to other ESX hosts in the cluster.
5. Click **Go To Pre-Check** to find the impact on the cluster if the object is removed or placed in maintenance mode.
6. Click **Remove** to remove the storage pool device.

1. Add a new device to the host.
   The host automatically detects the device.
2. If the host is unable to detect the device, perform a device rescan.
3. Claim a disk using the vSAN cluster > **Configure** > **vSAN** > **Disk Management**.

Replace a Capacity Device in vSAN OSA Cluster
You must replace a flash capacity device or a magnetic disk if you detect a failure or when you upgrade it.

- Verify that the storage controllers on the hosts are configured in passthrough mode and support the host-plug feature. If the storage controllers are configured in RAID 0 mode, see the vendor documentation for information about adding and removing devices.
- If you upgrade the capacity device, verify that the cluster contains enough space to migrate the data from the capacity device.

Before you physically remove the device from the host, you must manually delete the device from vSAN. When you unplug a capacity device without removing it from the vSAN cluster, the components on the disk are marked as absent. If the capacity device fails, the components on the disk are marked as degraded. When the number of failures of the object replica with the affected components exceeds the FTT value, the virtual machines on the disk become inaccessible. See Capacity Device Not Accessible in vSAN Cluster.

> **Note:**
> If your vSAN cluster uses deduplication and compression, you must remove the entire disk group from the cluster before you replace the device.

You can also watch the video about how to replace a failed capacity device in vSAN.

1. In the vSphere Client, navigate to the cluster.
2. On the **Configure** tab, click **Disk Management** under vSAN.
3. Select the flash capacity device or magnetic disk, and click **Remove Disk**.

> **Note:**
>
> You cannot remove a capacity device from the cluster with enabled deduplication and compression. You must remove the entire disk group. If you want to remove a disk group from a vSAN cluster with deduplication and compression enabled, see Add or Remove Disks with Deduplication and Compression Enabled.

4. In the Remove Disk dialog box, select **Full data migration** to transfer all the data available on the host to other ESX hosts in the cluster.
5. Click **Go To Pre-Check** to find the impact on the cluster if the object is removed or placed in maintenance mode.
6. Click **Remove** to remove the capacity device.

   You can use ESXCLI commands to remove a device from a host. For more information, see Remove a Device from a Host in vSAN Cluster by Using an ESXCLI Command. To troubleshoot, identify, and replace a failed disk, see Troubleshooting vSAN OSA disk issues and Identifying and replacing a failed disk.

1. Add a new device to the host.
   The host automatically detects the device.
2. If the host is unable to detect the device, perform a device rescan.

Replace a Storage Controller in vSAN OSA Cluster
You must replace a storage controller on a host if you detect a failure.

1. Place the host into maintenance mode and power down the host.
2. Replace the failed card.

   The replacement storage controller must have a supported firmware level listed in the *Broadcom Compatibility Guide* at https://compatibilityguide.broadcom.com/.
3. Power on the host.
4. Configure the card for passthrough mode. Refer to the vendor documentation for information about configuring the device.
5. Exit maintenance mode.

Remove a Device from a Host in vSAN Cluster by Using an ESXCLI Command
If you detect a failed storage device, a failed disk in a storage pool, or if you upgrade a device, you can manually remove it from a host by using an ESXCLI command.

Verify that the storage controllers on the hosts are configured in passthrough mode and support the host-plug feature.

If the storage controllers are configured in RAIA 0 mode, see the vendor documentation for information about adding and removing devices.

If you remove a flash caching device, vSAN deletes the disk group that is associated with the flash device and all its member devices. If you remove a physical disk from the storage pool, vSAN redistributes the data stored on that disk to the remaining disks within the pool.

1. Open an SSH connection to the ESX host.
2. Peform one of the following:
   - To identify the device ID of the failed device, run this command and learn the device ID from the output.

     ```
     esxcli vsan storage list
     ```
   - To identify the storage pool configuration, run this command.

     ```
     esxcli vsan storagepool list
     ```
3. Perform one of the following:
   - To remove the device from vSAN, run this command.

     ```
     esxcli vsan storage remove -d device_id
     ```
   - To remove the disk from vSAN storage pool, run this command.

     ```
     esxcli vsan storagepool remove --disk
     ```

   The following are the commands available for managing vSAN ESA cluster:

Table 1027: vSAN ESA Commands

Command	Description
`esxcli vsan storagepool add`	Add physical disk for vSAN usage.
`esxcli vsan storagepool list`	List vSAN storage pool configuration.
`esxcli vsan storagepool mount`	Mount vSAN disk from storage pool.
`esxcli vsan storagepool rebuild`	Rebuild vSAN storage pool disks.
`esxcli vsan storagepool remove`	Remove physical disk from storage pool. Requires one --disk or --uuid param.
`esxcli vsan storagepool unmount`	Unmount vSAN disk from storage pool.

1. Add a new device to the host.
   The host automatically detects the device.
2. If the host is unable to detect the device, perform a device rescan.